

MATH 132B: Convex Optimization.

October 2, 2026

Contents

0	How to scribe and proofread / <i>Bohan Zhou</i>	4
1	Lecture 1 on Definition of Convexity / <i>Bohan Zhou</i>	5
1.1	Lecture Notes	5
1.2	Exercises	6
1.3	More Examples	9
2	Lecture 2 on Minimizers of Convex functions / <i>Agustin Fitipaldi and Alexandra Aswani</i>	12
2.1	Lecture Notes	12
2.2	Exercises	13
2.3	More Examples	17
3	Lecture 3 on Gradient Descent Method / <i>Hannah Nguyen and Indhu Chandra</i>	19
3.1	Lecture Notes	19
3.1.1	Existence of minimizers	19
3.1.2	Gradient Descent and Gradient Flow	20
3.2	Exercises	22
3.3	More Examples	24
4	Lecture 4 on Convergence Analysis on Gradient Descent Method / <i>Pablo Gómez and Ritaja Subrahmanya</i>	27
4.1	Lecture Notes	27
4.1.1	Convergence Analysis of Gradient Flows	27
4.1.2	Convergence Analysis of constant stepsize Gradient Descent	27
4.1.3	Descent Properties By L -smoothness	29
4.2	Exercises	30
4.3	More Examples	30
5	Lecture 5 on Framework of Convergence Analysis and L-smoothness / <i>Sahand Ahmadi and Frank Reyes Felsner</i>	33
5.1	Lecture Notes	33
5.1.1	Convergence rate of fixed stepsize GD of convex and L -smooth function	33
5.1.2	L -Smoothness, Equivalent Characterizations, and Basic Consequences	34
5.2	Exercises	36
5.3	More Examples	37
6	Lecture 6 on L-smoothness and m-strongly convexity / <i>Charles Yang and Jacob Jurek</i>	41
6.1	Lecture Notes	41
6.1.1	First-Order Form	41
6.1.2	Second-Order Form	41
6.1.3	Zeroth-Order Form	41
6.1.4	L -smoothness on a geodesic space.	42
6.1.5	m -Strong Convexity and Duality with L -Smoothness	42
6.1.6	Equivalence of the 1.5th-Order and 1st-Order Definitions	43
6.1.7	Convergence Rate of Gradient Flow	44
6.2	Exercises	46
6.3	More Examples	47
7	Lecture Notes 7 on Gradient Flow, EVI and Gradient Descent / <i>Tao Xu, Andrew Wang</i>	51
7.1	Lecture Notes	51
7.2	Gradient Flow under Convexity	51

7.3	Gradient Flow under m -Strong Convexity	52
7.3.1	Function Value Decay for Gradient Flow	53
7.3.2	Useful Inequalities	54
7.3.3	General EVI and Moreau Envelope	54
7.3.4	Gradient Descent and the Discrete EVI	55
7.3.5	Using the Discrete EVI to Prove Convergence	56
7.4	Exercises	57
7.5	More Examples	57
8	Lecture Notes 8 on Stochastic Gradient Descent / <i>Peter Le and Ciara Nic Canna</i>	59
8.1	Lecture Notes	59
8.1.1	Stochastic Motivations	59
8.1.2	Stochastic Assumptions	60
8.1.3	Decent Lemmas	61
8.1.4	Corollary	62
8.2	Exercises	62
8.3	More Examples	64
9	Lecture Notes 9 on Subgradient / <i>Eric Wong, Don Tran</i>	67
9.1	Lecture Notes	67
9.2	Exercises	69
9.3	More Examples	69
10	Lecture Notes 10 on Proximal Operator / <i>Ben Thomas and Run Li</i>	74
10.1	Lecture Notes	74
10.1.1	From Regularization to the Proximal Operator	74
10.1.2	Proximal Operator	77
10.1.3	Proximal Update and Gradient Flow	77
10.2	Exercises	78
10.3	More Examples	78
11	Lecture Notes 11 on Proximal Operator and Moreau Envelope / <i>Jasper Luo, Cade Camacho</i>	81
11.1	Lecture Notes	81
11.2	Exercises	86
11.3	More Examples	87
12	Lecture Notes 12 on Proximal Gradient Method / <i>Yuxuan Zhao, Ben T.</i>	89
12.1	Lecture Notes	89
12.1.1	From Projected Gradient Descent to Proximal Gradient Method	89
12.1.2	Two Perspectives of Proximal Gradient Methods	90
12.1.3	Convergence Rate	91
12.2	More Examples	93
13	Lecture Notes 13 on Inexact Proximal Gradient Method / <i>Chenghan Li, Yuxuan Zhao.</i>	97
13.1	Lecture Notes	97
13.1.1	Two Sources of Inexactness	97
13.1.2	Approximate Optimality Condition for Inexact Proximal Updates	98
13.1.3	Connection with the Moreau Envelope	99
13.1.4	Convergence Rate of the Basic Inexact Proximal-Gradient Method	99
13.2	Exercises	100
13.3	More Examples	102
14	Lecture Notes 14 on Duality and Lagrangian / <i>Prashast Singh, Justin Zhou</i>	105
14.1	Lecture Notes	105
14.1.1	Primal-Dual in linear programming problem	105
14.1.2	General constrained optimization problem	106
14.1.3	Strong duality and duality gap	108
14.1.4	Main gain from duality	109
14.2	Exercises	110
14.3	More Examples	114
15	Lecture Notes 15 on Duality in the Space of Probability measures / <i>Ben Pearson, Bradley Nakagawa</i>	119
15.1	Lecture Notes	119
15.1.1	Second View of Lagrangian Duality	119

15.1.2 The Optimal Transport Problem	119
15.1.3 Dual Problem	120
15.2 Exercises	121
15.3 More Examples	122

0 How to scribe and proofread

Writing in $\text{\LaTeX}$ is more than an exercise in using $\text{\LaTeX}$! The experience of scribing and proofreading lecture notes is one of those takeaways of our classes.

Basic Rules

When you have finished typing (either you are a scribe or a proofreader, or provide solutions or more examples), **your code must compile without errors**. It is a good practice to do a compile and check every paragraph.

If you use some self-defined commands, add it to corresponding parts in `preamble_lecture_note.sty`. Please only use customized commands for those in high frequency. It is the best practice to mimic the codes where they look similarly to what you want.

Frequently Used Notations

$\iota_A : \Omega \mapsto \{0, 1\}$ (<code>\iota_A</code>) $\chi_A : \Omega \mapsto \{0, +\infty\}$ $\mathbb{1}_N$ (<code>\Id</code>) <code>id</code> (<code>\id</code>) $ \cdot , \ \cdot\ $ (<code>\abs{\cdot}</code> , <code>\norm{\cdot}</code>) $\langle x, y \rangle, \langle x, y \rangle$ (<code>\scal{x}{y}</code> or <code>\langle x, y \rangle</code>)	the indicator function. the characteristic function. a vector/matrix of ones. an identity operator. absolute value, norm. inner-product or dual pair.
--	--

Typical Examples for environments

Display math with label Multi-line math Multi-figure Minipage for formulas, figures etc.. List	(1.1) the proof to Theorem 1.4, Definition 1.7, Figure 1 Theorem 3.1 Theorem 3.1
--	--

1 Lecture 1 on Definition of Convexity

sec:lec1

Bohan summarizes: We start from the variant versions of definitions of convexity (of sets and functions), to compare their pros and cons, to understand their characterizations and key properties. One of main techniques is to “restrict” on a line by $(1-t)x + ty = x + t(y-x)$.

1.1 Lecture Notes

Definition 1.1 (Convex (Affine) combination). We say weights $\{a_i\}_{i=1}^m \subseteq \mathbb{R}$ are affine combination if $\sum_{i=1}^m a_i = 1$; they are convex combination if additionally $a_i \geq 0$ for every i .

Definition 1.2 (Convex Set). We say a set $C \subseteq \mathbb{R}^d$ is convex, if $x, y \in C$ implies that $(1-t)x + ty \in C$ for all $0 \leq t \leq 1$.

Example 1.1.1. Examples of convex sets: empty set, norm ball, hyperplane, halfspace, affine space.



Figure 1: Convex set. If you use some screenshot, try to use some high-resolution ones or use AI (like studio.creativefabrica.com) to improve them.

Example 1.1.2. There are some properties of convex set, such as separating hyperplane theorem, supporting hyperplane theorem, which are good to know (c.f. [BV04, Section 2.5]).

Definition 1.3 (0-order of convexity). We say a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if $\text{dom}(f) \subseteq \mathbb{R}^d$ is convex, and $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for all $t \in [0, 1]$ and $x, y \in \text{dom}(f)$. f is strictly convex if replacing $\leq$ by $<$ in (1.1). eq:0-convex (1.1)

Remark. This definition requires the minimal structure of the source space, i.e., replacing $\mathbb{R}^d$ by any space $(X, +)$. The Field Medalist Villani called this as “synthetic” points of view in [Vil09]. It is *more general, more stable* while *harder to check* due to being “global”.

Theorem 1.4 (Alternative characterization of 0-order definition). A function is convex if and only if its restriction to any line is convex.

A function is convex if and only if its epigraph $\text{epi}(f) = \{(x, a) \in \text{dom}(f) \times \mathbb{R} \mid f(x) \leq a\}$ is convex.

Proof. First, we parametrize any line by $\{x_0 + tv \mid t \in \mathbb{R}\}$ for any point x_0 and vector v , and want to show that $f(x)$ is convex if and only if $g(t) = f(x_0 + tv)$ is convex for any x_0, v .

“ $\Rightarrow$ ” is trivial. Now we show “ $\Leftarrow$ ”. To show that $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ holds for all $t \in [0, 1]$ and $x, y \in \text{dom}(f)$, note that $(1-t)x + ty = x + t(y-x)$, we just take $x_0 = x$ and $v = y-x$ in the definition of $g(t)$ being convex.

Second, assume f is convex, we want to show that

$$\text{if } (x_1, a_1), (x_2, a_2) \in \text{epi}(f), \text{ then } ((1-t)x_1 + tx_2, (1-t)a_1 + ta_2) \in \text{epi}(f).$$

By 0-order definition,

$$\begin{cases} f(x_1) \leq a_1 \\ f(x_2) \leq a_2 \end{cases} \implies f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2) \leq (1-t)a_1 + ta_2.$$

The reverse can be drawn from the same inequality above. ■

Definition 1.5 (1-order of convexity). If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable, then f is convex if $\text{dom}(f) \subseteq \mathbb{R}^d$ is convex, and $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in \text{dom}(f)$. eq:1-convex (1.2)

Remark. From the definition, x, y are exchangeable in positions, i.e., $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$ holds as well. Both equations state the same fact that the convex function lies above its tangent line at every point. Apparently, if $\nabla f(\bar{x}) = 0$ at some $\bar{x} \in \text{dom}(f)$, then $\bar{x}$ is the global minimizer. (We will discuss the other side.)

Definition 1.6 (2-order of convexity). If f is twice differentiable, then f is convex if $\text{dom}(f) \subseteq \mathbb{R}^d$ is convex and $\nabla^2 f(x) \succeq 0$ for any $x \in \text{dom}(f)$, i.e., $\nabla^2 f(x)$ is positive semi-definite.

Definition 1.7 ($1\frac{1}{2}$ -order of convexity). If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable, then f is convex if and only if $\nabla f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a monotone mapping, i.e., $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$ for any $x, y \in \text{dom}(f)$.

Proof. On one side, from 1-order of convexity

$$\begin{aligned} f(y) &\geq f(x) + \langle \nabla f(x), y - x \rangle \\ f(x) &\geq f(y) + \langle \nabla f(y), x - y \rangle, \end{aligned}$$

we sum them up and find that

$$0 \geq \langle \nabla f(x) - \nabla f(y), y - x \rangle.$$

On the other side, for fixed x, y , we define a curve $x + t(y - x)$, and define $g(t) = f(x + t(y - x))$, the chain rule yields that

$$g'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle,$$

and note that $g(0) = f(x)$, $g(1) = f(y)$, and $g'(t) \geq g'(0)$. Thus

$$g(1) = g(0) + \int_0^1 g'(t) dt \geq g(0) + \int_0^1 g'(0) dt = g(0) + g'(0),$$

which yields that $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$. ■

1.2 Exercises

Exercise. Show that halfspace $H = \{x \mid \langle x, a \rangle \leq b\}$ is a convex set.

Proof. *Drafted by Pablo, Proofread by Ritaja*

Let $x, y \in H$ and let $t \in [0, 1]$. By the linearity of the inner product, it follows that

$$\begin{aligned} \langle (1-t)x + ty, a \rangle &= (1-t)\langle x, a \rangle + t\langle y, a \rangle \\ &\leq (1-t)b + tb \\ &= b, \end{aligned}$$

as desired. ■

Exercise. Show that the set of positive semidefinite matrices is convex.

Proof. *Drafted by Ritaja, Proofread by Pablo*

Let $A, B \in \mathbb{R}^{d \times d}$ be positive semidefinite matrices and $t \in [0, 1]$. We first show that the convex sum is a symmetric matrix. By basic matrix properties,

$$\begin{aligned} (\lambda A + (1-\lambda)B)^T &= (\lambda A)^T + ((1-\lambda)B)^T \\ &= \lambda A^T + (1-\lambda)B^T \\ &= \lambda A + (1-\lambda)B. \end{aligned}$$

We now show that the convex sum satisfies $x^T M x \geq 0$ for all vectors $x \in \mathbb{R}^d$. By linearity,

$$x^T (\lambda A + (1-\lambda)B) x = \lambda x^T A x + (1-\lambda) x^T B x.$$

Since A, B are positive semi-definite we have $x^T A x \geq 0$, $x^T B x \geq 0$ for all $x \in \mathbb{R}^d$. Furthermore, $1 - \lambda, \lambda \geq 0$ by assumption. It follows that

$$\lambda x^T A x + (1-\lambda) x^T B x \geq 0.$$

Thus, the set of positive semi-definite matrices is convex. ■

Exercise. Show that for any set C (convex or not), its support function $I_C^*(x)$ is convex; if C is a convex set, then its indicator function $I_C(x)$ is convex.

$$I_C(x) = \begin{cases} 0 & x \in C; \\ +\infty & x \notin C. \end{cases} \quad \text{and} \quad I_C^*(x) = \sup_{y \in C} \langle x, y \rangle.$$

Proof. *Drafted by Sahand Ahmadi, Proofread by Frank Reyes Felsner*

Assume first that $C \neq \emptyset$. We will show that I_C^* is convex on $\mathbb{R}^d$.

Let $x_1, x_2 \in \mathbb{R}^d$ and let $\lambda \in [0, 1]$. Then

$$\begin{aligned} I_C^*(\lambda x_1 + (1 - \lambda)x_2) &= \sup_{y \in C} \langle \lambda x_1 + (1 - \lambda)x_2, y \rangle \\ &= \sup_{y \in C} \lambda \langle x_1, y \rangle + (1 - \lambda) \langle x_2, y \rangle. \end{aligned}$$

For every fixed $y \in C$, we have

$$\lambda \langle x_1, y \rangle + (1 - \lambda) \langle x_2, y \rangle \leq \lambda \sup_{z \in C} \langle x_1, z \rangle + (1 - \lambda) \sup_{z \in C} \langle x_2, z \rangle.$$

Taking the supremum over $y \in C$ on the left-hand side gives

$$I_C^*(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda I_C^*(x_1) + (1 - \lambda) I_C^*(x_2).$$

Hence I_C^* is convex.

(*Remark:* the above proof is not limited to a convex set C , since we do the convex combination in $\mathbb{R}^d$ only.)

Now assume that C is convex. We show that its indicator function

$$I_C(x) = \begin{cases} 0, & x \in C; \\ +\infty, & x \notin C \end{cases}$$

is convex.

Let $x_1, x_2 \in \mathbb{R}^d$ and $\lambda \in [0, 1]$. We must prove

$$I_C(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda I_C(x_1) + (1 - \lambda) I_C(x_2).$$

There are two cases.

Case 1: $x_1, x_2 \in C$. Since C is convex, $\lambda x_1 + (1 - \lambda)x_2 \in C$. Therefore

$$I_C(\lambda x_1 + (1 - \lambda)x_2) = 0,$$

while

$$\lambda I_C(x_1) + (1 - \lambda) I_C(x_2) = \lambda \cdot 0 + (1 - \lambda) \cdot 0 = 0.$$

So the desired inequality holds.

Case 2: At least one of x_1, x_2 is not in C .

Then at least one of $I_C(x_1), I_C(x_2)$ equals $+\infty$, and hence

$$\lambda I_C(x_1) + (1 - \lambda) I_C(x_2) = +\infty$$

(with the usual extended-real convention). Since $I_C(\lambda x_1 + (1 - \lambda)x_2) \leq +\infty$ always, the inequality again holds. Thus I_C is convex whenever C is convex. ■

Exercise. Given any vector $y \in \mathbb{R}^m$ and any matrix $A \in \mathbb{R}^{m \times n}$, show that $x \mapsto \|y - Ax\|_2^2$ is convex. We say L^2 -loss is a convex function(al).

Proof. *Drafted by Frank Reyes Felsner, Proofread by Hannah Nguyen*

Let us define the function f by

$$f(x) = \|y - Ax\|_2^2.$$

To prove that f is convex, let $x, z \in \mathbb{R}^n$ and $t \in [0, 1]$. Then

$$f(tx + (1 - t)z) = \|y - A(tx + (1 - t)z)\|_2^2 = \|t(y - Ax) + (1 - t)(y - Az)\|_2^2,$$

where we use the linearity of A . Next, define $u = y - Ax$ and $v = y - Az$. Then

$$f(tx + (1 - t)z) = \|tu + (1 - t)v\|_2^2 = (tu + (1 - t)v)^T (tu + (1 - t)v).$$

After expanding this expression and using the symmetry of the dot product, $u^T v = v^T u$, we obtain

$$(tu + (1 - t)v)^T (tu + (1 - t)v) = t^2 \|u\|_2^2 + (1 - t)^2 \|v\|_2^2 + 2t(1 - t) \langle u, v \rangle.$$

Our goal is to show that

$$f(tx + (1 - t)z) \leq tf(x) + (1 - t)f(z).$$

Since $f(x) = \|y - Ax\|_2^2$, and since $u = y - Ax$ and $v = y - Az$, we have

$$f(x) = \|u\|_2^2 \quad \text{and} \quad f(z) = \|v\|_2^2,$$

so

$$tf(x) + (1 - t)f(z) = t\|u\|_2^2 + (1 - t)\|v\|_2^2.$$

Now we must show that

$$t^2 \|u\|_2^2 + (1 - t)^2 \|v\|_2^2 + 2t(1 - t) \langle u, v \rangle \leq t\|u\|_2^2 + (1 - t)\|v\|_2^2.$$

Now, we group like terms and factor terms

$$\begin{aligned} (t^2 \|u\|_2^2 - t\|u\|_2^2) + ((1 - t)^2 \|v\|_2^2 - (1 - t)\|v\|_2^2) + 2t(1 - t) \langle u, v \rangle &= t(t - 1)\|u\|_2^2 - t(1 - t)\|v\|_2^2 + 2t(1 - t) \langle u, v \rangle \\ &= -t(1 - t)(\|u\|_2^2 + \|v\|_2^2 - 2 \langle u, v \rangle) \\ &= -t(1 - t)\|u - v\|_2^2 \leq 0. \end{aligned}$$

Thus, we have shown that

$$f(tx + (1 - t)z) \leq tf(x) + (1 - t)f(z).$$

Therefore, f is convex, and so

$$\|y - Ax\|_2^2 \text{ is convex in } x.$$

■

Exercise. If f is differentiable, show that 0-order definition of convexity is equivalent to 1-order definition of convexity.

Proof. *Drafted by Jacob Jurek, Proofread by Jasper Luo*

We prove both directions.

($\Rightarrow$) Assume f satisfies the 0-order definition of convexity. Fix $x, y \in C$ and let $t \in (0, 1]$. Since C is convex,

$$x + t(y - x) = (1 - t)x + ty \in C.$$

By the 0-order definition of convexity,

$$f(x + t(y - x)) = f((1 - t)x + ty) \leq (1 - t)f(x) + tf(y).$$

Subtracting $f(x)$ from both sides gives

$$f(x + t(y - x)) - f(x) \leq t(f(y) - f(x)).$$

Since $t > 0$, dividing by t yields

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

Taking the limit as $t \rightarrow 0^+$ and using differentiability of f at x , we get

$$\langle \nabla f(x), y - x \rangle \leq f(y) - f(x).$$

Therefore,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle,$$

which is the 1-order definition of convexity.

($\Leftarrow$) Assume f satisfies the 1-order definition of convexity. Let $x, y \in C$ and $t \in [0, 1]$. The cases $t = 0$ and $t = 1$ are trivial, so assume $t \in (0, 1)$.

Define

$$z = (1 - t)x + ty.$$

Since C is convex, $z \in C$. Applying the 1-order definition at the point z , first with x and then with y , gives

$$f(x) \geq f(z) + \langle \nabla f(z), x - z \rangle$$

$$f(y) \geq f(z) + \langle \nabla f(z), y - z \rangle.$$

Multiplying the first inequality by $(1 - t)$ and the second by t , then adding, we obtain

$$(1 - t)f(x) + tf(y) \geq f(z) + \langle \nabla f(z), (1 - t)(x - z) + t(y - z) \rangle.$$

Now,

$$(1 - t)(x - z) + t(y - z) = (1 - t)x + ty - z = z - z = 0.$$

Hence,

$$(1 - t)f(x) + tf(y) \geq f(z).$$

Since $z = (1 - t)x + ty$, this becomes

$$f((1 - t)x + ty) \leq (1 - t)f(x) + tf(y),$$

which is the 0-order definition of convexity.

Therefore, the 0-order and 1-order definitions of convexity are equivalent for differentiable functions. ■

Exercise. If f is twice differentiable, show that the 1-order definition of convexity is equivalent to 2-order definition of convexity.

Proof. *Proof by Alexandra Aswani, Proofread by Agustin Fitipaldi*

($\Rightarrow$) Assume f satisfies the 1-order definition of convexity. We show that

$$\nabla^2 f(x) \succeq 0$$

for every $x \in C$. Fix $x \in C$ and let $v \in \mathbb{R}^d$. Since f is twice differentiable then we know Taylor's theorem gives

$$f(x + tv) = f(x) + t\langle \nabla f(x), v \rangle + \frac{t^2}{2}v^T \nabla^2 f(x)v + o(t^2).$$

By the 1-order definition of convexity, we know $f(x + tv) \geq f(x) + \langle \nabla f(x), tv \rangle$. And substituting this with the Taylor expansion:

$$f(x) + t\langle \nabla f(x), v \rangle + \frac{t^2}{2}v^T \nabla^2 f(x)v + o(t^2) \geq f(x) + t\langle \nabla f(x), v \rangle.$$

Simplifying gives:

$$\frac{t^2}{2}v^T \nabla^2 f(x)v + o(t^2) \geq 0.$$

Since $t^2 > 0$ for $t \neq 0$, dividing by t^2 gives

$$\frac{1}{2}v^T \nabla^2 f(x)v + \frac{o(t^2)}{t^2} \geq 0.$$

Letting $t \rightarrow 0$, the remainder term vanishes and we get,

$$v^T \nabla^2 f(x)v \geq 0$$

for all $v \in \mathbb{R}^d$, which implies the 2-order definition of convexity.

($\Leftarrow$) Now let's assume f satisfies the 2-order definition of convexity, so $\nabla^2 f(x) \succeq 0$ for all $x \in C$. Let $x, y \in C$ and also define

$$g(t) = f(x + t(y - x)), \quad t \in [0, 1].$$

We also know by the chain rule,

$$g'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle,$$

and

$$g''(t) = (y - x)^T \nabla^2 f(x + t(y - x))(y - x).$$

Since $\nabla^2 f(x + t(y - x))$ is positive semi-definite, then

$$g''(t) \geq 0.$$

Thus g is convex on $[0, 1]$. Therefore,

$$g(1) \geq g(0) + g'(0).$$

Hence this proves f satisfies the 1-order definition of convexity. ■

Exercise. Show that if f is convex, then its sublevel sets $\{x \in \text{dom}(f) \mid f(x) \leq t\}$ are convex for any fixed $t \in \mathbb{R}$. Provide a counterexample that the converse is not true.

Proof. *Proof by Tao Xu, Proofread by Jacob Jurek*

Assume that f is convex. Fix $t \in \mathbb{R}$ and define the sublevel set

$$S_t = \{x \in \text{dom}(f) \mid f(x) \leq t\}$$

We show that S_t is convex.

Let $x, y \in S_t$ and let $\lambda \in [0, 1]$. Since $x, y \in S_t$, we have

$$f(x) \leq t \quad \text{and} \quad f(y) \leq t.$$

Since f is convex, $\text{dom}(f)$ is convex. Therefore $(1 - \lambda)x + \lambda y \in \text{dom}(f)$. By the 0-order definition of convexity,

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)t + \lambda t = t$$

Therefore,

$$(1 - \lambda)x + \lambda y \in S_t$$

Since $x, y \in S_t$ and $\lambda \in [0, 1]$ are arbitrary, S_t is convex.

Now we provide a counterexample which shows that the converse is not true. Consider

$$f(x) = x^3, \quad x \in \mathbb{R}.$$

For any fixed $t \in \mathbb{R}$, the sublevel set is

$$S_t = \{x \in \mathbb{R} \mid x^3 \leq t\}.$$

Since the function $x \mapsto x^3$ is increasing on $\mathbb{R}$, the inequality $x^3 \leq t$ is equivalent to $x \leq \sqrt[3]{t}$. Therefore,

$$S_t = (-\infty, \sqrt[3]{t}]$$

which is a convex subset of $\mathbb{R}$ for every $t \in \mathbb{R}$.

However, $f(x) = x^3$ is not convex on $\mathbb{R}$. Indeed, $f''(x) = 6x$. For $x < 0$, we have $f''(x) < 0$. Thus f is not convex on $\mathbb{R}$.

Therefore, a function may have convex sublevel sets for every $t \in \mathbb{R}$ without being convex. Hence the converse is not true. ■

1.3 More Examples

Bohan suggests to add more visualization examples about normal cones and supporting functions especially when the set C is nonconvex. Tao Xu constructs the following visualizations and proofreading is welcomed!

Example 1.3.1 (Support function of a nonconvex set in $\mathbb{R}$). Let

$$C = \{-1, 1\} \subseteq \mathbb{R}.$$

Its support function is

$$I_C^*(x) = \sup_{y \in C} \langle x, y \rangle = \max\{-x, x\} = |x|.$$

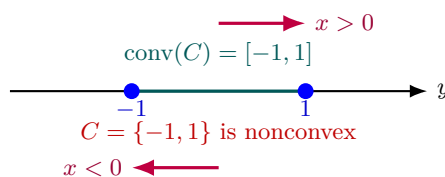


Figure 2: The nonconvex set $C = \{-1, 1\}$ and its convex hull. The support function is $I_C^*(x) = |x|$.

Explanation. This example shows that a support function can be convex even when the set itself is not convex. If

$x > 0$, the point $y = 1$ gives the largest value. If $x < 0$, the point $y = -1$ gives the largest value. Hence

$$I_C^*(x) = |x|$$

The missing interval between -1 and 1 does not affect the support function because the max value is still attained at the endpoints.

Example 1.3.2 (Support function of a nonconvex set in $\mathbb{R}^2$). Let

$$C = \{(-1, 0), (1, 0), (0, 1)\} \subseteq \mathbb{R}^2.$$

Its support function is

$$I_C^*(x) = \sup_{y \in C} \langle x, y \rangle, \quad x \in \mathbb{R}^2.$$

If $x = (x_1, x_2)$, then

$$I_C^*(x) = \max\{-x_1, x_1, x_2\}.$$

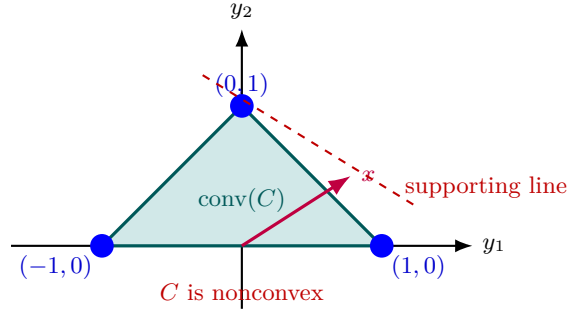


Figure 3: A nonconvex set C in $\mathbb{R}^2$ and its convex hull.

Explanation. The set C have only three points, So it's nonconvex. For each direction $x = (x_1, x_2)$, the support function compare the three values

$$-x_1, \quad x_1, \quad x_2$$

Thus

$$I_C^*(x) = \max\{-x_1, x_1, x_2\}$$

Example 1.3.3 (Normal cone of an interval in $\mathbb{R}$). Let

$$C = [-1, 1] \subseteq \mathbb{R}.$$

For $\bar{x} \in C$, the normal cone is

$$N_C(\bar{x}) = \{g \in \mathbb{R} \mid \langle g, y - \bar{x} \rangle \leq 0, \forall y \in C\}.$$

Thus,

$$N_C(\bar{x}) = \begin{cases} (-\infty, 0], & \bar{x} = -1, \\ \{0\}, & \bar{x} \in (-1, 1), \\ [0, \infty), & \bar{x} = 1. \end{cases}$$

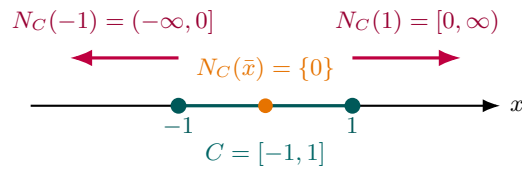


Figure 4: Normal cones of the interval $C = [-1, 1]$.

Explanation. At an interior point of $[-1, 1]$, we can move both left and right inside the set. So the only vector satisfy the normal cone inequality is $g = 0$. At the endpoints, the normal cone points outward. Therefore,

$$N_C(-1) = (-\infty, 0], \quad N_C(1) = [0, \infty]$$

This matches the intuition that boundary points have outward normal directions, while the interior points don't.

Example 1.3.4 (Normal cone of a square in $\mathbb{R}^2$). Let

$$C = [-1, 1]^2 \subseteq \mathbb{R}^2.$$

For a point $\bar{x} \in C$, the normal cone is

$$N_C(\bar{x}) = \{g \in \mathbb{R}^2 \mid \langle g, y - \bar{x} \rangle \leq 0, \forall y \in C\}.$$

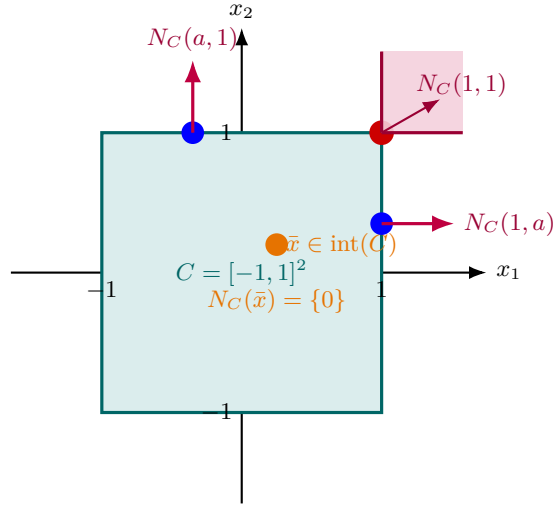


Figure 5: Normal cones of the square $C = [-1, 1]^2$.

For example,

$$N_C(1, 1) = \{g = (g_1, g_2) \in \mathbb{R}^2 \mid g_1 \geq 0, g_2 \geq 0\}.$$

At a point on the right edge $\bar{x} = (1, a)$ with $-1 < a < 1$,

$$N_C(1, a) = \{(g_1, 0) \in \mathbb{R}^2 \mid g_1 \geq 0\}.$$

Explanation. This example shows that how the normal cone depends on the position of $\bar{x}$. at an interior point, there is no outward direction, so

$$N_C(\bar{x}) = \{0\}$$

On an edge, the normal cone is a ray pointing outward. At the corner $(1, 1)$, two constraints are active, so the normal cone becomes

$$N_C(1, 1) = \{(g_1, g_2) \in \mathbb{R}^2 \mid g_1, g_2 \geq 0\}$$

Thus, the normal cone becomes larger at corners than at edges.

2 Lecture 2 on Minimizers of Convex functions

Indhu Chandra summarizes: We explore theorems and properties of convex functions. We introduce minimizers of convex functions.

Scribe Team: Scribed by Agustin Fitipaldi, Proofread by Alexandra Aswani.

2.1 Lecture Notes

Objective

To finish stating some of the ways we can determine convexity of a function, and then to state some powerful theorems that this convexity gives us.

We'd like a way to check convexity even in the highest of dimensions, however, we run into rather complex visualizations if we go much farther than $\mathbb{R}^3$. Domain restriction can come to our rescue here.

If we observe that any two points in a space $\mathbb{R}^n$ live on some line that connects them, we can reasonably check the condition of convexity on any point in a function whose domain is this $\mathbb{R}^n$ by *restricting it to these lines and checking the 1-dimensional convexity condition instead*.

Theorem 2.1 (Convexity via line restrictions). A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if and only if for any $x, y \in \mathbb{R}^d$, the function $g(t) = f(x + t(y - x))$ is also convex as a function of $t \in [0, 1]$.

Proof. ($\implies$) We want to show that if f is convex, then such a function $g(t)$ will also be convex. We know that

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

from the definition of convexity. We want to show that

$$g(\theta t_1 + (1 - \theta)t_2) \leq \theta g(t_1) + (1 - \theta)g(t_2).$$

Let $u = x + t_1(y - x)$ and $v = x + t_2(y - x)$, so that $g(t_1) = f(u)$ and $g(t_2) = f(v)$. Then

$$\begin{aligned} g(\theta t_1 + (1 - \theta)t_2) &= f(x + (\theta t_1 + (1 - \theta)t_2)(y - x)) \\ &= f(\theta(x + t_1(y - x)) + (1 - \theta)(x + t_2(y - x))) \\ &= f(\theta u + (1 - \theta)v) \\ &\leq \theta f(u) + (1 - \theta)f(v) \\ &= \theta g(t_1) + (1 - \theta)g(t_2) \end{aligned}$$

where the inequality follows directly from the convexity of f .

($\impliedby$) We want to show that if every such restriction to a line is convex, then f itself is convex. Fix any $x, y \in \mathbb{R}^n$ and $t \in [0, 1]$. Define

$$g(t) = f(x + t(y - x)).$$

By hypothesis, g is a convex function of the scalar t . Note that $g(0) = f(x)$ and $g(1) = f(y)$. Applying the definition of (scalar) convexity to g at the points 0 and 1, we get

$$g(t) \leq (1 - t)g(0) + t g(1)$$

which is exactly

$$f(x + t(y - x)) \leq (1 - t)f(x) + t f(y).$$

Since x, y , and t were arbitrary, f is convex. ■

Remark. This “restrict to a line” trick is a common and powerful technique: it lets us reduce questions about high-dimensional convexity to scalar convexity, which is far easier to work with. The same idea extends to Riemannian manifolds—if $f : M \rightarrow \mathbb{R}$ is a function on a manifold, we say f is convex with respect to some interpolation curve $\gamma(t)$ (with $\gamma(0) = x$ and $\gamma(1) = y$) if and only if the scalar function $t \mapsto f(\gamma(t))$ is convex.

Theorem 2.2 (First-order convexity $\Leftrightarrow$ monotone gradient). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable. Then f is convex in the first-order sense, i.e.

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \forall x, y,$$

if and only if ∇f is monotone, i.e.

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0 \quad \forall x, y.$$

Proof. ($\implies$) We assume the first-order condition holds. Since it holds for all x and y , we can write it in both orderings:

$$\begin{cases} f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle, \\ f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle. \end{cases}$$

Adding these two inequalities, the $f(x)$ and $f(y)$ terms cancel from both sides, leaving

$$\begin{aligned} 0 &\geq \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle \\ &= \langle \nabla f(x) - \nabla f(y), y - x \rangle. \end{aligned}$$

Negating both sides (and flipping the direction of $y - x$), we get

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

which is exactly the monotonicity of ∇f .

($\Leftarrow$) We assume ∇f is monotone and want to recover the first-order condition. We again use the line restriction: fix $x, y \in \mathbb{R}^n$ and define

$$g(t) = f(x + t(y - x)).$$

By the chain rule,

$$g'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle.$$

In particular, $g'(0) = \langle \nabla f(x), y - x \rangle$. Now, we claim that $g'(t) \geq g'(0)$ for all $t \in [0, 1]$. To see this, apply the monotonicity condition with the points x and $x + t(y - x)$:

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), (x + t(y - x)) - x \rangle \geq 0.$$

The second argument simplifies to $t(y - x)$, so

$$t \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \geq 0.$$

For $t \in (0, 1]$ we can divide by $t > 0$ to get

$$\langle \nabla f(x + t(y - x)), y - x \rangle \geq \langle \nabla f(x), y - x \rangle$$

which is exactly $g'(t) \geq g'(0)$. Now we integrate:

$$\begin{aligned} f(y) = g(1) &= g(0) + \int_0^1 g'(t) dt \\ &\geq g(0) + \int_0^1 g'(0) dt \\ &= g(0) + g'(0) \\ &= f(x) + \langle \nabla f(x), y - x \rangle \end{aligned}$$

which is the first-order convexity condition. ■

Properties of Convex Functions

With these characterizations in hand, we can now state one of the most practically important consequences of convexity.

Theorem 2.3 (Local minimizers are global). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex. If $\bar{x}$ is a local minimizer of f , then $\bar{x}$ is a global minimizer.

This is why convexity matters for optimization: many algorithms only guarantee convergence to a local minimum. *Convexity upgrades that to a global one for free.*

Recall that $\bar{x}$ is a local minimizer if there exists a neighborhood $V(\bar{x})$ such that $f(\bar{x}) \leq f(y)$ for all $y \in V(\bar{x})$.

Proof. We proceed by contradiction. Suppose there exists some x^* with $f(x^*) < f(\bar{x})$. Then by convexity, for any $t \in (0, 1)$,

$$\begin{aligned} f((1-t)\bar{x} + tx^*) &\leq (1-t)f(\bar{x}) + tf(x^*) \\ &< (1-t)f(\bar{x}) + tf(\bar{x}) \\ &= f(\bar{x}) \end{aligned}$$

where the strict inequality uses $f(x^*) < f(\bar{x})$. So the point $(1-t)\bar{x} + tx^*$ achieves a strictly lower function value than $\bar{x}$. But this point can be made arbitrarily close to $\bar{x}$: its distance is

$$\|(1-t)\bar{x} + tx^* - \bar{x}\| = t\|x^* - \bar{x}\| \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

So for sufficiently small $t > 0$, the point $(1-t)\bar{x} + tx^*$ lies inside $V(\bar{x})$ and has f -value strictly less than $f(\bar{x})$, contradicting the assumption that $\bar{x}$ is a local minimizer. ■

Remark. This result generalizes beyond $\mathbb{R}^n$. If $f : X \rightarrow \mathbb{R}$ is convex on any normed vector space X , local minimizers are still global. It even extends to Riemannian manifolds (M, d) , provided (M, d) is a *unique geodesic space*, meaning that for any two points $x, y \in M$ there is a unique geodesic $\gamma(t)$ with $\gamma(0) = x$, $\gamma(1) = y$, and

$$\frac{d(\gamma(t), \gamma(0))}{d(\gamma(1), \gamma(0))} = t$$

(i.e., γ has constant speed). If f is convex along geodesics in the sense that $t \mapsto f(\gamma(t))$ is a scalar convex function, then local minimizers of f on M are global as well.

2.2 Exercises

Example 2.2.1 (Proposed by Tao Xu, maximum of a family of affine functions is convex.). Consider

$$f(x) = \max_{i=1, \dots, m} \{a_i^\top x + b_i\},$$

where $a_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$. Show that f is convex. then explain why every local minimizer of f over a convex set is a global minimizer.

Proof. *Drafted by Justin Zhou, Proofread by Pablo Gómez*

Let $x, y \in \mathbb{R}^n$ and $t \in [0, 1]$. Then

$$\begin{aligned} f((1-t)x + ty) &= \max_{i=1, \dots, m} \{a_i^\top ((1-t)x + ty) + b_i\} \\ &= \max_{i=1, \dots, m} \{(1-t)(a_i^\top x + b_i) + t(a_i^\top y + b_i)\} \\ &\leq (1-t) \max_{i=1, \dots, m} \{a_i^\top x + b_i\} + t \max_{i=1, \dots, m} \{a_i^\top y + b_i\} \\ &= (1-t)f(x) + tf(y). \end{aligned}$$

Therefore f is convex.

Now let $C \subseteq \mathbb{R}^n$ be a convex set. Since f is convex, its restriction to C is also convex. Hence by the theorem that local minimizers of convex functions over convex sets are global minimizers, every local minimizer of f over C is a global minimizer. ■

Example 2.2.2 (Proposed by Tao Xu). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and differentiable. Show that if $\bar{x} \in \text{dom}(f)$ satisfies $\nabla f(\bar{x}) = 0$, then $\bar{x}$ is a global minimizer of f .

Proof. *Drafted by Justin Zhou, Proofread by Alexandra Aswani*

Let $y \in \text{dom}(f)$ be arbitrary. Since f is convex and differentiable on $\text{dom}(f)$, then it satisfies the first-order convexity inequality at $x = \bar{x}$

$$f(y) \geq f(\bar{x}) + \langle \nabla f(\bar{x}), y - \bar{x} \rangle.$$

By assumption, $\nabla f(\bar{x}) = 0$. Therefore,

$$f(y) \geq f(\bar{x}) + \langle 0, y - \bar{x} \rangle = f(\bar{x}).$$

Since this holds for every $y \in \text{dom}(f)$, $\bar{x}$ is a global minimizer of f . ■

Example 2.2.3 (Proposed by Tao Xu). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex. Prove that the set of all global minimizer

$$\arg \min_{x \in \text{dom}(f)} f(x)$$

is a convex set (possibly empty).

Proof. *Drafted by Alexandra Aswani, Proofread by Justin Zhou*

Let $S = \arg \min_{x \in \text{dom}(f)} f(x)$ be the set of all global minimizers of f . If $S = \emptyset$, then S is convex, so assume $S \neq \emptyset$. Let $x, y \in S$ and let $t \in [0, 1]$. Since x and y are global minimizers, we have

$$f(x) = f(y) = \min_{z \in \text{dom}(f)} f(z).$$

Since f is convex,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) = f(x).$$

However, since x is a global minimizer,

$$f(x) \leq f((1-t)x + ty).$$

Therefore,

$$f((1-t)x + ty) = f(x),$$

so $(1-t)x + ty$ is also a global minimizer. Hence,

$$(1-t)x + ty \in S.$$

Therefore this proves S is convex. ■

Example 2.2.4 (Proposed by Tao Xu). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be strictly convex, i.e., $f((1-t)x + ty) < (1-t)f(x) + tf(y)$. Prove that f has at most one global minimizer.

Proof. *Drafted by Justin Zhou, Proofread by Pablo Gómez*

Suppose for contradiction that f has two different global minimizers $x_1 \neq x_2$. Let

$$m = \min_{z \in \text{dom}(f)} f(z).$$

Then

$$f(x_1) = f(x_2) = m.$$

Since f is strictly convex, using $t = \frac{1}{2}$, we get

$$f\left(\frac{x_1 + x_2}{2}\right) < \frac{1}{2}f(x_1) + \frac{1}{2}f(x_2) = m.$$

This contradicts the minimality of m , so f has at most one global minimizer. ■

Example 2.2.5 (Proposed by Tao Xu). Give an explicit example of a differentiable but non-convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that has a strict local minimizer which is not a global minimizer. Sketch the graph (or describe its behavior) and explain precisely why the conclusion “local minimizer are global” fails without convexity.

Proof. *Drafted by Justin Zhou, Proofread by Peter Le*

Take $d = 1$ and consider

$$f(x) = x^3 - 3x.$$

This function is differentiable on $\mathbb{R}$. We have

$$f'(x) = 3x^2 - 3$$

and

$$f''(x) = 6x.$$

At $x = 1$,

$$f'(1) = 0$$

and

$$f''(1) = 6 > 0.$$

Therefore, $x = 1$ is a strict local minimizer.

However,

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} (x^3 - 3x) = -\infty.$$

So f has no global minimizer, and in particular $x = 1$ is not a global minimizer.

Also, f is not convex on $\mathbb{R}$, since

$$f''(x) = 6x < 0$$

for $x < 0$. The graph is a cubic curve with a local maximum at $x = -1$, a strict local minimum at $x = 1$, and the left tail going to $-\infty$. Thus the conclusion that local minimizers are global fails without convexity. ■

Example 2.2.6 (Proposed by Tao Xu). Let $C \subseteq \mathbb{R}^d$ be a convex set and let $f : C \rightarrow \mathbb{R}$ be convex. Suppose $\bar{x} \in C$ is a local minimizer of f over C (i.e., there exists a neighborhood $V(\bar{x})$ such that $f(\bar{x}) \leq f(y)$ for all $y \in V(\bar{x}) \cap C$). Prove that $\bar{x}$ is a global minimizer of f over C .

Proof. *Drafted by Justin Zhou, Proofread by Pablo Gómez*

Suppose for contradiction that $\bar{x}$ is not a global minimizer of f over C . Then there exists $x^* \in C$ such that

$$f(x^*) < f(\bar{x}).$$

For $t \in (0, 1)$, define

$$x_t = (1 - t)\bar{x} + tx^*.$$

Since C is convex and $\bar{x}, x^* \in C$, we have $x_t \in C$. By convexity of f ,

$$f(x_t) \leq (1 - t)f(\bar{x}) + tf(x^*).$$

Since $f(x^*) < f(\bar{x})$, this gives

$$f(x_t) < (1 - t)f(\bar{x}) + tf(\bar{x}) = f(\bar{x}).$$

Also,

$$\|x_t - \bar{x}\| = \|t(x^* - \bar{x})\| = t\|x^* - \bar{x}\| \xrightarrow{t \rightarrow 0} 0.$$

Then for sufficiently small $t > 0$ and any neighborhood $V(\bar{x})$, we have

$$x_t \in V(\bar{x}) \cap C.$$

But $f(x_t) < f(\bar{x})$ contradicts the local minimality of $f(\bar{x})$ over C . Hence $\bar{x}$ is a global minimizer of f over C . ■

Remark. If you compare this exercise with Theorem 2.3, you will find it as an extension from unbounded domains to a convex domain and the proof is identical. Two ingredients are important here:

- Convex function for a smaller value, and convex sets for closeness.
- A structure leads to neighborhood. See the remark of Theorem 2.3 as well.

Example 2.2.7 (Proposed by Tao Xu). Consider the function $f(x) = \max\{x, 0\}$ on $\mathbb{R}$. Determine all its global minimizer. Explain why the stationary point characterization and the “local minimizer are global” property do not apply to this function.

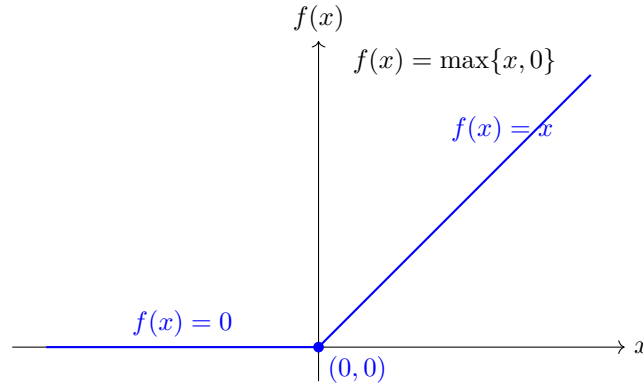
Proof. *Drafted by Peter Le, Proofread by Prashast Singh*

Rewriting as a Piecewise:

$$f(x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases}$$

By inspection, we can conclude that the smallest possible value this function may obtain is 0, where $x \leq 0$. For if $x > 0 \Rightarrow f(x) > 0$. Therefore:

$$\boxed{\arg \min f = (-\infty, 0]}$$



The stationary point characterization requires f to be convex and differentiable because it says $\bar{x}$ is a minimizer iff $\nabla f(\bar{x}) = 0$. It is clear from the above visualization that f is not differentiable at $x = 0$. The nonsmoothness of this function and lack of gradient at $x = 0$ means we may not apply our usual properties on it. ■

Example 2.2.8 (Proposed by Tao Xu). This example connects convex analysis with a standard optimization problem in statistics and machine learning. In logistic regression, one trains a classifier by choosing a parameter vector $x \in \mathbb{R}^d$ that makes the observed labels $y_i \in \{-1, 1\}$ compatible with the feature vectors $a_i \in \mathbb{R}^d$. A common training objective is the logistic loss

$$f(x) = \sum_{i=1}^n \log(1 + \exp(-y_i \langle a_i, x \rangle)).$$

Although this function comes from binary classification, the main point here is purely convex analytic.

1. Show that f is convex in x .
2. Under standard nondegeneracy assumptions on the data, f is in fact strictly convex. Use this to explain why the optimization landscape has no spurious local minima: every local minimizer of f over $\mathbb{R}^d$ is automatically a global minimizer.
3. Moreover, explain why any point $\bar{x}$ satisfying

$$\nabla f(\bar{x}) = 0$$

must be the unique global minimizer.

Hint: This is the standard loss function used in logistic regression, but no statistics background is needed. Treat it simply as a function of x . Use the definition of convexity from Lecture 1, together with Theorem 2.3 and Example 2.2.2 from this lecture.

Proof. Drafted by Bradley Nakagawa, Completed and Proofread by Run Li

1. Convexity of $f(x)$:

Consider the 1-order convexity definition

$$\begin{aligned} \nabla f(x) &= \sum_{i=1}^n -y_i a_i \frac{\exp(-y_i \langle a_i, x \rangle)}{1 + \exp(-y_i \langle a_i, x \rangle)} \\ \langle \nabla f(x), y - x \rangle &= \sum_{i=1}^n -y_i \frac{\exp(-y_i \langle a_i, x \rangle)}{1 + \exp(-y_i \langle a_i, x \rangle)} \langle a_i, y - x \rangle \end{aligned}$$

We want $f(x_1) \geq f(x_2) + \langle \nabla f(x_2), x_1 - x_2 \rangle$

$$f(x_1) = \sum_{i=1}^n \log(1 + \exp(-y_i \langle a_i, x_1 \rangle))$$

$$f(x_2) + \langle \nabla f(x_2), x_1 - x_2 \rangle = \sum_{i=1}^n \log(1 + \exp(-y_i \langle a_i, x_2 \rangle)) + \sum_{i=1}^n -y_i \frac{\exp(-y_i \langle a_i, x_2 \rangle)}{1 + \exp(-y_i \langle a_i, x_2 \rangle)} \langle a_i, x_1 - x_2 \rangle$$

f applied to the convex combination $(1-t)x_1 + tx_2$ where $t \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}^d$

$$f((1-t)x_1 + tx_2) = \sum_{i=1}^n \log(1 + \exp(-y_i \langle a_i, (1-t)x_1 + tx_2 \rangle))$$

We can use the properties of the inner product to separate it into two terms

$$\sum_{i=1}^n \log(1 + \exp(-y_i (\langle a_i, (1-t)x_1 \rangle + \langle a_i, tx_2 \rangle)))$$

Separating this into two exponentials multiplied by each other gives us

$$\sum_{i=1}^n \log(1 + e^{-y_i \langle a_i, (1-t)x_1 \rangle} e^{-y_i \langle a_i, tx_2 \rangle})$$

$$f((1-t)x_1 + tx_2) = \sum_{i=1}^n \log(1 + \exp(-y_i \langle a_i, (1-t)x_1 + tx_2 \rangle)) = \sum_{i=1}^n \log(1 + \exp(-y_i (\langle a_i, (1-t)x_1 \rangle + \langle a_i, tx_2 \rangle)))$$

Alternatively, computing the Hessian provides a more direct proof for the convexity of $f(x)$ (Definition 1.6):

We already have the gradient:

$$\nabla f(x) = \sum_{i=1}^n -y_i a_i p_i(x), \quad \text{where } p_i(x) = \frac{\exp(-y_i \langle a_i, x \rangle)}{1 + \exp(-y_i \langle a_i, x \rangle)} \in (0, 1).$$

Next, we compute the Hessian matrix $\nabla^2 f(x)$:

$$\nabla^2 f(x) = \sum_{i=1}^n y_i^2 a_i a_i^\top p_i(x)(1 - p_i(x)).$$

Since $y_i \in \{-1, 1\}$, $y_i^2 = 1$. Thus,

$$\nabla^2 f(x) = \sum_{i=1}^n p_i(x)(1 - p_i(x)) a_i a_i^\top.$$

For any arbitrary vector $v \in \mathbb{R}^d$, we evaluate the quadratic form:

$$v^\top \nabla^2 f(x) v = \sum_{i=1}^n p_i(x)(1 - p_i(x)) v^\top a_i a_i^\top v = \sum_{i=1}^n p_i(x)(1 - p_i(x)) \langle a_i, v \rangle^2 \geq 0.$$

Since $p_i(x) \in (0, 1)$, the scalar $p_i(x)(1 - p_i(x)) > 0$. Thus the sum of squares is always non-negative, showing $\nabla^2 f(x) \succeq 0$. This proves $f(x)$ is convex.

2. No spurious local minima:

Under standard nondegeneracy assumptions, the Hessian $\nabla^2 f(x)$ is strictly positive definite, making $f(x)$ strictly convex. Since f is convex, by **Theorem 2.3** (Local minimizers are global), any local minimizer of f is automatically a global minimizer. Therefore, the optimization landscape has no spurious local minima.

3. Unique global minimizer:

Suppose there is a point $\bar{x}$ satisfying $\nabla f(\bar{x}) = 0$. By **Example 2.2.2**, since f is convex and differentiable, this stationarity condition guarantees that $\bar{x}$ is a global minimizer. Furthermore, since f is strictly convex, it follows from **Example 2.2.4** that a strictly convex function can have at most one global minimizer. Thus, $\bar{x}$ must be the unique global minimizer. ■

2.3 More Examples

Example 2.3.1 (Proposed by Andrew Wang). Let $C \subseteq \mathbb{R}^d$ be a nonempty closed convex set, and let $y \in \mathbb{R}^d$. The projection of y onto C is defined by the following minimization problem:

$$P_C(y) = \operatorname{argmin}_{x \in C} \frac{1}{2} \|x - y\|^2.$$

This is a natural example of convex minimization. The function

$$q_y(x) = \frac{1}{2} \|x - y\|^2$$

is convex, and in fact strongly convex. Since the feasible set C is also convex, this is a convex optimization problem. Moreover, because q_y is strongly convex, the minimizer is unique. Thus the projection $P_C(y)$ is well-defined.

We now derive a useful characterization of this minimizer. Let

$$p = P_C(y).$$

Then $p \in C$, and for every $z \in C$,

$$\frac{1}{2} \|p - y\|^2 \leq \frac{1}{2} \|z - y\|^2.$$

Since C is convex, for every $z \in C$ and every $t \in [0, 1]$, the point

$$p + t(z - p)$$

also belongs to C . Hence the one-variable function

$$\phi(t) = \frac{1}{2} \|p + t(z - p) - y\|^2$$

has a minimum at $t = 0$ on the interval $[0, 1]$. Therefore its right derivative at 0 is nonnegative:

$$\phi'(0) = \langle p - y, z - p \rangle \geq 0.$$

Thus the projection satisfies the variational inequality

$$\langle p - y, z - p \rangle \geq 0 \quad \text{for all } z \in C.$$

Conversely, suppose that $p \in C$ satisfies

$$\langle p - y, z - p \rangle \geq 0 \quad \text{for all } z \in C.$$

Then for any $z \in C$,

$$\begin{aligned}\frac{1}{2}\|z - y\|^2 &= \frac{1}{2}\|p - y + z - p\|^2 \\ &= \frac{1}{2}\|p - y\|^2 + \langle p - y, z - p \rangle + \frac{1}{2}\|z - p\|^2 \\ &\geq \frac{1}{2}\|p - y\|^2.\end{aligned}$$

Therefore p is the minimizer of $q_y(x)$ over C , so $p = P_C(y)$.

This example shows that projection is not only a geometric operation, but also a convex minimization problem. The convexity of the feasible set and the strong convexity of the objective give a unique global minimizer. This viewpoint will also reappear later when projections are viewed as special cases of proximal operators.

3 Lecture 3 on Gradient Descent Method

sec:lec3

Sahand Ahmadi summarizes: We explore steepest descent search method for minimization problems. We discuss both iterative and analytical methods that use the negative gradient as a search direction. We also touch on what is necessary in order to guarantee the existence of minimizers.

Scribe Team: Hannah Nguyen and Indhu Chandra.

3.1 Lecture Notes

Before we discuss technical details about gradient descent methods, we recommend a helpful visualization tools for various gradient descent methods (https://github.com/lilipads/gradient_descent_viz).

3.1.1 Existence of minimizers

Theorem 3.1 (Minimizer of convex functions). Depending on which versions of convexity a function satisfies, the following holds:

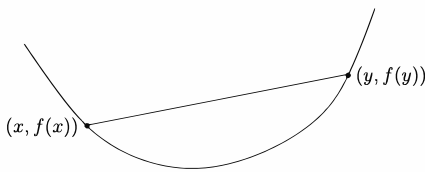
- i) $\bar{x}$ is the minimizer of a convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ if and only if $f(\bar{x}) \leq f((1-t)\bar{x} + ty)$ holds for any $y \in \text{dom}(f)$ and $t \in (0, 1)$.
- ii) For $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable and convex, $\bar{x}$ is its minimizer if and only if $\nabla f(\bar{x}) = 0$.

In general, for $f : C \rightarrow \mathbb{R}$ and $\bar{x}$ is an interior local minimizer and all partial derivatives of f exists at $\bar{x}$, then $\nabla f(\bar{x}) = 0$.

- iii) If $f : C \rightarrow \mathbb{R}$ is a convex and differentiable on the interior of C , then $\bar{x}$ is its minimizer if and only if $-\nabla f(\bar{x}) \in N_C(\bar{x})$, where the normal cone of C at $\bar{x}$

$$N_C(\bar{x}) = \left\{ g \in \mathbb{R}^d \mid \langle g, y - \bar{x} \rangle \leq 0, \text{ for all } y \in C. \right\}$$

Proof. Part i):



If $\bar{x}$ is a minimizer, thus $f(\bar{x}) \leq f(z)$ for all z . Taking $z = (1-t)\bar{x} + ty$, we complete the one side.

Assume $f(\bar{x}) \leq f((1-t)\bar{x} + ty)$ for any y , by the 0th order of definition (1.1), one has

$$f(\bar{x}) \leq f((1-t)\bar{x} + ty) \leq (1-t)f(\bar{x}) + tf(y)$$

$$\text{which yields that } f(\bar{x}) - (1-t)f(\bar{x}) \leq tf(y),$$

as $t > 0$, we conclude with $f(\bar{x}) \leq f(y)$.

Part ii):

$\Rightarrow$ Suppose $\bar{x}$ is a minimizer of f . Then for any direction d , one has $f(\bar{x}) \leq f(\bar{x} + td)$ for every $t > 0$. Since f is differentiable at $\bar{x}$, then

$$\lim_{t \rightarrow 0+} \frac{f(\bar{x} + td) - f(\bar{x})}{t} = \langle \nabla f(\bar{x}), d \rangle \geq 0.$$

In particular, taking $d = -\nabla f(\bar{x})$ yields that $-\|\nabla f(\bar{x})\|^2 \geq 0$. Thus the only choice is $\nabla f(\bar{x}) = 0$.

Remark: In this proof, we just use $\bar{x}$ must be an interior point whose gradient is well-defined. Thus the above proof can migrate to a function whose domain is not necessary $\mathbb{R}^d$.

$\Leftarrow$ Suppose $\nabla f(\bar{x}) = 0$. Because f is convex and differentiable, then, for all $y \in \mathbb{R}^d$ we can write it as:

$$f(y) \geq f(\bar{x}) + \langle \nabla f(\bar{x}), y - \bar{x} \rangle = f(\bar{x}) + \langle 0, y - \bar{x} \rangle = f(\bar{x})$$

Then, $f(y) \geq f(\bar{x})$ for all $y \in \mathbb{R}^d$, so $\bar{x}$ is a minimizer of f .

Part iii):

Since f is convex, $f(y) \geq f(\bar{x}) + \langle \nabla f(\bar{x}), y - \bar{x} \rangle \geq 0$ for all $x, y \in \mathbb{R}^d$.

Then $\bar{x}$ is a minimizer if and only if $\langle \nabla f(\bar{x}), y - \bar{x} \rangle \geq 0$, $\langle -\nabla f(\bar{x}), y - \bar{x} \rangle \geq 0$ for any $y \in C$. ■

Theorem 3.1 characterizes *what* a minimizer looks like, but it does not tell us whether one exists. The next two results address existence directly. Note that convexity plays no direct role here; the key assumptions are topological (compactness or coercivity). Before continuing, we will first define these properties in more detail and address their benefits.

Definition 3.2 (Closed Sets). A set E is closed if it contains all its limit points. A point x is a limit point of E if every neighborhood of x contains a point $x_0 \neq x$ and $x_0 \in E$. In other words, there exists a sequence $\{x_n\}$ for $x_n \in E$, where $x_n \rightarrow x$.

Remark. For example, $[0, 1]$ is a closed set (every point in this set is a limit point and is trivially contained in the set). However, $[0, 1)$ is not a closed set (1 is a limit point but $1 \notin [0, 1)$).

Definition 3.3 (Compactness). For a metric space X , a subset $E \subseteq X$ is compact if every open cover of E contains a finite subcover.

In other words, if $\{F_\alpha\}$ is an open cover of E , then there are finite indices $\alpha_1, \alpha_2, \dots, \alpha_n$ where

$$E \subset F_{\alpha_1} \cup F_{\alpha_2} \cup \dots \cup F_{\alpha_n}$$

In a finite-dimensional real space, we can make use of the following theorem:

Definition 3.4 (Heine-Borel). A set $E \subset \mathbb{R}^d$ is compact if and only if it is bounded and closed.

Remark. Not every closed set is necessarily bounded. For example, the set of all reals, $\mathbb{R}$, is a closed set, since you can find a convergent sequence to every real number in this set, but it is unbounded.

Then, we proceed to the Weierstrass Theorem, also commonly known as the Extreme Value Theorem:

Theorem 3.5 (Weierstrass Theorem). If f is continuous and $\text{dom}(f) \subseteq \mathbb{R}^d$ is compact, then a minimizer exists for $f(x)$.

Remark. Intuitively, coercivity ensures that the sublevel sets $\{x : f(x) \leq c\}$ are bounded. Combined with closedness and continuity, Weierstrass's theorem then applies to each sublevel set, guaranteeing a minimizer.

Proof. Since f is continuous on a compact domain, its image $f(x)$ is additionally compact. Then, by the Heine-Borel theorem, $f(x)$ is closed and bounded. Bounded implies the minima exists, and closed means that this minima can be attained. ■

Compactness is a strong assumption. When the domain is unbounded, coercivity serves as a substitute: it prevents minimizing sequences from escaping to infinity.

Definition 3.6 (Coercive). A function f is coercive if and only if

$$\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty$$

Remark. This means that when the distance between the input vector x and the origin increases, the value of f escapes to infinity (so that the minimizer won't escape to infinity).

Theorem 3.7 (Existence of Minimizer). If f is continuous and coercive function on a normed vector space, there exists a minimizer of $f(x)$ over any closed subset S .

Proof. Pick some $x_0 \in \text{dom}(S)$ and define a subset of S

$$T = \{x \in \text{dom}(S) | f(x) \leq f(x_0)\}$$

Coercivity implies that $f(x_0) < \infty$, since $x_0 < \infty$. Then, we have that T is a bounded and closed subset of S , which, by Heine-Borel, implies that T is also compact. Since f is continuous, by the Weierstrass Theorem, the minima of S exists and can be obtained. ■

3.1.2 Gradient Descent and Gradient Flow

Having established when minimizers exist, we now turn to *how* to find them algorithmically. The gradient points in the direction of the steepest ascent. Gradient descent uses the fact that the opposite direction is the steepest descent. The iteration is described as below:

$$x^{(k+1)} = x^{(k)} - \eta_k \nabla f(x^{(k)}),$$

where η_k is the step size and $\nabla f(x^{(k)})$ is the gradient evaluated at $x^{(k)}$. Ideally, $x^{(0)}, x^{(1)}, x^{(2)}, \dots, x^{(k+1)} \rightarrow \bar{x}$, where $\bar{x}$ is the minimizer. However, there are many conditions one need to check in order the sequence converges to the minimizer.

For example, the step size strategy:

$\eta_k = \lambda$	Constant (needs more assumptions on f)
$\eta_k = \frac{\lambda}{k}$	Decay (convergence possible but slower)
$\eta_k = \arg_{t \geq 0} f(x^{(k)} - t \nabla f(x^{(k)}))$	Exact Line Search
future	Backtracking

A key practical challenge is choosing η_k appropriately. The following figures illustrate the effect of step size on convergence behavior for a simple quadratic objective:

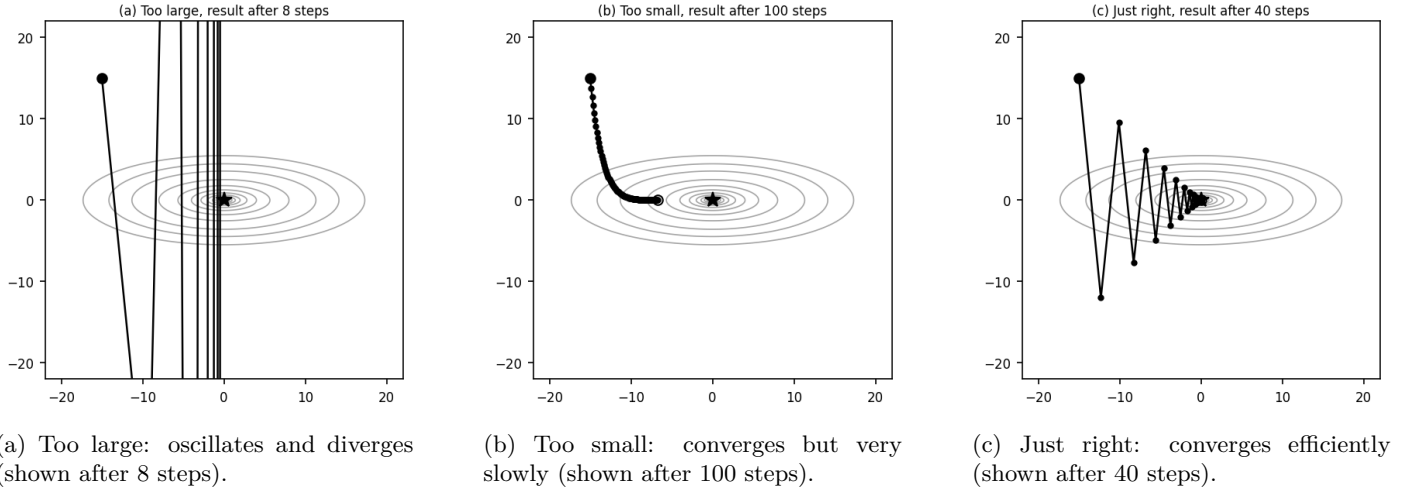


Figure 6: Effect of step size choice on gradient descent trajectories. Image sources: lecture handout.

Another example, the stopping criteria. From the first-derivative test, a frequently used criteria is

$$\|\nabla f(x^{(k)})\| \leq \epsilon$$

for a chosen tolerance ϵ .

In general, the continuous-in-time scheme is easier to analysis, thus we start with Gradient Flows. Since once reformulates the GD as

$$\frac{x^{(k+1)} - x^{(k)}}{\eta_k} = -\nabla f(x^{(k)}),$$

this is interpreted as the Forward Euler Scheme, taking the limit $\lim_{\eta_k \rightarrow 0} \frac{x^{(k+1)} - x^{(k)}}{\eta_k} = \dot{x}(t)$ yields

$$\dot{x}(t) = -\nabla f(x(t)).$$

Gradient descent as minimization of a quadratic surrogate. There is a second, illuminating way to derive the gradient descent update. By the second-order Taylor expansion,

$$f(y) \approx f(x^{(k)}) + \langle \nabla f(x^{(k)}), y - x^{(k)} \rangle + \frac{1}{2} \langle y - x^{(k)}, \text{Hess}(f)|_{x^{(k)}} (y - x^{(k)}) \rangle.$$

Approximating the Hessian by $\text{Hess}(f)|_{x^{(k)}} \approx \frac{1}{t} I$ (a scaled identity), the surrogate becomes

$$g(y) = f(x^{(k)}) + \langle \nabla f(x^{(k)}), y - x^{(k)} \rangle + \frac{1}{2t} \|y - x^{(k)}\|^2.$$

This is a quadratic $g(y) = a\|y - x^{(k)}\|^2 + \langle b, y \rangle + c$ with unique minimizer at $y_{\min} = -b/(2a)$. Minimizing over y gives

$$x^{(k+1)} = \underset{y}{\operatorname{argmin}} g(y) = x^{(k)} - t \nabla f(x^{(k)}).$$

This shows that one step of gradient descent with step size t is equivalent to minimizing a local quadratic upper bound on f . The approximation $\text{Hess}(f) \approx \frac{1}{t} I$ is the key heuristic justification: when f is L -smooth, the choice $t = \frac{1}{L}$ makes this approximation an actual upper bound, which is the basis of the descent lemma.

Theorem 3.8. We quote some basic facts about gradient flows from [San17], and will revisit these topics in the future.

- i) The classical ODE theory provides the existence and uniqueness of solution to the gradient flow with any initial data, under the assumption that the gradient of f is Lipschitz.
- ii) If X is an inner-product space, then the function value $f(x(t))$ along gradient flow is guaranteed decreasing,

$$\frac{d}{dt} f(x(t)) = \langle \nabla f(x(t)), \dot{x}(t) \rangle = -\|\nabla f(x(t))\|^2 \leq 0. \quad \text{eq:GF-value_descent} \quad (3.1)$$

Also, one has the energy dissipation identity

$$f(x(t_1)) - f(x(t_0)) = - \int_{t_0}^{t_1} \|\nabla f(x(t))\|^2 dt. \quad \text{eq:GF-EDI} \quad (3.2)$$

- iii) If f is additionally convex with minimizer $\bar{x}$, then the squared distance to the minimizer,

$$\text{distance}(t) := \frac{1}{2} \|x(t) - \bar{x}\|^2,$$

is also non-increasing. Indeed,

$$\begin{aligned} \frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 &= \langle x(t) - \bar{x}, \dot{x}(t) \rangle = \langle x(t) - \bar{x}, -\nabla f(x(t)) \rangle \\ &= \langle \nabla f(x(t)), \bar{x} - x(t) \rangle \leq f(\bar{x}) - f(x(t)) \leq 0, \end{aligned} \quad \text{eq:GF_dis_descent} \quad (3.3)$$

where the last inequality uses the first-order convexity condition $f(\bar{x}) \geq f(x) + \langle \nabla f(x), \bar{x} - x \rangle$, rearranged as $\langle \nabla f(x(t)), \bar{x} - x(t) \rangle \leq f(\bar{x}) - f(x(t))$.

Proof. Part (i) follows from the Picard–Lindelöf theorem: a Lipschitz right-hand side guarantees local existence and uniqueness of solutions to any ODE, and global existence follows from standard extensions.

Part (ii): Since f is differentiable and $x(t)$ is a solution of the gradient flow, the chain rule gives

$$\frac{d}{dt}f(x(t)) = \langle \nabla f(x(t)), \dot{x}(t) \rangle = \langle \nabla f(x(t)), -\nabla f(x(t)) \rangle = -\|\nabla f(x(t))\|^2 \leq 0,$$

which is (3.1). The energy dissipation identity (3.2) follows by integrating both sides over $[t_0, t_1]$:

$$f(x(t_1)) - f(x(t_0)) = \int_{t_0}^{t_1} \frac{d}{dt}f(x(t)) dt = - \int_{t_0}^{t_1} \|\nabla f(x(t))\|^2 dt.$$

Part (iii): By the chain rule applied to $\frac{1}{2}\|x(t) - \bar{x}\|^2$,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 = \langle x(t) - \bar{x}, \dot{x}(t) \rangle = \langle x(t) - \bar{x}, -\nabla f(x(t)) \rangle.$$

Since f is convex and differentiable, the first-order condition gives

$$f(\bar{x}) \geq f(x(t)) + \langle \nabla f(x(t)), \bar{x} - x(t) \rangle,$$

which rearranges to

$$\langle \nabla f(x(t)), \bar{x} - x(t) \rangle \leq f(\bar{x}) - f(x(t)).$$

Therefore,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 = \langle \nabla f(x(t)), \bar{x} - x(t) \rangle \leq f(\bar{x}) - f(x(t)) \leq 0,$$

where the last inequality holds because $\bar{x}$ is the minimizer, so $f(\bar{x}) \leq f(x(t))$ for all t . ■

Remark. Part (ii) holds under only a Lipschitz gradient assumption, while part (iii) additionally requires convexity. This is the first example where *descent in function value is easier to establish than descent in distance to the minimizer*—the latter requires strictly more structure. Part (iii) is also the first concrete **benefit of convexity**: it ensures that the gradient flow converges to the minimizer, even though analogous guarantees for gradient descent itself require more work (see follow-up lectures).

3.2 Exercises

Exercise. Recall for continuously differentiable function over $\mathbb{R}^d$, the first-order approximation is $f(x + d) \approx f(x) + \langle \nabla f(x), d \rangle + \mathcal{O}(\|d\|)$, thus we call d is a *descent direction* if $\langle \nabla f(x), d \rangle < 0$.

Prove that if d is a descent direction, there exists a maximal “stepsize” $\varepsilon > 0$, such that the descent is guaranteed in the sense of

$$f(x + td) < f(x) \quad \text{for all } t \in (0, \varepsilon].$$

Proof. Drafted by Jasper Luo, Proofread by Indhu Chandra

Since f is continuously differentiable, for fixed x and direction d , define

$$\phi(t) = f(x + td).$$

Then ϕ is differentiable, and by the chain rule,

$$\phi'(t) = \langle \nabla f(x + td), d \rangle.$$

In particular, because d is a descent direction, then

$$\phi'(0) = \langle \nabla f(x), d \rangle < 0,$$

Since ϕ' is continuous, there exists some $\varepsilon > 0$ such that

$$\phi'(t) < 0 \quad \text{for all } t \in (0, \varepsilon].$$

Therefore, ϕ is strictly decreasing on $(0, \varepsilon]$. Hence, for every $t \in (0, \varepsilon]$,

$$\phi(t) < \phi(0).$$

Substituting back $\phi(t) = f(x + td)$ and $\phi(0) = f(x)$, we obtain

$$f(x + td) < f(x) \quad \text{for all } t \in (0, \varepsilon].$$

Therefore, there exists a stepsize $\varepsilon > 0$ such that moving in the descent direction d guarantees a decrease in the function value. ■

Exercise. Prove a concrete example (with a visualization) that the sequence $\{x^{(k)}\}$ obtained from gradient descent algorithms may divert from the continuous curve created from the corresponding gradient flow.

Proof. Drafted by Peter Le, Proofread by Agustin Fitipaldi

Let

$$f(x) = \frac{1}{2}x^2$$

This is convex and differentiable. The minimizer is

$$\bar{x} = 0$$

Hence the gradient flow is

$$\dot{x}(t) = -\nabla f(x(t)) = -x(t)$$

Solving the differential equation yields $x(t) = Ce^{-t}$.

From an initial point $x(0) = 1 \Rightarrow C = 1$, the solution is $x(t) = e^{-t}$. This gives the continuous gradient flow. Now take $x^{(k+1)} = x^{(k)} - \eta_k \nabla f(x^{(k)})$ with constant step size $\eta = 3$ gives

$$x^{(k+1)} = x^{(k)} - 3x^{(k)}$$

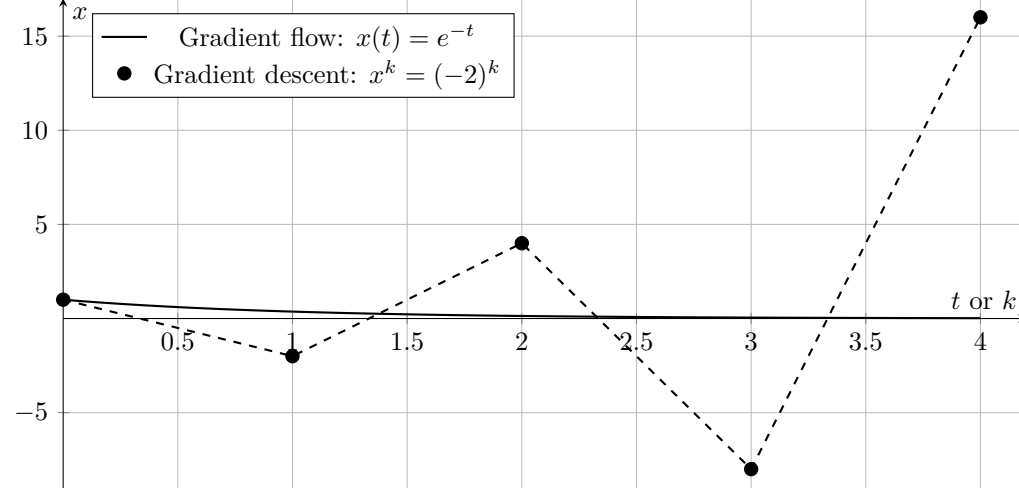
$$x^{(k+1)} = -2x^{(k)}$$

Starting from $x^{(0)} = 1$, this geometric sequence yields $x^{(k)} = (-2)^k$. The sequence $\{x^{(k)}\}$ does not follow $x(t) = e^{-t}$. It jumps back and forth, as it diverges from 0.

To sum everything up, even though the gradient flow converges, $x(t) = e^{-t} \rightarrow 0$; yet the sequence $\{x^{(k)}\}$ gives

$$|x^{(k)}| = 2^k \rightarrow \infty$$

This occurs because the step size is too large, making the method unstable. The following is a visualization of this



Exercise. Consider the quadratic function, $f : \mathbb{R}^d \rightarrow \mathbb{R}$ by $f(x) = \langle x, Ax \rangle + 2\langle b, x \rangle + c$, where A is a $d \times d$ positive definite matrix. Suppose we plan to use the gradient descent method to find its minimizer, with the exact line search method. Find the explicit formula for the stepsize selection.

Proof. Drafted by Alexandra Aswani, Proofread by Prashast Singh

For gradient descent, the equation has the form

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k).$$

Since $f(x) = \langle x, Ax \rangle + 2\langle b, x \rangle + c$, we have

$$\nabla f(x) = 2Ax + 2b.$$

Now let $g_k = \nabla f(x_k)$. Using exact line search method, we choose α_k to minimize

$$\phi(\alpha) = f(x_k - \alpha g_k).$$

Now let's compute

$$\begin{aligned} \phi(\alpha) &= \langle x_k - \alpha g_k, A(x_k - \alpha g_k) \rangle + 2\langle b, x_k - \alpha g_k \rangle + c \\ &= f(x_k) - \alpha \langle g_k, g_k \rangle + \alpha^2 \langle g_k, Ag_k \rangle \end{aligned}$$

Now let's take derivative by α :

$$\phi'(\alpha) = -\langle g_k, g_k \rangle + 2\alpha \langle g_k, Ag_k \rangle = 0,$$

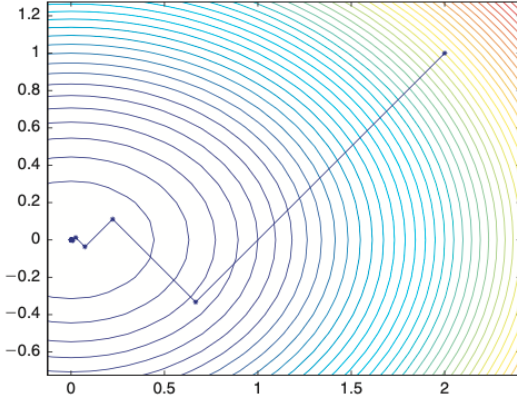
yields that

$$\alpha_k = \frac{\langle g_k, g_k \rangle}{2\langle g_k, Ag_k \rangle}.$$

Therefore we prove that the exact line search stepsize is

$$\alpha_k = \frac{\|\nabla f(x_k)\|_2^2}{2\langle \nabla f(x_k), A\nabla f(x_k) \rangle}.$$

Exercise. People always mention “zig-zag” effect on the gradient descent method. It roughly means that iterates of the gradient descent method show “zig-zag” searching path on the contour plot of the objective function.



The mathematical statement is as follows: given a continuously differentiable function f to minimize, let $\{x^{(k)}\}$ be the sequence generated by the gradient descent method with exact line search. Show that for any k ,

$$\langle x^{(k+2)} - x^{(k+1)}, x^{(k+1)} - x^{(k)} \rangle = 0,$$

i.e., the descent directions are normal.

Proof. Drafted by Agustin Fitipaldi, Proofread by Run Li

We have gradient descent with exact line search,

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)}), \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} - \alpha \nabla f(x^{(k)})),$$

and we want to show that consecutive steps are orthogonal,

$$\langle x^{(k+2)} - x^{(k+1)}, x^{(k+1)} - x^{(k)} \rangle = 0.$$

The two differences are nothing but scaled gradients. Reading straight off the update rule we have:

$$x^{(k+1)} - x^{(k)} = -\alpha_k \nabla f(x^{(k)}), \quad x^{(k+2)} - x^{(k+1)} = -\alpha_{k+1} \nabla f(x^{(k+1)}).$$

Substituting these and pulling the step lengths out to the front, the claim reduces to

$$\langle x^{(k+2)} - x^{(k+1)}, x^{(k+1)} - x^{(k)} \rangle = \alpha_{k+1} \alpha_k \langle \nabla f(x^{(k+1)}), \nabla f(x^{(k)}) \rangle, \quad (*)$$

so it is enough to show that consecutive gradients are orthogonal. This is where the line search comes in, the one ingredient not yet used.

Consider f along the search ray,

$$\phi(\alpha) = f(x^{(k)} - \alpha \nabla f(x^{(k)})).$$

The step length α_k minimizes ϕ , and provided $\nabla f(x^{(k)}) \neq 0$ (otherwise the method has already stopped) the minimizer is strictly interior, so $\phi'(\alpha_k) = 0$. By the chain rule,

$$\phi'(\alpha) = \langle \nabla f(x^{(k)} - \alpha \nabla f(x^{(k)})), -\nabla f(x^{(k)}) \rangle.$$

At $\alpha = \alpha_k$ the inner point is exactly $x^{(k+1)}$, so the stationarity condition becomes

$$0 = -\langle \nabla f(x^{(k+1)}), \nabla f(x^{(k)}) \rangle,$$

which is the orthogonality we were after.

Returning to (*),

$$\langle x^{(k+2)} - x^{(k+1)}, x^{(k+1)} - x^{(k)} \rangle = \alpha_{k+1} \alpha_k \cdot 0 = 0.$$

Geometrically the line search stops where the ray is tangent to a level set, and the next gradient leaves normal to that level set; the right angle between them, repeated, is the zig-zag. ■

3.3 More Examples

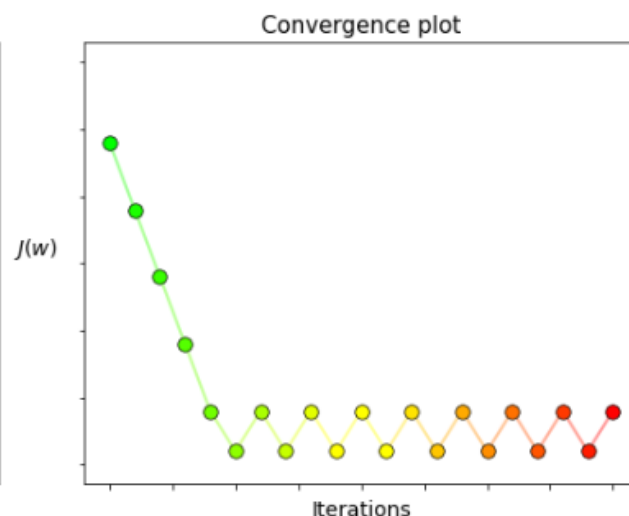
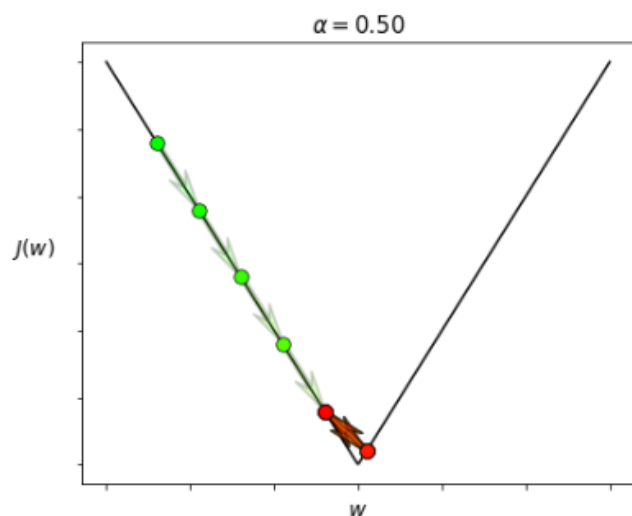
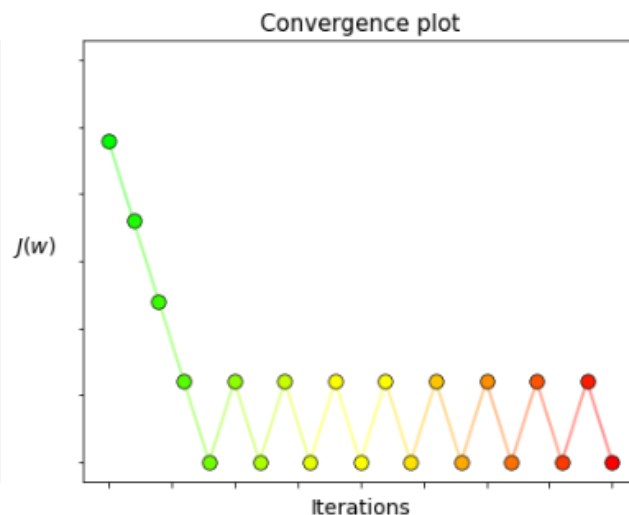
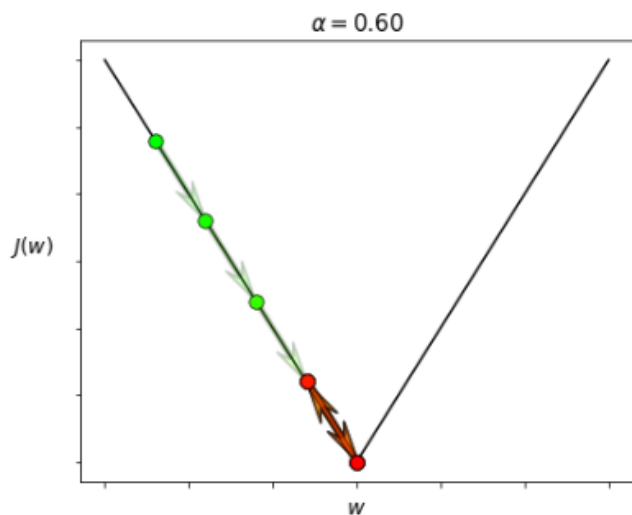
Example 3.3.1 (Proposed by Bohan Zhou). Throughout the course, we mainly focus on the fixed stepsize. Here is a link for variable stepsize (<https://perso.esiee.fr/~chierchg/optimization/content/07/scheduling.html>) with many helpful visualizations.

The *stepsize scheduling* is a technique to set additional decay parameters for stepsize beforehand, *without any feedback* throughout the optimization process. Some popular schedules are

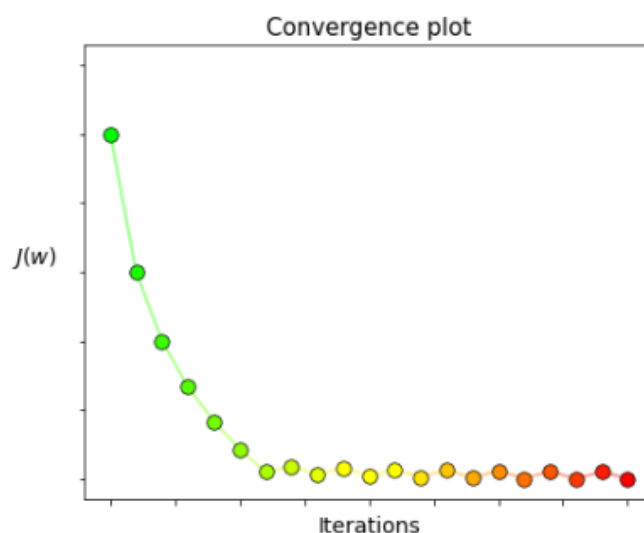
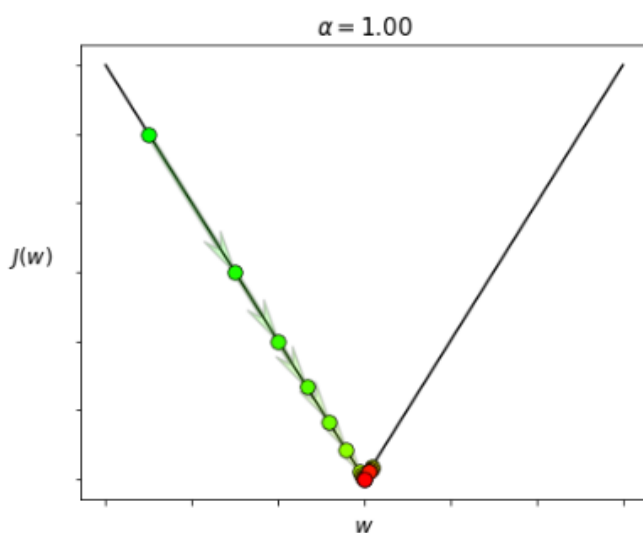
- Step decay every r iterations. $\eta_k = \eta_{\max} \lambda^{\lfloor (1+k)r \rfloor}$ for some $0 < \lambda < 1$.
- Harmonic decay $\eta_k = \frac{\eta_{\max}}{1+dk}$ with some progression parameter d .
- Exponential decay $\eta_k = \eta_{\max} e^{-dk}$ with some progression parameter d .
- Polynomial decay with a lower bound $\eta_k = \eta_{\min} + (\eta_{\max} - \eta_{\min}) \max(0, 1 - \frac{k}{r})^p$.
- Cosine annealing: cycling between $\eta_{\max}$ and $\eta_{\min}$.

When $\eta_k = \eta \neq 0$, the gradient descent update creates *pathological oscillations*, due to the nondifferentiability. The derivative or numerical finite difference does not vanish at the minimum.

For example, in the gradient descent to search the minimum of $f(x) = |x|$. A smaller step size won't necessarily provide a better solution (see below, the top figure uses a larger stepsize).



If we use the harmonic decay $\eta_k = \frac{\eta_0}{1+k}$, even start with large initial stepsize $\eta_0 = 1$, it provides a better approximation to the minimizer.



Example 3.3.2 (Proposed by Hannah Nguyen). Another choice of step size is the **backtracking line search**. At each iteration, the step size is adjusted based on the gradient and the current iterate. This allows for gradient descent to traverse functions with different levels of variability.

We would pick *smaller step sizes* to avoid overshooting in a *high variance region*, and larger step sizes to avoid undershooting in a *low variance region*. This gives us a good balance between the number of iterations and its ability to converge to the minima.

Before the $(k + 1)$ -th iteration, λ_k is updated by dividing itself by 2 (or some other number larger than 1) until the following inequality holds before proceeding with the next iteration of gradient descent:

$$f(x_k - \lambda_k \nabla f(x_k)) \leq f(x_k) - \frac{\lambda_k}{2} \|\nabla f(x_k)\|^2 \quad \text{eq:backtrack-cond (3.4)}$$

This inequality is derived from picking a suitable constant step size and the descent lemma covered in the following lecture. The benefit of this inequality is that the next iterate $f(x_{k+1})$ is guaranteed to decrease by a constant $\frac{\lambda_k}{2} \|\nabla f(x_k)\|^2$ from the previous iterate $f(x_k)$.

For example the 1D case, starting with Taylor's Theorem:

$$f(x+t) = f(x) + tf'(x) + \int_x^{x+t} (x+t-s)f''(s)ds$$

Assume $|f''(s)| \leq L$:

$$f(x+t) \leq f(x) + tf'(x) + \frac{t^2}{2}L$$

Let $x = x_k$ and $t = -\lambda f'(x_k)$:

$$\begin{aligned} f(x_{k+1}) &= f(x_k - \lambda f'(x_k)) \leq f(x_k) - \lambda f'(x_k)f'(x_k) + \frac{1}{2}L[\lambda f'(x_k)]^2 \\ &= f(x_k) - [f'(x_k)]^2(\lambda - \frac{L}{2}\lambda^2). \end{aligned}$$

Then, to gain (learning) from the method:

$$\lambda - \frac{L}{2}\lambda^2 > 0 \implies 0 < \lambda < \frac{2}{L}$$

where the best gain comes from when $\lambda - \frac{L}{2}\lambda^2$ is maximal, occurring at $\lambda = \frac{1}{L}$.

As a result, an **effective gradient descent method** is:

$$x_{k+1} = x_k - \lambda f'(x_k) = x_k - \frac{1}{L} f'(x_k) = x_k - \frac{1}{\max |f''(x)|} f'(x_k)$$

and earlier results implies:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} [f'(x_k)]^2$$

Now, substitute $x_{k+1} = x_k - \lambda_k \nabla f(x_k)$, $L = \frac{1}{\lambda_k}$, and $[f'(x_k)]^2 = \|\nabla f(x_k)\|^2$ for a general function, we get the original inequality:

$$f(x_k - \lambda_k \nabla f(x_k)) \leq f(x_k) - \frac{\lambda_k}{2} \|\nabla f(x_k)\|^2$$

On a well-conditioned quadratic function, backtracking line search doesn't affect the iterations compared to constant step size. For a poorly-conditioned quadratic function, in a region with lower curvature, the step sizes increases, so these functions benefit from backtracking line search. Non-convex functions also benefit here as well.

The equations are derived from the following lecture notes: https://skim.math.msstate.edu/LectureNotes/Machine_Learning_Lecture.pdf

4 Lecture 4 on Convergence Analysis on Gradient Descent Method

Hannah Nguyen summarizes: We explore the convergence of gradient flows, gradient descent with constant stepsize, and convex and L-smooth f . We are also introduced to the definition of L-Lipschitz and the descent lemma, which gives additional useful properties for the gradient descent algorithm.

Scribe Team: Pablo Gómez and Ritaja Subrahmanya.

4.1 Lecture Notes

Our main focus today is on the convergence of gradient descent, specifically how convexity helps us to analyze the convergence of gradient descent with a fixed step size.

For the following, we will work with the formulation of gradient descent given by

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$$

for some constant step size η .

4.1.1 Convergence Analysis of Gradient Flows

First we examine the continuous case, with gradient flow given by $\dot{x}(t) = -\nabla f(x(t))$. In this situation we take the limit $\eta \rightarrow 0$. Recall from (3.3) that if f is convex, we have that

$$\frac{d}{dt} \left[\frac{1}{2} \|x(t) - \bar{x}\|^2 \right] \leq f(\bar{x}) - f(x(t)) \leq 0. \quad \text{eq:GF-EVI (GF EVI)}$$

Here $\bar{x}$ denotes the minimizer of f , hence the inequality tells us that the distance to the minimizer decreases in time. This gives a *descent property (without rate)* with respect to the distance to a minimizer.

Rearranging the expression and integrating both sides with respect to t yields

$$\begin{aligned} \int_0^T [f(x(t)) - f(\bar{x})] dt &\leq \int_0^T \frac{d}{dt} \left[-\frac{1}{2} \|x(t) - \bar{x}\|^2 \right] dt \\ &= -\frac{1}{2} \|x(T) - \bar{x}\|^2 + \frac{1}{2} \|x(0) - \bar{x}\|^2. \end{aligned}$$

Consider the left hand side. Notice that the integrand is decreasing as $x(t)$ approaches $\bar{x}$ (i.e. as t increases), hence

$$\int_0^T f(x(t)) - f(\bar{x}) dt \leq \int_0^T f(x(t)) - f(\bar{x}) dt.$$

On the left we are integrating on a constant, yielding

$$[f(x(T)) - f(\bar{x})]T \leq -\frac{1}{2} \|x(T) - \bar{x}\|^2 + \frac{1}{2} \|x(0) - \bar{x}\|^2 \leq \frac{1}{2} \|x(0) - \bar{x}\|^2.$$

This is our first convergence result for the function value in gradient flows.

Theorem 4.1. For a convex differentiable function f , gradient flow converges to a minimizer at a rate of $\mathcal{O}(\frac{1}{T})$. Specifically, the function values satisfy

$$f(x(T)) - f(\bar{x}) \leq \frac{\|x(0) - \bar{x}\|^2}{2T}.$$

Notice that the above theorem tells us that the the function converges at a rate of $\frac{1}{T}$.

Remark. Notice in this proof convexity was helpful to give us the important bound. The expression that we derived was called the **dissipation inequality** or **Evolution Variational Inequality**.

4.1.2 Convergence Analysis of constant stepsize Gradient Descent

Now we return to Gradient Descent: $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$. There are two notions of convergence that we are looking at. The first is convergence with respect to the distance from an input $x^{(k)}$ to $\bar{x}$, where we take

$$\|x^{(k)} - \bar{x}\| \propto \varphi(k)$$

and hope to show that $\varphi(k)$ is decreasing. The second is in terms of function values, where we consider

$$|f(x^{(k)}) - f(\bar{x})| = f(x^{(k)}) - f(\bar{x}) \propto \psi(k)$$

and again want $\psi(k)$ to be a decreasing function.

Similar to our proof for gradient flows, we note that we can produce a sequence of inequalities where

$$\begin{aligned} f(x^{(k)}) - f(\bar{x}) &\leq f(x^{(k-1)}) - f(\bar{x}) + \mathbf{err}(k-1) \\ f(x^{(k-1)}) - f(\bar{x}) &\leq f(x^{(k-2)}) - f(\bar{x}) + \mathbf{err}(k-2) \\ &\vdots \\ f(x^{(2)}) - f(\bar{x}) &\leq f(x^{(1)}) - f(\bar{x}) + \mathbf{err}(1) \end{aligned}$$

Then summing up the values notice that many of the terms cancel to yield

$$f(x^{(k)}) - f(\bar{x}) \leq f(x^{(1)}) - f(\bar{x}) + \sum \mathbf{err}(i).$$

However, until we understand the behavior of $\sum \mathbf{err}(i)$ we cannot say anything concrete about the convergence of f . To remedy this, we will impose a new condition on f . To find the convergence rate for gradient descent of fixed step size, we introduce an assumption on the gradient ∇f .

Definition 4.2 (gradient is L -Lipschitz). A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ whose gradient is L -Lipschitz if it is continuously differentiable and $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.

An equivalent characterization is the following.

Proposition 4.1.1. A function whose gradient is L -Lipschitz if and only if for any x, y we have

$$f(x) + \langle \nabla f(x), y - x \rangle - \frac{L}{2} \|y - x\|^2 \leq f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Proof. “ $\implies$ ”: note that $f(y) - f(x) = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt$, thus

$$\begin{aligned} |f(y) - f(x) - \langle \nabla f(x), y - x \rangle| &= \left| \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle dt \right| \\ &\leq \int_0^1 |\langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle| dt \\ &\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \|y - x\| dt \quad \text{Cauchy-Schwarz} \\ &\leq \int_0^1 L \|y - x\| dt = \frac{L}{2} \|y - x\|^2 \quad \text{gradient is } L\text{-Lipschitz.} \end{aligned}$$

“ $\impliedby$ ”: this side is slightly tricky and introduces some new class of function properties.

Let $g(x) = f(x) - \langle \nabla f(y), x \rangle$. Note that $\nabla g(y) = 0$ (some issues here), i.e., y minimizes g . In particular, $g(y) \leq g(x - \frac{1}{L} \nabla g(x))$. Since

■

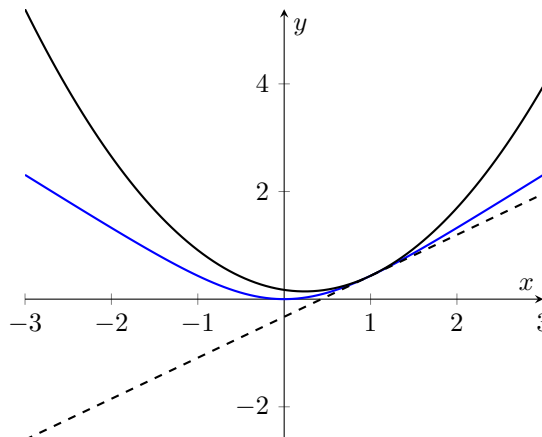
Recall from last lecture that the candidates provided by gradient descent satisfy

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - \eta \nabla f(x^{(k)}) \\ &= \underset{y}{\operatorname{argmin}} \left\{ f(x^{(k)}) + \langle \nabla f(x^{(k)}), y - x^{(k)} \rangle + \frac{1}{2\eta} \|y - x^{(k)}\|^2 \right\}. \end{aligned}$$

Furthermore we note that if f is convex and L -smooth we have

$$f(x) + \langle \nabla f(x), y - x \rangle \leq f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Notice that the left side is the supporting tangent plane and the right side is the same tangent plane with the addition of some quadratic function.



Considering the above image, we are looking at some function f whose graph is given by the blue line. Let the dashed line denote the tangent plane at a point $(x, f(x))$. For each y we look at the distance between the point and the point where the tangent plane occurs. We see that as we go away from the minima, the tangent is no longer an accurate model and hence we need to add an additional penalty quadratic term. We thus see that the black curve denotes the upper bound for this function.

Remark. This technique is called **quadratic regularization**, and in the future we will look at this further through the notion of **proximal operators**. This is called regularization because the distance function "adds convexity" (in the form of a quadratic term) to our original function, making the resulting function more convex than the original. This is done with the hope that this new function helps us in our optimization.

4.1.3 Descent Properties By L-smoothness

subsub:descent-Lsmooth

Let's analyze and understand these inequalities

$$f(x) + \langle \nabla f(x), y - x \rangle \leq f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2$$

in the context of gradient descent. Substituting $y = x^{(k+1)}$ and $x = x^{(k)}$ yields

$$\begin{aligned} f(x^{(k+1)}) &\leq f(x^{(k)}) + \langle \nabla f(x^{(k)}), -\eta \nabla f(x^{(k)}) \rangle + \frac{L}{2} \|\eta \nabla f(x^{(k)})\|^2 \\ &= f(x^{(k)}) + \left(\frac{L}{2} \eta^2 - 1 \right) \|\nabla f(x^{(k)})\|^2. \end{aligned} \quad \text{eq:descent-Lsmooth-eta (4.1)}$$

Lemma 4.3 (Descent Lemma). If f is convex, L -smooth, and continuously differentiable, then the following hold:

- (1) If $\eta \in (0, \frac{2}{L})$ then we have that $f(x^{(k+1)}) < f(x^{(k)})$ because $(\frac{L}{2} \eta^2 - 1) < 0$. This means that each step ensures descent (unless $\|\nabla f\| = 0$, in which case we have reached the global minimizer).
- (2) $\eta = \frac{1}{L}$ then for each update $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2L} \|\nabla f(x^{(k)})\|^2$.

The second term here is called the descent progress coming from the L -smoothness. For general $\eta \in (0, \frac{1}{L}]$ we have the same result as (2) with $\frac{\eta}{2}$ in place of $\frac{1}{2L}$.

Remark. It might also be possible to loosen the conditions on the convexity of f to a local notion of convexity. It might be a fun exercise to see if similar results hold with these weaker conditions.

We now want to look at the convergence rate for gradient descent of convex, L -smooth functions.

Lemma 4.4. For f convex and L -smooth $\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2 \leq 2\eta(f(\bar{x}) - f(x^{k+1}))$ where $\eta \in (0, \frac{1}{L}]$.

Proof. We want to show that a decay function φ satisfying $f(x^{(k)}) - f(\bar{x}) \propto \varphi(k)$ is "nicely" bounded.

First we start by noting that

$$\|x^{(k+1)} - \bar{x}\| = \langle x^{(k)} - \bar{x} - \eta \nabla f(x^{(k)}), x^{(k)} - \bar{x} - \eta \nabla f(x^{(k)}) \rangle = \|x^{(k)} - \bar{x}\|^2 - 2\eta \langle x^{(k)} - \bar{x}, \nabla f(x^{(k)}) \rangle + \eta^2 \|\nabla f(x^{(k)})\|^2$$

We then consider the bounds on each term. Note that since $\eta \leq \frac{1}{L}$ it follows that by the Descent Lemma, $\eta^2 \|\nabla f(x^{(k)})\|^2 \leq 2\eta(f(x^{(k)}) - f(x^{(k+1)}))$.

Furthermore, since f is convex, we have that

$$-2\eta \langle x^{(k)} - \bar{x}, \nabla f(x^{(k)}) \rangle = 2\eta \langle \bar{x} - x^{(k)}, \nabla f(x^{(k)}) \rangle = 2\eta(f(\bar{x}) - f(x^{(k)}))$$

Hence, combining these bounds, the result follows.

Now, taking the summation over all of the terms, and noticing that the sum is telescoping we find that

$$f(x^{(k+1)}) - f(\bar{x}) \leq \frac{1}{2\eta k} \|x^{(1)} - \bar{x}\|^2.$$

■

We note that since $\bar{x}$ is the minimizer we have $\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2 \leq 0$. We see that this is showing us decay in terms of distance. However, we note that looking at the above lemma, we see that this is not sufficient to yield the convergence rate.

Rearranging our expression we get that

$$f(x^{(k+1)}) - f(\bar{x}) \leq \frac{1}{2\eta} [\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2]$$

Considering the expression for each x^i and summing the terms, we have that

$$\sum_{i=1}^K [f(x^{(i)}) - f(\bar{x})] \leq \frac{1}{2\eta} [\|x^{(1)} - \bar{x}\|^2 - \|x^{(K)} - \bar{x}\|^2]$$

We see that the left side is the discretized version of the conditions for gradient flows and the right side remains

identical. Thus, we can apply the same ideas as we did previously to yield

$$\sum_{i=1}^K [f(x^{(i)}) - f(\bar{x})] \leq \frac{1}{2\eta} [\|x^{(1)} - \bar{x}\|^2 - \|x^K - \bar{x}\|^2] \leq \frac{1}{2\eta} \|x^1 - \bar{x}\|^2$$

Since $[f(x^{(i)}) - f(\bar{x})]$ is decreasing, it follows that

$$K[f(x^{(K)}) - f(\bar{x})] \leq \sum_{i=1}^K [f(x^{(i)}) - f(\bar{x})]$$

Thus, it follows that

$$f(x^{(K)}) - f(\bar{x}) \leq \frac{\|x^1 - \bar{x}\|^2}{2\eta K}$$

where we note that t is the step size and K is the number of iterations.

Hence, we see gradient descent applied to f has a convergence rate of $\mathcal{O}(\frac{1}{tK})$

4.2 Exercises

Exercise. In lecture (see also Lemma 4.3), we have shown two types of descent lemmas in terms of objective values, assuming f is convex and L -smooth. Let $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$.

- For any $\eta \in (0, \frac{2}{L})$, $f(x^{(k+1)}) < f(x^{(k)})$ provided with $\|\nabla f(x^{(k)})\| \neq 0$.
- When $\eta = \frac{1}{L}$, $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2L} \|\nabla f(x^{(k)})\|^2$.

In fact, one can show that for any $\eta \in (0, \frac{1}{L}]$, it holds $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$.

Give a second thought why one cannot extend this inequality beyond that interval, i.e., $[\frac{1}{L}, \frac{2}{L})$.

Proof. Drafted by Indhu Chandra, Proofread by Sahand Ahmadi

Recall that L -smoothness gives the following upper bound (the *descent lemma*):

$$f(x^{(k+1)}) \leq f(x^{(k)}) + \langle \nabla f(x^{(k)}), x^{(k+1)} - x^{(k)} \rangle + \frac{L}{2} \|x^{(k+1)} - x^{(k)}\|^2.$$

Substituting $x^{(k+1)} - x^{(k)} = -\eta \nabla f(x^{(k)})$, the right-hand side becomes

$$f(x^{(k)}) - \eta \|\nabla f(x^{(k)})\|^2 + \frac{L\eta^2}{2} \|\nabla f(x^{(k)})\|^2 = f(x^{(k)}) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x^{(k)})\|^2,$$

so we obtain the general bound (4.1)

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x^{(k)})\|^2.$$

For this to imply $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$, we need

$$\eta \left(1 - \frac{L\eta}{2}\right) \geq \frac{\eta}{2}.$$

Since $\eta > 0$, dividing both sides by η gives

$$1 - \frac{L\eta}{2} \geq \frac{1}{2} \iff \frac{L\eta}{2} \leq \frac{1}{2} \iff \eta \leq \frac{1}{L}.$$

Why the inequality fails on $(\frac{1}{L}, \frac{2}{L})$. For $\eta \in (\frac{1}{L}, \frac{2}{L})$ the coefficient $1 - \frac{L\eta}{2}$ lies in $(0, \frac{1}{2})$, so (4.1) still guarantees descent (since $\eta(1 - \frac{L\eta}{2}) > 0$), but the descent amount $\eta(1 - \frac{L\eta}{2})$ is *strictly less than* $\frac{\eta}{2}$. The sharper bound $\frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$ therefore no longer holds.

Concretely, at $\eta = \frac{1}{L}$ both sides of the required inequality are equal:

$$\eta \left(1 - \frac{L\eta}{2}\right) \Big|_{\eta=1/L} = \frac{1}{L} \cdot \frac{1}{2} = \frac{1}{2L} = \frac{\eta}{2} \Big|_{\eta=1/L}.$$

For any $\eta > \frac{1}{L}$, the left side drops below $\frac{\eta}{2}$, so the tighter bound $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$ is violated. This is why the interval cannot be extended beyond $\frac{1}{L}$. ■

4.3 More Examples

Example 4.3.1. (Proposed by Alexandra Aswani.) **The effect of conditioning on gradient descent.**

In Lecture 4, we proved that for a convex and L -smooth function, then gradient descent produces a sequence whose objective values decrease and converge to the global minimum. However, these results do not explain why some convex optimization problems are solved much faster than others in practice.

A key idea in optimization is that the geometry of the objective function can strongly affect the behavior of gradient descent. In particular, quadratic functions with elongated level sets often lead to slower convergence than functions with circular level sets. This phenomenon is closely related to the condition number of the Hessian matrix and is

discussed in Boyd and Vandenberghe's *Convex Optimization* (Chapter 9). To illustrate this, consider the two convex quadratic functions

$$f_1(x_1, x_2) = \frac{1}{2}(x_1^2 + x_2^2)$$

and

$$f_2(x_1, x_2) = \frac{1}{2}(x_1^2 + 100x_2^2).$$

Both functions have the same minimizer

$$\bar{x} = (0, 0),$$

and both satisfy the assumptions of Lemma 4.3 and Lemma 4.4. The Hessians are

$$\nabla^2 f_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$\nabla^2 f_2 = \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix}.$$

The condition numbers are therefore

$$\kappa(\nabla^2 f_1) = 1, \quad \kappa(\nabla^2 f_2) = 100.$$

Starting from the same initial point, for example

$$x^{(0)} = (1, 1),$$

and applying gradient descent with a fixed stepsize, the iterates for f_1 move directly toward the minimizer. In contrast, the iterates for f_2 tend to zig-zag across the narrow valley formed by its elongated level sets, resulting in slower convergence.

This example demonstrates that convexity and L -smoothness alone do not completely determine the practical performance of gradient descent. Although Lemma 4.3 guarantees descent and Lemma 4.4 guarantees convergence for both functions, the condition number of the Hessian can significantly influence how quickly the algorithm reaches the minimizer.

Reference. Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004, Chapter 9.

Example 4.3.2 (Proposed by Tao Xu). (Limitations of Gradient Descent) This example shows what can go wrong when gradient descent leaves the safe world of convex functions. Consider the simple non-convex function

$$f(x) = x^4 - 3x^2 + 1, \quad x \in \mathbb{R},$$

whose gradient is

$$\nabla f(x) = 4x^3 - 6x.$$

The function has two valleys and one hilltop, so the behavior of gradient descent can depend strongly on where we start and how large a step we take.

We run the gradient descent iteration

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}),$$

where $\eta > 0$ is the stepsize. Try several initial points, such as

$$x^{(0)} = -1.5, \quad x^{(0)} = 0.5,$$

and compare the results for different choices of η .

Here is the graph of $f(x)$:

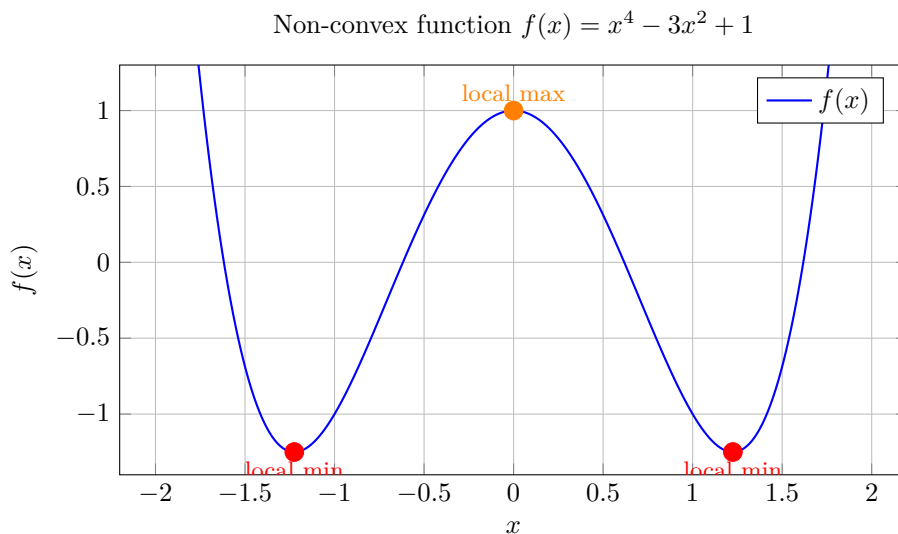


Figure 7: Graph of the non-convex function $f(x) = x^4 - 3x^2 + 1$. It has two local minima and one local maximum.

What do you observe? Does gradient descent always move to the same minimizer? Can a large stepsize make the iteration oscillate or diverge? How does this differ from the convex case, where the geometry is much more predictable?

Example 4.3.3. (Proposed by Peter Le). (Gradient Descent on Simple Quadratic) This example uses a simple function to illustrate properties of the Gradient Descent in an easy-to-understand way. Consider

$$f(x) = \frac{1}{2}x^2$$

By inspection, the minimizer is clearly

$$\bar{x} = 0$$

Now we check for the assumptions. The gradient and Hessian are

$$\nabla f(x) = x$$

$$\nabla^2 f(x) = 1$$

$f(x)$ is clearly convex, L -smooth, and continuously differentiable

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$$

$$|x - y| = 1|x - y|$$

So in this case $L = 1$.

Now we can explicitly apply Gradient Descent. Use $\eta = \frac{1}{2}$, then the each update is the following

$$x^{(k+1)} = x^{(k)} - \frac{1}{2}x^{(k)}$$

Let's take the initial point to be

$$x^{(0)}$$

Then

$x^{(0)} = 4$	$f(x^{(0)}) = 8$
$x^{(1)} = 2$	$f(x^{(0)}) = 2$
$x^{(2)} = 1$	$f(x^{(0)}) = 0.5$
$x^{(3)} = 0.5$	$f(x^{(0)}) = 0.125$
$x^{(4)} = 0.25$	$f(x^{(0)}) = 0.03125$
$x^{(5)} = 0.125$	$f(x^{(0)}) = 0.0078125$

Remark. We can observe that each iteration moves towards zero, accompanied by a decrease in $f(x)$ after each step. Which perfectly illustrates Lemma 4.3. Now we want to show $\|\nabla f\| = 0$ happens at our minimizer to indicate we have arrived at our solution.

Check

$$\nabla f(\bar{x}) = 0$$

$$\nabla f(0) = 0$$

Indeed, we have arrived at the theorized result. By Lemma 4.3, $\bar{x} = 0$ is our global minimizer.

5 Lecture 5 on Framework of Convergence Analysis and L -smoothness

Charles Yang Summarizes: This section explores an alternative proof for deriving the convergence rate of the gradient descent method with fixed step size.

Scribe Team: Sahand Ahmadi and Frank Reyes Felsner.

5.1 Lecture Notes

5.1.1 Convergence rate of fixed stepsize GD of convex and L -smooth function

Our goal is to derive a discrete EVI for gradient descent and then use it to obtain a convergence rate. Throughout, we assume that $\bar{x} \in \arg \min f$ and that the iterates are given by

$$x^{k+1} = x^k - t\eta \nabla f(x^k).$$

The roadmap for the discussion is

convex problem + gradient descent $\implies$ $\left. \begin{array}{l} \text{variant of EVI} \\ L\text{-smoothness (Alternative assumptions)} \end{array} \right\} \implies \text{oracle EVI} \implies \text{convergence rate}.$

The first derivation produces a *variant of EVI*, namely a distance-suboptimality inequality with an additional residual term involving $\|\nabla f(x^k)\|^2$. Smoothness then controls this residual term and upgrades the estimate to the *oracle EVI*, i.e. the one-step inequality that directly yields the rate after summation.

Step 1: Convexity relates suboptimality to a first-order term. By convexity,

$$f(x^k) - f(\bar{x}) \leq \langle \nabla f(x^k), x^k - \bar{x} \rangle.$$

We now rewrite the right-hand side in terms of squared distances. To do so, we use the identity

$$\langle u, v \rangle = \frac{1}{2} (\|u\|^2 + \|v\|^2 - \|u - v\|^2).$$

Since $x^{k+1} - x^k = -\eta \nabla f(x^k)$, we have $\nabla f(x^k) = \frac{1}{\eta}(x^k - x^{k+1})$. Hence

$$\begin{aligned} f(x^k) - f(\bar{x}) &\leq \langle \nabla f(x^k), x^k - \bar{x} \rangle \\ &= \left\langle \frac{1}{\eta}(x^k - x^{k+1}), x^k - \bar{x} \right\rangle \\ &= \frac{1}{2\eta} [\|x^k - x^{k+1}\|^2 + \|x^k - \bar{x}\|^2 - \|x^k - x^{k+1} - x^k + \bar{x}\|^2] \\ &= \frac{1}{2\eta} (\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2) + \frac{1}{2\eta} \|\eta \nabla f(x^k)\|^2. \end{aligned}$$

The first term represents distance decrease toward the minimizer, while the second term is a residual term. Thus we obtain the following *variant of EVI*:

$$f(x^k) - f(\bar{x}) \leq \frac{1}{2\eta} (\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2) + \frac{\eta}{2} \|\nabla f(x^k)\|^2. \quad (\text{Convex GD-EVI}) \quad \text{eq:GD-EVI}$$

Step 2: L -smoothness gives a descent estimate for function values. Recall that if f is L -smooth, then

$$f(y) \leq \underbrace{f(x) + \langle \nabla f(x), y - x \rangle}_{\text{tangent plane / first-order model}} + \underbrace{\frac{L}{2} \|y - x\|^2}_{\text{quadratic correction for curvature}}.$$

In other words, f lies below its tangent plane plus a quadratic error term, as shown in Figure 8. This geometric picture shows that the tangent approximation alone may underestimate the function value, so the quadratic correction term is needed to control the curvature.

This implies the descent lemma for function values. Indeed, applying this inequality with $x = x^k$ and $y = x^{k+1} = x^k - \eta \nabla f(x^k)$ gives

$$y - x = -\eta \nabla f(x^k).$$

Therefore,

$$\begin{aligned} f(x^{k+1}) &\leq f(x^k) + \langle \nabla f(x^k), -\eta \nabla f(x^k) \rangle + \frac{L}{2} \|\eta \nabla f(x^k)\|^2 \\ &= f(x^k) - \eta \|\nabla f(x^k)\|^2 + \frac{L\eta^2}{2} \|\nabla f(x^k)\|^2 \\ &= f(x^k) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x^k)\|^2. \end{aligned}$$

Please also refer to Lemma 4.3 when picking $\eta \in (0, \frac{1}{L}]$.

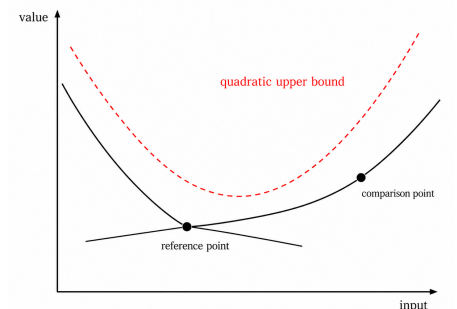


Figure 8: *Jasper Luo creates:* Geometric illustration of L -smoothness.

Step 3: Control the residual term. Please see the exercise below to control the residual term, when assuming f is L -Lipschitz. For the convergence-rate proof when assuming f is L -smooth, however, we use the stronger smoothness estimate obtained in Step 2:

$$\frac{\eta}{2} \|\nabla f(x^k)\|^2 \leq f(x^k) - f(x^{k+1}).$$

Using L -smoothness, the variant of EVI upgrades to the oracle EVI:

$$f(x^k) - f(\bar{x}) \leq \frac{1}{2\eta} \left(\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2 \right) + f(x^k) - f(x^{k+1}).$$

Canceling $f(x^k)$ from both sides yields the *oracle EVI*:

$$f(x^{k+1}) - f(\bar{x}) \leq \frac{1}{2\eta} \left(\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2 \right).$$

This one-step oracle inequality is the only input needed for the telescoping argument, and therefore serves as the bridge from the algorithm to the convergence rate. Rewriting it gives

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq f(\bar{x}) - f(x^{k+1}).$$

This may be interpreted as saying that the Forward Euler scheme of squared distance to $\bar{x}$ is controlled by the function-value gap.

Taking $\eta \rightarrow 0$, we recover the gradient flow inequality (**GF EVI**)

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)) \leq 0.$$

Thus in continuous time, the squared distance to $\bar{x}$ is nonincreasing, and the EVI explains how this distance decay is tied to objective suboptimality.

Step 4: Use the oracle EVI to obtain the convergence rate. Summing the oracle EVI over $i = 0, \dots, k$ gives

$$\begin{aligned} \sum_{i=0}^k (f(x^{i+1}) - f(\bar{x})) &\leq \frac{1}{2\eta} \sum_{i=0}^k \left(\|x^i - \bar{x}\|^2 - \|x^{i+1} - \bar{x}\|^2 \right) \\ &= \frac{\|x^0 - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2}{2\eta} \\ &\leq \frac{\|x^0 - \bar{x}\|^2}{2\eta}. \end{aligned}$$

Since $f(x^{i+1})$ is nonincreasing, it follows that

$$(k+1)(f(x^{k+1}) - f(\bar{x})) \leq \sum_{i=0}^k (f(x^{i+1}) - f(\bar{x})),$$

and hence

$$f(x^{k+1}) - f(\bar{x}) \leq \frac{\|x^0 - \bar{x}\|^2}{2\eta(k+1)}.$$

Therefore,

$$f(x^k) - f(\bar{x}) = \mathcal{O}\left(\frac{1}{k}\right).$$

Remark. The takeaway is that EVI combines distance and functional values. In particular, EVI is essential for the convergence rate.

Theorem 5.1 (Convergence rate for fixed stepsize GD of convex and L -smooth function). Assume f is convex and L -smooth, for any stepsize $\eta \in (0, \frac{1}{L}]$, the fixed stepsize gradient descent scheme $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$ has the convergence rate $\mathcal{O}(\frac{1}{k})$, more precisely,

$$f(x^{(k)}) - f(\bar{x}) \leq \frac{\|x^0 - \bar{x}\|^2}{2\eta k}.$$

5.1.2 L -Smoothness, Equivalent Characterizations, and Basic Consequences

Our goal is to explain the meaning of L -smoothness, prove its equivalent formulations, and record the consequences that will later be used in the analysis of gradient descent. Throughout, the main point is that L -smoothness controls the curvature of f : the function cannot bend upward too sharply, and its first-order approximation incurs at most a quadratic error. This differs L -Lipschitz gradient Proposition 4.1.1 which controls both sides of curvatures.

The overall structure is

definition of L -smooth $\iff$ directional gradient inequality

$\implies$ useful estimates at a stationary point $\implies$ connection with L -Lipschitz gradient.

For convex differentiable functions, this should be viewed together with the first-order lower bound from convexity and the monotonicity of the gradient map. In that sense, convexity gives the lower side of the first-order model, while smoothness gives the upper side. In this sense, L -smoothness is equivalent to L -Lipschitz gradient for any convex functions.

Definition 5.2. We say that f is L -smooth ($L > 0$ is a good setting for convex functions) if

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2$$

and this is equivalent to

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L \|y - x\|^2.$$

The first inequality says that the tangent plane at x , with a quadratic correction term measuring curvature, gives an upper model for $f(y)$. In other words, the graph of f stays below its first-order Taylor approximation plus an error of size at most $\frac{L}{2} \|y - x\|^2$. The second inequality says that the gradient cannot vary too rapidly in the direction $y - x$. These two statements encode the same regularity condition.

For comparison, recall a similar equivalence for convexity:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \iff \langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0.$$

The first inequality says that for a convex differentiable function, the tangent plane is a global lower bound. The second says that the gradient map is monotone. Thus convexity provides a lower first-order estimate, while L -smoothness provides an upper first-order estimate with quadratic correction.

Proof. Assume

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

To derive the second formulation, we write the same inequality with x and y exchanged. This is natural because the desired conclusion involves the symmetric quantity $\nabla f(y) - \nabla f(x)$.

Swapping x and y ,

$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2} \|x - y\|^2 = f(y) - \langle \nabla f(y), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Adding the two inequalities gives^a

$$0 \leq \langle \nabla f(x) - \nabla f(y), y - x \rangle + L \|y - x\|^2,$$

hence

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L \|y - x\|^2.$$

■

^aThe key point is that after adding the two estimates, the function values cancel and only the gradient difference remains. Therefore the upper Taylor bound directly yields a control on how much the gradient can change along the direction joining x and y .

Proof. Assume

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L \|y - x\|^2 \quad \forall x, y.$$

We now recover the usual L -smoothness inequality. The standard idea is to reduce the problem to one variable by restricting f to the line segment joining x and y . Fix x, y and define

$$\phi(t) := f(x + t(y - x)), \quad t \in [0, 1].$$

Then by the chain rule,

$$\phi'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle.$$

Applying the assumption with y replaced by $x + t(y - x)$,

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), t(y - x) \rangle \leq Lt^2 \|y - x\|^2.$$

For $t > 0$, divide by t :

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \leq Lt \|y - x\|^2.$$

Thus

$$\phi'(t) - \phi'(0) \leq Lt \|y - x\|^2.$$

Integrating from 0 to 1,

$$\phi(1) - \phi(0) - \phi'(0) \leq \int_0^1 Lt \|y - x\|^2 dt = \frac{L}{2} \|y - x\|^2.$$

Since

$$\phi(1) = f(y), \quad \phi(0) = f(x), \quad \phi'(0) = \langle \nabla f(x), y - x \rangle,$$

we obtain

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

So the directional bound on the gradient contains exactly the information needed to recover the upper Taylor estimate. This proves the equivalence between the two formulations of L -smoothness. ■

We now record some important consequences that follow from the smoothness inequality.

Theorem 5.3 (Property for minimizers). Suppose that $\bar{x}$ is a minimizer, i.e., $f(\bar{x}) = 0$ and $f(\bar{x}) \leq f(y)$ for any y . Then

$$f(y) \leq f(\bar{x}) + \frac{L}{2} \|y - \bar{x}\|^2.$$

In particular for the GD scheme,

$$f(x^{k+1}) \leq f(x^k) - \frac{t}{2} \|\nabla f(x^k)\|^2, \text{ for } t \in \left(0, \frac{1}{L}\right].$$

Consequently, a common inequality $f(\bar{x}) \leq f(x^{k+1}) \leq f(x^k) - \frac{1}{2L} \|\nabla f(x^k)\|^2$ for the choice $t = \frac{1}{L}$. Taking $x^k = y$, this yields

$$f(y) \geq f(x^{k+1}) + \frac{1}{2L} \|\nabla f(y)\|^2 \geq f(\bar{x}) + \frac{1}{2L} \|\nabla f(y)\|^2,$$

and therefore

$$\frac{1}{2L} \|\nabla f(y)\|^2 \leq f(y) - f(\bar{x}) \leq \frac{L}{2} \|y - \bar{x}\|^2. \quad \text{eq:L-smooth-minimizer (5.1)}$$

This theorem essentially is a result from Subsubsection 4.1.3. The inequality (5.1) is especially useful because it connects three quantities: gradient size, objective gap, and distance to the stationary point. The left inequality says that suboptimality dominates the squared gradient norm, while the right inequality says that suboptimality is itself controlled by squared distance to $\bar{x}$.

The definition Definition 4.2 for a function whose gradient is L -Lipschitz, speaks of “two sides of L -smoothness”. When f is convex and L -smooth,

$$0 \leq f(y) - [f(x) + \langle \nabla f(x), y - x \rangle] \leq \frac{L}{2} \|y - x\|^2.$$

The left inequality is the convexity lower bound which is stronger than the one derived from L -Lipschitz gradient, and the right inequality is the smoothness upper bound.

Theorem 5.4.

gradient is L -Lip $\implies L$ -smooth.

Moreover, if f is convex, then

gradient is L -Lip $\iff L$ -smooth.

Proof. Here we provide an alternative proof based on their alternative definitions. To prove the forward implication, assume f is gradient L -Lip, then

$$\begin{aligned} \langle \nabla f(y) - \nabla f(x), y - x \rangle &\leq \|\nabla f(y) - \nabla f(x)\| \|y - x\| \\ &\leq L \|y - x\| \|y - x\| = L \|y - x\|^2. \end{aligned}$$

The first inequality is Cauchy–Schwarz, and the second uses the L -Lipschitz property of the gradient. Hence we recover

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L \|y - x\|^2,$$

which is equivalent to L -smoothness by the result proved earlier. $\blacksquare$

Remark. The main conclusion is that L -smoothness can be understood from several compatible viewpoints: as an upper Taylor bound, as a directional control on the gradient, and, in the convex setting, as one half of the two-sided first-order estimate that governs the geometry of gradient descent.

5.2 Exercises

Exercise. The framework of convergence analysis can be applied onto many different settings.

Assume that function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, differentiable and L -Lipschitz. That is $\|\nabla f(x)\| \leq L$.

Let $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$. Show that

$$f\left(\frac{1}{K} \sum_{i=1}^K x^{(i)}\right) - f(\bar{x}) \leq \frac{\|x_1 - \bar{x}\|^2}{2\eta K} + \frac{\eta L^2}{2}.$$

Hint: for convex functions, the Jensen’s inequality implies that $f(\frac{1}{K} \sum_{i=1}^K x^{(i)}) \leq \frac{1}{K} \sum_{i=1}^K f(x^{(i)})$.

Question: why don’t you have the simple $f(x^{(K)})$ on the left-hand side in the convergence rate?

Proof. Drafted by Sahand Ahmadi, Proofread by Alexandra Aswani

Define the averaged iterate

$$\bar{x}^{(K)} := \frac{1}{K} \sum_{i=1}^K x^{(i)}.$$

We start from the gradient descent update

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}).$$

For any minimizer $\bar{x} \in \arg \min f$, expand the squared distance:

$$\begin{aligned} \|x^{(k+1)} - \bar{x}\|^2 &= \|x^{(k)} - \eta \nabla f(x^{(k)}) - \bar{x}\|^2 \\ &= \|x^{(k)} - \bar{x}\|^2 - 2\eta \langle \nabla f(x^{(k)}), x^{(k)} - \bar{x} \rangle + \eta^2 \|\nabla f(x^{(k)})\|^2. \end{aligned}$$

Rearranging the equation gives

$$2\eta \langle \nabla f(x^{(k)}), x^{(k)} - \bar{x} \rangle = \|x^{(k)} - \bar{x}\|^2 - \|x^{(k+1)} - \bar{x}\|^2 + \eta^2 \|\nabla f(x^{(k)})\|^2.$$

Since f is convex and differentiable,

$$f(x^{(k)}) - f(\bar{x}) \leq \langle \nabla f(x^{(k)}), x^{(k)} - \bar{x} \rangle.$$

Therefore,

$$2\eta (f(x^{(k)}) - f(\bar{x})) \leq \|x^{(k)} - \bar{x}\|^2 - \|x^{(k+1)} - \bar{x}\|^2 + \eta^2 \|\nabla f(x^{(k)})\|^2.$$

Using the assumption that f is L -Lipschitz, i.e.

$$\|\nabla f(x^{(k)})\| \leq L,$$

we obtain

$$2\eta (f(x^{(k)}) - f(\bar{x})) \leq \|x^{(k)} - \bar{x}\|^2 - \|x^{(k+1)} - \bar{x}\|^2 + \eta^2 L^2.$$

Summing from $k = 1$ to K yields

$$\begin{aligned} 2\eta \sum_{k=1}^K (f(x^{(k)}) - f(\bar{x})) &\leq \sum_{k=1}^K (\|x^{(k)} - \bar{x}\|^2 - \|x^{(k+1)} - \bar{x}\|^2) + \sum_{k=1}^K \eta^2 L^2 \\ &= \|x^{(1)} - \bar{x}\|^2 - \|x^{(K+1)} - \bar{x}\|^2 + K\eta^2 L^2 \\ &\leq \|x^{(1)} - \bar{x}\|^2 + K\eta^2 L^2. \end{aligned}$$

Divide by $2\eta K$:

$$\frac{1}{K} \sum_{k=1}^K (f(x^{(k)}) - f(\bar{x})) \leq \frac{\|x^{(1)} - \bar{x}\|^2}{2\eta K} + \frac{\eta L^2}{2}.$$

Now apply Jensen's inequality for the convex function f :

$$f\left(\frac{1}{K} \sum_{k=1}^K x^{(k)}\right) \leq \frac{1}{K} \sum_{k=1}^K f(x^{(k)}).$$

Hence

$$\begin{aligned} f(\bar{x}^{(K)}) - f(\bar{x}) &= f\left(\frac{1}{K} \sum_{k=1}^K x^{(k)}\right) - f(\bar{x}) \\ &\leq \frac{1}{K} \sum_{k=1}^K f(x^{(k)}) - f(\bar{x}) \\ &= \frac{1}{K} \sum_{k=1}^K (f(x^{(k)}) - f(\bar{x})) \\ &\leq \frac{\|x^{(1)} - \bar{x}\|^2}{2\eta K} + \frac{\eta L^2}{2}. \end{aligned}$$

This proves

$$f\left(\frac{1}{K} \sum_{i=1}^K x^{(i)}\right) - f(\bar{x}) \leq \frac{\|x^{(1)} - \bar{x}\|^2}{2\eta K} + \frac{\eta L^2}{2}.$$

■

Why do we not get a simple $f(x^{(K)})$ on the left-hand side? Because here we only assume that f is convex and L -Lipschitz, not L -smooth. Under these weaker assumptions, the individual function values $f(x^{(k)})$ don't need to decrease monotonically along the gradient descent iterates, so we cannot directly bound the last iterate $f(x^{(K)})$. What the proof controls is the *average* of the function values,

$$\frac{1}{K} \sum_{k=1}^K f(x^{(k)}),$$

and then Jensen's inequality transfers this bound to the function value at the averaged iterate

$$\frac{1}{K} \sum_{k=1}^K x^{(k)}.$$

So in the nonsmooth/Lipschitz setting, the natural convergence guarantee is for the *ergodic average*, not necessarily for the last iterate.

5.3 More Examples

Example 5.3.1 (Proposed by Jasper Luo, A quadratic example comparing GD with convex smooth and strongly-convex rates). This example gives a concrete visualization of how the stepsize affects gradient descent for a convex and L -smooth function. It also compares the observed convergence with the rate estimate in Theorem 5.1 and with the faster rate available when the function is additionally strongly convex.

Consider the simple convex function

$$f(x) = \frac{1}{2}x^2, \quad x \in \mathbb{R}.$$

Then

$$\nabla f(x) = x, \quad f''(x) = 1.$$

Therefore, f is convex and 1-smooth. Hence $L = 1$, and the unique minimizer is

$$\bar{x} = 0.$$

Moreover, since $f''(x) = 1$, the function is also 1-strongly convex. Thus in this example we have

$$L = 1, \quad \mu = 1.$$

The gradient descent iteration is

$$x^{k+1} = x^k - \eta \nabla f(x^k) = x^k - \eta x^k = (1 - \eta)x^k.$$

Thus

$$x^k = (1 - \eta)^k x^0.$$

Since $f(\bar{x}) = 0$, the function-value gap is

$$f(x^k) - f(\bar{x}) = \frac{1}{2}(x^k)^2 = \frac{1}{2}(1 - \eta)^{2k}(x^0)^2.$$

The distance to the minimizer is

$$\|x^k - \bar{x}\| = |x^k| = |1 - \eta|^k |x^0|.$$

Now take $x^0 = 2$. Theorem 5.1 gives, for every $\eta \in (0, 1/L] = (0, 1]$,

$$f(x^k) - f(\bar{x}) \leq \frac{\|x^0 - \bar{x}\|^2}{2\eta k} = \frac{2}{\eta k}.$$

Therefore, for the two reference stepsizes,

$$\eta = 0.5 : \quad f(x^k) - f(\bar{x}) \leq \frac{4}{k},$$

and

$$\eta = 1 : \quad f(x^k) - f(\bar{x}) \leq \frac{2}{k}.$$

Because here $f(x) = \frac{1}{2}x^2$, this function-value estimate also gives a corresponding distance estimate:

$$\|x^k - \bar{x}\| \leq \frac{\|x^0 - \bar{x}\|}{\sqrt{\eta k}} = \frac{2}{\sqrt{\eta k}}.$$

Thus the distance reference curves coming from Theorem 5.1 are

$$\eta = 0.5 : \quad \|x^k - \bar{x}\| \leq \frac{2}{\sqrt{0.5k}},$$

and

$$\eta = 1 : \quad \|x^k - \bar{x}\| \leq \frac{2}{\sqrt{k}}.$$

However, this quadratic has more structure than just convexity and L -smoothness. Since it is also μ -strongly convex with $\mu = 1$, we can compare the observed decay with the faster theoretical rate for μ -strongly convex and L -smooth functions. For $\eta \in (0, 1/L]$, this rate has the geometric form

$$f(x^k) - f(\bar{x}) \leq (1 - \eta\mu)^k (f(x^0) - f(\bar{x})).$$

In the present example, $x^0 = 2$ and

$$f(x^0) - f(\bar{x}) = 2.$$

Therefore,

$$\eta = 0.5 : \quad f(x^k) - f(\bar{x}) \leq 2(0.5)^k.$$

For $\eta = 1$, the strongly-convex estimate gives zero after one step, which agrees with the exact iteration:

$$x^1 = (1 - 1)x^0 = 0.$$

For the distance to the minimizer, the corresponding geometric reference for this quadratic example is

$$\|x^k - \bar{x}\| \leq |1 - \eta\mu|^k \|x^0 - \bar{x}\|.$$

Since $\mu = 1$ and $x^0 = 2$, this gives

$$\eta = 0.5 : \quad \|x^k - \bar{x}\| \leq 2(0.5)^k,$$

and for $\eta = 1$ the distance becomes zero after one step.

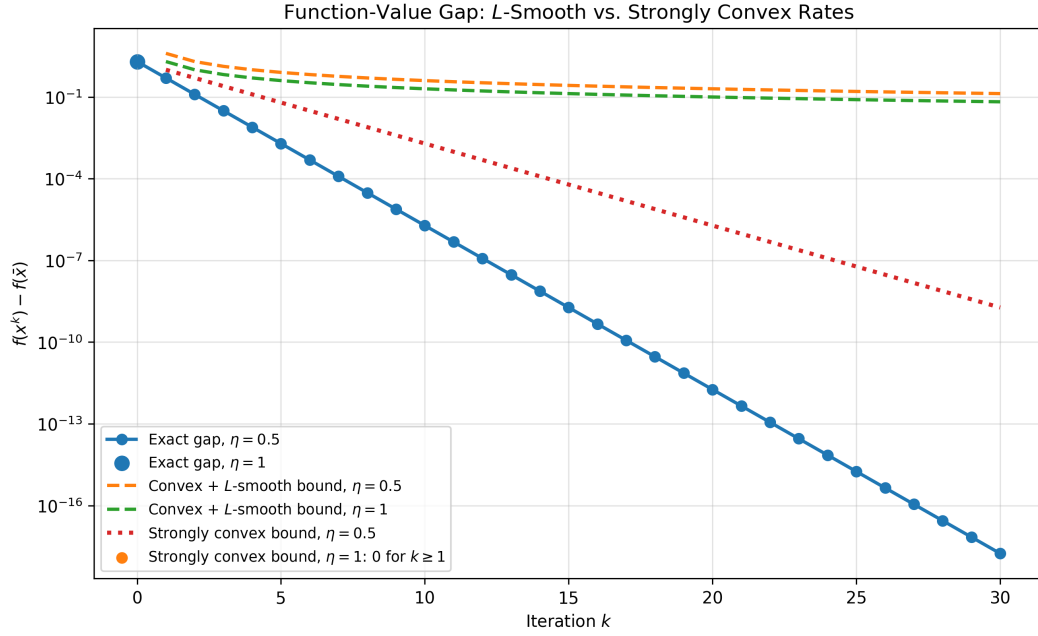


Figure 9: *Jasper Luo creates*: Function-value convergence for gradient descent on $f(x) = \frac{1}{2}x^2$ from $x^0 = 2$. The solid curve shows the exact gap for $\eta = 0.5$, and the marker at $k = 0$ records the $\eta = 1$ case, which reaches the minimizer after one step. The dashed curves show the convex and L -smooth Theorem 5.1 reference bounds $\|x^0 - \bar{x}\|^2/(2\eta k)$ for $\eta = 0.5$ and $\eta = 1$. The dotted curve shows the faster strongly-convex reference bound $(1 - \eta\mu)^k(f(x^0) - f(\bar{x}))$ for $\eta = 0.5$, with $\mu = 1$.

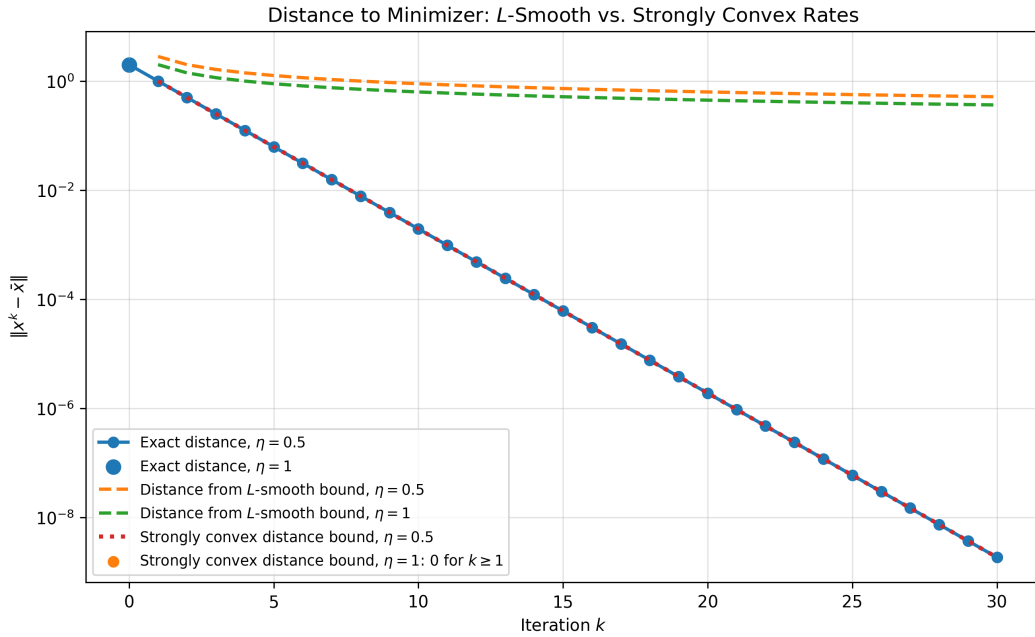


Figure 10: *Jasper Luo creates*: Distance-to-minimizer convergence for the same quadratic example. The solid curve shows the exact distance for $\eta = 0.5$, and the marker at $k = 0$ records the $\eta = 1$ case, which reaches the minimizer after one step. The dashed curves show the distance references derived from the convex and L -smooth Theorem 5.1 bound, while the dotted curve shows the faster strongly-convex distance reference $|1 - \eta\mu|^k\|x^0 - \bar{x}\|$ for $\eta = 0.5$, with $\mu = 1$.

This example shows that the rate in Theorem 5.1 is a general upper bound, not necessarily a tight description of the exact behavior of a particular quadratic. For $\eta = 0.5$, the exact function-value error decays as

$$f(x^k) - f(\bar{x}) = 2 \cdot 0.25^k,$$

which is much faster than the theorem's $\mathcal{O}(1/k)$ reference bound. The strongly-convex reference bound

$$2(0.5)^k$$

is also geometric, so it captures the faster decay regime better than the convex and L -smooth estimate alone, although it is still an upper bound rather than the exact equality. For $\eta = 1$, the method reaches the minimizer in one step:

$$x^1 = 0.$$

Thus the exact error is zero for all $k \geq 1$, while Theorem 5.1 still gives the valid but conservative bound $2/k$. This also explains why the lecture uses the safe stepsize condition

$$\eta \in \left(0, \frac{1}{L}\right].$$

For this particular quadratic function, convergence actually holds for the larger range $0 < \eta < 2/L$. However, the condition $\eta \leq 1/L$ is a clean and safe assumption that guarantees the descent estimate used in the convergence proof:

$$f(x^{k+1}) \leq f(x^k) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x^k)\|^2.$$

Takeaways. The L -smoothness constant controls the upper curvature of the objective, while strong convexity provides additional curvature from below. Theorem 5.1 gives a universal $\mathcal{O}(1/k)$ upper bound for convex and L -smooth functions, but this quadratic example is also strongly convex, so a faster geometric convergence rate is available. The plots compare these two theoretical descriptions with the exact behavior of gradient descent.

6 Lecture 6 on L -smoothness and m -strongly convexity

Alexandra Aswani Summarizes:: This lecture explains how L -smoothness, a bound on how fast gradients change, connects to different definitions of convexity.

Scribe Team: Charles Yang, Jacob Jurek.

6.1 Lecture Notes

In this section, we establish a connection between L -smoothness and m -strongly convexity, then link it back to previously learned definitions of convexity.

Definition 6.1. If gradient ∇f is L -Lipshitz, then

$$\|\nabla f(y) - \nabla f(x)\| \leq L\|x - y\| \Leftrightarrow |f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2}\|x - y\|^2$$

Remark. Applying the Cauchy-Schwarz inequality to the Lipschitz bound gives the 1.5-*th order* smoothness inequality:

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L\|y - x\|^2.$$

Separately, the Lipschitz condition is equivalent to the quadratic bound

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2}\|y - x\|^2.$$

6.1.1 First-Order Form

Definition 6.2 (L -smooth, first-order form). A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth if

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2 \quad \forall x, y,$$

or equivalently, $\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L\|y - x\|^2$ for all x, y .

Remark. A key bridge to the zeroth-order form: f is L -smooth if and only if the map

$$g(y) := \frac{L}{2}\|y\|^2 - f(y)$$

is *convex*. This lets us transfer every convexity characterization to a smoothness characterization by applying it to g .

6.1.2 Second-Order Form

Definition 6.3 (L -smooth, second-order form). If f is twice differentiable, then f is L -smooth if and only if

$$LI - \nabla^2 f(y) \succeq 0 \quad \forall y,$$

i.e. $LI - \nabla^2 f(y)$ is positive semidefinite, or equivalently

$$\lambda_{\max}(\nabla^2 f(y)) \leq L \quad \forall y.$$

Remark. The largest eigenvalue of the Hessian is uniformly bounded above by L . Compare with m -strong convexity, where the smallest eigenvalue is bounded *below* by m (see Section ??).

6.1.3 Zeroth-Order Form

We derive the zeroth-order form by exploiting the equivalence “ f is L -smooth $\Leftrightarrow g(y) = \frac{L}{2}\|y\|^2 - f(y)$ is convex.”

Derivation. Define $g(y) = \frac{L}{2}\|y\|^2 - f(y)$, so that $\nabla g(y) = Ly - \nabla f(y)$.

Step 1: Smoothness implies convexity of g .

Starting from the 1.5-*th order* smoothness inequality and rearranging:

$$\begin{aligned} \langle \nabla f(y) - \nabla f(x), y - x \rangle &\leq L\|y - x\|^2 = \langle L(y - x), y - x \rangle \\ \Rightarrow \langle [Ly - \nabla f(y)] - [Lx - \nabla f(x)], y - x \rangle &\geq 0 \\ \Rightarrow \langle \nabla g(y) - \nabla g(x), y - x \rangle &\geq 0, \end{aligned}$$

which is the 1.5-*th order* condition for convexity of g .

Step 2: Apply zeroth-order convexity to g .

Since g is convex, $g((1-t)x + ty) \leq (1-t)g(x) + tg(y)$. Substituting the definition of g :

$$\frac{L}{2}\|(1-t)x + ty\|^2 - f((1-t)x + ty) \leq (1-t)\left[\frac{L}{2}\|x\|^2 - f(x)\right] + t\left[\frac{L}{2}\|y\|^2 - f(y)\right].$$

Step 3: Rearrange.

Using the identity $\|(1-t)x + ty\|^2 = (1-t)\|x\|^2 + t\|y\|^2 - t(1-t)\|x - y\|^2$, the $\frac{L}{2}(\dots)$ terms simplify and we obtain:

$$\begin{aligned} f((1-t)x + ty) &\geq (1-t)f(x) + tf(y) + \frac{L}{2} \left[\|(1-t)x + ty\|^2 - (1-t)\|x\|^2 - t\|y\|^2 \right] \\ &\geq (1-t)f(x) + tf(y) - L \frac{t(1-t)}{2} \|x - y\|^2. \end{aligned}$$

■

Definition 6.4 (L -smooth, zeroth-order form). f is L -smooth if and only if for all x, y and all $t \in [0, 1]$,

$$f((1-t)x + ty) \geq (1-t)f(x) + tf(y) - L \frac{t(1-t)}{2} \|x - y\|^2.$$

Remark. Convexity gives an *upper* bound on f along a chord (Jensen's inequality); L -smoothness gives a matching *lower* bound, up to a quadratic error controlled by L . Together they “sandwich” f between two parabolas.

6.1.4 L -smoothness on a geodesic space.

The zeroth-order form is the only one of the four definitions that makes sense without a linear structure, since it only references f -values along a curve. This leads naturally to a generalization to non-Euclidean settings.

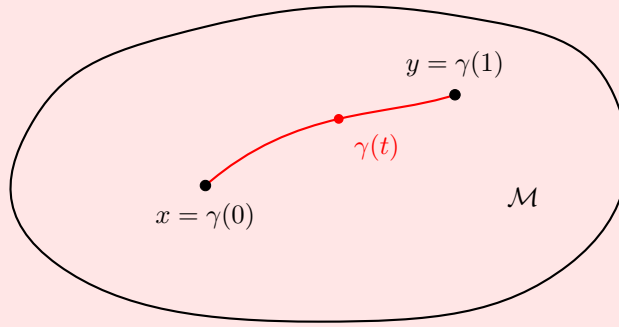


Figure 11: A geodesic $\gamma : [0, 1] \rightarrow \mathcal{M}$ connecting x and y .

Definition 6.5 (L -smooth along geodesics). Let $\mathcal{M}$ be a geodesic space and let $\gamma : [0, 1] \rightarrow \mathcal{M}$ be a constant-speed geodesic with $\gamma(0) = x$ and $\gamma(1) = y$. A function $f : \mathcal{M} \rightarrow \mathbb{R}$ is L -smooth along geodesics if

$$f(\gamma(t)) \geq (1-t)f(x) + tf(y) - L \frac{t(1-t)}{2} d^2(x, y)$$

for all $x, y \in \mathcal{M}$ and all $t \in [0, 1]$.

Example 6.1.1. Let $\mathcal{M}$ be the “space of poems” (with some metric $d(A_1, A_2)$ making it a geodesic space). Consider two lines from Tagore,

$$\begin{aligned} A_1 &= \text{“To see a world in a grain of sand”,} \\ A_2 &= \text{“Hold infinity in the palm of your hand”,} \end{aligned}$$

and let $f : \text{poem} \mapsto \# \text{ of bookmarks}$, with $f(A_1) = 10\text{k}$ and $f(A_2) = 15\text{k}$. One can then ask whether this f is L -smooth along geodesics on the space of poems — an illustration that the geodesic definition can be considered on very general spaces, not only manifolds.

Remark. The four definitions of L -smoothness (Lipschitz gradient; first-order quadratic upper bound; second-order Hessian bound; zeroth-order reverse-Jensen inequality) are mutually equivalent in the Euclidean setting. Only the zeroth-order form survives to geodesic spaces, which is why it becomes the “primary” definition in non-Euclidean optimization.

6.1.5 m -Strong Convexity and Duality with L -Smoothness

Definition 6.6 (m -strongly convex, $m > 0$). A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is m -strongly convex if

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq m \|y - x\|^2 > 0 \quad \forall x, y.$$

Equivalently, any of the following hold:

- (i) **First-order:** $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2$.
- (ii) **Second-order (if f twice differentiable):** $\nabla^2 f(y) \succeq mI$, i.e. $\lambda_{\min}(\nabla^2 f(y)) \geq m$ for all y .
- (iii) **Zeroth-order:** $y \mapsto f(y) - \frac{m}{2} \|y\|^2$ is convex.
- (iv) **Chord bound:** $f((1-t)x + ty) \leq (1-t)f(x) + tf(y) - m \frac{t(1-t)}{2} d^2(x, y)$.

Remark. The two notions are *mirror images*: every inequality sign is flipped. The highlighted $\geq$ (strong convexity) and $\leq$ (smoothness) are the only structural difference.

	m -strongly convex	L -smooth
1.5th-order	$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq m \ y - x\ ^2$	$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L \ y - x\ ^2$
1st-order	$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \ y - x\ ^2$	$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \ y - x\ ^2$
2nd-order	$\lambda_{\min}(\nabla^2 f) \geq m$	$\lambda_{\max}(\nabla^2 f) \leq L$
0th-order	$f(y) - \frac{m}{2} \ y\ ^2$ convex	$\frac{L}{2} \ y\ ^2 - f(y)$ convex

6.1.6 Equivalence of the 1.5th-Order and 1st-Order Definitions

Theorem 6.7. The following statements are equivalent:

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq m \|y - x\|^2,$$

and

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2.$$

Proof.

($\Leftarrow$)

Assume

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2.$$

Swapping x and y gives

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{m}{2} \|y - x\|^2.$$

Adding the two inequalities,

$$0 \geq \langle \nabla f(x), y - x \rangle + \langle \nabla f(y), x - y \rangle + m \|y - x\|^2.$$

Since

$$\langle \nabla f(y), x - y \rangle = -\langle \nabla f(y), y - x \rangle,$$

we obtain

$$0 \geq -\langle \nabla f(y) - \nabla f(x), y - x \rangle + m \|y - x\|^2,$$

which implies

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq m \|y - x\|^2.$$

($\Rightarrow$)

Assume

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq m \|y - x\|^2.$$

Write

$$(1 - t)x + ty = x + t(y - x),$$

and define

$$g(t) = f(x + t(y - x)).$$

Then

$$g'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle.$$

Applying strong convexity to the pair

$$x, \quad x + t(y - x),$$

gives

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), t(y - x) \rangle \geq mt^2 \|y - x\|^2.$$

Dividing by $t > 0$,

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \geq mt\|y - x\|^2,$$

or equivalently,

$$g'(t) \geq g'(0) + mt\|y - x\|^2.$$

Integrating from 0 to 1,

$$\begin{aligned} g(1) - g(0) &= \int_0^1 g'(t) dt \\ &\geq \int_0^1 (g'(0) + mt\|y - x\|^2) dt \\ &= g'(0) + \frac{m}{2}\|y - x\|^2. \end{aligned}$$

Substituting

$$g(1) = f(y), \quad g(0) = f(x), \quad g'(0) = \langle \nabla f(x), y - x \rangle,$$

yields

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2}\|y - x\|^2. \quad \blacksquare$$

Remark. The proof is exactly analogous to the corresponding equivalence proof for convexity and L -smoothness. The only difference is the additional quadratic term involving m .

6.1.7 Convergence Rate of Gradient Flow

We first introduce Grönwall's Lemma, which is necessary to derive the convergence rate of gradient flow under m -strongly convexity.

Lemma 6.8 (Grönwall). Suppose $u : [0, \infty) \rightarrow \mathbb{R}$ is differentiable and satisfies

$$u'(t) \leq -au(t)$$

for some constant $a > 0$. Then

$$u(t) \leq u(0)e^{-at} \quad \forall t \geq 0.$$

Proof. Define

$$w(t) = e^{at}u(t).$$

Differentiating,

$$\begin{aligned} w'(t) &= ae^{at}u(t) + e^{at}u'(t) \\ &= e^{at}(u'(t) + au(t)). \end{aligned}$$

Since

$$u'(t) \leq -au(t),$$

we have

$$u'(t) + au(t) \leq 0.$$

Therefore

$$w'(t) \leq 0,$$

which means $w(t)$ is non-increasing. Hence

$$w(t) \leq w(0).$$

Substituting back,

$$e^{at}u(t) \leq u(0),$$

and multiplying both sides by e^{-at} yields

$$u(t) \leq u(0)e^{-at}. \quad \blacksquare$$

Remark. The equality case

$$u'(t) = -au(t)$$

has solution

$$u(t) = u(0)e^{-at},$$

so Grönwall's Lemma states that any function satisfying

$$u'(t) \leq -au(t)$$

must decay at least as fast as the corresponding exponential.

Now consider the gradient flow differential equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Assume that f is m -strongly convex and let

$$\bar{x} = \arg \min_x f(x).$$

We study the evolution of the squared distance to the minimizer. Define

$$V(t) = \frac{1}{2} \|x(t) - \bar{x}\|^2.$$

Differentiating,

$$\begin{aligned} V'(t) &= \langle x(t) - \bar{x}, \dot{x}(t) \rangle \\ &= \langle x(t) - \bar{x}, -\nabla f(x(t)) \rangle \\ &= \langle \bar{x} - x(t), \nabla f(x(t)) \rangle. \end{aligned}$$

Since $\bar{x}$ minimizes f ,

$$\nabla f(\bar{x}) = 0.$$

Applying the first-order form of strong convexity with $x = x(t)$ and $y = \bar{x}$ gives

$$f(\bar{x}) \geq f(x(t)) + \langle \nabla f(x(t)), \bar{x} - x(t) \rangle + \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Rearranging,

$$\langle \bar{x} - x(t), \nabla f(x(t)) \rangle \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Because $\bar{x}$ is the minimizer,

$$f(\bar{x}) - f(x(t)) \leq 0.$$

Therefore

$$V'(t) \leq -\frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Using

$$V(t) = \frac{1}{2} \|x(t) - \bar{x}\|^2,$$

we obtain

$$V'(t) \leq -mV(t).$$

Applying Grönwall's lemma with $u(t) = V(t)$ and $a = m$ yields

$$V(t) \leq V(0)e^{-mt}.$$

Substituting the definition of V ,

$$\frac{1}{2} \|x(t) - \bar{x}\|^2 \leq \frac{1}{2} \|x(0) - \bar{x}\|^2 e^{-mt}.$$

Hence

$$\boxed{\|x(t) - \bar{x}\| \leq \|x(0) - \bar{x}\| e^{-mt/2}.}$$

Equivalently,

$$\boxed{\|x(t) - \bar{x}\|^2 \leq \|x(0) - \bar{x}\|^2 e^{-mt}.}$$

This is the first convergence-rate theorem for gradient flow: under m -strong convexity, the distance to the minimizer decays exponentially fast.

6.2 Exercises

Example 6.2.1 (Adding a small quadratic term removes a flat direction. Proposed by Ben Thomas). Here is a simple example showing why one often adds a small quadratic regularization term. Suppose we have one equation but two unknowns. Let

$$A = \begin{pmatrix} 1 & 1 \end{pmatrix},$$

and consider

$$f(x) = \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} (x_1 + x_2 - b)^2.$$

This function only cares about the sum $x_1 + x_2$. It cannot distinguish between points such as $(1, 1)$, $(2, 0)$, or $(3, -1)$, as long as the sum is the same.

The Hessian is

$$\nabla^2 f = A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

This matrix has eigenvalues

$$2 \quad \text{and} \quad 0.$$

The zero eigenvalue corresponds to the direction

$$(1, -1),$$

because moving in that direction changes x_1 and x_2 in opposite ways, so the sum $x_1 + x_2$ does not change. Therefore f is convex, but it is not strongly convex. There is a flat direction.

Indeed, every point satisfying

$$x_1 + x_2 = b$$

minimizes f . So the minimizer is not unique. Geometrically, the bottom of the function is not a single point. It is a whole line.

Now add a small quadratic penalty:

$$f_\mu(x) = \frac{1}{2} \|Ax - b\|^2 + \frac{\mu}{2} \|x\|^2, \quad \mu > 0.$$

Then

$$\nabla^2 f_\mu = A^T A + \mu I = \begin{pmatrix} 1 + \mu & 1 \\ 1 & 1 + \mu \end{pmatrix}.$$

The eigenvalues are now

$$2 + \mu \quad \text{and} \quad \mu.$$

So the zero eigenvalue has been lifted up to μ . The flat direction is no longer flat.

By the second-order characterization, this means

$$f_\mu \text{ is } \mu\text{-strongly convex}$$

and

$$f_\mu \text{ is } (2 + \mu)\text{-smooth}.$$

Thus

$$m = \mu, \quad L = 2 + \mu,$$

and the condition number is

$$\kappa = \frac{L}{m} = \frac{2 + \mu}{\mu}.$$

The minimizer is now unique. Solving the first-order condition gives

$$\nabla f_\mu(x) = 0,$$

so

$$(A^T A + \mu I) \bar{x} = A^T b.$$

Hence

$$\bar{x} = (A^T A + \mu I)^{-1} A^T b.$$

For example, if $b = 2$ and $\mu = 1$, then

$$A^T A + \mu I = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad A^T b = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \bar{x} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

which gives

$$\bar{x} = \begin{pmatrix} 2/3 \\ 2/3 \end{pmatrix}.$$

The main point is that the original least-squares problem had a flat valley. Adding

$$\frac{\mu}{2} \|x\|^2$$

tilts that valley upward in every direction. The smallest Hessian eigenvalue goes from 0 to μ , so the problem becomes strongly convex and the minimizer becomes unique. At the same time, the choice of μ affects the conditioning:

$$\kappa = \frac{2 + \mu}{\mu}.$$

A very small μ barely changes the original problem, but it also gives a large condition number. A larger μ improves conditioning, but changes the problem more. This is the basic tradeoff behind Tikhonov regularization.

6.3 More Examples

Example 6.3.1 (Proposed by Peter Le). The main idea of this example is to illustrate how convexity and L-smoothness bound the graph of f to ensure that gradient descent works effectively.

Consider

$$f(x) = \frac{1}{2}x^2$$

Computing the gradient and the Hessian gives

$$\nabla f(x) = x \quad \nabla^2 f(x) = 1$$

So for every x ,

$$0.5 \leq \nabla^2 f(x) \leq 1.5$$

This means that f is both:

$$0.5\text{-strongly convex}$$

and

$$1.5\text{-smooth}$$

Therefore, by the first-order form of strong convexity,

$$f(y) \geq f(x) + f'(x)(y - x) + \frac{0.5}{2}(y - x)^2$$

And by the first-order form of L-smoothness,

$$f(y) \leq f(x) + f'(x)(y - x) + \frac{1.5}{2}(y - x)^2$$

Thus the function is "sandwiched" between these 2 opposite bounds

$$f(x) + f'(x)(y - x) + \frac{0.5}{2}(y - x)^2 \leq f(y) \leq f(x) + f'(x)(y - x) + \frac{1.5}{2}(y - x)^2$$

Below is an illustration of these 2 bounds with the objective function:

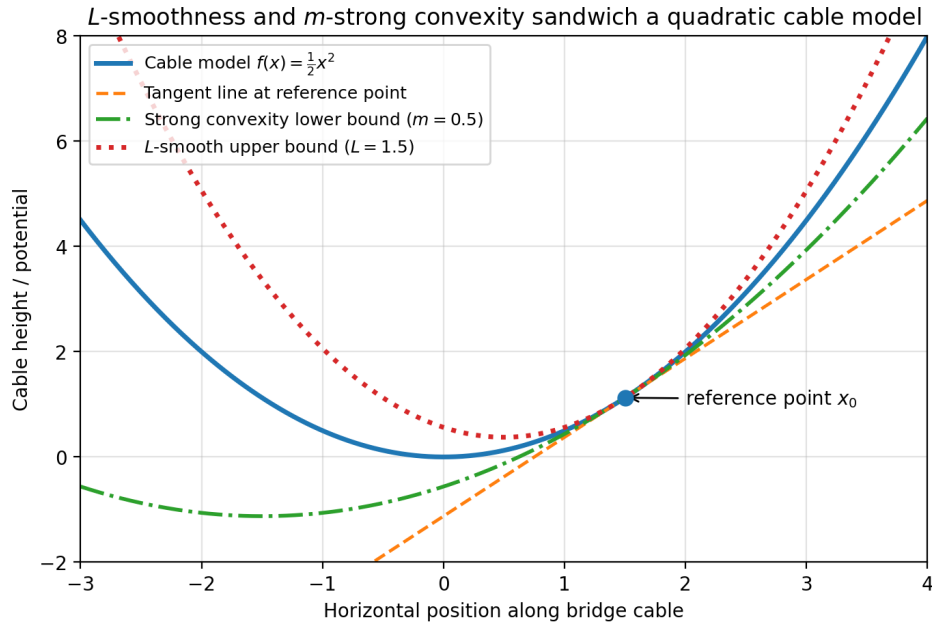


Figure 12: Comparison of our function, the convexity lower bound (green line), and the L-smooth upper bound (red line) on a plot. This illustrates how $f(x)$ is "sandwiched" given that it meets both conditions. This ensures acceptable performance when gradient descent is applied. Both of these conditions are powerful because they ensure that $f(x)$ behaves "nicely" enough to use gradient descent on.

What happens if these conditions are not met?

Consider Case 1: (Not Convex/Too Flat)

$$g(x) = x^4$$

Then

$$g'(x) = 4x^3 \quad g''(x) = 12x^2$$

Notice that

$$g''(0) = 0$$

Therefore there does not exist a constant $m > 0$ for every x such that:

$$g''(x) \geq m$$

In other words, $g(x)$ is not convex enough and $\nabla f(x) = 4x^3$ becomes extremely small as we approach the minimum. This results in tiny gradient descent updates near the minimizer, and thus the algorithm moves very slowly.

Case I. Too Flat (Not Strongly Convex)

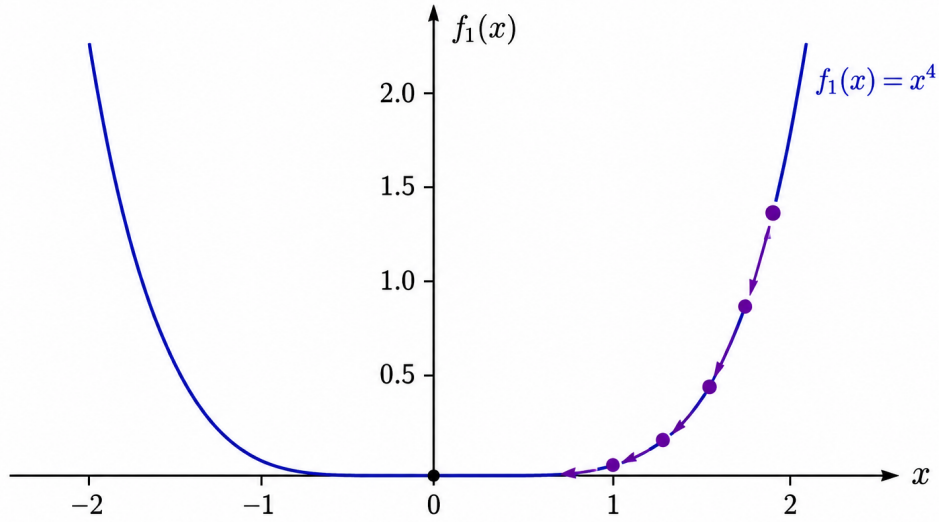


Figure 13: Plot of the function x^4 . We notice the flat area as we approach the minimum. When gradient descent is applied, the algorithm moves very slowly in this region because the gradient is so small (the function changes a little bit). This results in poor performance. Convexity ensures gradient descent keeps making meaningful progress.

Finally consider Case 2: (Not L-smooth/Too Sharp)

$$h(x) = e^{x^2}$$

Then

$$h'(x) = 2xe^{x^2} \quad h''(x) = (2 + 4x^2)e^{x^2}$$

Since as

$$|x| \rightarrow \infty$$

Gives

$$h''(x) \rightarrow \infty$$

We get that there does not exist a finite constant L such that:

$$h''(x) \leq L$$

Therefore $h(x)$ is not L-smooth. The consequence is that the gradient changes rapidly, often resulting in overshoot of the minimizer, and making convergence unstable.

Case 2. Too Sharp (Not L -Smooth)

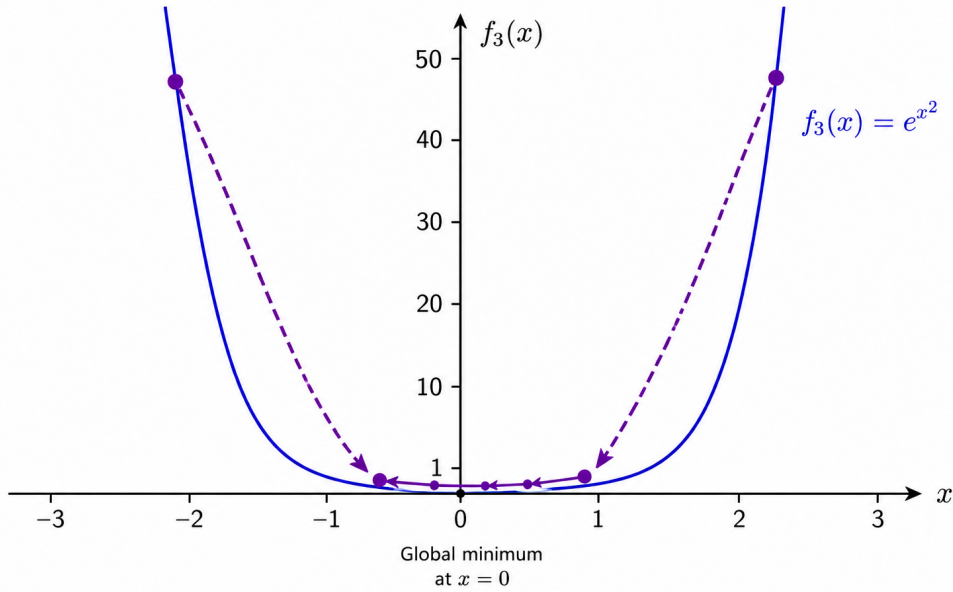


Figure 14: Plot of the function e^{x^2} . We observe that the function is extremely steep. This implies large function changes that would result in large gradient descent jumps, causing overshooting unless small step sizes are used. Either way, this too results in poor performance. L -smoothness ensures that gradient descent doesn't jump too far.

Takeaway: m -strong convexity and L -smoothness are powerful conditions we use in optimization to ensure adequate performance for gradient descent. Convexity gives a lower quadratic bound, while L -smoothness imposes an upper quadratic bound; that together, "sandwiches" the objective function, preventing it from being too sharp or too flat. This ensures that the function bends nicely towards the minimizer and guarantees acceptable performance under gradient descent.

Example 6.3.2 (Proposed by Bradley Nakagawa). From the previous example, we know that m -strong convexity gives a lower bound and L -smoothness gives an upper bound for a function. Therefore, if we know that a function is m -strongly convex and L -smooth with $m = L$, we can find an explicit formula for our objective function, up to a constant vector.

Suppose we have that $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable, m -strongly convex, and L -smooth with $m = L$.

From m -strong convexity, we get that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq \frac{m}{2} \|y - x\|^2$$

From L -smoothness, we get that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq \frac{L}{2} \|y - x\|^2$$

Since $m = L$, we also know that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq \frac{m}{2} \|y - x\|^2$$

So

$$\frac{m}{2} \|y - x\|^2 \leq \langle \nabla f(y) - \nabla f(x), y - x \rangle \leq \frac{m}{2} \|y - x\|^2$$

This means that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle = \frac{m}{2} \|y - x\|^2$$

We can rewrite this as

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle = \frac{m}{2} \langle y - x, y - x \rangle$$

Since this must hold true for all x and all y ,

$$\nabla f(y) - \nabla f(x) = \frac{m}{2} (y - x)$$

So

$$\nabla f(y) - \frac{m}{2} y = \nabla f(x) - \frac{m}{2} x$$

This also must hold true for all x and all y . By holding x constant and moving y through the entire domain of f , we can get that

$$\nabla f(y) - \frac{m}{2} y = \nabla f(x) - \frac{m}{2} x = c$$

where $c \in \mathbb{R}^d$ is a constant vector.

Therefore,

$$\nabla f(x) = \frac{m}{2} x + c$$

So

$$f(x) = \frac{m}{4} \|x\|^2 + \langle c, x \rangle + k$$

where $k \in \mathbb{R}$ is another constant vector.

In general, we do not have functions where $m = L$. However, the difference between m and L roughly corresponds to how much our function is allowed to deviate from functions of the form

$$\frac{m}{4}\|x\|^2 + \langle c, x \rangle + k$$

With a small difference between m and L , we can be sure that our function will have roughly similar behavior to that of the function above.

7 Lecture Notes 7 on Gradient Flow, EVI and Gradient Descent

Jasper Luo summarizes: This lecture explains how the evolution variational inequality gives convergence estimates for gradient flow and gradient descent under convexity, L -smoothness, and m -strong convexity.

Scribe Team: Scribed by Tao Xu, Andrew Wang

7.1 Lecture Notes

In this section, we study gradient flow and gradient descent from the point of view of the evolution variational inequality (EVI). The main goal is to understand how convexity, L -smoothness, and m -strong convexity imply convergence rates.

Our goal is to revisit gradient flow and gradient descent from the perspective of the evolution variational inequality (EVI), and then use this viewpoint to derive convergence estimates. Throughout the lecture, the guiding idea is that the same EVI structure appears in both continuous time and discrete time.

The roadmap for the discussion is

$$\left. \begin{array}{l} \text{convexity / strong convexity} \implies \text{EVI} \implies \left. \begin{array}{l} \text{gradient flow estimates} \\ \text{Moreau envelope interpretation} \\ \text{discrete EVI for gradient descent} \end{array} \right\} \implies \text{convergence rates.} \end{array} \right\}$$

In the continuous-time setting, the EVI gives estimates for gradient flow directly. In the discrete-time setting, gradient descent satisfies an analogous inequality, but an additional residual term appears. The main task is then to use L -smoothness and a suitable stepsize to control this extra term.

Throughout this lecture, let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable, and let

$$\bar{x} \in \arg \min f$$

denote a minimizer whenever it exists. The gradient flow is

$$\dot{x}(t) = -\nabla f(x(t)),$$

whereas gradient descent is

$$x^{k+1} = x^k - \eta \nabla f(x^k),$$

with stepsize $\eta > 0$.

We recall the two structural assumptions used later. The function f is m -strongly convex if, for all $x, y \in \mathbb{R}^d$,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2.$$

It is L -smooth if, for all $x, y \in \mathbb{R}^d$,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

The main cases considered in this lecture can be summarized as follows.

Case	Assumptions	Main inequality	Main consequence
Gradient flow, convex case	f is convex	Continuous EVI without a strong convexity term	Distance–function value control
Gradient flow, strongly convex case	f is m -strongly convex	EVI with the term $-\frac{m}{2} \ x - y\ ^2$	Exponential convergence
Moreau envelope viewpoint	General EVI with arbitrary comparison point y	Integrated EVI	Moreau-type upper bound
Gradient descent	f is L -smooth and m -strongly convex	Discrete EVI with an extra residual term	Linear convergence

7.2 Gradient Flow under Convexity

Theorem 7.1 (EVI for gradient flow under convexity). Assume that f is differentiable and convex, and let $\bar{x} \in \arg \min f$. Suppose that $x(t)$ solves the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)).$$

Consider the gradient flow

$$\dot{x}(t) = -\nabla f(x(t))$$

let $\bar{x}$ be a minimizer of f . Assume that f is convex. By convexity,

$$f(\bar{x}) \geq f(x(t)) + \langle \nabla f(x(t)), \bar{x} - x(t) \rangle$$

Equivalently,

$$\langle \nabla f(x(t)), x(t) - \bar{x} \rangle \geq f(x(t)) - f(\bar{x})$$

Now, compute

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 = \langle x(t) - \bar{x}, \dot{x}(t) \rangle$$

Using the gradient flow equation,

$$\dot{x}(t) = -\nabla f(x(t))$$

we get

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 = -\langle x(t) - \bar{x}, \nabla f(x(t)) \rangle$$

Therefore, by convexity,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t))$$

Proposition 7.2.1 (EVI under convexity). Assume that f is differentiable and convex, and let $x(t)$ solve the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then, for every fixed $y \in \mathbb{R}^d$,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - y\|^2 \leq f(y) - f(x(t)).$$

In particular, choosing $y = \bar{x}$ gives

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)).$$

This inequality means that the squared distance to the minimizer decreases whenever $x(t)$ is not already optimal.

7.3 Gradient Flow under m -Strong Convexity

Theorem 7.2 (EVI and distance decay under m -strong convexity). Assume that f is differentiable and m -strongly convex, and let $\bar{x} \in \arg \min f$. Suppose that $x(t)$ solves the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Consequently,

$$\|x(t) - \bar{x}\|^2 \leq e^{-mt} \|x(0) - \bar{x}\|^2.$$

Assume now that f is m -strongly convex. Then for all x, y ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2$$

Take

$$x = x(t), \quad y = \bar{x}$$

then

$$f(\bar{x}) \geq f(x(t)) + \langle \nabla f(x(t)), \bar{x} - x(t) \rangle + \frac{m}{2} \|x(t) - \bar{x}\|^2$$

Rearranging,

$$-\langle \nabla f(x(t)), x(t) - \bar{x} \rangle \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2$$

Since

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 = -\langle \nabla f(x(t)), x(t) - \bar{x} \rangle,$$

we obtain

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Proposition 7.3.1 (EVI under m -strong convexity). Assume that f is differentiable and m -strongly convex, and let $x(t)$ solve the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then, for every fixed $y \in \mathbb{R}^d$,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - y\|^2 \leq f(y) - f(x(t)) - \frac{m}{2} \|x(t) - y\|^2.$$

In particular, choosing $y = \bar{x}$ gives

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Since $\bar{x}$ is a minimizer,

$$f(\bar{x}) - f(x(t)) \leq 0.$$

Therefore,

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq -\frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Let us

$$D(t) = \frac{1}{2} \|x(t) - \bar{x}\|^2.$$

Then

$$D'(t) \leq -mD(t).$$

By Gronwall's inequality,

$$D(t) \leq D(0)e^{-mt}.$$

Hence,

$$\frac{1}{2} \|x(t) - \bar{x}\|^2 \leq \frac{1}{2} \|x(0) - \bar{x}\|^2 e^{-mt}.$$

Equivalently,

$$\|x(t) - \bar{x}\|^2 \leq \|x(0) - \bar{x}\|^2 e^{-mt}.$$

Remark. This is a distance decay estimate. To convert it into a function value decay estimate, one may additionally use L -smoothness:

$$f(x(t)) - f(\bar{x}) \leq \frac{L}{2} \|x(t) - \bar{x}\|^2.$$

7.3.1 Function Value Decay for Gradient Flow

Theorem 7.3 (Function value decay for gradient flow). Assume that f is differentiable and m -strongly convex, and let $\bar{x} \in \arg \min f$. Suppose that $x(t)$ solves the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then

$$f(x(t)) - f(\bar{x}) \leq [f(x(0)) - f(\bar{x})] e^{-2mt}.$$

We now prove a sharper convergence rate for

$$f(x(t)) - f(\bar{x}).$$

Along the gradient flow,

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then

$$\frac{d}{dt} [f(x(t)) - f(\bar{x})] = \langle \nabla f(x(t)), \dot{x}(t) \rangle.$$

Substituting the gradient flow equation,

$$\frac{d}{dt} [f(x(t)) - f(\bar{x})] = -\|\nabla f(x(t))\|^2.$$

If f is m -strongly convex, then

$$\|\nabla f(x(t))\|^2 \geq 2m [f(x(t)) - f(\bar{x})].$$

Therefore,

$$\frac{d}{dt} [f(x(t)) - f(\bar{x})] \leq -2m [f(x(t)) - f(\bar{x})].$$

Let us

$$E(t) = f(x(t)) - f(\bar{x}).$$

Then

$$E'(t) \leq -2mE(t).$$

By Gronwall's inequality,

$$E(t) \leq E(0)e^{-2mt}.$$

Thus,

$$f(x(t)) - f(\bar{x}) \leq [f(x(0)) - f(\bar{x})] e^{-2mt}.$$

Remark. The value gap of the function decays at a rate e^{-2mt} along the continuous-time gradient flow. This is obtained directly from the identity

$$\frac{d}{dt}f(x(t)) = -\|\nabla f(x(t))\|^2.$$

7.3.2 Useful Inequalities

Assume f is convex, L -smooth and $\bar{x}$ is a minimizer. Then

$$\frac{1}{2L}\|\nabla f(y)\|^2 \leq f(y) - f(\bar{x}) \leq \frac{L}{2}\|y - \bar{x}\|^2.$$

If f is m -strongly convex, then

$$\frac{m}{2}\|y - \bar{x}\|^2 \leq f(y) - f(\bar{x}) \leq \frac{1}{2m}\|\nabla f(y)\|^2.$$

Explanation. The inequality of L -smoothness gives an upper quadratic control on the value of the function. The m -strong convexity inequality gives a lower quadratic control. Together, they imply that the function value gap, the distance to the minimizer, and the gradient norm are comparable quantities.

7.3.3 General EVI and Moreau Envelope

Proposition 7.3.2 (Moreau-type upper bound from EVI). Assume that f is differentiable and m -strongly convex, and let $x(t)$ solve the gradient flow equation

$$\dot{x}(t) = -\nabla f(x(t)).$$

Then

$$f(x(t)) \leq \inf_y \left\{ f(y) + \frac{m}{2(e^{mt} - 1)} \|x(0) - y\|^2 \right\}.$$

Equivalently, if

$$\lambda = \frac{e^{mt} - 1}{m},$$

then

$$f(x(t)) \leq M_{\lambda f}(x(0)).$$

For any fixed y , the strongly convex EVI gives

$$\frac{d}{dt} \frac{1}{2} \|x(t) - y\|^2 \leq f(y) - f(x(t)) - \frac{m}{2} \|x(t) - y\|^2.$$

Move the strong convexity term to the left:

$$\frac{d}{dt} \frac{1}{2} \|x(t) - y\|^2 + m \frac{1}{2} \|x(t) - y\|^2 \leq f(y) - f(x(t)).$$

Multiplying both sides by e^{mt} ,

$$\frac{d}{dt} \left[e^{mt} \frac{1}{2} \|x(t) - y\|^2 \right] \leq e^{mt} [f(y) - f(x(t))].$$

Integrating from 0 to t gives the following.

$$\frac{e^{mt}}{2} \|x(t) - y\|^2 - \frac{1}{2} \|x(0) - y\|^2 \leq \int_0^t e^{ms} [f(y) - f(x(s))] ds.$$

Since $f(x(t))$ is decreasing along the gradient flow, for $0 \leq s \leq t$,

$$f(x(s)) \geq f(x(t)).$$

Therefore,

$$f(y) - f(x(s)) \leq f(y) - f(x(t)).$$

Hence,

$$\int_0^t e^{ms} [f(y) - f(x(s))] ds \leq \int_0^t e^{ms} [f(y) - f(x(t))] ds.$$

Thus,

$$\int_0^t e^{ms} [f(y) - f(x(t))] ds = \frac{e^{mt} - 1}{m} [f(y) - f(x(t))].$$

Therefore,

$$\frac{e^{mt}}{2} \|x(t) - y\|^2 - \frac{1}{2} \|x(0) - y\|^2 \leq \frac{e^{mt} - 1}{m} [f(y) - f(x(t))].$$

Rearranging,

$$f(x(t)) \leq f(y) + \frac{m}{2(e^{mt} - 1)} \|x(0) - y\|^2 - \frac{me^{mt}}{2(e^{mt} - 1)} \|x(t) - y\|^2.$$

Dropping the nonpositive term,

$$-\frac{me^{mt}}{2(e^{mt} - 1)} \|x(t) - y\|^2 \leq 0,$$

we get

$$f(x(t)) \leq f(y) + \frac{m}{2(e^{mt} - 1)} \|x(0) - y\|^2 \quad \text{for any } y.$$

Taking the infimum over y ,

$$f(x(t)) \leq \inf_y \left\{ f(y) + \frac{m}{2(e^{mt} - 1)} \|x(0) - y\|^2 \right\}.$$

Let

$$\lambda = \frac{e^{mt} - 1}{m}.$$

Then

$$\frac{1}{2\lambda} = \frac{m}{2(e^{mt} - 1)}.$$

Definition 7.4 (Proximal map). The proximal map of λf is defined by

$$\text{prox}_{\lambda f}(x) = \arg \min_y \left\{ f(y) + \frac{1}{2\lambda} d^2(y, x) \right\}.$$

Definition 7.5 (Moreau envelope). The Moreau envelope of λf is defined by

$$M_{\lambda f}(x) = \min_y \left\{ f(y) + \frac{1}{2\lambda} d^2(y, x) \right\}.$$

Thus,

$$M_{\lambda f} : X \rightarrow \mathbb{R}.$$

With

$$\lambda = \frac{e^{mt} - 1}{m},$$

the previous estimate becomes

$$f(x(t)) \leq M_{\lambda f}(x(0)).$$

Remark. The inequality is an upper bound, not an equality. The EVI implies

$$f(x(t)) \leq M_{\lambda f}(x(0)),$$

not

$$f(x(t)) = M_{\lambda f}(x(0)).$$

7.3.4 Gradient Descent and the Discrete EVI

Theorem 7.6 (Discrete EVI for gradient descent). Assume that f is differentiable and m -strongly convex, and let $\bar{x} \in \arg \min f$. Consider gradient descent

$$x^{k+1} = x^k - \eta \nabla f(x^k), \quad \eta > 0.$$

Then

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq f(\bar{x}) - f(x^k) - \frac{m}{2} \|x^k - \bar{x}\|^2 + \frac{1}{2\eta} \|x^{k+1} - x^k\|^2.$$

Now consider gradient descent:

$$x^{k+1} = x^k - \eta \nabla f(x^k).$$

Assume f is m -strongly convex and L -smooth. By m -strong convexity,

$$f(\bar{x}) \geq f(x^k) + \langle \nabla f(x^k), \bar{x} - x^k \rangle + \frac{m}{2} \|x^k - \bar{x}\|^2.$$

Equivalently,

$$f(x^k) - f(\bar{x}) \leq \langle \nabla f(x^k), x^k - \bar{x} \rangle - \frac{m}{2} \|x^k - \bar{x}\|^2.$$

Since

$$x^{k+1} = x^k - \eta \nabla f(x^k),$$

we have

$$\nabla f(x^k) = \frac{1}{\eta}(x^k - x^{k+1}).$$

Therefore,

$$\langle \nabla f(x^k), x^k - \bar{x} \rangle = \frac{1}{\eta} \langle x^k - x^{k+1}, x^k - \bar{x} \rangle.$$

By the cosine law identity,

$$2 \langle x^k - x^{k+1}, x^k - \bar{x} \rangle = \|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2 + \|x^k - x^{k+1}\|^2.$$

Therefore,

$$\langle \nabla f(x^k), x^k - \bar{x} \rangle = \frac{1}{2\eta} (\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2 + \|x^k - x^{k+1}\|^2).$$

Substituting this into the strong convexity inequality gives

$$f(x^k) - f(\bar{x}) \leq \frac{1}{2\eta} (\|x^k - \bar{x}\|^2 - \|x^{k+1} - \bar{x}\|^2 + \|x^k - x^{k+1}\|^2) - \frac{m}{2} \|x^k - \bar{x}\|^2.$$

Rearranging,

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq f(\bar{x}) - f(x^k) - \frac{m}{2} \|x^k - \bar{x}\|^2 + \frac{1}{2\eta} \|x^{k+1} - x^k\|^2.$$

Definition 7.7 (Discrete EVI). For gradient descent, the discrete EVI is

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq f(\bar{x}) - f(x^k) - \frac{m}{2} \|x^k - \bar{x}\|^2 + \frac{1}{2\eta} d^2(x^k, x^{k+1}).$$

The last term

$$\frac{1}{2\eta} d^2(x^k, x^{k+1})$$

is the extra term caused by discretization.

Compare this with the continuous EVI:

$$\frac{d}{dt} \frac{1}{2} \|x(t) - \bar{x}\|^2 \leq f(\bar{x}) - f(x(t)) - \frac{m}{2} \|x(t) - \bar{x}\|^2.$$

Remark. The discrete EVI is not simply the direct discretization of the continuous EVI, because the discrete version contains the additional term

$$\frac{1}{2\eta} d^2(x^k, x^{k+1}).$$

7.3.5 Using the Discrete EVI to Prove Convergence

Theorem 7.8 (Linear convergence of gradient descent). Assume that f is L -smooth and m -strongly convex, and let $\bar{x} \in \arg \min f$. Consider gradient descent

$$x^{k+1} = x^k - \eta \nabla f(x^k),$$

with stepsize

$$0 < \eta \leq \frac{1}{L}.$$

Then

$$\|x^k - \bar{x}\|^2 \leq (1 - m\eta)^k \|x^0 - \bar{x}\|^2.$$

Consequently,

$$f(x^k) - f(\bar{x}) \leq \frac{L}{2} (1 - m\eta)^k \|x^0 - \bar{x}\|^2.$$

From the discrete EVI,

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq f(\bar{x}) - f(x^k) - \frac{m}{2} \|x^k - \bar{x}\|^2 + \frac{1}{2\eta} \|x^{k+1} - x^k\|^2.$$

We want to show

$$f(\bar{x}) - f(x^k) + \frac{1}{2\eta} \|x^{k+1} - x^k\|^2 \leq 0.$$

Since

$$x^{k+1} - x^k = -\eta \nabla f(x^k),$$

we get

$$\frac{1}{2\eta} \|x^{k+1} - x^k\|^2 = \frac{\eta}{2} \|\nabla f(x^k)\|^2.$$

By L -smoothness,

$$\frac{1}{2L} \|\nabla f(x^k)\|^2 \leq f(x^k) - f(\bar{x}).$$

If

$$0 < \eta \leq \frac{1}{L},$$

then

$$\frac{\eta}{2} \|\nabla f(x^k)\|^2 \leq \frac{1}{2L} \|\nabla f(x^k)\|^2.$$

Therefore,

$$\frac{1}{2\eta} \|x^{k+1} - x^k\|^2 \leq f(x^k) - f(\bar{x}).$$

Thus,

$$f(\bar{x}) - f(x^k) + \frac{1}{2\eta} \|x^{k+1} - x^k\|^2 \leq 0.$$

Substituting this into the discrete EVI,

$$\frac{\|x^{k+1} - \bar{x}\|^2 - \|x^k - \bar{x}\|^2}{2\eta} \leq -\frac{m}{2} \|x^k - \bar{x}\|^2.$$

Hence,

$$\frac{1}{2\eta} \|x^{k+1} - \bar{x}\|^2 \leq \left(\frac{1}{2\eta} - \frac{m}{2} \right) \|x^k - \bar{x}\|^2.$$

Multiplying by 2η gives

$$\|x^{k+1} - \bar{x}\|^2 \leq (1 - m\eta) \|x^k - \bar{x}\|^2.$$

Since $m \leq L$ and $0 < \eta \leq 1/L$,

$$0 \leq 1 - m\eta < 1.$$

Iterating,

$$\|x^k - \bar{x}\|^2 \leq (1 - m\eta)^k \|x^0 - \bar{x}\|^2.$$

Equivalently,

$$\|x^k - \bar{x}\| \leq (1 - m\eta)^{k/2} \|x^0 - \bar{x}\|.$$

Using L -smoothness,

$$f(x^k) - f(\bar{x}) \leq \frac{L}{2} \|x^k - \bar{x}\|^2.$$

Therefore,

$$f(x^k) - f(\bar{x}) \leq \frac{L}{2} (1 - m\eta)^k \|x^0 - \bar{x}\|^2.$$

Remark. Thus gradient descent has linear convergence under m -strong convexity and L -smoothness:

$$\|x^k - \bar{x}\|^2 = O\left((1 - m\eta)^k\right),$$

and

$$f(x^k) - f(\bar{x}) = O\left((1 - m\eta)^k\right).$$

7.4 Exercises

7.5 More Examples

Example 7.5.1 (*Proposed by Indhu Chandra*, Strong convexity upgrades $\mathcal{O}(1/k)$ to a linear rate). In this lecture, we discuss the convergence analysis for m -strongly and L -smooth function. Here we provide additional perspective from the condition number.

Recall that $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is m -strongly convex ($\mu > 0$) if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2 \quad \forall x, y.$$

Compare this with the L -smoothness upper bound:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Together, the function is sandwiched between two quadratics of curvature m and L . The ratio

$$\kappa := \frac{L}{m} \geq 1$$

is called the *condition number*. It measures how elongated the level sets of f are: when $\kappa = 1$, the level sets are perfectly round and gradient descent converges in one step (scaled appropriately); when $\kappa \gg 1$, the level sets are highly elongated and convergence is slow.

Concrete function. Consider the diagonal quadratic on $\mathbb{R}^2$:

$$f(x_1, x_2) = \frac{m}{2} x_1^2 + \frac{L}{2} x_2^2.$$

This function is m -strongly convex and L -smooth (the Hessian is $\text{diag}(m, L)$), with $\bar{x} = (0, 0)$ and $\kappa = L/m$. The gradient descent update decouples:

$$x_1^{k+1} = (1 - \eta m) x_1^k, \quad x_2^{k+1} = (1 - \eta L) x_2^k.$$

Choosing the *optimal stepsize* $\eta^* = \frac{2}{m+L}$ balances the two coordinates, giving contraction factor

$$\rho := \frac{\kappa - 1}{\kappa + 1} < 1,$$

and the exact function-value gap satisfies

$$f(x^k) - f(\bar{x}) \leq \rho^{2k} f(x^0).$$

This is a **linear** (geometric) convergence rate: the error decreases by a fixed factor ρ^2 at each step, far faster than the $\mathcal{O}(1/k)$ guaranteed under convexity alone.

Comparison with the convex-only rate. Under only L -smoothness (no strong convexity), the lecture proved

$$f(x^k) - f(\bar{x}) \leq \frac{\|x^0 - \bar{x}\|^2}{2\eta k} = \mathcal{O}\left(\frac{1}{k}\right).$$

Strong convexity upgrades this to $\mathcal{O}(\rho^{2k})$. For a target accuracy ε , the iteration counts compare as:

$$\text{Convex + smooth: } k = \mathcal{O}\left(\frac{1}{\varepsilon}\right), \quad \text{Strongly convex: } k = \mathcal{O}\left(\kappa \log \frac{1}{\varepsilon}\right).$$

The logarithmic dependence on $1/\varepsilon$ is the key gain from strong convexity, and the condition number κ now controls how many iterations are needed.

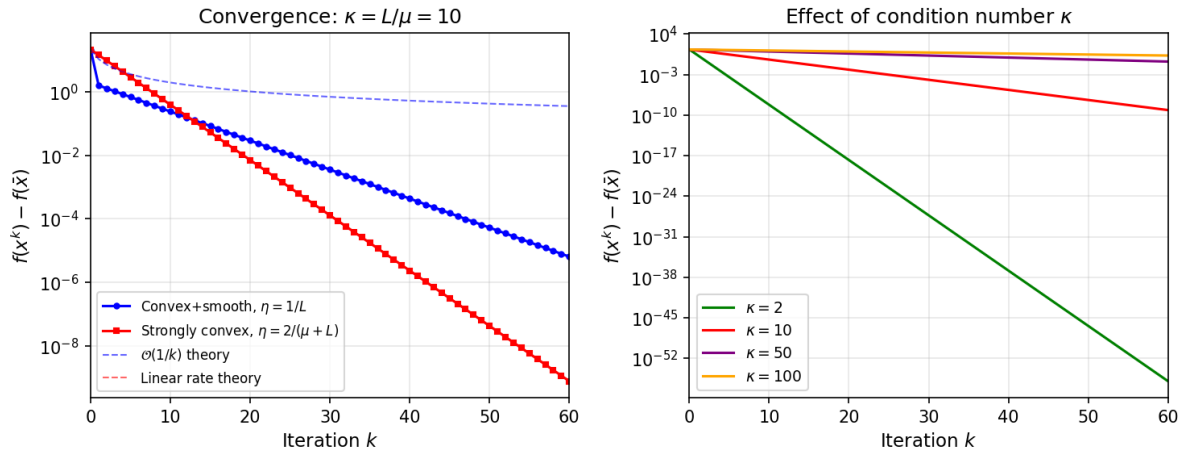


Figure. *Left:* convergence of GD on the diagonal quadratic with $\kappa = 10$. The convex-only rate (blue, $\eta = 1/L$) follows the $\mathcal{O}(1/k)$ dashed curve, while the strongly convex rate (red, $\eta = 2/(m + L)$) follows the dashed linear-rate curve on the log scale. *Right:* effect of the condition number κ on the linear rate. As κ increases from 2 to 100, convergence degrades substantially, showing that a large condition number is the main obstacle to fast optimization.

Take-home messages.

1. **Strong convexity changes the convergence regime.** The rate goes from sublinear $\mathcal{O}(1/k)$ to linear $\mathcal{O}(\rho^{2k})$. On a log scale, convergence becomes a straight line.
2. **The condition number $\kappa = L/m$ is the key quantity.** A well-conditioned problem (κ close to 1) converges in very few steps; a poorly conditioned one ($\kappa \gg 1$) may converge slowly even with the optimal stepsize. Preconditioning methods (such as Newton's method or quasi-Newton methods) are motivated precisely by reducing the effective κ .
3. **The optimal stepsize $\eta^* = 2/(m + L)$ is not $1/L$.** Under strong convexity, one can afford a larger step than $1/L$, and the optimal choice balances the two extreme eigenvalues of the Hessian. This is a preview of how second-order information improves first-order methods.

8 Lecture Notes 8 on Stochastic Gradient Descent

Ritaja Subrahmanya summarizes: This lecture explores the method of stochastic gradient descent which is an extension of the standard gradient descent algorithm. The motivations for the method and the framework for implementing the algorithm were discussed. Finally, a comparison between stochastic gradient descent and traditional gradient descent was discussed.

Scribe Team: Scribed by Peter Le, Proofread by Ciara Nic Canna.

8.1 Lecture Notes

8.1.1 Stochastic Motivations

The focus of this Lecture is the **Stochastic Gradient Descent** (SGD). The idea of this variant of the classic gradient descent method is to combat the expensive nature of direct computation of the gradient. In situations where the cost of deriving the gradient is too high or is simply infeasible, the stochastic gradient descent method provides a faster, less computationally heavy solution.

Examples where we consider this method include problems where we work in a domain with a large dimension, have a very large dataset, or have missing/polluted data.

Definition 8.1. General framework for Stochastic Gradient Descent

$$x^{(k+1)} = x^{(k)} - \eta_k g(x^{(k)}; \xi^{(k)})$$

where:

$g(x^{(k)}; \xi^{(k)})$: = a substitute for the gradient

$\xi^{(k)}$: = random variable/seed/noise

Remark. The goal is to pick a random gradient estimate $g(x^{(k)}; \xi^{(k)})$ that roughly approximates the gradient $\nabla f(x^{(k)})$. We typically want to use SGD in situations where the stochastic gradient is “good enough” or where a certain amount of error is acceptable. SGD is generally not as accurate as true gradient descent. When deciding when SGD is appropriate, consider whether gradient computation is feasible and whether the potential error is acceptable.

Example 8.1.1. Empirical Risk Minimization Problem

$$\min_{x \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m f_i(x)$$

where

- the number of data samples, m , is extremely large;
- Each f_i corresponds to one data sample.

To compute the gradient $\nabla f(x) = \frac{1}{m} \sum_{i=1}^m \nabla f_i(x)$, the cost per iteration to compute the gradient is $O(md)$. This becomes very expensive as m increases.

So when we have a large number of dataset we can alternatively, approach the problem using SGD to reduce this cost to $O(d)$. Pick a random $i_j \in \{1, 2, \dots, m\}$. For the next iteration, we have the update formula is given by:

$$x^{(k+1)} = x^{(k)} - \eta_k \nabla f_{i_j}(x^{(k)})$$

Note that instead of computing m gradients, we pick a random index among our data samples and compute only that gradient to determine the next iteration. This is why the cost per iteration becomes $O(d)$ using the SGD approach.

The main idea is to – in lieu of fully computing the gradient – pick a random seed to estimate the true gradient for the next iteration. Even though SGD requires more iterations mathematically, each iteration is much cheaper, especially in large data sets where direct gradient calculations are infeasible.

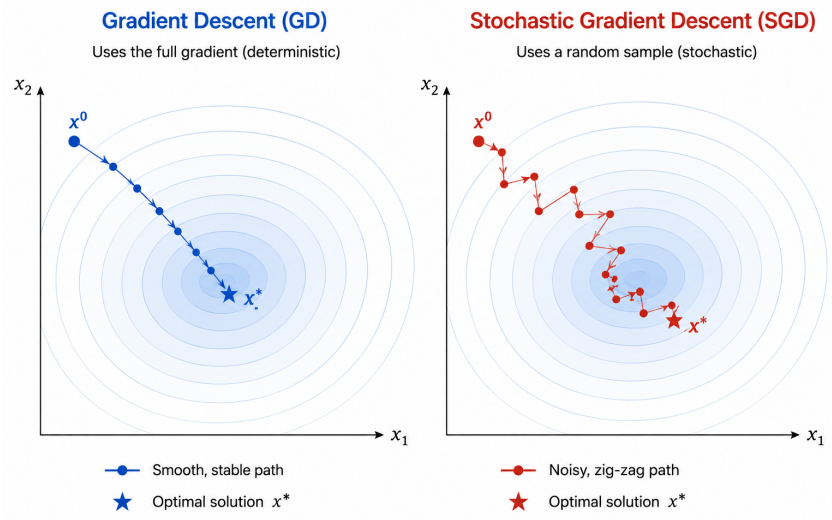


Figure 15: *Peter Le creates with Chat GPT*: Sample Optimization Path Illustration – GD vs SGD

Remark. We can observe from the above illustration that using true GD gives us a comparatively smoother path to the optimal solution. Though it can be more computationally expensive, full gradient descent often gives a "better" route. SGD gives a more noisy path to the solution. Since the gradient is approximate with a random point, the path will not be as "perfect". However, if less accuracy is acceptable, SGD will reach the solution while requiring less resources.

Example 8.1.2 (Machine Learning Motivation). A common application of SGD arises in machine learning. The average error (or loss) of a machine learning model across an entire training dataset is defined:

$$\text{Loss}(\theta) = \frac{1}{|D_{\text{train}}|} \sum_{(x,y) \in D_{\text{train}}} \text{Loss}(x, y, \theta)$$

Where,

- D_{train} is a very large training dataset.
- $\theta \in \mathbb{R}^m$ (very large).

Each (x, y) is a pair of an input, x , and output, y . Each $\text{Loss}(x, y, \theta)$ is the loss from one training sample.

In modern machine learning problems, both the number of parameters and the size of the dataset can become extremely large. As a result, computing the full gradient $\nabla \text{Loss}(\theta)$ may be computationally infeasible.

Furthermore:

1. the dataset may not be available all at once (which is an online learning),
2. some data may be missing or corrupted,
3. treating all data equally can lead to overfitting.

SGD addresses these issues by approximating the full gradient using only a small random subset of the data at each iteration.

8.1.2 Stochastic Assumptions

subsub:sto-assump

Some common assumptions in SGD:

1. Unbiased: $\mathbb{E}[g(x^{(k)}, \xi^{(k)})] = \nabla f(x^{(k)})$
2. Bounded Variance:
 - (a) $\mathbb{E}[\|g(x^{(k)}, \xi^{(k)}) - \nabla f(x^{(k)})\|^2] \leq \sigma^2$
 - (b) $\mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2] \leq G$

Remark. Our first assumption says that on average, the stochastic gradient points in the correct direction. Although each iteration is not forced to point in the correct direction, we assume that the trend goes toward the optimal solution. Though we allow noise, it may not be arbitrarily large, and therefore, we bound the error with the second assumption. Two versions of bounded variance prevent SGD from becoming unstable.

Definition 8.2. Expectation in the Unbiased Assumption

We define $\mathbb{E}[g(x^{(k)}, \xi^{(k)})] = \nabla f(x^{(k)})$ as:

$$\int g(x^{(k)}, \xi^{(k)}) d\mathbb{P}(\xi) = \nabla f(x^{(k)}) \quad (\text{continuous case}) \quad (8.1)$$

$$\frac{1}{N} \sum_{\xi} g(x^{(k)}, \xi^{(k)}) = \nabla f(x^{(k)}) \quad (\text{discrete case}) \quad (8.2)$$

Definition 8.3. Variance in the Bounded Variance Assumption

Recall $\text{Var}(\xi) = \mathbb{E}[\|\xi - \mathbb{E}[\xi]\|^2] = \mathbb{E}[\|\xi\|^2] - (\mathbb{E}[\xi])^2$, thus

$$\mathbb{E}\|g(x^{(k)}, \xi^{(k)}) - \nabla f(x^{(k)})\|^2 = \mathbb{E}\|g(x^{(k)}, \xi^{(k)})\|^2 - \|\nabla f(x^{(k)})\|^2 \quad (8.3)$$

8.1.3 Decent Lemmas

Theorem 8.4 (GD descent lemma). If $\eta \in (0, \frac{1}{L}]$ and a differentiable $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth, then the gradient descent update $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$ satisfies

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2.$$

Proof. Recall from 6.2, since f is L -smooth and then

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|x - y\|^2, \quad \forall x, y \in \mathbb{R}^d.$$

Substituting $x = x^{(k)}$ $y = x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$, we get

$$\begin{aligned} f(x^{(k+1)}) &\leq f(x^{(k)}) + \langle \nabla f(x^{(k)}), x^{(k)} - \eta \nabla f(x^{(k)}) - x^{(k)} \rangle + \frac{L}{2} \|x^{(k)} - (x^{(k)} - \eta \nabla f(x^{(k)}))\|^2 \\ &\leq f(x^{(k)}) - \eta \langle \nabla f(x^{(k)}), \nabla f(x^{(k)}) \rangle + \frac{L\eta^2}{2} \|\nabla f(x^{(k)})\|^2 \end{aligned}$$

By simplifying the inner product,

$$f(x^{(k+1)}) \leq f(x^{(k)}) + \left(\frac{L\eta^2}{2} - \eta \right) \|\nabla f(x^{(k)})\|^2$$

If $\eta \in (0, \frac{1}{L}]$, then $\frac{L\eta^2}{2} - \eta \leq \frac{\eta}{2} - \eta = -\frac{\eta}{2}$,

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{\eta}{2} \|\nabla f(x^{(k)})\|^2$$

■

Remark. Each iteration decreases the function value unless the gradient is zero.

Theorem 8.5 (SGD descent lemme). If $\eta \in (0, \frac{1}{L})$, $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth, and $g(x^{(k)}, \xi^{(k)})$ is unbiased, then

$$\frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}[\|\nabla f(x^{(i)})\|^2] \leq \frac{2(f(x_0) - f(\bar{x}))}{\eta k} + \sigma^2$$

Proof. Take the following for the SGD approach

$$x^{(k+1)} = x^{(k)} - \eta g(x^{(k)}, \xi^{(k)}) \quad \leftarrow \quad x^{(k+1)} = x^{(k+1)}(\xi^{(k)})$$

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \eta \langle \nabla f(x^{(k)}), g(x^{(k)}, \xi^{(k)}) \rangle + \frac{L\eta^2}{2} \|g(x^{(k)}, \xi^{(k)})\|^2$$

Take the Expectation on both sides

$$\mathbb{E}(f(x^{(k+1)})) \leq f(x^{(k)}) - \eta \|\nabla f(x^{(k)})\|^2 + \frac{L\eta^2}{2} \mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2]$$

Recalling 2a) from the Stochastic Assumption (Sec Subsubsection 8.1.2)

$$\mathbb{E}[\|g(x^{(k)}, \xi^{(k)}) - \nabla f(x^{(k)})\|^2] \leq \sigma^2$$

$$\mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2] - \|\nabla f(x^{(k)})\|^2 \leq \sigma^2$$

Plug the above into $\mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2]$

$$\boxed{\mathbb{E}[f(x^{(k+1)})] \leq f(x^{(k)}) - \left(\frac{L\eta^2}{2} - \eta \right) \|\nabla f(x^{(k)})\|^2 + \frac{L\eta^2}{2} \sigma^2}$$

Take $\eta \leq \frac{1}{L}$

$$\leq f(x^{(k)}) - \frac{1}{2} \eta \|\nabla f(x^{(k)})\|^2 + \frac{1}{2} \eta \sigma^2$$

$$\begin{aligned}
\|\nabla f(x^{(k)})\|^2 &\leq \frac{2(f(x^{(k)}) - \mathbb{E}(f(x^{(k+1)})))}{\eta} + \sigma^2 \\
&\leq \frac{2(f(x^{(k)}) - f(\bar{x}))}{\eta} + \sigma^2 \\
\frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}[\|\nabla f(x^{(i)})\|^2] &\leq \frac{2(f(x_0) - f(\bar{x}))}{\eta k} + \sigma^2
\end{aligned}$$

■

Theorem 8.6.

8.1.4 Corollary

$$\frac{1}{k} \sum_{i=0}^{k-1} \mathbb{E}[\|\nabla f(x^{(i)})\|^2] \geq \min_i \mathbb{E}[\|\nabla f(x^{(i)})\|^2]$$

And at least one iteration $j \in \{0, 1, \dots, k\}$

$$\mathbb{E}[\|\nabla f(x^{(j)})\|^2] \leq O\left(\frac{1}{k}\right)$$

Furthermore, f is L -smooth and convex

$$\begin{aligned}
f(x^{(k)}) &\leq f(y) - \langle \nabla f(x^{(k)}), y - x^{(k)} \rangle \\
&= f(y) - \langle \mathbb{E}[g(x^{(k)}, \xi^{(k)})], y - x^{(k)} \rangle \\
&= f(y) - \mathbb{E}[\langle g(x^{(k)}, \xi^{(k)}), y - x^{(k)} \rangle] \quad (x^{(k+1)} = x^{(k)} - \eta g(x^{(k)}, \xi^{(k)})) \\
&= f(y) - \frac{1}{\eta} \mathbb{E}[\langle x^{(k)} - x^{(k+1)}, y - x^{(k)} \rangle] \\
&= f(y) + \mathbb{E}\left[\frac{1}{2\eta} (\|x^{(k)} - y\|^2 - \|x^{(k+1)} - y\|^2 + d^2(x^{(k)}, x^{(k+1)}))\right]
\end{aligned}$$

Descent Lemma in the Function Value

$$\begin{aligned}
f(x^{(k)}) &\leq f(\bar{x}) + \mathbb{E}\left[\frac{\|x^{(k)} - \bar{x}\|^2 - \|x^{(k+1)} - \bar{x}\|^2}{2\eta}\right] + \frac{1}{2\eta} \mathbb{E}[d^2(x^{(k)}, x^{(k+1)})] \\
\sum_{i=0}^{k-1} f(x^{(i)}) &\leq k f(\bar{x}) + \frac{1}{2\eta} \|x^0 - \bar{x}\|^2 + \frac{\eta^2}{2\eta} \sum_{i=0}^{k-1} \mathbb{E}[\|g(x^{(i)}, \xi^{(i)})\|^2] \\
&\quad (\text{We define } G = \mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2])
\end{aligned}$$

Take 2b) from Stochastic Assumption (Sec Subsubsection 8.1.2)

$$\frac{1}{k} \sum_{i=0}^{k-1} f(x^{(i)}) \leq f(\bar{x}) + \frac{1}{2\eta k} \|x^{(0)} - \bar{x}\|^2 + \frac{\eta}{2} G$$

Take $\eta = \sqrt{kG}$ and $\frac{1}{2\eta k} = \frac{\eta}{2} G$

$$= f(\bar{x}) + \frac{\sqrt{G}}{2\sqrt{k}} (\|x^{(0)} - \bar{x}\|^2 + 1)$$

$$\frac{1}{k} \sum_{i=0}^{k-1} (f(x^{(i)}) - f(\bar{x})) \leq \frac{\sqrt{G}}{2\sqrt{k}} (\|x^{(0)} - \bar{x}\|^2 + 1)$$

Therefore, at least one iteration $j \in \{1, 2, \dots, k\}$ we have

$$f(x^{(j)}) - f(\bar{x}) \leq O\left(\frac{1}{\sqrt{k}}\right)$$

Hence, SGD is slower but cheaper than traditional GD.

8.2 Exercises

We have not yet discussed the convergence analysis for a diminishing stepsize. In fact, the same techniques still apply.

Exercise. Let $f(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ be m -strongly convex and the iterate scheme be $x^{(k+1)} = x^{(k)} - \eta_k g(x^{(k)}, \xi^{(k)})$. Assume that $g(x^{(k)}, \xi^{(k)})$ is unbiased and has bounded second moments, i.e.,

$$\begin{aligned}
\mathbb{E}[g(x^{(k)}, \xi^{(k)})] &= \nabla f(x^{(k)}), \\
\mathbb{E}\left[\|g(x^{(k)}, \xi^{(k)})\|^2\right] &\leq \sigma^2.
\end{aligned}$$

Show that

$$\mathbb{E}\left[\|x^{(k+1)} - \bar{x}\|^2\right] \leq (1 - 2m\eta_k) \mathbb{E}\left[\|x^{(k)} - \bar{x}\|^2\right] + \eta_k^2 \sigma^2.$$

eq:SGD-iter (8.4)

Proof. *Drafted by Run Li, Proofread by Charles Yang*

Recall the SGD update rule: $x^{(k+1)} = x^{(k)} - \eta_k g(x^{(k)}, \xi^{(k)})$. Subtracting the minimizer $\bar{x}$ from both sides and taking the squared L_2 norm, we have:

$$\|x^{(k+1)} - \bar{x}\|^2 = \|x^{(k)} - \bar{x} - \eta_k g(x^{(k)}, \xi^{(k)})\|^2 = \|x^{(k)} - \bar{x}\|^2 - 2\eta_k \langle x^{(k)} - \bar{x}, g(x^{(k)}, \xi^{(k)}) \rangle + \eta_k^2 \|g(x^{(k)}, \xi^{(k)})\|^2.$$

Taking the expectation conditioned on $x^{(k)}$:

$$\mathbb{E}[\|x^{(k+1)} - \bar{x}\|^2 \mid x^{(k)}] = \|x^{(k)} - \bar{x}\|^2 - 2\eta_k \langle x^{(k)} - \bar{x}, \mathbb{E}[g(x^{(k)}, \xi^{(k)}) \mid x^{(k)}] \rangle + \eta_k^2 \mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2 \mid x^{(k)}].$$

By the unbiasedness assumption, $\mathbb{E}[g(x^{(k)}, \xi^{(k)}) \mid x^{(k)}] = \nabla f(x^{(k)})$. By the bounded variance assumption, $\mathbb{E}[\|g(x^{(k)}, \xi^{(k)})\|^2 \mid x^{(k)}] \leq \sigma^2$.

Thus:

$$\mathbb{E}[\|x^{(k+1)} - \bar{x}\|^2 \mid x^{(k)}] \leq \|x^{(k)} - \bar{x}\|^2 - 2\eta_k \langle x^{(k)} - \bar{x}, \nabla f(x^{(k)}) \rangle + \eta_k^2 \sigma^2.$$

Since f is m -strongly convex, the monotonicity of the gradient yields $\langle \nabla f(x^{(k)}) - \nabla f(\bar{x}), x^{(k)} - \bar{x} \rangle \geq m\|x^{(k)} - \bar{x}\|^2$. Since $\bar{x}$ is the global minimizer, $\nabla f(\bar{x}) = 0$, which gives $\langle \nabla f(x^{(k)}), x^{(k)} - \bar{x} \rangle \geq m\|x^{(k)} - \bar{x}\|^2$. Substituting this into our inequality:

$$\mathbb{E}[\|x^{(k+1)} - \bar{x}\|^2 \mid x^{(k)}] \leq \|x^{(k)} - \bar{x}\|^2 - 2m\eta_k \|x^{(k)} - \bar{x}\|^2 + \eta_k^2 \sigma^2 = (1 - 2m\eta_k) \|x^{(k)} - \bar{x}\|^2 + \eta_k^2 \sigma^2.$$

Taking the full expectation on both sides yields the desired result:

$$\mathbb{E}[\|x^{(k+1)} - \bar{x}\|^2] \leq (1 - 2m\eta_k) \mathbb{E}[\|x^{(k)} - \bar{x}\|^2] + \eta_k^2 \sigma^2.$$

■

Exercise. Under the assumptions of previous exercise, take $\eta_k = \frac{\theta}{k+1}$ for some $\theta > \frac{1}{2m}$, show that

$$\mathbb{E}[\|x^{(k)} - \bar{x}\|^2] \leq \frac{\max\left\{\frac{2\theta^2\sigma^2}{2m\theta-1}, \|x_0 - \bar{x}\|^2\right\}}{k+1}.$$

Ask yourself why one needs $\eta_k \propto \frac{1}{k}$; what if $\eta_k \propto \frac{1}{\sqrt{k}}$?

Proof. *Drafted by Run Li, Proofread by Eric Wong.*

We proceed by mathematical induction on k . Let $a_k = \mathbb{E}[\|x^{(k)} - \bar{x}\|^2]$ and $C = \max\left\{\frac{2\theta^2\sigma^2}{2m\theta-1}, a_0\right\}$. We aim to show $a_k \leq \frac{C}{k+1}$ for all $k \geq 0$.

Base case ($k = 0$). We have $a_0 \leq C = \frac{C}{0+1}$, which holds trivially by definition of C .

Inductive step. Assume $a_k \leq \frac{C}{k+1}$ for some $k \geq 0$. We show $a_{k+1} \leq \frac{C}{k+2}$. From the previous exercise,

$$a_{k+1} \leq (1 - 2m\eta_k) a_k + \eta_k^2 \sigma^2.$$

Substituting $\eta_k = \frac{\theta}{k+1}$ and the inductive hypothesis $a_k \leq \frac{C}{k+1}$:

$$a_{k+1} \leq \left(1 - \frac{2m\theta}{k+1}\right) \frac{C}{k+1} + \frac{\theta^2\sigma^2}{(k+1)^2} = \frac{C(k+1-2m\theta) + \theta^2\sigma^2}{(k+1)^2}.$$

It suffices to show this is at most $\frac{C}{k+2}$, i.e.,

$$\frac{C(k+1-2m\theta) + \theta^2\sigma^2}{(k+1)^2} \leq \frac{C}{k+2}.$$

Multiplying both sides by the strictly positive quantity $(k+1)^2(k+2)$:

$$(k+2)[C(k+1-2m\theta) + \theta^2\sigma^2] \leq C(k+1)^2.$$

Expanding both sides and subtracting $C(k^2 + 2k + 1)$:

$$C[(k+2)(1-2m\theta) - 1] + \theta^2\sigma^2(k+2) \leq 0.$$

Dividing by $k+2 > 0$:

$$C\left(1 - 2m\theta - \frac{1}{k+2}\right) + \theta^2\sigma^2 \leq 0,$$

or equivalently,

$$C\left(2m\theta - 1 + \frac{1}{k+2}\right) \geq \theta^2\sigma^2.$$

Since $\theta > \frac{1}{2m}$ we have $2m\theta - 1 > 0$, so

$$C\left(2m\theta - 1 + \frac{1}{k+2}\right) > C(2m\theta - 1) \geq \frac{2\theta^2\sigma^2}{2m\theta - 1} \cdot (2m\theta - 1) = 2\theta^2\sigma^2 \geq \theta^2\sigma^2.$$

The required inequality holds, so $a_{k+1} \leq \frac{C}{k+2}$, completing the induction.

Regarding the stepsize question. The choice $\eta_k \propto \frac{1}{k}$ is governed by two classical conditions for stochastic approximation: $\sum_k \eta_k = \infty$ (ensuring the iterates can travel arbitrarily far and reach the minimizer) and $\sum_k \eta_k^2 < \infty$ (ensuring the accumulated variance from stochastic noise remains bounded). The schedule $\eta_k = \frac{\theta}{k+1}$ satisfies both: the harmonic series diverges, while $\sum \frac{1}{(k+1)^2} < \infty$. If instead $\eta_k \propto \frac{1}{\sqrt{k}}$, then $\sum \eta_k^2 \propto \sum \frac{1}{k} = \infty$, so the second condition fails. The variance term no longer stays bounded, and the $\mathcal{O}(1/k)$ rate for strongly convex objectives cannot be achieved; one obtains only $\mathcal{O}(1/\sqrt{k})$ convergence instead. ■

8.3 More Examples

Example 8.3.1 (Proposed by Sahand Ahmadi, A toy SGD model: noisy paths, but convergence in expectation). To connect the theory of Lecture 8 with an explicit experiment, consider the empirical risk minimization problem

$$f(x) = \frac{1}{2}(f_1(x) + f_2(x)), \quad x \in \mathbb{R},$$

where

$$f_1(x) = \frac{1}{2}(x-1)^2, \quad f_2(x) = \frac{1}{2}(x+1)^2.$$

Then

$$f(x) = \frac{1}{4}((x-1)^2 + (x+1)^2) = \frac{1}{2}x^2 + \frac{1}{2},$$

so the unique minimizer is

$$\bar{x} = 0.$$

The full gradient is

$$\nabla f(x) = x.$$

Thus, gradient descent with fixed stepsize η takes the form

$$x_{\text{GD}}^{(k+1)} = x_{\text{GD}}^{(k)} - \eta x_{\text{GD}}^{(k)} = (1 - \eta)x_{\text{GD}}^{(k)}.$$

Since f is 1-smooth, Lecture 8 suggests taking a step size satisfying $\eta \leq 1/L = 1$. For example, with $\eta = 0.5$, gradient descent becomes

$$x_{\text{GD}}^{(k+1)} = \frac{1}{2}x_{\text{GD}}^{(k)},$$

so the iterates contract smoothly to 0, see left figure in Figure 16.

Now consider SGD. At each step, choose either f_1 or f_2 with equal probability and use the corresponding sample gradient:

$$g(x, \xi) = \begin{cases} x - 1, & \text{with probability } \frac{1}{2}, \\ x + 1, & \text{with probability } \frac{1}{2}. \end{cases}$$

Then the SGD update is

$$x^{(k+1)} = x^{(k)} - \eta_k g(x^{(k)}, \xi^{(k)}).$$

Why this matches the stochastic assumptions. First,

$$\mathbb{E}[g(x, \xi)] = \frac{1}{2}(x-1) + \frac{1}{2}(x+1) = x = \nabla f(x),$$

so the stochastic gradient is unbiased. Also,

$$g(x, \xi) - \nabla f(x) \in \{-1, 1\},$$

hence

$$\mathbb{E}[(g(x, \xi) - \nabla f(x))^2] = 1.$$

So this toy model has bounded variance with $\sigma^2 = 1$, exactly as assumed in Lecture 8.

Fixed-step SGD. Take the same constant stepsize $\eta = 0.5$. Then

$$x^{(k+1)} = x^{(k)} - 0.5 g(x^{(k)}, \xi^{(k)}).$$

Because $\mathbb{E}[g(x^{(k)}, \xi^{(k)}) \mid x^{(k)}] = x^{(k)}$, we obtain

$$\mathbb{E}[x^{(k+1)} \mid x^{(k)}] = x^{(k)} - 0.5 x^{(k)} = 0.5 x^{(k)}.$$

Taking expectation again gives

$$\mathbb{E}[x^{(k+1)}] = 0.5 \mathbb{E}[x^{(k)}].$$

Therefore,

$$\mathbb{E}[x^{(k)}] = (0.5)^k x^{(0)} \rightarrow 0.$$

So even though the individual SGD iterates are noisy, the method still approaches the minimizer in expectation, see middle figure in Figure 16.

Harmonic decay. Motivated by the stepsize-scheduling example, we may also try a decreasing step size such as

$$\eta_k = \frac{0.8}{k+1}.$$

Then the noise term is gradually damped as k increases. In practice, this produces a path that still fluctuates, but the oscillations become smaller and the iterates settle much closer to the minimizer, see right figure in Figure 16.

Toy example: $f(x) = \frac{1}{2}x^2 + \frac{1}{2}$, start at $x^{(0)} = 3$

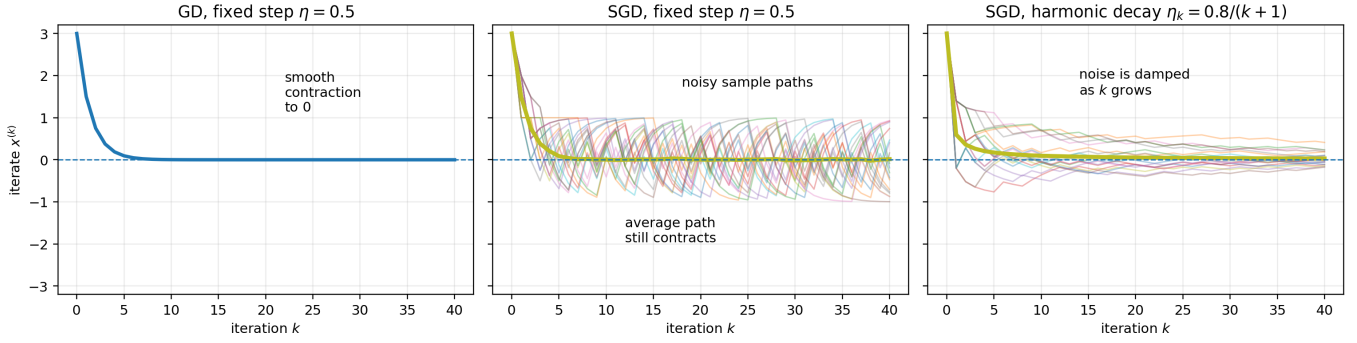


Figure 16: Comparison of GD and SGD on the toy objective $f(x) = \frac{1}{2}x^2 + \frac{1}{2}$, starting from $x^{(0)} = 3$. Left: gradient descent with fixed step $\eta = 0.5$ contracts smoothly to the minimizer. Middle: fixed-step SGD is noisy, but its average path still moves toward 0. Right: harmonic decay $\eta_k = 0.8/(k+1)$ reduces the noise over time and gives a more stable approach to the minimizer.

Takeaway. This example gives a concrete picture of the main message of Lecture 8. Gradient descent is smoother because it uses the exact gradient. SGD is noisier because each step uses only a random gradient estimate, but if the estimate is unbiased and has controlled variance, the method still moves in the correct direction on average. A decreasing step size improves the practical behavior by damping the noise while keeping the computational advantages of SGD.

Example 8.3.2 (Proposed by Ben Pearson, SGD escapes saddle points where GD stalls). Throughout this course we have analyzed SGD as a noisy approximation to GD, with the noise being an unfortunate cost of computational savings. This example shows the opposite: on non-convex functions, *the noise in SGD is a feature*. It allows SGD to escape saddle points where GD stalls indefinitely.

Setup. Consider the non-convex function

$$f(x_1, x_2) = x_1^2 - x_2^2,$$

which has a saddle point at the origin: $\nabla f(0, 0) = 0$, yet $(0, 0)$ is neither a local minimum nor a local maximum.

We model the stochastic gradient as follows. At each step, instead of computing $\nabla f(x) = (2x_1, -2x_2)$ exactly, SGD uses a noisy estimate

$$g(x, \xi) = \nabla f(x) + \xi, \quad \xi \sim \mathcal{N}(0, \sigma^2 I),$$

where ξ is independent Gaussian noise with variance σ^2 in each coordinate. This satisfies the unbiased assumption $\mathbb{E}[g(x, \xi)] = \nabla f(x)$ and the bounded variance assumption $\mathbb{E}[\|g - \nabla f\|^2] = 2\sigma^2$.

What GD does at a saddle. Suppose we initialize at $x^{(0)} = (\delta, 0)$ for a small $\delta > 0$, nearly at the saddle. The GD update with stepsize η is

$$x_1^{(k+1)} = x_1^{(k)} - \eta \cdot 2x_1^{(k)} = (1 - 2\eta)x_1^{(k)}, \quad x_2^{(k+1)} = x_2^{(k)} - \eta \cdot (-2x_2^{(k)}) = (1 + 2\eta)x_2^{(k)}.$$

Starting from $x^{(0)} = (\delta, 0)$, the x_2 component remains exactly zero for all k , since $x_2^{(0)} = 0$ propagates as $x_2^{(k)} = (1 + 2\eta)^k \cdot 0 = 0$. GD is therefore confined to the x_1 -axis, where f is strictly convex, and converges toward the origin:

$$x^{(k)} = ((1 - 2\eta)^k \delta, 0) \rightarrow (0, 0).$$

Starting from the neighborhood of the saddle point, GD finds the saddle point and stops there, reporting $\nabla f = 0$ as if it had found a minimizer. It has no mechanism to detect that the origin is not a minimum.

What SGD does at a saddle. SGD initialized at $(\delta, 0)$ takes the update

$$x_1^{(k+1)} = (1 - 2\eta)x_1^{(k)} - \eta\xi_1^{(k)}, \quad x_2^{(k+1)} = (1 + 2\eta)x_2^{(k)} - \eta\xi_2^{(k)}.$$

The x_2 component is no longer pinned to zero. Even if $x_2^{(0)} = 0$, the first noise injection gives $x_2^{(1)} = -\eta\xi_2^{(0)}$, which is nonzero with probability one. Once x_2 is nonzero, the factor $(1 + 2\eta) > 1$ amplifies it at every subsequent step. The x_2 direction is unstable: any perturbation away from zero grows geometrically. The noise from SGD provides exactly the perturbation needed to ensure we do not get stuck in the saddle point.

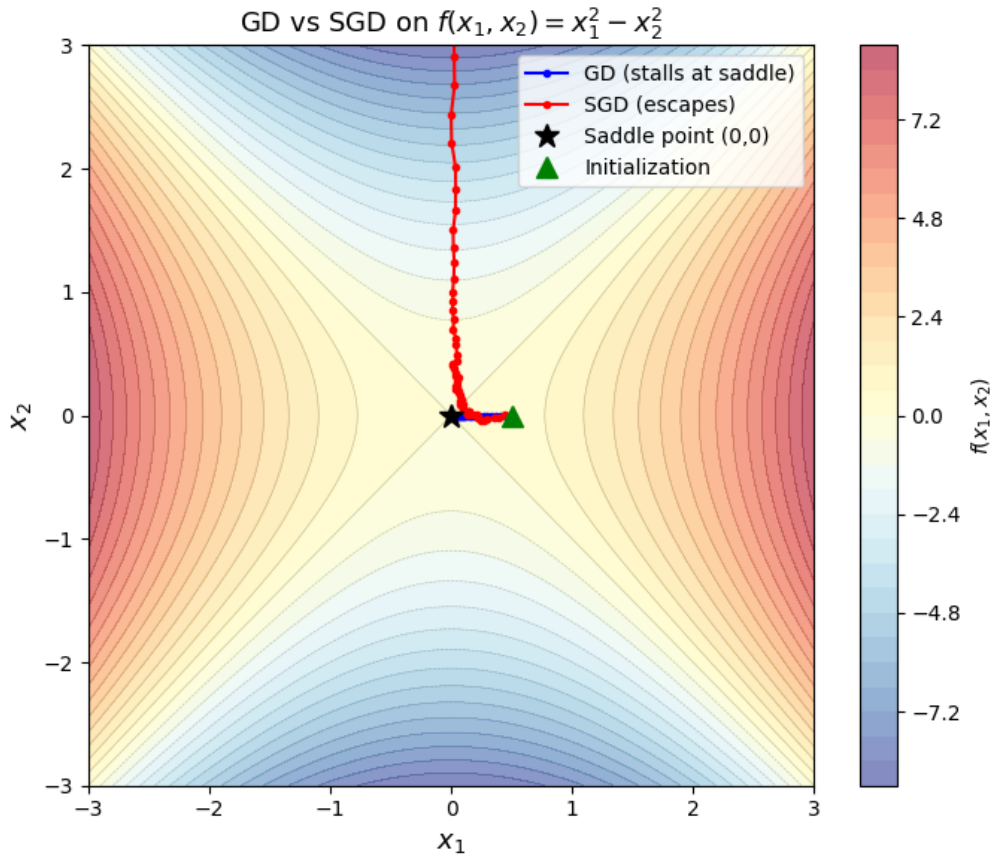


Figure 17: Trajectories of GD and SGD on $f(x_1, x_2) = x_1^2 - x_2^2$, both initialized at $(\delta, 0)$ near the saddle at the origin. GD (blue) converges to the saddle and stalls. SGD (red) receives noise in the x_2 direction, which the negative curvature amplifies, causing escape. Contour lines show level sets of f ; darker shading indicates lower values.

Takeaway. This example demonstrates that while the noise in SGD is often viewed as an unfortunate computational trade-off, it serves as a crucial feature rather than a bug when optimizing non-convex landscapes. If Gradient Descent is initialized on a subspace where the gradient along an unstable direction is zero, it trap it at the saddle point indefinitely. Stochastic Gradient Descent breaks this limitation because its random perturbations knock the iterates off the strict saddle subspace. Once perturbed, the negative curvature of the landscape amplifies the motion, allowing SGD to successfully escape where GD stalls. This ability to escape local minima as well as its computational efficiency makes SGD highly utilized in the field of machine learning

9 Lecture Notes 9 on Subgradient

Peter Le summarizes: This lecture we explore 0th-order methods of optimization that doesn't require the gradient. This becomes necessary when working with nonsmooth objective functions, where every point may not be differentiable. To work with them, we introduced to the subgradient and subdifferential. Subgradients effectively generalize the gradient by defining a supporting line or slope that stays under the function. The subdifferential is collection of subgradients into a set. Though subgradients do not replace traditional gradients, they make it possible to tackle convex, nonsmooth functions.

Scribe Team: Eric Wong, Don Tran

9.1 Lecture Notes

The goal of this lecture is to develop tools that remain effective even when classical gradient-based methods fail, that is, when the objective function is nonsmooth.

Consider the absolute value function $f(x) = |x|$, a simple convex but nonsmooth function. At $x = 0$, f is not differentiable, so the gradient does not exist. Without a well-defined gradient, standard first-order methods cannot be applied directly. This forces us to resort to derivative-free (0th-order) optimization methods, which are often computationally expensive in high dimensions and typically come with weaker convergence guarantees compared to gradient-based methods.

Another setting where smooth methods break down is when f is non-convex but lies close to some convex differentiable function. In such cases, we would like to keep the spirit of gradient-based reasoning while relaxing the definition just enough to handle the “weak” nonconvexity, while preserving the fundamental properties (such as optimality conditions and calculus rules) that make gradients useful.

To address nonsmooth problems (whether the function is convex or non-convex), we introduce the concepts of **subgradients and the subdifferential**. These generalizations allow us to extend many fundamental results from smooth convex optimization. In particular, they **enable us to formulate necessary (and, in the convex case, sufficient) optimality conditions**.

Definition 9.1 (subgradient in $\mathbb{R}^d$). For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, a vector $g \in \mathbb{R}^d$ is called a **subgradient** of f at $x \in \text{dom}(f)$ if

$$f(y) \geq f(x) + \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^d.$$

Definition 9.2 (subgradient, Duality Pair). For a function $f : \mathbb{E} \rightarrow \mathbb{R}$ with dual pairing $\langle \cdot, \cdot \rangle : \mathbb{E} \times \mathbb{E}^* \rightarrow \mathbb{R}$, a vector $g \in \mathbb{E}^*$ is called a **subgradient** of f at $x \in \text{dom}(f)$ if

$$f(y) \geq f(x) + \langle g, y - x \rangle_{\text{dual}} \quad \forall y \in \text{dom}(f).$$

Example 9.1.1 (Probability measures and smooth functions). Let $\mathbb{E} = \{\text{probability measures on } X\}$, so that $\rho \in \mathbb{E}$ iff $\int_X d\rho = 1$. Let $\mathbb{E}^* = \{\text{smooth functions on } X\}$, so that $g \in \mathbb{E}^*$ iff $g \in C^\infty(X)$. The natural dual pairing $\langle \cdot, \cdot \rangle : \mathbb{E} \times \mathbb{E}^* \rightarrow \mathbb{R}$ is given by

$$\langle \rho, g \rangle = \int_X g(x) d\rho(x),$$

which is the integration pairing supplied by the Riesz Representation Theorem.

The subgradient inequality $f(y) \geq f(x) + \langle g, y - x \rangle$ can be read geometrically: the right-hand side, viewed as a function of y , lies entirely below f and agrees with f at $y = x$. When f is non-convex or has a kink at x , there can be many such g , each one “supporting” f at x from below. This non-uniqueness motivates collecting all subgradients into a single set-valued object.

Definition 9.3 (subdifferential). The **subdifferential** of f at x , denoted $\partial f(x)$, is the set-valued map

$$\partial f : \mathbb{R}^d \rightarrow A \subseteq \mathbb{R}^d, \quad x \mapsto \partial f(x) = \{g \in \mathbb{R}^d : g \text{ is a subgradient of } f \text{ at } x\}.$$

Remark (Sum rule). There can be multiple subgradients at a single point x . When x is a relative interior point of $\text{dom}(f) \cap \text{dom}(g)$, the subdifferential satisfies the sum rule

$$\underbrace{\partial(f+g)}_{\text{set}} = \underbrace{\partial f}_{\text{set}} + \underbrace{\partial g}_{\text{set}}.$$

That is, for any $h \in \partial(f+g)$ we can write $h = h_1 + h_2$ with $h_1 \in \partial f$ and $h_2 \in \partial g$. Note this is a *set operation*, not a vector or scalar operation.

Proposition 9.1.2 (Basic properties of the subdifferential). Please refer to [Bec17, Section 3 Subgradient] for subgradients calculus.

1. If f is differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$ is single-valued.

2. If $f : \mathbb{E} \rightarrow (-\infty, +\infty]$ but $f \not\equiv +\infty$ everywhere, then $\partial f(x)$ is a closed (limit) and convex (combination / average / geodesic) set.

Theorem 9.4 (Fermat's optimality condition). Let $f : \mathbb{E} \rightarrow \mathbb{R}$ be convex. Then

$$\bar{x} = \arg \min_{x \in \mathbb{E}} f(x) \Leftrightarrow 0 \in \partial f(\bar{x}).$$

Proof. ($\Leftarrow$) If $0 \in \partial f(\bar{x})$, then by definition of subgradient

$$f(x) \geq f(\bar{x}) + \langle 0, x - \bar{x} \rangle = f(\bar{x}) \quad \forall x \in \mathbb{E},$$

so $\bar{x}$ is a global minimizer.

($\Rightarrow$) Conversely, if $\bar{x}$ minimizes f , then $f(x) \geq f(\bar{x}) = f(\bar{x}) + \langle 0, x - \bar{x} \rangle$ for all x , so 0 satisfies the subgradient inequality and $0 \in \partial f(\bar{x})$. ■

Example 9.1.3 (Constrained minimization via indicator functions). Suppose we want to describe the minimizer of the constrained problem

$$\min_{x \in C} f(x).$$

Recalling the indicator function

$$I_C(x) = \begin{cases} 0 & x \in C, \\ +\infty & x \notin C, \end{cases}$$

this is equivalent to the unconstrained problem

$$\min_x f(x) + I_C(x).$$

Assume f is convex and differentiable. By Fermat's optimality condition together with the sum rule,

$$\bar{x} = \arg \min f(x) + I_C(x) \Leftrightarrow 0 \in \partial[f(\bar{x}) + I_C(\bar{x})] = \partial f(\bar{x}) + \partial I_C(\bar{x}) = \{\nabla f(\bar{x})\} + \partial I_C(\bar{x}).$$

Equivalently,

$$-\nabla f(\bar{x}) \in \partial I_C(\bar{x}),$$

which is our criterion to test whether $\bar{x}$ is a minimizer (see also Theorem 3.1). Moreover, $\partial I_C(\bar{x}) = N_C(\bar{x})$, the normal cone (see exercise below), so the condition reads

$$-\nabla f(\bar{x}) \in N_C(\bar{x}).$$

Example 9.1.4 (ℓ_1 norm). Let $f(x) = \|x\|_1 = \sum_{i=1}^n |x_i|$. Since the subdifferential of a separable sum decomposes coordinatewise, $\partial f(x) = \sum_{i=1}^n \partial |x_i|$, where for each scalar component

$$\partial |x_i| = \begin{cases} \{+1\} & x_i > 0, \\ [-1, +1] & x_i = 0, \\ \{-1\} & x_i < 0. \end{cases}$$

At $x = 0$, the subdifferential is the ℓ_∞ unit ball $[-1, 1]^n$, which is consistent with the general fact for norms shown next.

Example 9.1.5 (ℓ_2 norm). Let $f(x) = \|x\|_2$. Then

$$\partial \|x\|_2 = \begin{cases} \frac{x}{\|x\|_2} & x \neq 0, \\ \{y \mid \|y\|_2 \leq 1\} & x = 0. \end{cases}$$

Away from the origin, $\|\cdot\|_2$ is differentiable with the displayed gradient. At the origin, any vector in the unit ball is a valid subgradient, again recovering the general norm result below.

Example 9.1.6 (Subdifferential of an arbitrary norm at 0). Let $f : \mathbb{E} \rightarrow \mathbb{R}$, defined by the map $x \mapsto \|x\|$, to be any norm. Then the subdifferential at the origin is the closed unit ball of the dual norm:

$$\partial f(0) = \{g \in \mathbb{E}^* : \|g\|_* \leq 1\}.$$

Proof. Let $g \in \mathbb{E}^*$ and let $\langle \cdot, \cdot \rangle$ denote the dual pair. Recall the definition of the dual norm,

$$\|g\|_* = \sup_{\substack{x \in \mathbb{E} \\ \|x\|=1}} \langle x, g \rangle = \sup_{\substack{x \in \mathbb{E} \\ \|x\| \leq 1}} \langle x, g \rangle = \sup_{\substack{x \in \mathbb{E} \\ x \neq 0}} \frac{\langle x, g \rangle}{\|x\|}.$$

By the definition of subgradient,

$$g \in \partial f(0) \Leftrightarrow f(y) \geq f(0) + \langle g, y - 0 \rangle \quad \forall y \in \text{dom}(f),$$

which simplifies (since $f(0) = \|0\| = 0$) to

$$\|y\| \geq \langle g, y \rangle \quad \forall y \in \mathbb{E}.$$

We show this is equivalent to $\|g\|_* \leq 1$.

($\Leftarrow$) Assume $\|g\|_* \leq 1$. By the Cauchy–Schwarz inequality for duality,

$$\langle g, y \rangle \leq \|g\|_* \|y\| \leq \|y\|,$$

so the subgradient inequality holds.
 $(\Rightarrow)$ Assume $\|y\| \geq \langle g, y \rangle$ for all y . Then

$$\|g\|_* = \sup_{y \neq 0} \frac{\langle g, y \rangle}{\|y\|} \leq 1.$$

■

9.2 Exercises

Exercise. In Section 1 and Section 3, we introduce the indicator function $I_C(x)$ and normal cone function $N_C(x)$ respectively. Show that for any nonempty set, $\partial I_C(x) = N_C(x)$ for any $x \in S$.

Proof. Drafted by Ben Pearson, Proofread by Tao Xu

Recall the definitions. The indicator function of a nonempty set $C \subseteq \mathbb{R}^d$ is

$$I_C(x) = \begin{cases} 0 & x \in C, \\ +\infty & x \notin C, \end{cases}$$

the subdifferential is

$$\partial I_C(x) = \{g \in \mathbb{R}^d : I_C(y) \geq I_C(x) + \langle g, y - x \rangle \forall y \in \mathbb{R}^d\},$$

and the normal cone at $x \in C$ is

$$N_C(x) = \{g \in \mathbb{R}^d : \langle g, y - x \rangle \leq 0 \forall y \in C\}.$$

If $x \notin C$, then $I_C(x) = +\infty$, so $\partial I_C(x) = \emptyset$, and by convention $N_C(x) = \emptyset$ as well. It therefore suffices to consider $x \in C$.

$(\subseteq)$ Let $g \in \partial I_C(x)$, so that

$$I_C(y) \geq I_C(x) + \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^d.$$

Since $x \in C$ we have $I_C(x) = 0$, so the inequality becomes

$$I_C(y) \geq \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^d.$$

For any $y \in C$, we have $I_C(y) = 0$, hence

$$\langle g, y - x \rangle \leq 0 \quad \forall y \in C.$$

This is exactly the definition of $g \in N_C(x)$, so $g \in N_C(x)$.

$(\supseteq)$ Let $g \in N_C(x)$, so that

$$\langle g, y - x \rangle \leq 0 \quad \forall y \in C.$$

We must verify the subgradient inequality $I_C(y) \geq I_C(x) + \langle g, y - x \rangle$ for all $y \in \mathbb{R}^d$. Since $x \in C$, we have $I_C(x) = 0$, so this reduces to

$$I_C(y) \geq \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^d.$$

There are two cases:

- If $y \in C$: then $I_C(y) = 0$ and $\langle g, y - x \rangle \leq 0$ by assumption, so $I_C(y) = 0 \geq \langle g, y - x \rangle$. ✓
- If $y \notin C$: then $I_C(y) = +\infty \geq \langle g, y - x \rangle$. ✓

In both cases the subgradient inequality holds, so $g \in \partial I_C(x)$.

Since both inclusions hold, $\partial I_C(x) = N_C(x)$ for all $x \in C$.

■

9.3 More Examples

Example 9.3.1 (Proposed by Run Li). In classical mechanics, dry friction (Coulomb friction) presents a classic non-smooth phenomenon. When two surfaces slide against each other (relative velocity $v \neq 0$), the kinetic friction force is a constant magnitude. However, when the surfaces are stationary relative to each other ($v = 0$), the static friction force does not have a single fixed formula. It adjusts within a closed interval to balance any applied external force. Traditional smooth calculus fails to provide a single unified equation for both states. The subdifferential provides a natural mathematical framework to resolve this.

Optimization Formulation. We can model friction using a dissipation potential function. Let μ be the friction coefficient and F_n be the normal force. The dissipation potential with respect to the relative velocity $v \in \mathbb{R}$ is defined as

$$D(v) = \mu F_n |v|$$

In variational mechanics, the friction force F_{fric} is formally defined as the negative subdifferential of the dissipation potential with respect to velocity (the negative sign ensures the force opposes motion):

$$F_{\text{fric}} \in -\partial D(v)$$

Derivation using Subgradients. We apply the subdifferential calculus for the absolute value function. The function $D(v)$ is convex. We evaluate its subdifferential in three cases.

When $v > 0$, the system is sliding forward. The function $D(v)$ is differentiable and its subdifferential is $\partial D(v) = \{\mu F_n\}$. Thus, $F_{\text{fric}} = -\mu F_n$.

When $v < 0$, the system is sliding backward. The subdifferential is $\partial D(v) = \{-\mu F_n\}$. Thus, $F_{\text{fric}} = \mu F_n$.

When $v = 0$, the system is stationary. The function $D(v)$ is non-differentiable at this point. Recalling the property of the absolute value function where $\partial|v|_{v=0} = [-1, 1]$, we scale it by μF_n . Since the interval is symmetric, its negation remains the same:

$$-\partial D(0) = [-\mu F_n, \mu F_n]$$

Connection to Fermat's Optimality Condition. Suppose the system is subject to an external applied force F_{ext} . The effective energy functional of the system is given by $E(v) = D(v) - F_{\text{ext}}v$. By Fermat's optimality condition (Theorem 9.4), the system reaches a stable equilibrium (minimizes energy) at rest ($v = 0$) if and only if $0 \in \partial E(0)$. Utilizing the sum rule for subdifferentials, we obtain:

$$0 \in \partial D(0) - F_{\text{ext}} \implies F_{\text{ext}} \in \partial D(0) = [-\mu F_n, \mu F_n]$$

At equilibrium, Newton's first law requires $F_{\text{fric}} + F_{\text{ext}} = 0$, which perfectly aligns with our formulation:

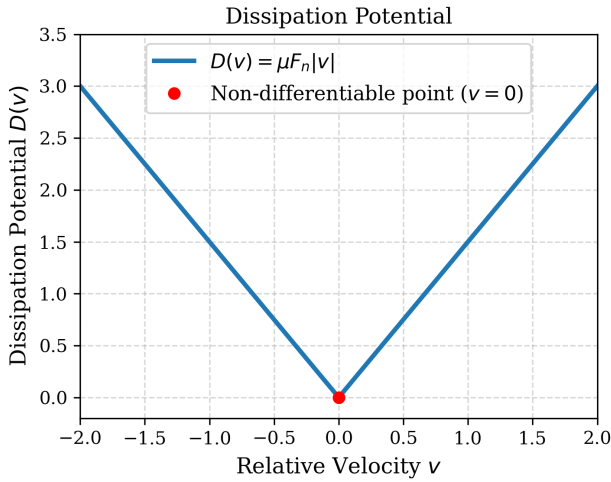
$$F_{\text{fric}} = -F_{\text{ext}} \in -\partial D(0).$$

Physical Interpretation and ℓ_1 -Norm Sparsity. Notice the structural equivalence between the dissipation potential $D(v) = \mu F_n |v|$ and the 1D weighted ℓ_1 -norm regularization. Just as the ℓ_1 -norm penalty in optimization forces parameter sparsity (exact zero solutions) due to its wide subdifferential interval at the origin, dry friction physically forces the velocity to be exactly zero, creating the phenomenon of macroscopic rest.

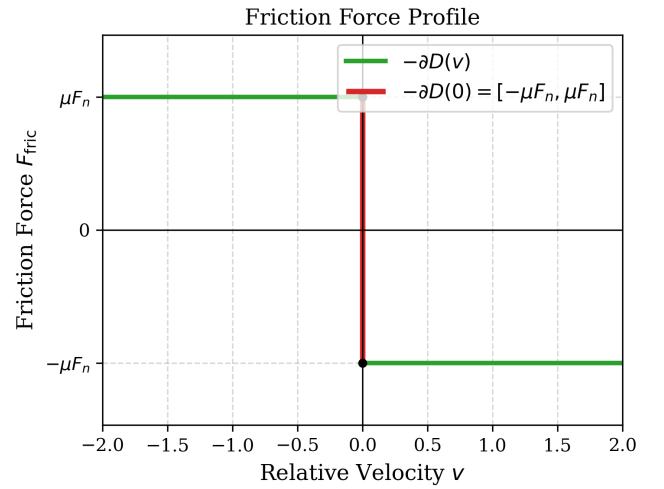
Furthermore, the set-valued inclusion $F_{\text{fric}} \in -\partial D(v)$ elegantly replaces the cumbersome piecewise formulation found in traditional mechanics:

$$F_{\text{fric}} = \begin{cases} -\mu F_n \operatorname{sgn}(v) & \text{if } v \neq 0 \\ -F_{\text{ext}} \quad (\text{where } |F_{\text{ext}}| \leq \mu F_n) & \text{if } v = 0 \end{cases}$$

with a single, mathematically unified relationship.



(a) Dissipation potential $D(v)$ with a sharp corner at $v = 0$.



(b) The set-valued negative subdifferential $-\partial D(v)$ representing friction force.

Figure 18: Geometric relationship between the non-smooth dissipation potential and the multi-valued friction force via subgradients.

Example 9.3.2 (Proposed by Ben Thomas. Nonnegative least squares: a minimizer where the gradient is not zero.). For an unconstrained differentiable problem, Fermat's condition is simply

$$\nabla f(\bar{x}) = 0.$$

For a constrained problem, this is no longer the right condition. If the feasible set is C , then the correct condition is

$$-\nabla f(\bar{x}) \in N_C(\bar{x}).$$

The normal cone keeps track of the directions in which the constraint prevents us from moving.

Consider the nonnegative least squares problem

$$\min_{x \geq 0} f(x) = \frac{1}{2} \|Ax - b\|^2, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

First, ignore the constraint. Then the usual least-squares solution is

$$x_{\text{unc}} = (A^\top A)^{-1} A^\top b.$$

Here

$$A^\top A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad A^\top b = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

so

$$x_{\text{unc}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

This is not feasible, since the first coordinate is negative.

The constraint $x \geq 0$ forces us to stay in the nonnegative quadrant. Since the unconstrained solution wants $x_1 < 0$, we try the boundary $x_1 = 0$. On that boundary,

$$x = \begin{pmatrix} 0 \\ x_2 \end{pmatrix}.$$

Then

$$Ax = \begin{pmatrix} x_2 \\ 0 \end{pmatrix},$$

so the objective becomes

$$f(0, x_2) = \frac{1}{2} \left\| \begin{pmatrix} x_2 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} \right\|^2 = \frac{1}{2} ((x_2 - 2)^2 + 1).$$

This is minimized when

$$x_2 = 2.$$

Thus the natural candidate is

$$\bar{x} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Now we check optimality using the normal cone. Since

$$\nabla f(x) = A^\top (Ax - b) = A^\top Ax - A^\top b,$$

we get

$$\nabla f(\bar{x}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

So

$$\nabla f(\bar{x}) \neq 0.$$

This does not mean $\bar{x}$ is not a minimizer. It only means that the usual unconstrained condition is not the right one here.

At $\bar{x} = (0, 2)$, the first coordinate is on the boundary and the second coordinate is in the interior. Therefore, for

$$C = \{x : x_1 \geq 0, x_2 \geq 0\},$$

the normal cone is

$$N_C(\bar{x}) = \{g : g_1 \leq 0, g_2 = 0\}.$$

But

$$-\nabla f(\bar{x}) = -\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix},$$

and this vector lies in $N_C(\bar{x})$. Hence

$$-\nabla f(\bar{x}) \in N_C(\bar{x}),$$

so $\bar{x} = (0, 2)$ satisfies the constrained Fermat condition.

The point is that the gradient does not have to vanish at a constrained minimizer. In this example,

$$\nabla f(\bar{x}) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The second component is zero because $x_2 > 0$, so we are free to move in that coordinate. But the first component is positive because the function would like to decrease by moving in the negative x_1 -direction. That move is forbidden by the constraint $x_1 \geq 0$. The normal cone condition records exactly this: the only descent direction left is blocked by the boundary.

Equivalently, writing the constraint using the indicator function I_C , Fermat's condition is

$$0 \in \partial(f + I_C)(\bar{x}) = \nabla f(\bar{x}) + N_C(\bar{x}),$$

or

$$-\nabla f(\bar{x}) \in N_C(\bar{x}),$$

using $\partial I_C = N_C$ from Example 9.1.3.

Example 9.3.3 (*Proposed by Andrew Wang*). The hinge loss is a standard nonsmooth convex loss function used in support vector machines. In one dimension, it is defined by

$$\ell(z) = \max\{0, 1 - z\}.$$

This function is convex because it is the maximum of two affine functions:

$$0 \quad \text{and} \quad 1 - z.$$

However, it is not differentiable at $z = 1$, where the two affine pieces meet.

We first compute the subdifferential of ℓ . If $z < 1$, then the active piece is $1 - z$, so

$$\partial \ell(z) = \{-1\}.$$

If $z > 1$, then the active piece is 0, so

$$\partial \ell(z) = \{0\}.$$

At $z = 1$, both affine pieces are active. The possible slopes are -1 and 0 , and the subdifferential is the convex hull of these two values:

$$\partial \ell(1) = [-1, 0].$$

Therefore,

$$\partial\ell(z) = \begin{cases} \{-1\}, & z < 1, \\ [-1, 0], & z = 1, \\ \{0\}, & z > 1. \end{cases}$$

Now consider a binary classification data point (a_i, y_i) , where

$$a_i \in \mathbb{R}^d, \quad y_i \in \{-1, 1\}.$$

For a linear classifier with parameter $w \in \mathbb{R}^d$ and intercept $b \in \mathbb{R}$, the margin is

$$z_i = y_i(\langle w, a_i \rangle + b).$$

The corresponding hinge loss is

$$\phi_i(w, b) = \max\{0, 1 - y_i(\langle w, a_i \rangle + b)\}.$$

Using the subdifferential of ℓ and the affine change of variables

$$z_i = y_i(\langle w, a_i \rangle + b),$$

we obtain the following subgradient with respect to (w, b) :

$$\partial\phi_i(w, b) = \begin{cases} \{(-y_i a_i, -y_i)\}, & y_i(\langle w, a_i \rangle + b) < 1, \\ \{(-\theta y_i a_i, -\theta y_i) : \theta \in [0, 1]\}, & y_i(\langle w, a_i \rangle + b) = 1, \\ \{(0, 0)\}, & y_i(\langle w, a_i \rangle + b) > 1. \end{cases}$$

This formula has a simple interpretation. If the data point has margin greater than 1, then it contributes no subgradient:

$$\partial\phi_i(w, b) = \{(0, 0)\}.$$

If the margin is less than 1, then the data point violates the desired margin condition and contributes the subgradient

$$(-y_i a_i, -y_i).$$

If the margin is exactly 1, then the loss is nonsmooth and the subdifferential is a whole line segment between these two possibilities.

For example, the regularized support vector machine objective can be written as

$$F(w, b) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max\{0, 1 - y_i(\langle w, a_i \rangle + b)\},$$

where $C > 0$. The term $\frac{1}{2} \|w\|^2$ is smooth, while the hinge losses are convex but nonsmooth. Therefore, subgradients allow us to apply first-order optimization ideas even though the objective is not differentiable everywhere.

This example illustrates the main idea of this lecture: subgradients extend gradient-based reasoning to nonsmooth convex functions. At smooth points of the hinge loss, the subdifferential contains a single slope. At the kink $z = 1$, the subdifferential contains multiple possible slopes, which records the nonsmooth geometry of the function.

Example 9.3.4 (*Proposed by Bradley Nakagawa*). If a function is subdifferentiable (that is, it has a nonempty subdifferential at every point in its domain), it is convex.

Since every point has a nonempty subdifferential, there is therefore at least one supporting hyperplane for the epigraph of the function at every point in the domain. Because the function values are the boundary of the epigraph, this means that the epigraph has supporting hyperplanes all along its boundary. Thus, the epigraph is a convex set. Therefore, the function is convex.

We can now ask: Is the converse true? If a function is convex, then is it subdifferentiable on its whole domain?

The converse is not true in general. For example, consider the function

$$f(x) = \begin{cases} +\infty & x < 0 \\ -\sqrt{x} & x \geq 0 \end{cases}$$

This function is convex by the 0-th order definition of convexity.

Case 1: If both $x, y < 0$ and $t \in [0, 1]$, then

$$f((1-t)x + ty) = +\infty$$

and

$$(1-t)f(x) + tf(y) = f(x) + t(f(y) - f(x)) = +\infty$$

So

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

Case 2: If both $x, y \geq 0$ and $t \in [0, 1]$, then

$$f((1-t)x + ty) = -\sqrt{(1-t)x + ty}$$

and

$$(1-t)f(x) + tf(y) = (1-t)(-\sqrt{x}) + t(-\sqrt{y})$$

Since we know that both of the above expressions are negative, we know that their squares will have the opposite

relation as the original expressions. We can determine their relationship by subtracting them from each other.

$$(1-t)x + ty - ((1-t)^2x + 2(1-t)t\sqrt{xy} + t^2y)$$

Expanding and rearranging this gives us

$$(1-t)x - (1-2t+t^2)x + ty - t^2y - 2(1-t)t\sqrt{xy}$$

Then we get

$$(t-t^2)x + (t-t^2)y - 2(t-t^2)\sqrt{xy}$$

Dividing by $(t-t^2)$, which is positive since $t \in [0, 1]$, gives us

$$x + y - 2\sqrt{xy} = (\sqrt{x} - \sqrt{y})^2 \geq 0$$

This means that

$$(f((1-t)x + ty))^2 - ((1-t)f(x) + tf(y))^2 \geq 0$$

and so

$$(f((1-t)x + ty))^2 \geq ((1-t)f(x) + tf(y))^2$$

Thus,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

Case 3: WLOG, if $x < 0$, $y \geq 0$, and $t \in [0, 1]$, then there are two possibilities: If $t = 1$, then

$$f((1-t)x + ty) = f(y)$$

and

$$(1-t)f(x) + tf(y) = f(y)$$

If $t \neq 1$, then $1-t \geq 0$, so

$$(1-t)f(x) + tf(y) = +\infty$$

So

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

Despite this function being convex, it does not have a nonempty subdifferential at $x = 0$.

We can prove this by contradiction. Suppose $g \in \partial f(0)$. Then for all y ,

$$f(y) \geq f(0) + g(y-0)$$

This must also hold true for $y > 0$, so

$$-\sqrt{y} \geq gy$$

Therefore,

$$g \leq -\frac{1}{\sqrt{y}} < 0$$

This must hold for all $y > 0$. So let $y = \frac{1}{4g^2} > 0$. Then

$$g \leq -\frac{1}{\sqrt{\frac{1}{4g^2}}} = -2|g|$$

This is not possible, so the subdifferential of f must be empty.

Remark. Although we cannot guarantee that the subdifferential is nonempty for the entirety of the domain of a convex function, we still can guarantee its existence in the interior of the domain by the supporting hyperplane theorem (mentioned in Example 1.1.2).

10 Lecture Notes 10 on Proximal Operator

Jacob Jurek summarizes: This lecture discussed two perspectives of the proximal operator: one from regularization, where it arises naturally in problems like LASSO and soft-thresholding to encourage sparsity, and the other from gradient flow, where the proximal update corresponds to the backward Euler discretization of gradient descent. sec:lec10

Scribe Team: Ben Thomas and Run Li.

Please refer to [Bec17, Section 6, The Proximal Operator] for proximal operator calculus.

10.1 Lecture Notes

Our goal in this lecture is to bridge the concept of regularization, specifically how it induces sparsity, with the numerical tools used to solve non-smooth optimization problems. We will see how the closed-form solution of the LASSO problem naturally motivates a general mathematical tool called the proximal operator, which in turn provides a stable backward Euler discretization for gradient flow.

The roadmap for the discussion is as follows.

Regression + Regularization $\implies$ Proximal Operator;

Proximal Operator $\implies$ Gradient Flow.

10.1.1 From Regularization to the Proximal Operator

The goal of regression is to find the best parameter β that fits the data $(x_1, x_2, \dots, x_d, y)$ in the least-squared error sense.

For data points $(x^{(i)}, y^{(i)})$, $i = 1, \dots, m$, ordinary least squares solves

$$\beta = \arg \min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2,$$

where y is the output vector, X is the data matrix, and β is the parameter vector. And the simple regression case expands to

$$\frac{1}{2} \sum_{i=1}^m |y^{(i)} - \beta_0 - \beta_1 x^{(i)}|^2,$$

When the regression problem is underdetermined, there may be many possible solutions. In this case we add a regularization/filter term to

1. encode user-specific preferences,
2. help with computation,
3. prefer simpler or sparse solutions.

The general regularized regression problem is

$$\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda R(\beta),$$

where λ is the regularization weight.

Remark. A large λ makes the model sensitive to the user's preferences; a small λ means those preferences are not a high priority.

Now consider several choices for $R(\beta)$.

ℓ_0 regularization.

$$\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_0,$$

where $\|\beta\|_0$ counts the number of nonzero entries of $\beta = (\beta_0, \beta_1, \dots, \beta_d)$. This directly prefers sparsity: more zeros in β means a smaller $\|\beta\|_0$.

ℓ_1 regularization: LASSO (Least Absolute Shrinkage and Selection Operator).

$$\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1.$$

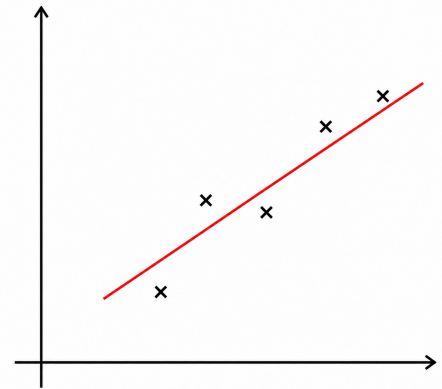
This is the LASSO problem. It still encourages sparsity, and is generally more computationally manageable than ℓ_0 .

ℓ_2 regularization.

$$\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2,$$

where $\|\beta\|_2$ is the standard vector length. This is convex and differentiable, but does not usually force coefficients to be exactly zero.

Geometry of ℓ_1 and ℓ_2 . For ℓ_1 , the unit ball is $\|\beta\|_1 = |\beta_1| + |\beta_2| \leq 1$ (a diamond).



For ℓ_2 , the unit ball is $\|\beta\|_2 = \sqrt{\beta_1^2 + \beta_2^2} \leq 1$ (a disk).

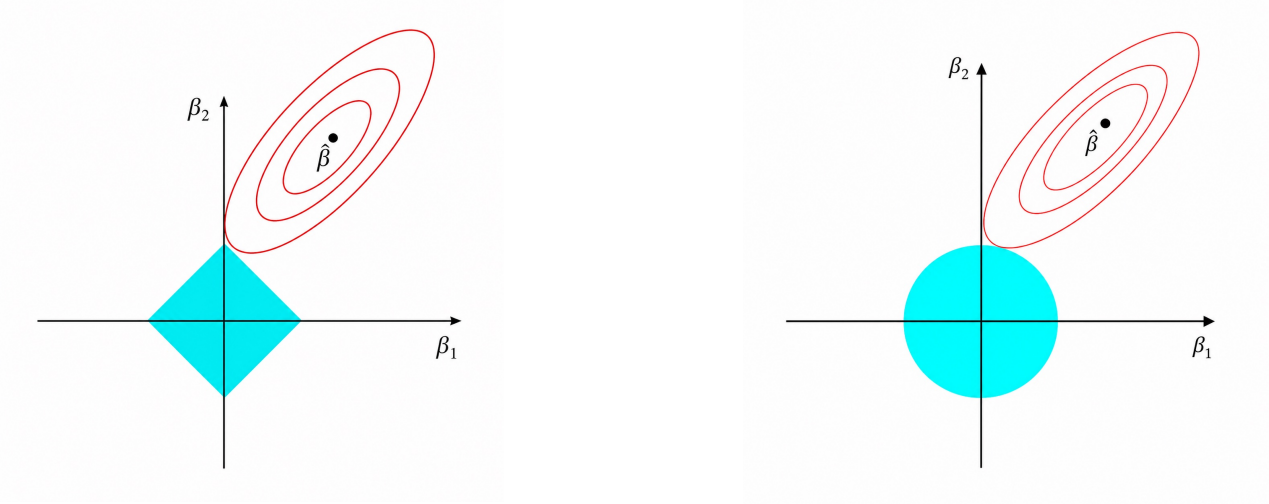


Figure 19: Unit balls for regularizers. Left: The ℓ_1 unit ball (diamond). Right: The ℓ_2 unit ball (disk).

Remark. The ℓ_1 ball has corners, so there is a higher probability that a corner becomes the touching point with the level sets of the least-squares objective. This is why ℓ_1 regularization is more likely to produce sparse solutions.

The regularized and constrained forms are equivalent:

$$\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \iff \min_{\|\beta\|_1 \leq t} \frac{1}{2} \|y - X\beta\|_2^2,$$

and similarly for ℓ_2 .

To further illustrate how the geometry of the unit ball influences sparsity, we can also consider the q -norm with $q < 1$, where

$$\|\beta\|_q = (|\beta_1|^q + \dots + |\beta_d|^q)^{1/q}.$$

The shape has sharper corners, but $\|\beta\|_q$ is non-convex.

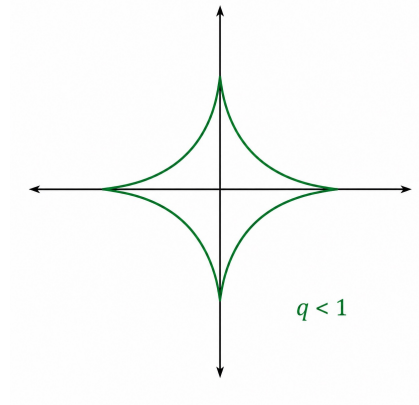


Figure 20: Unit ball for q -norm ($q < 1$).

Example 10.1.1 (LASSO and Soft-Thresholding). As established geometrically, the ℓ_1 penalty induces sparsity precisely because of its sharp corners. However, these corners also imply that the LASSO objective function is non-differentiable at points where $\beta_i = 0$. Therefore, instead of setting a standard gradient to zero, we must use the subdifferential and apply Fermat's condition.

To find the LASSO solution

$$\arg \min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1,$$

we apply Fermat's condition:

$$\mathbf{0} \in -X^T(y - X\beta) + \lambda \partial \|\beta\|_1,$$

which gives

$$X^T(y - X\beta) \in \lambda \partial \|\beta\|_1.$$

For the ℓ_1 -norm,

$$\|\beta\|_1 = \sum_{i=1}^d |\beta_i|, \quad \partial \|\beta\|_1 = \sum_{i=1}^d \partial |\beta_i|,$$

and coordinate-wise,

$$\partial|\beta_i| = \begin{cases} 1, & \beta_i > 0, \\ -1, & \beta_i < 0, \\ [-1, 1], & \beta_i = 0. \end{cases}$$

Thus an element of $\partial\|\beta\|_1$ is a vector $\nu = (\nu_1, \dots, \nu_d)$ with $\nu_i \in \partial|\beta_i|$. See Beck's textbook on subdifferential calculus for more details.

In general, the coupled data matrix X prevents an explicit closed-form solution for the LASSO problem. However, to gain intuition and derive an exact formula, we consider the decoupled orthogonal case.

Special case $X = I$. The LASSO problem becomes the soft-thresholding problem

$$\min_{\beta} \frac{1}{2} \|y - \beta\|^2 + \lambda \|\beta\|_1.$$

By Fermat's condition, the vector optimality condition is $y - \beta \in \lambda \partial\|\beta\|_1$. Because both the squared ℓ_2 distance and the ℓ_1 norm completely separate across dimensions, this vector inclusion decouples into independent scalar inclusions for each coordinate:

$$y_i - \beta_i \in \lambda \partial|\beta_i|.$$

This decoupled condition can be solved coordinate-wise by analyzing the three cases of the subdifferential:

- **Case 1** ($\beta_i > 0$): Here, $\partial|\beta_i| = 1$. The condition becomes $y_i - \beta_i = \lambda$, which yields $\beta_i = y_i - \lambda$. Since we assumed $\beta_i > 0$, this implies $y_i - \lambda > 0$, hence $y_i > \lambda$.
- **Case 2** ($\beta_i < 0$): Here, $\partial|\beta_i| = -1$. The condition becomes $y_i - \beta_i = -\lambda$, yielding $\beta_i = y_i + \lambda$. Since we assumed $\beta_i < 0$, this implies $y_i + \lambda < 0$, hence $y_i < -\lambda$.
- **Case 3** ($\beta_i = 0$): Here, $\partial|\beta_i| = [-1, 1]$. The condition becomes $y_i - 0 \in \lambda[-1, 1]$, which translates directly to $y_i \in [-\lambda, \lambda]$.

Combining these three cases yields the exact piecewise formula for the soft-thresholding operator:

$$\beta_i = \begin{cases} y_i - \lambda, & y_i > \lambda, \\ y_i + \lambda, & y_i < -\lambda, \\ 0, & y_i \in [-\lambda, \lambda]. \end{cases}$$

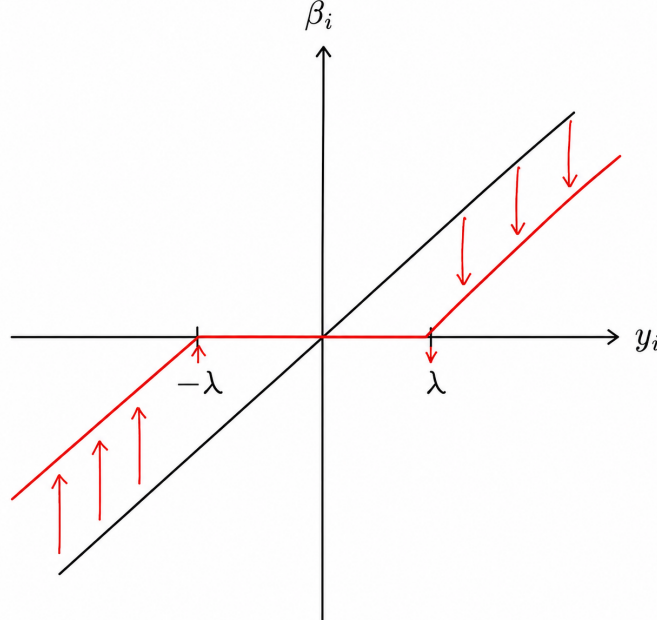


Figure 21: The soft-thresholding operator. The red line represents the optimal β_i as a function of the input y_i . The flat region where $\beta_i = 0$ for $y_i \in [-\lambda, \lambda]$ demonstrates the sparsity-inducing property of the ℓ_1 penalty.

So soft-thresholding takes input $(y_1, \dots, y_d)$ and produces sparse optimal output $(\beta_1, \dots, \beta_d)$.

Observe the underlying structure of this soft-thresholding objective: we minimized a non-smooth penalty function (the ℓ_1 norm) plus a quadratic distance to a fixed point y . This specific algebraic structure is a prime example of a much broader mathematical concept. By generalizing this structure, replacing the ℓ_1 norm with a general function f and introducing a scale parameter η , we arrive at the proximal operator.

10.1.2 Proximal Operator

For a general proper, closed, and convex function f , instead of directly solving $\min_x f(x)$, we define the proximal operator by

$$\mathbf{Prox}_{\eta f}(y) = \arg \min_x \left\{ f(x) + \frac{1}{2\eta} \|x - y\|^2 \right\},$$

where y is fixed and x is the variable. So

$$\mathbf{Prox}_{\eta f} : y \mapsto \bar{x},$$

where

$$\bar{x} \in \arg \min_x \left\{ f(x) + \frac{1}{2\eta} \|x - y\|^2 \right\} = \arg \min_x \left\{ \eta f(x) + \frac{1}{2} \|x - y\|^2 \right\}.$$

Remark. The quadratic term $\frac{1}{2\eta} \|x - y\|^2$ acts as a regularizer: it makes the objective $\frac{1}{\eta}$ -strongly convex and differentiable in x , regardless of whether f has those properties.

Theorem 10.1. If $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is a proper, closed, and convex function, then

$$\min_x \left\{ f(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}$$

has a unique solution, regardless of y .

Proof. Let

$$g(x; y) = f(x) + \frac{1}{2\eta} \|x - y\|^2.$$

Since

$$g(x; y) - \frac{1}{\eta} \cdot \frac{1}{2} \|x - y\|^2 = f(x)$$

is convex, g is $\frac{1}{\eta}$ -strongly convex in x . Hence the minimizer is unique. ■

10.1.3 Proximal Update and Gradient Flow

If f is convex on a domain $D \subseteq \mathbb{R}^d$, then $\mathbf{Prox}_{\eta f}$ is well-defined on all of $\mathbb{R}^d$:

$$\mathbf{Prox}_{\eta f}(y) = \arg \min_{x \in D} \left\{ f(x) + \frac{1}{2\eta} \|y - x\|^2 \right\}, \quad y \in \mathbb{R}^d.$$

(We postpone the proof to the next lecture.)

With the operator properly defined for a single input y , we can construct an iterative optimization algorithm by repeatedly applying it. Instead of a fixed y , we feed the current iterate x^k into the operator to generate the next iterate x^{k+1} . This recursive application is known as the proximal update:

$$x^{k+1} = \mathbf{Prox}_{\eta f}(x^k) = \arg \min_x \left\{ f(x) + \frac{1}{2\eta} \|x - x^k\|^2 \right\}.$$

This discrete update scheme is deeply connected to continuous-time gradient flow. To see this connection, we examine the first-order optimality condition of the update step. Since f is convex,

$$0 \in \partial f(y) + \frac{1}{\eta} (y - x^k) \Big|_{y=x^{k+1}} = \partial f(x^{k+1}) + \frac{1}{\eta} (x^{k+1} - x^k),$$

where $\partial f(x^{k+1})$ is set-valued. Hence

$$\frac{1}{\eta} (x^{k+1} - x^k) \in -\partial f(x^{k+1}).$$

Further assume that f is differentiable, so $\partial f(x^{k+1}) = \{\nabla f(x^{k+1})\}$. Then

$$\frac{1}{\eta} (x^{k+1} - x^k) = -\nabla f(x^{k+1}).$$

This discrete update equation is precisely the **backward Euler scheme** for numerical integration. By taking the continuous-time limit as the step size $\eta \rightarrow 0$, this difference equation recovers the continuous gradient flow ODE:

$$\dot{x} = -\nabla f(x).$$

Table 3: Comparison of Forward and Backward Euler Schemes

Backward Euler	Forward Euler
Implicit	Explicit
Slow	Fast
Stable for moderate stepsizes	Unstable unless stepsize is small
Proximal method / minimizing movement	Gradient descent

One of the most powerful aspects of the proximal update (referred to earlier as the minimizing movement scheme) is that its formulation does not explicitly require gradients or a vector space structure. It only relies on function evaluations and a distance measure. Consequently, this framework naturally extends to general metric spaces.

Let $F : (X, d) \rightarrow \mathbb{R}$ be a function defined on a metric space (X, d) . Then the proximal update step becomes

$$x^{k+1} \in \arg \min_{x \in X} \left\{ F(x) + \frac{1}{2\eta} d^2(x, x^k) \right\}.$$

This directly generalizes the Euclidean proximal update

$$x^{k+1} = \arg \min_x \left\{ f(x) + \frac{1}{2\eta} \|x - x^k\|^2 \right\},$$

simply by replacing the squared Euclidean norm $\|x - x^k\|^2$ with the squared metric distance $d^2(x, x^k)$.

10.2 Exercises

Exercise (Proposed by Chenghan Li, Proofread by Eric Wong). Compute the proximal operator of the function

$$f(x) = \lambda|x|, \quad x \in \mathbb{R},$$

where $\lambda > 0$. In other words, find

$$\text{Prox}_{\eta f}(z) = \arg \min_{x \in \mathbb{R}} \left\{ \lambda|x| + \frac{1}{2\eta} (x - z)^2 \right\}.$$

Proof. We solve the minimization problem

$$\min_{x \in \mathbb{R}} \left\{ \lambda|x| + \frac{1}{2\eta} (x - z)^2 \right\}.$$

Since the objective is convex (a sum of a convex and a strictly convex function), the subgradient optimality condition $0 \in \partial g(x)$ is necessary and sufficient for a global minimum. Since the absolute value is not differentiable at $x = 0$, we consider three cases.

If $x > 0$, then $|x| = x$. The objective becomes

$$\lambda x + \frac{1}{2\eta} (x - z)^2.$$

Taking the derivative and setting it equal to zero gives

$$\lambda + \frac{1}{\eta} (x - z) = 0.$$

Thus

$$x = z - \eta\lambda.$$

This case is valid when $x > 0$, so we need $z > \eta\lambda$.

If $x < 0$, then $|x| = -x$. The objective becomes

$$-\lambda x + \frac{1}{2\eta} (x - z)^2.$$

Taking the derivative and setting it equal to zero gives

$$-\lambda + \frac{1}{\eta} (x - z) = 0.$$

Thus

$$x = z + \eta\lambda.$$

This case is valid when $x < 0$, so we need $z < -\eta\lambda$.

If neither of the above conditions holds, i.e., $|z| \leq \eta\lambda$, then the minimizer is $x = 0$. We verify this via the subgradient optimality condition

$$0 \in \lambda \partial|x| + \frac{1}{\eta} (x - z).$$

At $x = 0$, we have $\partial|0| = [-1, 1]$, and $\frac{1}{\eta} (0 - z) = -\frac{z}{\eta}$, so the condition becomes

$$0 \in \lambda[-1, 1] - \frac{z}{\eta},$$

which is equivalent to $\frac{z}{\eta} \in \lambda[-1, 1]$, i.e., $|z| \leq \eta\lambda$. This confirms that $x = 0$ is the global minimizer in this case. Therefore,

$$\text{Prox}_{\eta f}(z) = \begin{cases} z - \eta\lambda, & z > \eta\lambda, \\ 0, & |z| \leq \eta\lambda, \\ z + \eta\lambda, & z < -\eta\lambda. \end{cases}$$

This is the soft-thresholding operator. ■

10.3 More Examples

Example 10.3.1 (Proposed by Ben Thomas: Overview of the graphical LASSO (Friedman–Hastie–Tibshirani, 2008)). For jointly Gaussian variables, the inverse covariance tells us which variables are conditionally related, and estimating it sparsely gives a composite optimization problem like the ones studied in Lecture 12.

Model. $x^{(1)}, \dots, x^{(n)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma), \Sigma \succ 0,$

$$S = \frac{1}{n} \sum_{k=1}^n x^{(k)} x^{(k)\top}, \quad \Theta = \Sigma^{-1}, \quad \Theta_{ij} = 0 \Leftrightarrow x_i \perp\!\!\!\perp x_j \mid x_{\text{others}}.$$

A nonzero off-diagonal entry Θ_{ij} is a conditional-dependence edge (not a causal one); the graph is determined by the off-diagonal support of Θ .

Problem. The MLE maximizes $\log \det \Theta - \text{tr}(S\Theta)$:

$$\nabla (\log \det \Theta - \text{tr}(S\Theta)) = \Theta^{-1} - S = 0 \implies \hat{\Theta}_{\text{MLE}} = S^{-1}.$$

The estimator is generally dense and may be unstable. Moreover, S is singular whenever the data matrix has rank less than p , which necessarily occurs when $n < p$. The graphical LASSO solves the penalized problem:

$$\min_{\Theta \succ 0} \underbrace{-\log \det \Theta + \text{tr}(S\Theta)}_{f(\Theta) \text{ smooth, convex on } \Theta \succ 0} + \underbrace{\lambda \|\Theta\|_1}_{h(\Theta) \text{ convex, nonsmooth}}.$$

Here $-\log \det \Theta$ is a barrier ($\rightarrow +\infty$ as $\Theta \rightarrow \partial\{\succ 0\}$) and $\text{tr}(S\Theta)$ is linear in Θ .

Algorithm (block coordinate descent on $W = \Theta^{-1}$, not proximal-gradient). KKT, with $\Gamma \in \partial \|\Theta\|_1$:

$$0 \in -\Theta^{-1} + S + \lambda \partial \|\Theta\|_1 \Leftrightarrow W - S - \lambda \Gamma = 0, \quad \Gamma_{ij} \in \partial |\Theta_{ij}| = \begin{cases} \{\text{sign}(\Theta_{ij})\}, & \Theta_{ij} \neq 0, \\ [-1, 1], & \Theta_{ij} = 0. \end{cases}$$

Diagonal: $\Gamma_{ii} = 1 \Rightarrow W_{ii} = S_{ii} + \lambda$ (off-diagonal-only convention: $W_{ii} = S_{ii}$). Partition with the active column last,

$$\Theta = \begin{pmatrix} \Theta_{11} & \theta_{12} \\ \theta_{12}^\top & \theta_{22} \end{pmatrix}, \quad W = \begin{pmatrix} W_{11} & w_{12} \\ w_{12}^\top & w_{22} \end{pmatrix}, \quad S = \begin{pmatrix} S_{11} & s_{12} \\ s_{12}^\top & s_{22} \end{pmatrix},$$

and take the (1, 2) blocks of $W - S - \lambda \Gamma = 0$ and of $W\Theta = I$:

$$w_{12} - s_{12} - \lambda \gamma_{12} = 0, \quad W_{11}\theta_{12} + w_{12}\theta_{22} = 0 \implies w_{12} = W_{11}\beta, \quad \beta := -\frac{\theta_{12}}{\theta_{22}}.$$

Since $\theta_{22} > 0$, $\text{sign}(\theta_{12}) = -\text{sign}(\beta)$, so $\gamma_{12} \in -\partial \|\beta\|_1$ and substitution gives a LASSO:

$$0 \in W_{11}\beta - s_{12} + \lambda \partial \|\beta\|_1 \Leftrightarrow \hat{\beta} = \arg \min_{\beta \in \mathbb{R}^{p-1}} \frac{1}{2} \beta^\top W_{11} \beta - s_{12}^\top \beta + \lambda \|\beta\|_1.$$

Coordinate descent, $A := W_{11}$, $r_j := s_{12,j} - \sum_{k \neq j} A_{jk} \beta_k$:

$$A_{jj}\beta_j - r_j + \lambda \partial |\beta_j| \ni 0 \implies \beta_j = \frac{S_\lambda(r_j)}{A_{jj}}, \quad S_\lambda(u) = \text{sign}(u) \max\{|u| - \lambda, 0\},$$

the proximal operator of $\lambda |\cdot|$ (Lecture 12). Sweep through the columns until the updates converge, and then recover the precision-matrix estimate as

$$\hat{\Theta} = \hat{W}^{-1}.$$

2×2 case with an off-diagonal-only penalty. $S = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$, off-diagonal penalty:

$$W = \begin{pmatrix} 1 & S_\lambda(c) \\ S_\lambda(c) & 1 \end{pmatrix}, \quad \Theta_{12} = \frac{-S_\lambda(c)}{1 - S_\lambda(c)^2}, \quad \Theta_{12} = 0 \Leftrightarrow |c| \leq \lambda.$$

E.g. $c = 0.5$: $\lambda = 0.3 \Rightarrow \Theta_{12} = \frac{-0.2}{0.96} \approx -0.208$ (edge kept); $\lambda = 0.6 \Rightarrow \Theta = I$ (edge dropped).

Numerical illustration. Using the off-diagonal-only penalty, consider the standardized Wisconsin breast-cancer data, with $n = 569$ and $p = 10$, so that S is the correlation matrix. Declaring an edge whenever $|\hat{\Theta}_{ij}| > 10^{-4}$, at $\lambda = 0.20$ the unregularized estimate S^{-1} has all 45 possible edges, with entries as large as approximately ± 1700 , while the graphical LASSO retains 29 edges. The resulting graph approximately suggests a size group and a shape group (Figure 22).

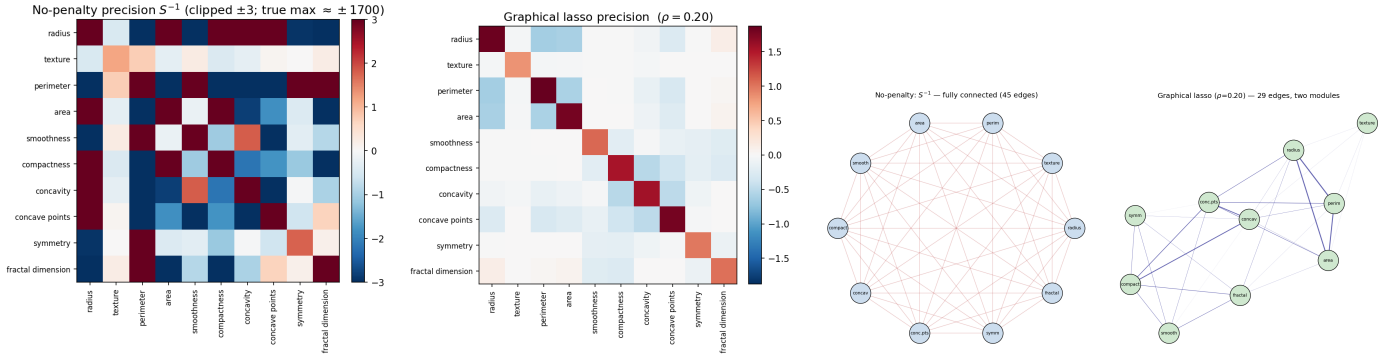


Figure 22: Graphical LASSO on the standardized breast-cancer data. Left: S^{-1} (clipped to ± 3 ; true max $\approx \pm 1700$), dense. Middle: graphical LASSO ($\lambda = 0.20$), weak entries zeroed. Right: graphs— S^{-1} fully connected (45 edges) vs. graphical LASSO (29 edges).

Example 10.3.2 (Proposed by Cade Camacho: Proximal Operator in Denoising). Suppose we observe a noisy signal $y \in \mathbb{R}^d$ and want to recover a cleaner version. A common way to do this is to solve

$$\min_x \frac{1}{2} \|x - y\|^2 + \lambda R(x),$$

where the first term keeps x close to the data y , and the second term encourages some structure such as smoothness. This can be rewritten as

$$x^* = \mathbf{Prox}_{\lambda R}(y)$$

So the proximal operator takes a noisy input y and returns a cleaned version.

The proximal operator is balancing two things:

- staying close to the observed data y ,
- following a simple structure given by $R(x)$.

If $R(x) = \|x\|_1$, then the proximal operator becomes soft-thresholding:

$$\mathbf{Prox}_{\lambda \|\cdot\|_1}(y) = S_\lambda(y)$$

This means that small values in y are treated as noise and set to zero. Large values are shrunk towards zero and kept.

Example:

Let

$$y = (3, -1, 0.5, 2) \in \mathbb{R}^4$$

be a simple noisy signal. We want to recover a cleaner version x by solving the proximal problem

$$\min_x \frac{1}{2} \|x - y\|^2 + \lambda \|x\|_1, \quad \lambda = 1.$$

This corresponds to applying the proximal operator of the ℓ_1 norm:

$$x^* = \mathbf{Prox}_{\lambda \|\cdot\|_1}(y),$$

which acts coordinate-wise via soft-thresholding.

We compute:

$$x_i^* = \text{sign}(y_i) \max\{|y_i| - 1, 0\}.$$

Evaluating each entry:

$$\begin{aligned} x_1^* &= 3 - 1 = 2, \\ x_2^* &= 0 \quad (\text{since } |-1| - 1 = 0), \\ x_3^* &= 0 \quad (\text{since } 0.5 < 1), \\ x_4^* &= 2 - 1 = 1. \end{aligned}$$

Therefore:

$$x^* = (2, 0, 0, 1).$$

The proximal operator has

- removed noisy components,
- reduced larger entries toward zero,
- produced a sparse representation of the signal.

Therefore, the proximal operator acts as a denoising map, suppressing values below a threshold and preserving stronger components.

11 Lecture Notes 11 on Proximal Operator and Moreau Envelope

Justin Zhou summarizes: We explored the Moreau envelope and proximal operator as tools for making convex optimization problems easier to study and solve. The Moreau envelope regularizes a function by adding a quadratic penalty, while the proximal operator finds the point that balances staying close to the input with lowering the function value. The lecture also explains that projection is a special case of the proximal operator, and that minimizers of a function are exactly the fixed points of its proximal operator.

Scribe Team: Jasper Luo, Cade Camacho

11.1 Lecture Notes

The Moreau envelope and proximal operator. Let $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper closed convex function and let $\eta > 0$. The *Moreau envelope* of f is defined by

$$M_{\eta f}(x) = \inf_{y \in \text{dom}(f)} \left\{ f(y) + \frac{1}{2\eta} \|y - x\|^2 \right\}. \quad (11.1)$$

The quadratic term $\frac{1}{2\eta} \|y - x\|^2$ regularizes the original function. For each fixed x , the function

$$y \mapsto f(y) + \frac{1}{2\eta} \|y - x\|^2$$

is $\frac{1}{\eta}$ -strongly convex in y . Hence the minimizer is unique. This motivates the definition of the *proximal operator*:

$$\mathbf{Prox}_{\eta f}(x) = \arg \min_{y \in \text{dom}(f)} \left\{ f(y) + \frac{1}{2\eta} \|y - x\|^2 \right\}. \quad (11.2)$$

For a proper closed convex function f , both $\mathbf{Prox}_{\eta f}$ and $M_{\eta f}$ are defined on all of $\mathbb{R}^d$, although the minimization itself is taken over $\text{dom}(f)$.

The proximal operator can also be interpreted as a backward Euler step for the gradient flow. When f is differentiable, gradient descent uses the forward Euler update

$$x_{k+1} = x_k - \eta \nabla f(x_k).$$

In contrast, the proximal update is

$$x_{k+1} = \mathbf{Prox}_{\eta f}(x_k) = \arg \min_{y \in \text{dom}(f)} \left\{ f(y) + \frac{1}{2\eta} \|y - x_k\|^2 \right\},$$

which is implicit in the next point x_{k+1} and therefore corresponds to a backward Euler discretization.

Projection as a special case. The projection onto a closed convex set $C \subseteq \mathbb{R}^d$ is

$$\mathbf{Proj}_C(x) = \arg \min_{y \in C} \frac{1}{2} \|y - x\|^2.$$

Using the indicator function

$$I_C(y) = \begin{cases} 0, & y \in C, \\ +\infty, & y \notin C, \end{cases}$$

we can rewrite this constrained problem as

$$\mathbf{Proj}_C(x) = \arg \min_{y \in \mathbb{R}^d} \left\{ I_C(y) + \frac{1}{2} \|y - x\|^2 \right\}.$$

Therefore,

$$\mathbf{Prox}_{I_C}(x) = \mathbf{Proj}_C(x).$$

Thus projection is a special case of the proximal operator.

We now recall two useful classes of maps. A map $T : V \rightarrow V$ is *nonexpansive* if

$$\|T(x) - T(y)\| \leq \|x - y\| \quad \forall x, y \in V.$$

It is *firmly nonexpansive* if

$$\|T(x) - T(y)\|^2 \leq \langle T(x) - T(y), x - y \rangle \quad \forall x, y \in V.$$

This should not be confused with m -strong convexity in Definition 6.6. Firm nonexpansiveness immediately implies nonexpansiveness by Cauchy–Schwarz:

$$\|T(x) - T(y)\|^2 \leq \langle T(x) - T(y), x - y \rangle \leq \|T(x) - T(y)\| \|x - y\|.$$

Theorem 11.1 (Projection is firmly nonexpansive). Let $C \subseteq \mathbb{R}^d$ be closed and convex. Then $\mathbf{Proj}_C$ is firmly nonexpansive:

$$\|\mathbf{Proj}_C(x) - \mathbf{Proj}_C(y)\|^2 \leq \langle \mathbf{Proj}_C(x) - \mathbf{Proj}_C(y), x - y \rangle.$$

Consequently, $\mathbf{Proj}_C$ is nonexpansive.

Proof. Let

$$p = \mathbf{Proj}_C(x), \quad q = \mathbf{Proj}_C(y).$$

By the first-order optimality condition for projection onto a convex set,

$$\langle x - p, z - p \rangle \leq 0 \quad \forall z \in C,$$

and

$$\langle y - q, z - q \rangle \leq 0 \quad \forall z \in C.$$

Taking $z = q$ in the first inequality and $z = p$ in the second gives

$$\langle x - p, q - p \rangle \leq 0, \quad \langle y - q, p - q \rangle \leq 0.$$

Equivalently,

$$\langle x - p, p - q \rangle \geq 0, \quad \langle q - y, p - q \rangle \geq 0.$$

Now decompose $x - y$ as

$$x - y = (x - p) + (p - q) + (q - y).$$

Then

$$\begin{aligned} \langle x - y, p - q \rangle &= \langle x - p, p - q \rangle + \|p - q\|^2 + \langle q - y, p - q \rangle \\ &\geq \|p - q\|^2. \end{aligned}$$

Therefore,

$$\|\mathbf{Proj}_C(x) - \mathbf{Proj}_C(y)\|^2 = \|p - q\|^2 \leq \langle p - q, x - y \rangle,$$

which proves firm nonexpansiveness. Nonexpansiveness follows from Cauchy–Schwarz. $\blacksquare$

The proximal operator as a resolvent. A second viewpoint connects the proximal operator with the subdifferential. For a proper closed convex function f ,

$$\mathbf{Prox}_{\eta f} = (I + \eta \partial f)^{-1}. \quad (11.3)$$

Here $(I + \eta \partial f)^{-1}$ is a resolvent of the set-valued map ∂f .

Theorem 11.2 (Resolvent form of the proximal map). For every $x \in \mathbb{R}^d$,

$$p = \mathbf{Prox}_{\eta f}(x) \iff x \in p + \eta \partial f(p).$$

Equivalently,

$$\mathbf{Prox}_{\eta f}(x) = (I + \eta \partial f)^{-1}(x).$$

Proof. Let $p = \mathbf{Prox}_{\eta f}(x)$. By the definition of the proximal operator, p minimizes

$$y \mapsto f(y) + \frac{1}{2\eta} \|y - x\|^2.$$

The first-order optimality condition is

$$0 \in \partial f(p) + \frac{1}{\eta}(p - x).$$

Hence

$$\frac{x - p}{\eta} \in \partial f(p),$$

or equivalently,

$$x \in p + \eta \partial f(p).$$

This is exactly the condition that

$$p \in (I + \eta \partial f)^{-1}(x).$$

Therefore,

$$\mathbf{Prox}_{\eta f}(x) = (I + \eta \partial f)^{-1}(x). \quad \blacksquare$$

Formally, the resolvent expression is analogous to

$$(I + \eta \partial f)^{-1} \simeq I - \eta \partial f,$$

just as

$$\frac{1}{1 + \lambda x} \simeq 1 - \lambda x \quad \text{when } |\lambda x| \ll 1.$$

This analogy explains why the proximal step can be viewed as an implicit version of a gradient step.

Firm nonexpansiveness of the proximal map. The projection theorem above is a special case of a more general property.

Theorem 11.3 (The proximal map is firmly nonexpansive). Let f be proper, closed, and convex. Then $\mathbf{Prox}_{\eta f}$ is firmly nonexpansive:

$$\|\mathbf{Prox}_{\eta f}(x) - \mathbf{Prox}_{\eta f}(y)\|^2 \leq \langle \mathbf{Prox}_{\eta f}(x) - \mathbf{Prox}_{\eta f}(y), x - y \rangle.$$

Consequently, $\mathbf{Prox}_{\eta f}$ is nonexpansive.

Proof. Let

$$p = \mathbf{Prox}_{\eta f}(x), \quad q = \mathbf{Prox}_{\eta f}(y).$$

By Theorem 11.2,

$$\frac{x-p}{\eta} \in \partial f(p), \quad \frac{y-q}{\eta} \in \partial f(q).$$

Since f is convex, ∂f is monotone. Therefore,

$$\left\langle \frac{x-p}{\eta} - \frac{y-q}{\eta}, p-q \right\rangle \geq 0.$$

Multiplying by η gives

$$\langle x-y-(p-q), p-q \rangle \geq 0.$$

Hence

$$\langle x-y, p-q \rangle \geq \|p-q\|^2.$$

This is precisely the firm nonexpansiveness inequality. ■

Minimizers and fixed points. A natural question is how the minimizers of f are related to the proximal operator.

Theorem 11.4 (Minimizers and fixed points of the proximal operator). A point $\bar{x}$ is a minimizer of f if and only if $\bar{x}$ is a fixed point of the proximal operator:

$$\bar{x} \in \arg \min f \iff \bar{x} = \mathbf{Prox}_{\eta f}(\bar{x}). \quad (11.4)$$

Proof. Assume first that $\bar{x} \in \arg \min f$. Then $0 \in \partial f(\bar{x})$. Hence

$$\bar{x} \in \bar{x} + \eta \partial f(\bar{x}),$$

so by the resolvent formulation,

$$\mathbf{Prox}_{\eta f}(\bar{x}) = \bar{x}.$$

Conversely, suppose that $\mathbf{Prox}_{\eta f}(\bar{x}) = \bar{x}$. By the optimality condition for the proximal operator,

$$\frac{\bar{x} - \bar{x}}{\eta} = 0 \in \partial f(\bar{x}).$$

For a convex function, $0 \in \partial f(\bar{x})$ is equivalent to $\bar{x}$ being a global minimizer. Therefore,

$$\bar{x} \in \arg \min f. \quad \blacksquare$$

This motivates the fixed point iteration

$$x_{k+1} = T(x_k),$$

where the goal is to find a point $\bar{x}$ satisfying $\bar{x} = T(\bar{x})$. A standard sufficient condition for convergence is that T is a contraction: there exists $\lambda < 1$ such that

$$\|T(x) - T(y)\| \leq \lambda \|x - y\| \quad \forall x, y.$$

If T is a contraction and $\bar{x} = T(\bar{x})$, then

$$\begin{aligned} \|x_{k+1} - \bar{x}\| &= \|T(x_k) - T(\bar{x})\| \\ &\leq \lambda \|x_k - \bar{x}\| \\ &\leq \lambda^2 \|x_{k-1} - \bar{x}\| \\ &\leq \dots \\ &\leq \lambda^k \|x_1 - \bar{x}\|. \end{aligned}$$

Since $\lambda^k \rightarrow 0$, the iterates converge to $\bar{x}$.

For the proximal operator, however, a strict contraction estimate usually does not hold. Instead, $\mathbf{Prox}_{\eta f}$ is firmly nonexpansive. This gives a useful averaged iteration:

$$x^{k+1} = (1-\alpha)x^k + \alpha \mathbf{Prox}_{\eta f}(x^k), \quad \alpha \in (0, 1). \quad (11.5)$$

Theorem 11.5 (Averaged firmly nonexpansive iteration). Let $T : V \rightarrow V$ be firmly nonexpansive and let

$$S = (1-\alpha)I + \alpha T, \quad \alpha \in (0, 1).$$

If $\bar{x} \in \text{Fix}(T)$, then for every $x \in V$,

$$\|Sx - \bar{x}\|^2 \leq \|x - \bar{x}\|^2 - \alpha(1-\alpha)\|Tx - x\|^2.$$

In particular, the averaged iteration $x^{k+1} = Sx^k$ is Fejér monotone with respect to the fixed point set.

Proof. Since $\bar{x} \in \text{Fix}(T)$, we have $T\bar{x} = \bar{x}$. Write

$$Sx - \bar{x} = (1-\alpha)(x - \bar{x}) + \alpha(Tx - \bar{x}).$$

Using the identity

$$\begin{aligned} \|(1-\alpha)a + \alpha b\|^2 &= (1-\alpha)\|a\|^2 + \alpha\|b\|^2 - \alpha(1-\alpha)\|a-b\|^2, \\ \text{with } a = x - \bar{x} \text{ and } b = Tx - \bar{x}, \text{ we get} \\ \|Sx - \bar{x}\|^2 &= (1-\alpha)\|x - \bar{x}\|^2 + \alpha\|Tx - \bar{x}\|^2 - \alpha(1-\alpha)\|x - Tx\|^2. \end{aligned}$$

Since T is firmly nonexpansive, it is nonexpansive. Therefore,

$$\|Tx - \bar{x}\| = \|Tx - T\bar{x}\| \leq \|x - \bar{x}\|.$$

Substituting this into the previous display yields

$$\|Sx - \bar{x}\|^2 \leq \|x - \bar{x}\|^2 - \alpha(1-\alpha)\|Tx - x\|^2.$$

This is the claimed contraction-type estimate. It is not a strict contraction in general, but it gives a decrease in the distance to the fixed point up to the residual term $\|Tx - x\|^2$. ■

Basic properties of the Moreau envelope. We now record the main properties of $M_{\eta f}$.

Theorem 11.6 (Minimum value and minimizers). The Moreau envelope preserves the minimum value and the set of minimizers of f . More precisely, if $\arg \min f \neq \emptyset$, then

$$\min_{x \in \mathbb{R}^d} M_{\eta f}(x) = \min_{x \in \text{dom}(f)} f(x),$$

and

$$\arg \min_{x \in \mathbb{R}^d} M_{\eta f}(x) = \arg \min_{x \in \text{dom}(f)} f(x).$$

Proof. Let

$$m = \min_{x \in \text{dom}(f)} f(x).$$

For any x and any y ,

$$f(y) + \frac{1}{2\eta}\|x - y\|^2 \geq m.$$

Taking the infimum over y gives

$$M_{\eta f}(x) \geq m.$$

On the other hand, if $\bar{x} \in \arg \min f$, then choosing $y = \bar{x}$ gives

$$M_{\eta f}(\bar{x}) \leq f(\bar{x}) + \frac{1}{2\eta}\|\bar{x} - \bar{x}\|^2 = m.$$

Thus $\min M_{\eta f} = m$.

Now suppose $x \in \arg \min M_{\eta f}$. Let $p = \mathbf{Prox}_{\eta f}(x)$. Then

$$m = M_{\eta f}(x) = f(p) + \frac{1}{2\eta}\|x - p\|^2.$$

Since $f(p) \geq m$ and the quadratic term is nonnegative, both inequalities must be equalities. Hence $f(p) = m$ and $\|x - p\| = 0$, so $x = p \in \arg \min f$. This proves

$$\arg \min M_{\eta f} \subseteq \arg \min f.$$

The reverse inclusion follows from the calculation above with $x = \bar{x} \in \arg \min f$. Therefore,

$$\arg \min M_{\eta f} = \arg \min f. \quad \blacksquare$$

Theorem 11.7 (Convexity of the Moreau envelope). If f is convex, then $M_{\eta f}$ is convex.

Proof. Let $x_1, x_2 \in \mathbb{R}^d$ and $t \in [0, 1]$. Set

$$x_t = (1-t)x_1 + tx_2, \quad p_i = \mathbf{Prox}_{\eta f}(x_i) \quad (i = 1, 2),$$

and

$$p_t = (1-t)p_1 + tp_2.$$

By the definition of the Moreau envelope,

$$M_{\eta f}(x_t) \leq f(p_t) + \frac{1}{2\eta}\|x_t - p_t\|^2.$$

Since f is convex and the squared norm is convex,

$$\begin{aligned} f(p_t) + \frac{1}{2\eta}\|x_t - p_t\|^2 &\leq (1-t)f(p_1) + tf(p_2) \\ &\quad + \frac{1-t}{2\eta}\|x_1 - p_1\|^2 + \frac{t}{2\eta}\|x_2 - p_2\|^2 \\ &= (1-t)M_{\eta f}(x_1) + tM_{\eta f}(x_2). \end{aligned}$$

Therefore,

$$M_{\eta f}(x_t) \leq (1-t)M_{\eta f}(x_1) + tM_{\eta f}(x_2),$$

so $M_{\eta f}$ is convex. ■

Theorem 11.8 (Differentiability of the Moreau envelope). Let f be proper, closed, and convex. Then $M_{\eta f}$ is differentiable on $\mathbb{R}^d$, and

$$\nabla M_{\eta f}(x) = \frac{1}{\eta} (x - \mathbf{Prox}_{\eta f}(x)). \quad (11.6)$$

Moreover, $\nabla M_{\eta f}$ is $\frac{1}{\eta}$ -Lipschitz.

Proof. For fixed x , the minimizer in the definition of the Moreau envelope is unique because the objective is strongly convex in y . Therefore, by differentiating the minimized expression with respect to x at the unique minimizer $p = \mathbf{Prox}_{\eta f}(x)$, we obtain

$$\nabla M_{\eta f}(x) = \frac{1}{\eta}(x - p) = \frac{1}{\eta} (x - \mathbf{Prox}_{\eta f}(x)).$$

It remains to prove the Lipschitz estimate. Let

$$p = \mathbf{Prox}_{\eta f}(x), \quad q = \mathbf{Prox}_{\eta f}(y).$$

Using the gradient formula above,

$$\nabla M_{\eta f}(x) - \nabla M_{\eta f}(y) = \frac{1}{\eta} [(x - y) - (p - q)].$$

By firm nonexpansiveness of the proximal map,

$$\|p - q\|^2 \leq \langle p - q, x - y \rangle.$$

Therefore,

$$\begin{aligned} \|(x - y) - (p - q)\|^2 &= \|x - y\|^2 + \|p - q\|^2 - 2 \langle x - y, p - q \rangle \\ &\leq \|x - y\|^2 - \|p - q\|^2 \\ &\leq \|x - y\|^2. \end{aligned}$$

Hence

$$\|\nabla M_{\eta f}(x) - \nabla M_{\eta f}(y)\| \leq \frac{1}{\eta} \|x - y\|.$$

Thus $M_{\eta f}$ is $\frac{1}{\eta}$ -smooth. ■

Theorem 11.9 (Strong convexity of the Moreau envelope). If f is μ -strongly convex with $\mu > 0$, then $M_{\eta f}$ is

$$\frac{\mu}{1 + \eta\mu}$$

-strongly convex.

Proof. Let $x_1, x_2 \in \mathbb{R}^d$, $t \in [0, 1]$, and

$$x_t = (1 - t)x_1 + tx_2.$$

Set

$$p_i = \mathbf{Prox}_{\eta f}(x_i) \quad (i = 1, 2), \quad p_t = (1 - t)p_1 + tp_2.$$

As in the proof of convexity,

$$M_{\eta f}(x_t) \leq f(p_t) + \frac{1}{2\eta} \|x_t - p_t\|^2.$$

Since f is μ -strongly convex,

$$f(p_t) \leq (1 - t)f(p_1) + tf(p_2) - \frac{\mu}{2}t(1 - t)\|p_1 - p_2\|^2.$$

Also,

$$\|x_t - p_t\|^2 = \|(1 - t)(x_1 - p_1) + t(x_2 - p_2)\|^2$$

implies

$$\begin{aligned} \|x_t - p_t\|^2 &\leq (1 - t)\|x_1 - p_1\|^2 + t\|x_2 - p_2\|^2 \\ &\quad - t(1 - t)\|(x_1 - p_1) - (x_2 - p_2)\|^2. \end{aligned}$$

Combining the two estimates gives

$$\begin{aligned} M_{\eta f}(x_t) &\leq (1 - t)M_{\eta f}(x_1) + tM_{\eta f}(x_2) \\ &\quad - \frac{t(1 - t)}{2} \left[\mu\|p_1 - p_2\|^2 + \frac{1}{\eta} \|(x_1 - x_2) - (p_1 - p_2)\|^2 \right]. \end{aligned}$$

For any vector a and d , the minimum of

$$\mu\|a\|^2 + \frac{1}{\eta}\|d - a\|^2$$

over a is

$$\frac{\mu}{1 + \eta\mu} \|d\|^2.$$

Taking $a = p_1 - p_2$ and $d = x_1 - x_2$, we obtain

$$\mu \|p_1 - p_2\|^2 + \frac{1}{\eta} \|(x_1 - x_2) - (p_1 - p_2)\|^2 \geq \frac{\mu}{1 + \eta\mu} \|x_1 - x_2\|^2.$$

Therefore,

$$M_{\eta f}(x_t) \leq (1 - t)M_{\eta f}(x_1) + tM_{\eta f}(x_2) - \frac{1}{2} \frac{\mu}{1 + \eta\mu} t(1 - t) \|x_1 - x_2\|^2.$$

This is exactly $\frac{\mu}{1 + \eta\mu}$ -strong convexity of $M_{\eta f}$. ■

Remark. If f is only convex, $M_{\eta f}$ is convex and differentiable, but it is not necessarily strongly convex. Strong convexity of the Moreau envelope requires additional curvature from f itself. The proximal objective in the definition of the proximal operator, however, is always $\frac{1}{\eta}$ -strongly convex in the minimization variable y .

Summary. The proximal operator and Moreau envelope are closely connected:

$$\mathbf{Prox}_{\eta f} = (I + \eta \partial f)^{-1}, \quad \nabla M_{\eta f}(x) = \frac{1}{\eta} (x - \mathbf{Prox}_{\eta f}(x)).$$

The proximal map is firmly nonexpansive, and minimizers of f are exactly fixed points of $\mathbf{Prox}_{\eta f}$. The Moreau envelope preserves minimizers, is convex when f is convex, is differentiable even when f is nonsmooth, and becomes strongly convex when f is strongly convex.

11.2 Exercises

Exercise. Try to use the definition to prove that Moreau Envelope $M_{\eta f}(y) = \max_{x \in D} f(x) + \frac{1}{2\eta} \|x - y\|^2$ is convex, provided with a convex f .

Proof. Drafted by Don Tran, Proofread by Tao Xu

We prove the statement as written: for a nonempty set $D \subseteq \mathbb{R}^d$ and a convex function $f : D \rightarrow \mathbb{R}$, the function

$$M_{\eta f}(y) = \max_{x \in D} \left\{ f(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}$$

is convex in y , provided D is such that the maximum is finite for every $y \in \mathbb{R}^d$ (e.g. D bounded).^a

The key observation is that we do not need to argue at the level of optimizers at all, since convexity follows directly from the structure of the objective as a function of y .

Step 1: pointwise convexity in y . For each fixed $x \in D$, define

$$\phi_x(y) := f(x) + \frac{1}{2\eta} \|x - y\|^2.$$

As a function of y , the first term $f(x)$ is a constant, and the second term $\frac{1}{2\eta} \|x - y\|^2$ is convex in y (the squared norm is convex, and convexity is preserved under translation). Hence ϕ_x is convex in y for every $x \in D$.

Step 2: pointwise supremum of convex functions is convex. Recall the general fact: if $\{\phi_x\}_{x \in D}$ is any family of convex functions on $\mathbb{R}^d$, then

$$\Phi(y) := \sup_{x \in D} \phi_x(y)$$

is convex. Indeed, for $y_1, y_2 \in \mathbb{R}^d$ and $t \in [0, 1]$,

$$\begin{aligned} \Phi((1 - t)y_1 + ty_2) &= \sup_{x \in D} \phi_x((1 - t)y_1 + ty_2) \\ &\leq \sup_{x \in D} \{(1 - t)\phi_x(y_1) + t\phi_x(y_2)\} && \text{(convexity of each } \phi_x) \\ &\leq (1 - t) \sup_{x \in D} \phi_x(y_1) + t \sup_{x \in D} \phi_x(y_2) && \text{(sup of a sum } \leq \text{ sum of sups)} \\ &= (1 - t)\Phi(y_1) + t\Phi(y_2). \end{aligned}$$

Step 3: apply to our setting. Since $M_{\eta f}(y) = \max_{x \in D} \phi_x(y)$ is exactly such a pointwise supremum of convex functions of y , $M_{\eta f}$ is convex. ■

^aIf $D = \mathbb{R}^d$ the supremum is $+\infty$ since the quadratic blows up as $\|x\| \rightarrow \infty$, so some such restriction on D is needed for the object to be well-defined.

Example 11.2.1 (Proposed by Ben Thomas). Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex. Show that the Moreau envelope $M_{\eta f}$ is differentiable with

$$\nabla M_{\eta f}(x) = \frac{1}{\eta} (x - \mathbf{Prox}_{\eta f}(x)).$$

In particular, conclude that $\nabla M_{\eta f}(x) = 0$ if and only if $x = \mathbf{Prox}_{\eta f}(x)$, i.e., x is a minimizer of f .

Proof. Drafted by Ben Thomas, Proofread by Ciara Nic Canna

Denote $p = \mathbf{Prox}_{\eta f}(x)$. By definition,

$$M_{\eta f}(x) = f(p) + \frac{1}{2\eta} \|p - x\|^2.$$

Since the quadratic term $\frac{1}{2\eta}\|p-x\|^2$ is differentiable in x with gradient $\frac{1}{\eta}(x-p)$, and p is the minimizer so contributes zero first-order variation by the envelope theorem, we obtain

$$\nabla M_{\eta f}(x) = \frac{1}{\eta}(x - \text{Prox}_{\eta f}(x)).$$

Finally, $\nabla M_{\eta f}(x) = 0$ if and only if $x = \text{Prox}_{\eta f}(x)$, which by Theorem 11.4 holds if and only if $x \in \arg \min f$. ■

11.3 More Examples

Example 11.3.1 (Proposed by *Don Tran*: The proximal operator is firmly non-expansive). Above, it is noted that the backward Euler (proximal) scheme is “stable even for moderate step-size,” while the forward Euler scheme (gradient descent) is “unstable unless the step-size is small.” Here we work out the property of the proximal operator that is responsible for this: $\text{Prox}_{\eta f}$ is *firmly non-expansive*, which is strong enough to guarantee that simply iterating the operator can never blow up. This follows [Bec17, Chapter 6].

Recall that an operator $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is *non-expansive* (equivalently 1-Lipschitz) if $\|T(x) - T(y)\| \leq \|x - y\|$ for all x, y , and *firmly non-expansive* if it satisfies the stronger inequality

$$\|T(x) - T(y)\|^2 \leq \langle x - y, T(x) - T(y) \rangle.$$

Firm non-expansiveness implies non-expansiveness: applying Cauchy–Schwarz to the right-hand side gives $\|T(x) - T(y)\|^2 \leq \|x - y\| \|T(x) - T(y)\|$, and dividing by $\|T(x) - T(y)\|$ recovers the 1-Lipschitz bound. The converse fails, which makes it a stronger condition.

Theorem 11.10 (Firm non-expansiveness of the prox). Let $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be proper, closed, and convex, and let $\eta > 0$. Then

$$\|\text{Prox}_{\eta f}(x) - \text{Prox}_{\eta f}(y)\|^2 \leq \langle x - y, \text{Prox}_{\eta f}(x) - \text{Prox}_{\eta f}(y) \rangle.$$

Proof. Write $u = \text{Prox}_{\eta f}(x)$ and $v = \text{Prox}_{\eta f}(y)$. From the optimality condition for the proximal update derived in lecture,

$$\frac{1}{\eta}(x - u) \in \partial f(u), \quad \frac{1}{\eta}(y - v) \in \partial f(v).$$

The sub-differential of a convex function is monotone, so for these two subgradients,

$$\left\langle \frac{1}{\eta}(x - u) - \frac{1}{\eta}(y - v), u - v \right\rangle \geq 0.$$

Multiplying by $\eta > 0$ and expanding the left-hand side gives $\langle (x - y) - (u - v), u - v \rangle \geq 0$, which rearranges to

$$\langle x - y, u - v \rangle \geq \|u - v\|^2.$$

This is exactly the firm non-expansiveness inequality. ■

The optimality condition has a useful rephrasing. Rearranging $\frac{1}{\eta}(x - u) \in \partial f(u)$ gives $x \in (I + \eta \partial f)(u)$, that is,

$$\text{Prox}_{\eta f} = (I + \eta \partial f)^{-1}.$$

This inverse is called the *resolvent* of ∂f . So the backward Euler step would be $x^{k+1} = (I + \eta \nabla f)^{-1} x^k$, which means each proximal update applies the resolvent. The fact that resolvents of monotone operators are firmly non-expansive makes the scheme unconditionally stable.

This is what explains the stability of backward Euler step. The minimizers of f are exactly the fixed points of $\text{Prox}_{\eta f}$, since $0 \in \partial f(x^*)$ is equivalent to $\text{Prox}_{\eta f}(x^*) = x^*$. The proximal point method $x^{k+1} = \text{Prox}_{\eta f}(x^k)$ is therefore a fixed-point iteration, and because the prox is non-expansive, the distance to any minimizer cannot grow:

$$\|x^{k+1} - x^*\| = \|\text{Prox}_{\eta f}(x^k) - \text{Prox}_{\eta f}(x^*)\| \leq \|x^k - x^*\|,$$

and this holds for *any* $\eta > 0$. Compare this with gradient descent, $x^{k+1} = x^k - \eta \nabla f(x^k)$: the map $I - \eta \nabla f$ is non-expansive only when η is small relative to the curvature (for L -smooth f , when $\eta \leq 2/L$). Too large a stepsize makes it expansive, and the iterates diverge, which leads to instability.

Takeaway. Non-expansiveness is the property that explains why the backward Euler (proximal) scheme is stable even for moderate step-size, while the forward Euler scheme (gradient descent) is unstable for larger sizes. The proximity term $\frac{1}{2\eta}\|x - y\|^2$ does more than make each subproblem well-posed (Theorem 10.1); it forces the resulting operator to be firmly non-expansive, so iterating it can never increase the distance to a solution, whatever the stepsize. Gradient descent gets no such guarantee, which is the structural reason implicit schemes trade a harder per-step subproblem

for unconditional stability.

Example 11.3.2 (Proposed by Chenghan Li, Moreau envelope of the absolute value function). Consider the convex function

$$f(x) = |x|, \quad x \in \mathbb{R}.$$

For $\eta > 0$, the Moreau envelope of f is defined by

$$f_\eta(z) = \min_{x \in \mathbb{R}} \left\{ |x| + \frac{1}{2\eta}(x - z)^2 \right\}.$$

From the soft-thresholding formula, the minimizer is

$$\text{Prox}_{\eta f}(z) = \begin{cases} z - \eta, & z > \eta, \\ 0, & |z| \leq \eta, \\ z + \eta, & z < -\eta. \end{cases}$$

We compute the value of the Moreau envelope in three cases.

If $z > \eta$, then the minimizer is $x = z - \eta$. Hence

$$f_\eta(z) = |z - \eta| + \frac{1}{2\eta}((z - \eta) - z)^2 = z - \eta + \frac{\eta}{2} = z - \frac{\eta}{2}.$$

If $z < -\eta$, then the minimizer is $x = z + \eta$. Hence

$$f_\eta(z) = |z + \eta| + \frac{1}{2\eta}((z + \eta) - z)^2 = -(z + \eta) + \frac{\eta}{2} = -z - \frac{\eta}{2}.$$

If $|z| \leq \eta$, then the minimizer is $x = 0$. Hence

$$f_\eta(z) = |0| + \frac{1}{2\eta}(0 - z)^2 = \frac{z^2}{2\eta}.$$

Therefore,

$$f_\eta(z) = \begin{cases} z - \frac{\eta}{2}, & z > \eta, \\ \frac{z^2}{2\eta}, & |z| \leq \eta, \\ -z - \frac{\eta}{2}, & z < -\eta. \end{cases}$$

This example shows that the Moreau envelope smooths the nonsmooth function $f(x) = |x|$. The original function $|x|$ has a sharp corner at 0, while its Moreau envelope replaces the corner by the quadratic function $\frac{z^2}{2\eta}$ on the interval $[-\eta, \eta]$.

12 Lecture Notes 12 on Proximal Gradient Method

Don Tran summarizes: We study composite convex optimization of the form $\min f(x) + h(x)$ with f convex, smooth, and h convex, possibly non-smooth, a setting that captures both constrained optimization and sparsity-promoting regularization. We introduce projected gradient descent as a special case, derive the proximal gradient iteration, and interpret the algorithm as a forward-backward Euler discretization of gradient flow. We then establish optimality conditions via sub-differential calculus and operator splitting, and analyze convergence rates via an EVI-style descent lemma and the Bregman divergence, recovering the standard sublinear and linear rates of gradient descent without requiring the full objective to be smooth.

Scribe Team: Yuxuan Zhao, Ben T.

12.1 Lecture Notes

We consider the standard composite convex optimization problem

$$\min_{x \in \mathbb{R}^d} F(x) = f(x) + h(x), \quad \text{eq:lec12-composite (12.1)}$$

under the following assumptions:

- $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and L -smooth;
- $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is proper, closed, convex, and possibly non-differentiable.

The main point of this framework is that the smooth part f can be treated by an explicit gradient step, while the nonsmooth part h can be treated by a proximal step. This allows us to use first-order methods even when the full objective $F = f + h$ is not smooth.

Example 12.1.1 (Constrained optimization as a composite problem). The constrained problem

$$\min_{x \in C} f(x)$$

is equivalent to the unconstrained composite problem

$$\min_{x \in \mathbb{R}^d} f(x) + I_C(x),$$

where I_C is the indicator function of C :

$$I_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases}$$

When C is convex, I_C is convex. However, I_C is generally not differentiable. Thus constrained optimization fits naturally into the composite form above.

12.1.1 From Projected Gradient Descent to Proximal Gradient Method

We begin with projected gradient descent for the constrained problem

$$\min_{x \in C} f(x).$$

The iteration first takes an ordinary gradient descent step and then projects back onto the constraint set:

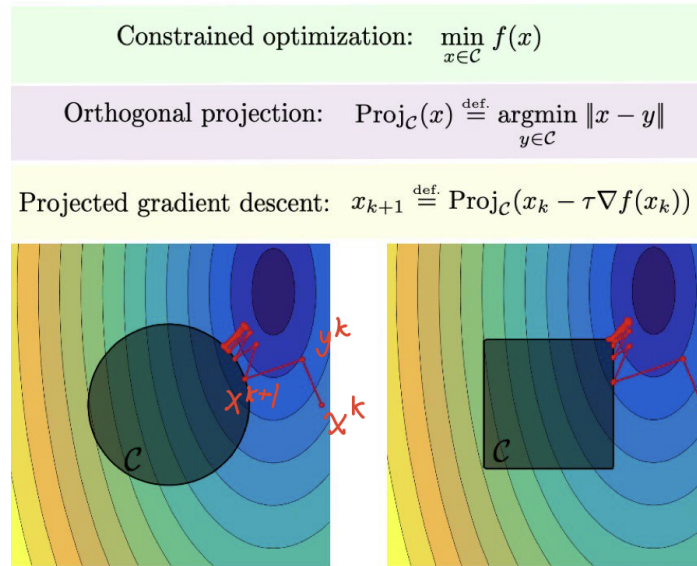
$$\begin{cases} y^k = x^k - \eta \nabla f(x^k), \\ x^{k+1} = \text{Proj}_C(y^k). \end{cases}$$

Equivalently,

$$x^{k+1} = \text{Proj}_C(x^k - \eta \nabla f(x^k)). \quad \text{eq:lec12-pgd (12.2)}$$

The intermediate point y^k may fail to belong to C , so the projection step restores feasibility.

Remark. Projected gradient descent is simple to state and relatively easy to analyze. However, the projection step can be expensive when the constraint set C has complicated geometry. This motivates replacing the projection operator by the more general proximal operator.



For the composite problem, the proximal gradient method is defined by

$$x^{k+1} = \mathbf{Prox}_{\eta h} \left(x^k - \eta \nabla f(x^k) \right). \quad \text{eq:lec12-prox-grad (12.3)}$$

This should be viewed as a generalization of projected gradient descent. Indeed, if $h = I_C$, then

$$\mathbf{Prox}_{\eta I_C} = \mathbf{Proj}_C,$$

and the proximal gradient update becomes exactly projected gradient descent.

Thus the proximal gradient method has the following structure:

$$x^0 \longrightarrow x^1 \longrightarrow \dots \longrightarrow x^k \longrightarrow x^{k+1} \longrightarrow \dots \longrightarrow \bar{x}.$$

The goal is to show that the iterates approach a minimizer $\bar{x}$ of $F = f + h$, under appropriate assumptions.

12.1.2 Two Perspectives of Proximal Gradient Methods

There are two useful ways to understand the proximal gradient update.

1. Fermat optimality and splitting. A point $\bar{x}$ minimizes $F = f + h$ if and only if it satisfies Fermat's optimality condition

$$0 \in \partial F(\bar{x}).$$

Since f is differentiable, the subdifferential condition becomes

$$0 \in \nabla f(\bar{x}) + \partial h(\bar{x}). \quad \text{eq:lec12-opt-cond (12.4)}$$

Multiplying this optimality condition by $\eta > 0$, adding and subtracting $\bar{x}$, gives

$$\bar{x} - \eta \nabla f(\bar{x}) \in \bar{x} + \eta \partial h(\bar{x}).$$

In operator notation,

$$(I - \eta \nabla f)(\bar{x}) \in (I + \eta \partial h)(\bar{x}).$$

Applying the inverse operator gives

$$\bar{x} = (I + \eta \partial h)^{-1} (I - \eta \nabla f)(\bar{x}).$$

Recall that

$$\mathbf{Prox}_{\eta h} = (I + \eta \partial h)^{-1}.$$

Therefore a minimizer $\bar{x}$ satisfies the fixed-point relation

$$\bar{x} = \mathbf{Prox}_{\eta h} (\bar{x} - \eta \nabla f(\bar{x})). \quad \text{eq:lec12-fixed-point (12.5)}$$

The proximal gradient method is obtained by iterating this fixed-point map.

Remark. In Euclidean space, the proximal operator of a proper, closed, convex function is single-valued. In more general settings, such as spaces of probability measures, the analogous resolvent may require additional care.

2. Forward-backward splitting. A second interpretation comes from gradient flow. Suppose for a moment that h is differentiable. Then the gradient flow for $F = f + h$ is

$$\dot{x}(t) = -\nabla f(x(t)) - \nabla h(x(t)).$$

Using a mixed discretization, we treat f explicitly and h implicitly:

$$\frac{x^{k+1} - x^k}{\eta} = -\nabla f(x^k) - \nabla h(x^{k+1}).$$

Rearranging gives

$$x^{k+1} + \eta \nabla h(x^{k+1}) = x^k - \eta \nabla f(x^k).$$

Thus

$$x^{k+1} = (I + \eta \nabla h)^{-1} (I - \eta \nabla f)(x^k). \quad \text{eq:lec12-forward-backward (12.6)}$$

This is called a forward-backward splitting method:

- the forward step $I - \eta \nabla f$ treats the smooth part f explicitly;
- the backward step $(I + \eta \nabla h)^{-1}$ treats the second part h implicitly.

When h is nonsmooth, ∇h is replaced by the subdifferential ∂h , and the backward step becomes

$$(I + \eta \partial h)^{-1} = \mathbf{Prox}_{\eta h}.$$

Therefore the nonsmooth forward-backward update is exactly the proximal gradient update.

12.1.3 Convergence Rate

We now study the convergence of proximal gradient descent. Throughout this section, we use the stepsize

$$\eta = \frac{1}{L}.$$

Thus the iteration is

$$x^{k+1} = \mathbf{Prox}_{\frac{1}{L}h} \left(x^k - \frac{1}{L} \nabla f(x^k) \right).$$

The key estimate is a variant of the EVI inequality adapted to the proximal gradient update. This is the main inequality from which descent and convergence rates follow.

Theorem 12.1 (Proximal-gradient EVI). Assume f is convex and L -smooth, and h is proper, closed, and convex. Let

$$F(x) = f(x) + h(x).$$

For a given point y , define

$$y^+ = \mathbf{Prox}_{\frac{1}{L}h} \left(y - \frac{1}{L} \nabla f(y) \right).$$

Then, for every $x \in \mathbb{R}^d$,

$$F(y^+) - F(x) \leq \frac{L}{2} \|x - y\|^2 - \frac{L}{2} \|x - y^+\|^2 - g(x, y), \quad \text{eq:lec12-evi (12.7)}$$

where

$$g(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle. \quad \text{eq:lec12-bregman (12.8)}$$

Moreover,

$$g(x, y) \geq 0$$

by convexity of f .

Proof. By definition of the proximal operator,

$$y^+ = \arg \min_z \left\{ h(z) + \frac{L}{2} \left\| z - \left(y - \frac{1}{L} \nabla f(y) \right) \right\|^2 \right\}.$$

Expanding the square and ignoring constants independent of z , this is equivalent to

$$y^+ = \arg \min_z \left\{ h(z) + \langle \nabla f(y), z - y \rangle + \frac{L}{2} \|z - y\|^2 \right\}.$$

The first-order optimality condition for this convex minimization problem is

$$0 \in \partial h(y^+) + \nabla f(y) + L(y^+ - y).$$

Hence

$$-\nabla f(y) - L(y^+ - y) \in \partial h(y^+).$$

Using the subgradient inequality for h , for every x ,

$$h(x) \geq h(y^+) + \langle -\nabla f(y) - L(y^+ - y), x - y^+ \rangle.$$

Rearranging gives

$$h(y^+) - h(x) \leq \langle \nabla f(y), x - y^+ \rangle + L \langle y^+ - y, x - y^+ \rangle.$$

Since f is L -smooth,

$$f(y^+) \leq f(y) + \langle \nabla f(y), y^+ - y \rangle + \frac{L}{2} \|y^+ - y\|^2.$$

Adding the previous two estimates, and then subtracting $f(x)$, gives

$$\begin{aligned} F(y^+) - F(x) &\leq f(y) - f(x) + \langle \nabla f(y), x - y \rangle \\ &\quad + L \langle y^+ - y, x - y^+ \rangle + \frac{L}{2} \|y^+ - y\|^2. \end{aligned}$$

The first three terms are

$$f(y) - f(x) + \langle \nabla f(y), x - y \rangle = -g(x, y).$$

Therefore

$$F(y^+) - F(x) \leq -g(x, y) + L \langle y^+ - y, x - y^+ \rangle + \frac{L}{2} \|y^+ - y\|^2.$$

Now use the identity

$$\|x - y\|^2 = \|x - y^+\|^2 + 2 \langle x - y^+, y^+ - y \rangle + \|y^+ - y\|^2.$$

It follows that

$$L\langle y^+ - y, x - y^+ \rangle + \frac{L}{2}\|y^+ - y\|^2 = \frac{L}{2}\|x - y\|^2 - \frac{L}{2}\|x - y^+\|^2.$$

Substituting this into the previous inequality proves the desired EVI inequality.

Finally, since f is convex,

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle.$$

Therefore

$$g(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle \geq 0.$$

■

Remark. The term $g(x, y)$ is the Bregman divergence associated with f . It measures the difference between $f(x)$ and the first-order Taylor approximation of f at y . It is not generally a distance, because it need not be symmetric and need not satisfy the triangle inequality.

Proposition 12.1.2 (Descent lemma). Assume f is convex and L -smooth, and h is proper, closed, and convex. Let

$$F(x) = f(x) + h(x), \quad x^{k+1} = \mathbf{Prox}_{\frac{1}{L}h} \left(x^k - \frac{1}{L} \nabla f(x^k) \right).$$

Then

$$F(x^{k+1}) \leq F(x^k).$$

eq:lec12-descent
(12.9)

Moreover, if $\bar{x} \in \arg \min F$, then

$$\|x^{k+1} - \bar{x}\| \leq \|x^k - \bar{x}\|.$$

eq:lec12-fejer
(12.10)

Proof. Apply the proximal-gradient EVI with

$$y = x^k, \quad y^+ = x^{k+1}.$$

Then for every x ,

$$F(x^{k+1}) - F(x) \leq \frac{L}{2}\|x - x^k\|^2 - \frac{L}{2}\|x - x^{k+1}\|^2 - g(x, x^k).$$

First choose $x = x^k$. Since $g(x^k, x^k) = 0$, we obtain

$$F(x^{k+1}) - F(x^k) \leq -\frac{L}{2}\|x^{k+1} - x^k\|^2 \leq 0.$$

This proves the objective descent estimate.

Next choose $x = \bar{x}$, where $\bar{x} \in \arg \min F$. Then

$$F(x^{k+1}) - F(\bar{x}) \leq \frac{L}{2}\|\bar{x} - x^k\|^2 - \frac{L}{2}\|\bar{x} - x^{k+1}\|^2 - g(\bar{x}, x^k).$$

Since $F(x^{k+1}) - F(\bar{x}) \geq 0$ and $g(\bar{x}, x^k) \geq 0$, we get

$$0 \leq \frac{L}{2} \left(\|\bar{x} - x^k\|^2 - \|\bar{x} - x^{k+1}\|^2 \right).$$

Therefore

$$\|\bar{x} - x^{k+1}\|^2 \leq \|\bar{x} - x^k\|^2,$$

which proves the claimed Fejér monotonicity estimate. ■

Remark. The descent lemma is useful because $F = f + h$ need not be L -smooth. Ordinary gradient descent applied directly to F would require smoothness of the full objective. In proximal gradient descent, only the smooth part f is handled by a gradient step, while the nonsmooth part h is handled by the proximal operator.

Exercise. Check what is required for $F = f + h$ to be L -smooth, given that f is convex and L -smooth, and h is convex.

Proof. Since both f and h are convex, $F = f + h$ is convex. However, F is not necessarily smooth because h need not be differentiable. For example,

$$h(x) = \lambda \|x\|_1$$

is convex but not differentiable at points where some coordinate is zero. Hence $f + h$ is not smooth in general.

To make F smooth, we need an additional smoothness assumption on h . If h is differentiable and L_h -smooth, then

$$\nabla F(x) = \nabla f(x) + \nabla h(x).$$

Therefore,

$$\begin{aligned} \|\nabla F(x) - \nabla F(y)\| &\leq \|\nabla f(x) - \nabla f(y)\| + \|\nabla h(x) - \nabla h(y)\| \\ &\leq (L + L_h)\|x - y\|. \end{aligned}$$

Thus F is $(L + L_h)$ -smooth. In the twice-differentiable case, F is L_F -smooth if and only if

$$\nabla^2 F(x) \preceq L_F I \quad \text{for all } x.$$

■

Theorem 12.2 (Sublinear convergence rate). Assume f is convex and L -smooth, and h is proper, closed, and convex. Let

$$x^{k+1} = \mathbf{Prox}_{\frac{1}{L}h} \left(x^k - \frac{1}{L} \nabla f(x^k) \right).$$

If $\bar{x} \in \arg \min F$, then

$$F(x^k) - F(\bar{x}) \leq \frac{L\|x^0 - \bar{x}\|^2}{2k}. \quad \text{eq:lec12-sublinear (12.11)}$$

Therefore,

$$F(x^k) - F(\bar{x}) = \mathcal{O}\left(\frac{1}{k}\right).$$

Proof. Using the proximal-gradient EVI with

$$y = x^i, \quad y^+ = x^{i+1}, \quad x = \bar{x},$$

we obtain

$$F(x^{i+1}) - F(\bar{x}) \leq \frac{L}{2}\|x^i - \bar{x}\|^2 - \frac{L}{2}\|x^{i+1} - \bar{x}\|^2 - g(\bar{x}, x^i).$$

Since $g(\bar{x}, x^i) \geq 0$,

$$F(x^{i+1}) - F(\bar{x}) \leq \frac{L}{2}(\|x^i - \bar{x}\|^2 - \|x^{i+1} - \bar{x}\|^2). \quad \text{eq:lec12-onestep (12.12)}$$

Summing the previous one-step estimate from $i = 0$ to $k - 1$, the right-hand side telescopes:

$$\sum_{i=0}^{k-1} (F(x^{i+1}) - F(\bar{x})) \leq \frac{L}{2}(\|x^0 - \bar{x}\|^2 - \|x^k - \bar{x}\|^2) \leq \frac{L}{2}\|x^0 - \bar{x}\|^2.$$

By the descent lemma, the sequence $F(x^k)$ is nonincreasing. Hence

$$F(x^k) - F(\bar{x}) \leq \frac{1}{k} \sum_{i=0}^{k-1} (F(x^{i+1}) - F(\bar{x})).$$

Combining this with the telescoping bound gives the desired sublinear rate. ■

Remark. This matches the standard convergence rate for gradient descent on a smooth convex function:

$$f(x^k) - f(\bar{x}) \leq \frac{L\|x^0 - \bar{x}\|^2}{2k}.$$

The difference is that proximal gradient descent obtains the same order of convergence without requiring the full objective $F = f + h$ to be smooth.

Theorem 12.3 (Linear convergence under strong convexity). Assume f is m -strongly convex and L -smooth, and h is proper, closed, and convex. Let

$$x^{k+1} = \text{Prox}_{\frac{1}{L}h} \left(x^k - \frac{1}{L} \nabla f(x^k) \right).$$

If $\bar{x} \in \arg \min F$, then

$$\|x^k - \bar{x}\|^2 \leq \left(1 - \frac{m}{L}\right)^k \|x^0 - \bar{x}\|^2. \quad \text{eq:lec12-linear (12.13)}$$

Proof. Since f is m -strongly convex,

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{m}{2}\|x - y\|^2.$$

Therefore,

$$g(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle \geq \frac{m}{2}\|x - y\|^2.$$

Apply the proximal-gradient EVI with

$$y = x^k, \quad y^+ = x^{k+1}, \quad x = \bar{x}.$$

Then, using the strong convexity lower bound for g ,

$$\begin{aligned} F(x^{k+1}) - F(\bar{x}) &\leq \frac{L}{2}\|x^k - \bar{x}\|^2 - \frac{L}{2}\|x^{k+1} - \bar{x}\|^2 - g(\bar{x}, x^k) \\ &\leq \frac{L}{2}\|x^k - \bar{x}\|^2 - \frac{L}{2}\|x^{k+1} - \bar{x}\|^2 - \frac{m}{2}\|x^k - \bar{x}\|^2. \end{aligned}$$

Since $F(x^{k+1}) - F(\bar{x}) \geq 0$, we get

$$0 \leq \frac{L-m}{2}\|x^k - \bar{x}\|^2 - \frac{L}{2}\|x^{k+1} - \bar{x}\|^2.$$

Thus

$$\|x^{k+1} - \bar{x}\|^2 \leq \left(1 - \frac{m}{L}\right) \|x^k - \bar{x}\|^2. \quad \text{eq:lec12-linear-contract (12.14)}$$

Iterating this contraction inequality yields the desired linear rate. ■

12.2 More Examples

Gabriel Peyre's twitter is very motivated. A good visualization on projected gradient descent can be found in (<https://x.com/gabrielpeyre/status/1536211765922390016>). Many algorithms have been visualized in his series of twitter "Oldies but goldies".

Example 12.2.1 (Proposed by Chenghan Li, Proximal gradient method for LASSO). Consider the LASSO problem

$$\min_{x \in \mathbb{R}^d} F(x) = \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1,$$

where $A \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$, and $\lambda > 0$. This is a composite convex optimization problem

$$F(x) = f(x) + h(x),$$

where

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2, \quad h(x) = \lambda \|x\|_1.$$

Here f is smooth and convex, while h is convex but not differentiable. The gradient of the smooth part is

$$\nabla f(x) = A^T(Ax - b).$$

Therefore, the proximal gradient update becomes

$$x^{k+1} = \text{Prox}_{\eta h} \left(x^k - \eta \nabla f(x^k) \right) = \text{Prox}_{\eta \lambda \|\cdot\|_1} \left(x^k - \eta A^T(Ax^k - b) \right).$$

From the soft-thresholding formula,

$$\text{Prox}_{\eta \lambda \|\cdot\|_1}(z) = S_{\eta \lambda}(z),$$

where $S_{\eta \lambda}$ is applied coordinatewise:

$$S_{\eta \lambda}(z_i) = \begin{cases} z_i - \eta \lambda, & z_i > \eta \lambda, \\ 0, & |z_i| \leq \eta \lambda, \\ z_i + \eta \lambda, & z_i < -\eta \lambda. \end{cases}$$

Thus the proximal gradient method for LASSO is

$$x^{k+1} = S_{\eta \lambda} \left(x^k - \eta A^T(Ax^k - b) \right).$$

This example shows the main idea of proximal gradient descent. We first take a usual gradient descent step for the smooth loss

$$\frac{1}{2} \|Ax - b\|_2^2,$$

and then apply the proximal operator of the nonsmooth regularizer

$$\lambda \|x\|_1.$$

In the LASSO case, the proximal step is soft-thresholding. It shrinks each coordinate toward zero and sends small coordinates exactly to zero, which explains why LASSO promotes sparse solutions.

Example 12.2.2 (Proposed by Eric Wong, Proximal Gradient vs. Subgradient Descent on LASSO, Proofread by Frank Reyes Felsner). We compare the two methods on the two-dimensional LASSO problem

$$\min_{x \in \mathbb{R}^2} F(x) = \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1,$$

with $A \in \mathbb{R}^{m \times 2}$, $b \in \mathbb{R}^m$, and $\lambda > 0$.

Since $F = f + h$ is not differentiable along the coordinate axes, plain gradient descent is not directly applicable. A subgradient method substitutes some $g^k \in \partial \|x^k\|_1$ for the missing gradient, but subgradient steps do not guarantee descent at every iterate, and a common theoretical choice is a diminishing step-size such as $\eta_k = \eta / \sqrt{k+1}$, which gives the standard $O(1/\sqrt{k})$ convergence rate for convex non-smooth optimization. The two update rules are:

- **Subgradient descent** ($\eta_k = \eta / \sqrt{k+1}$):

$$x^{k+1} = x^k - \eta_k \left(A^T(Ax^k - b) + \lambda g^k \right), \quad g^k \in \partial \|x^k\|_1.$$

- **Proximal gradient descent** ($\eta = 1/L$, $L = \|A\|_2^2$):

$$x^{k+1} = S_{\eta \lambda} \left(x^k - \eta A^T(Ax^k - b) \right).$$

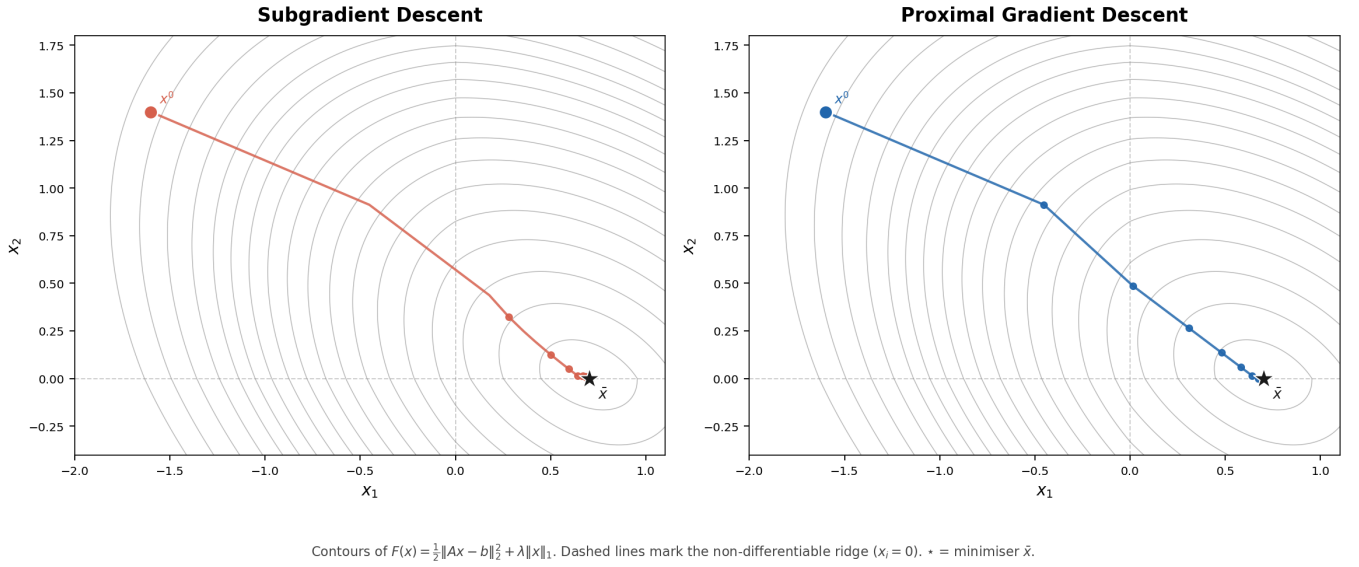


Figure 23: Iterates of subgradient descent (red, 120 steps) and proximal gradient descent (blue, 40 steps) on the two-dimensional LASSO objective. Contours of F are shown in gray; dashed lines mark the non-differentiable ridge $\{x_i = 0\}$; $\star$ denotes the minimiser $\bar{x}$.

Figure 23 shows the iterates of both methods starting from the same x^0 . Three features stand out:

1. *Oscillation.* The subgradient iterates bounce across the non-differentiable ridge $\{x_i = 0\}$, while the proximal gradient iterates converge smoothly.
2. *Exact sparsification.* Soft-thresholding snaps coordinates precisely to zero in finite steps; the subgradient method only approaches zero asymptotically.
3. *Fejér monotonicity.* By Proposition 12.1.2, the proximal gradient iterates satisfy $\|x^{k+1} - \bar{x}\| \leq \|x^k - \bar{x}\|$ at every step, visible as a smooth inward spiral toward $\bar{x}$.

Example 12.2.3 (Proposed by Andrew Wang). Consider the elastic net regression problem

$$\min_{x \in \mathbb{R}^d} F(x) = \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1 + \frac{\gamma}{2} \|x\|_2^2,$$

where $A \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$, $\lambda > 0$, and $\gamma > 0$. This problem is similar to LASSO, but it also includes the additional quadratic regularization term

$$\frac{\gamma}{2} \|x\|_2^2.$$

We can write the objective in the composite form

$$F(x) = f(x) + h(x),$$

where

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2 + \frac{\gamma}{2} \|x\|_2^2, \quad h(x) = \lambda \|x\|_1.$$

Here f is smooth and convex, while h is convex but not differentiable. The gradient of the smooth part is

$$\nabla f(x) = A^T(Ax - b) + \gamma x.$$

Moreover, ∇f is Lipschitz continuous with Lipschitz constant

$$L = \|A^T A\|_2 + \gamma.$$

Therefore, for a stepsize $0 < \eta \leq 1/L$, the proximal gradient update is

$$x^{k+1} = \text{Prox}_{\eta h} \left(x^k - \eta \nabla f(x^k) \right).$$

Substituting the expression for $\nabla f(x^k)$, we obtain

$$x^{k+1} = \text{Prox}_{\eta \lambda \|\cdot\|_1} \left(x^k - \eta \left(A^T(Ax^k - b) + \gamma x^k \right) \right).$$

Using the soft-thresholding formula,

$$\text{Prox}_{\eta \lambda \|\cdot\|_1}(z) = S_{\eta \lambda}(z),$$

where $S_{\eta \lambda}$ is applied coordinatewise:

$$S_{\eta \lambda}(z_i) = \begin{cases} z_i - \eta \lambda, & z_i > \eta \lambda, \\ 0, & |z_i| \leq \eta \lambda, \\ z_i + \eta \lambda, & z_i < -\eta \lambda. \end{cases}$$

Thus the proximal gradient method for elastic net regression is

$$x^{k+1} = S_{\eta \lambda} \left(x^k - \eta \left(A^T(Ax^k - b) + \gamma x^k \right) \right).$$

This example extends the LASSO example. In LASSO, the smooth part only contains the least-squares loss

$$\frac{1}{2}\|Ax - b\|_2^2.$$

For elastic net regression, the additional term

$$\frac{\gamma}{2}\|x\|_2^2$$

is smooth, so it is included in the gradient step. The nonsmooth part is still the ℓ^1 -regularizer, so the proximal step is still given by soft-thresholding. Therefore, proximal gradient descent naturally handles elastic net regression by combining a gradient step for the smooth least-squares and ridge terms with a proximal step for the nonsmooth sparsity term.

13 Lecture Notes 13 on Inexact Proximal Gradient Method

Cade Camacho summarizes: In this lecture, we explain the inexact proximal gradient method for minimizing functions that are composed of both smooth and nonsmooth parts. This method allows for small errors in the proximal updates while guaranteeing approximate optimality within ϵ . We show that this algorithm converges at the expected $O(1/k)$ rate provided that the errors are controlled.

Scribe Team: Chenghan Li, Yuxuan Zhao.

13.1 Lecture Notes

Throughout this lecture, unless otherwise stated, we consider the composite minimization problem

$$\min_{x \in \mathbb{R}^d} f(x) := g(x) + h(x),$$

where $g : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, differentiable, and has L -Lipschitz continuous gradient, and $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is a proper lower semi-continuous convex function. The function h is allowed to be nonsmooth. We also assume that $f = g + h$ attains its minimum at some point $x^* \in \mathbb{R}^d$ whenever a convergence statement is made.

The exact proximal-gradient method is

$$x^{k+1} = \text{prox}_{\eta h} \left(x^k - \eta \nabla g(x^k) \right),$$

where

$$\text{prox}_{\eta h}(y) = \arg \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

Equivalently, if

$$y^k = x^k - \eta \nabla g(x^k),$$

then

$$x^{k+1} = \arg \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y^k\|^2 \right\}.$$

Thus each proximal-gradient step contains an inner optimization problem. In many applications, this inner problem cannot be solved exactly, or solving it exactly may be too expensive. This motivates the inexact proximal-gradient method.

13.1.1 Two Sources of Inexactness

There are two common sources of error.

First, the gradient of g may be computed inexactly:

$$\nabla g(x^k) \rightsquigarrow \nabla g(x^k) + e_k,$$

where $e_k \in \mathbb{R}^d$ denotes the gradient error.

Second, the proximal problem may only be solved approximately. Instead of requiring x^{k+1} to be the exact minimizer of the proximal objective, we only require it to be close to the minimum value.

Definition 13.1 (Inexact proximal-gradient update). Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and differentiable, and let $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper lower semi-continuous convex function. Fix a stepsize $\eta > 0$, an iterate $x^k \in \mathbb{R}^d$, a gradient error vector $e_k \in \mathbb{R}^d$, and an accuracy level $\varepsilon_k \geq 0$. Define

$$y^k = x^k - \eta(\nabla g(x^k) + e_k).$$

We say that x^{k+1} is an inexact proximal-gradient update with error ε_k if

$$h(x^{k+1}) + \frac{1}{2\eta} \|x^{k+1} - y^k\|^2 \leq \varepsilon_k + \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y^k\|^2 \right\}.$$

In words, x^{k+1} does not need to solve the proximal subproblem exactly. Its proximal objective value is allowed to be at most ε_k above the optimal value.

Definition 13.2 (ε -minimizer). Let $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper convex function, and assume that f attains its minimum on $\mathbb{R}^d$. Let $\varepsilon \geq 0$. A point $\hat{x} \in \text{dom}(f)$ is called an ε -minimizer of f if

$$f(\hat{x}) \leq \varepsilon + \min_{x \in \mathbb{R}^d} f(x).$$

Equivalently,

$$f(\hat{x}) - \min_{x \in \mathbb{R}^d} f(x) \leq \varepsilon.$$

When $\varepsilon = 0$, this reduces to the usual notion of a minimizer.

Recall that for a proper convex function $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$, the ordinary subdifferential is defined by

$$\partial f(\hat{x}) = \left\{ u \in \mathbb{R}^d : f(z) \geq f(\hat{x}) + \langle u, z - \hat{x} \rangle \text{ for all } z \in \mathbb{R}^d \right\}.$$

The ε -subdifferential relaxes this condition by allowing an error of size ε .

Definition 13.3 (ε -subdifferential). Let $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper convex function. Let $\hat{x} \in \text{dom}(f)$ and let $\varepsilon \geq 0$. The ε -subdifferential of f at $\hat{x}$ is

$$\partial_\varepsilon f(\hat{x}) = \left\{ u \in \mathbb{R}^d : f(z) \geq f(\hat{x}) + \langle u, z - \hat{x} \rangle - \varepsilon \text{ for all } z \in \mathbb{R}^d \right\}.$$

Remark. The ordinary subdifferential is the special case $\varepsilon = 0$:

$$\partial_0 f(\hat{x}) = \partial f(\hat{x}).$$

For $\varepsilon > 0$, the affine lower support is allowed to lie ε below the function value.

Theorem 13.4 (ε -minimizer and ε -subdifferential). Let $f : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper convex function, and assume that f attains its minimum on $\mathbb{R}^d$. Let $\hat{x} \in \text{dom}(f)$ and let $\varepsilon \geq 0$. Then

$$\hat{x} \text{ is an } \varepsilon\text{-minimizer of } f \iff 0 \in \partial_\varepsilon f(\hat{x}).$$

Proof. By definition,

$$0 \in \partial_\varepsilon f(\hat{x})$$

if and only if

$$f(z) \geq f(\hat{x}) + \langle 0, z - \hat{x} \rangle - \varepsilon \quad \text{for all } z \in \mathbb{R}^d.$$

Since $\langle 0, z - \hat{x} \rangle = 0$, this is equivalent to

$$f(z) \geq f(\hat{x}) - \varepsilon \quad \text{for all } z \in \mathbb{R}^d.$$

Equivalently,

$$f(\hat{x}) \leq f(z) + \varepsilon \quad \text{for all } z \in \mathbb{R}^d.$$

Taking the minimum over z , we get

$$f(\hat{x}) \leq \min_{z \in \mathbb{R}^d} f(z) + \varepsilon.$$

Therefore $\hat{x}$ is an ε -minimizer of f .

The converse follows by reversing the same argument. If

$$f(\hat{x}) \leq \min_{z \in \mathbb{R}^d} f(z) + \varepsilon,$$

then for every $z \in \mathbb{R}^d$,

$$f(\hat{x}) \leq f(z) + \varepsilon,$$

which implies

$$f(z) \geq f(\hat{x}) - \varepsilon = f(\hat{x}) + \langle 0, z - \hat{x} \rangle - \varepsilon.$$

Hence $0 \in \partial_\varepsilon f(\hat{x})$. ■

Example 13.1.1 (ε -Subdifferential of a Quadratic Function). Let $\eta > 0$, fix $y \in \mathbb{R}^d$, and define the convex differentiable quadratic function

$$q(x) = \frac{1}{2\eta} \|x - y\|^2.$$

Since q is differentiable,

$$\nabla q(x) = \frac{1}{\eta}(x - y).$$

The ordinary subdifferential is

$$\partial q(x) = \left\{ \frac{1}{\eta}(x - y) \right\}.$$

For the ε -subdifferential, with $\varepsilon \geq 0$, the subgradient is allowed to deviate from the exact gradient. In fact,

$$\partial_\varepsilon q(x) = \left\{ \frac{1}{\eta}(x - y + \xi) : \|\xi\| \leq \sqrt{2\eta\varepsilon} \right\}.$$

Equivalently,

$$u \in \partial_\varepsilon q(x) \iff \left\| u - \frac{1}{\eta}(x - y) \right\| \leq \sqrt{\frac{2\varepsilon}{\eta}}.$$

Remark. Thus the ε -subdifferential of a quadratic function is not a single vector. It is a ball around the exact gradient. The radius is of order $\sqrt{\varepsilon}$.

13.1.2 Approximate Optimality Condition for Inexact Proximal Updates

The previous theorem lets us translate inexact proximal updates into approximate first-order optimality conditions.

Lemma 13.5 (Approximate optimality condition for inexact proximal updates). Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex and differentiable, and let $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper lower semi-continuous convex function. Fix $\eta > 0$, an iterate $x^{k-1} \in \mathbb{R}^d$, a gradient error vector $e_k \in \mathbb{R}^d$, and an accuracy level $\varepsilon_k \geq 0$. Suppose that $x^k \in \text{dom}(h)$ is an ε_k -optimal solution of the proximal subproblem

$$\min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \left\| x - \left(x^{k-1} - \eta(\nabla g(x^{k-1}) + e_k) \right) \right\|^2 \right\}.$$

Then there exists a vector $r_k \in \mathbb{R}^d$ satisfying

$$\|r_k\| \leq \sqrt{\frac{2\varepsilon_k}{\eta}},$$

such that

$$\frac{1}{\eta}(x^{k-1} - x^k) - \nabla g(x^{k-1}) - e_k - r_k \in \partial_{\varepsilon_k} h(x^k).$$

Proof. Define

$$y^{k-1} = x^{k-1} - \eta(\nabla g(x^{k-1}) + e_k).$$

The proximal subproblem is

$$\min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y^{k-1}\|^2 \right\}.$$

Since x^k is an ε_k -minimizer of this convex proximal objective, the previous theorem gives

$$0 \in \partial_{\varepsilon_k} \left(h(\cdot) + \frac{1}{2\eta} \|\cdot - y^{k-1}\|^2 \right) (x^k).$$

Using the ε -subdifferential of the quadratic term, there exists a perturbation vector $\xi_k \in \mathbb{R}^d$ such that

$$\|\xi_k\| \leq \sqrt{2\eta\varepsilon_k}$$

and

$$-\frac{1}{\eta}(x^k - y^{k-1} + \xi_k) \in \partial_{\varepsilon_k} h(x^k).$$

Set

$$r_k = \frac{\xi_k}{\eta}.$$

Then

$$\|r_k\| \leq \sqrt{\frac{2\varepsilon_k}{\eta}}.$$

Substituting the definition of y^{k-1} gives

$$\frac{1}{\eta}(x^{k-1} - x^k) - \nabla g(x^{k-1}) - e_k - r_k \in \partial_{\varepsilon_k} h(x^k).$$

■

Remark. In the exact case, $\varepsilon_k = 0$, $e_k = 0$, and $r_k = 0$. The condition becomes

$$\frac{1}{\eta}(x^{k-1} - x^k) - \nabla g(x^{k-1}) \in \partial h(x^k),$$

which is the usual optimality condition for the proximal-gradient method.

13.1.3 Connection with the Moreau Envelope

Let $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper lower semi-continuous convex function and let $\eta > 0$. The Moreau envelope of h with parameter η is

$$M_{\eta h}(y) = \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

The proximal operator returns a minimizer of this problem:

$$\text{prox}_{\eta h}(y) = \arg \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

Thus an inexact proximal update can also be written as

$$h(x^k) + \frac{1}{2\eta} \|x^k - y^{k-1}\|^2 \leq M_{\eta h}(y^{k-1}) + \varepsilon_k.$$

That is, the proximal objective value at x^k is within ε_k of the Moreau envelope value.

13.1.4 Convergence Rate of the Basic Inexact Proximal-Gradient Method

We now state a convergence rate for the basic inexact proximal-gradient method. In this theorem we use the step size

$$\eta = \frac{1}{L}.$$

The update takes the form

$$x^k = \text{prox}_{h/L} \left(x^{k-1} - \frac{1}{L} (\nabla g(x^{k-1}) + e_k) \right),$$

where the proximal problem is solved only up to error ε_k .

Theorem 13.6 (Convergence rate under convexity). Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex, differentiable, and have L -Lipschitz continuous gradient. Let $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper lower semi-continuous convex function. Define $f = g + h$, and assume that f attains its minimum at some $x^* \in \mathbb{R}^d$. Let $\{e_i\}_{i \geq 1} \subset \mathbb{R}^d$ be gradient error vectors and let $\{\varepsilon_i\}_{i \geq 1}$ be nonnegative proximal accuracy levels. Suppose that, for each $i \geq 1$, x^i is an ε_i -optimal solution of the proximal subproblem with stepsize $1/L$ and gradient error e_i . Then, for all $k \geq 1$,

$$f\left(\frac{1}{k} \sum_{i=1}^k x^i\right) - f(x^*) \leq \frac{L}{2k} \left(\|x^0 - x^*\| + 2A_k + \sqrt{2B_k} \right)^2,$$

where

$$A_k = \sum_{i=1}^k \left(\frac{\|e_i\|}{L} + \sqrt{\frac{2\varepsilon_i}{L}} \right), \quad B_k = \sum_{i=1}^k \frac{\varepsilon_i}{L}.$$

Proof sketch. The proof has two main steps.

First, using the approximate optimality condition from the previous lemma, one derives a discrete EVI-type inequality:

$$f(x^i) - f(x^*) \leq \frac{L}{2} \left(\|x^{i-1} - x^*\|^2 - \|x^i - x^*\|^2 \right) + \varepsilon_i + \left(\|e_i\| + \sqrt{2L\varepsilon_i} \right) \|x^i - x^*\|.$$

The first term telescopes after summing over $i = 1, \dots, k$.

Second, the error term still contains $\|x^i - x^*\|$. To control this quantity, one applies a discrete Gronwall-type argument. This gives a uniform bound on the iterates in terms of

$$\|x^0 - x^*\|, \quad A_k, \quad B_k.$$

Substituting this bound back into the summed EVI inequality gives

$$\sum_{i=1}^k \left(f(x^i) - f(x^*) \right) \leq \frac{L}{2} \left(\|x^0 - x^*\| + 2A_k + \sqrt{2B_k} \right)^2.$$

Finally, by convexity of f ,

$$f\left(\frac{1}{k} \sum_{i=1}^k x^i\right) \leq \frac{1}{k} \sum_{i=1}^k f(x^i).$$

Dividing by k gives the desired result. ■

Remark. If there are no errors, then $e_i = 0$ and $\varepsilon_i = 0$ for all i . Hence

$$A_k = B_k = 0,$$

and the bound becomes

$$f\left(\frac{1}{k} \sum_{i=1}^k x^i\right) - f(x^*) \leq \frac{L}{2k} \|x^0 - x^*\|^2,$$

which recovers the usual $O(1/k)$ convergence rate.

Remark. The same $O(1/k)$ convergence rate is preserved if the accumulated errors remain bounded. For example, it is enough that

$$\sum_{i=1}^{\infty} \|e_i\| < \infty, \quad \sum_{i=1}^{\infty} \sqrt{\varepsilon_i} < \infty.$$

A sufficient condition is

$$\|e_i\| = O\left(\frac{1}{i^{1+\delta}}\right), \quad \sqrt{\varepsilon_i} = O\left(\frac{1}{i^{1+\delta}}\right)$$

for some $\delta > 0$.

13.2 Exercises

Exercise (Moreau envelope formulation of the inexact proximal step). Let $h : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ be a proper lower semi-continuous convex function, let $\eta > 0$, and define the Moreau envelope of h by

$$M_{\eta h}(y) = \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

Fix $y \in \mathbb{R}^d$ and let $\varepsilon \geq 0$. Suppose $\hat{x} \in \text{dom}(h)$ satisfies the inexact proximal condition

$$h(\hat{x}) + \frac{1}{2\eta} \|\hat{x} - y\|^2 \leq \varepsilon + \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

Rewrite this condition in terms of the Moreau envelope. Then show that there exists a vector $r \in \mathbb{R}^d$ with

$$\|r\| \leq \sqrt{\frac{2\varepsilon}{\eta}}$$

such that

$$\frac{1}{\eta}(y - \hat{x}) - r \in \partial_\varepsilon h(\hat{x}).$$

Proof. *Proof by Chenghan Li, proofread by Don Tran*

By the definition of the Moreau envelope,

$$M_{\eta h}(y) = \min_{x \in \mathbb{R}^d} \left\{ h(x) + \frac{1}{2\eta} \|x - y\|^2 \right\}.$$

Therefore the inexact proximal condition is exactly

$$h(\hat{x}) + \frac{1}{2\eta} \|\hat{x} - y\|^2 \leq M_{\eta h}(y) + \varepsilon.$$

This is the Moreau envelope formulation: the proximal objective value at $\hat{x}$ is within ε of the Moreau envelope value. Now define the convex function

$$F(x) = h(x) + \frac{1}{2\eta} \|x - y\|^2.$$

The inequality above says that $\hat{x}$ is an ε -minimizer of F . By the equivalence between ε -minimizers and ε -subdifferentials (Theorem 13.4),

$$0 \in \partial_\varepsilon F(\hat{x}).$$

Using the standard ε -subdifferential sum rule in the weak form

$$\partial_\varepsilon(h + q)(\hat{x}) \subseteq \partial_\varepsilon h(\hat{x}) + \partial_\varepsilon q(\hat{x}),$$

where

$$q(x) = \frac{1}{2\eta} \|x - y\|^2,$$

there exist

$$u \in \partial_\varepsilon h(\hat{x}), \quad v \in \partial_\varepsilon q(\hat{x})$$

such that

$$0 = u + v.$$

For the quadratic term q , we have

$$\partial_\varepsilon q(\hat{x}) = \left\{ \frac{1}{\eta}(\hat{x} - y + \xi) : \|\xi\| \leq \sqrt{2\eta\varepsilon} \right\}.$$

Thus for some $\xi \in \mathbb{R}^d$ satisfying

$$\|\xi\| \leq \sqrt{2\eta\varepsilon},$$

we have

$$v = \frac{1}{\eta}(\hat{x} - y + \xi).$$

Since $0 = u + v$, it follows that

$$u = -v = \frac{1}{\eta}(y - \hat{x}) - \frac{\xi}{\eta}.$$

Set

$$r = \frac{\xi}{\eta}.$$

Then

$$\|r\| = \frac{1}{\eta} \|\xi\| \leq \frac{1}{\eta} \sqrt{2\eta\varepsilon} = \sqrt{\frac{2\varepsilon}{\eta}}.$$

Therefore

$$\frac{1}{\eta}(y - \hat{x}) - r = u \in \partial_\varepsilon h(\hat{x}).$$

This is the desired approximate optimality condition. ■

Example 13.2.1. A common example is the LASSO problem

$$\min_x \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1.$$

Here

$$g(x) = \frac{1}{2} \|Ax - b\|^2, \quad h(x) = \lambda \|x\|_1.$$

The function g is smooth and convex, while h is convex but nonsmooth. The proximal-gradient method applies naturally because the nonsmooth part is handled by the proximal operator.

Example 13.2.2. If the proximal operator of h does not have a simple closed form, we may solve the proximal subproblem approximately. The error tolerance ε_k controls how accurately the inner proximal problem is solved. The convergence theorem shows that the outer method can still converge at the usual $O(1/k)$ rate, provided the errors decrease sufficiently fast.

13.3 More Examples

Example 13.3.1 (Proposed by Frank Reyes Felsner).

Why Inexactness Is Not a Problem

The exact proximal-gradient update is

$$x^{k+1} = \text{prox}_{\eta h} \left(x^k - \eta \nabla g(x^k) \right).$$

In practice, we may have two sources of error:

$$\nabla g(x^k) \rightsquigarrow \nabla g(x^k) + e_k,$$

and the proximal subproblem may only be solved approximately.

The convergence theorem does not require the errors to be zero. It requires their accumulated effect to be finite:

$$\sum_{k=1}^{\infty} \|e_k\| < \infty, \quad \sum_{k=1}^{\infty} \sqrt{\varepsilon_k} < \infty.$$

So inexactness is not a problem when the errors shrink fast enough:

$$\boxed{\text{controlled errors do not overwhelm convergence.}}$$

Worked Example: A Simple Composite Problem

Consider

$$f(x) = g(x) + h(x), \quad g(x) = \frac{1}{2}(x-4)^2, \quad h(x) = |x|.$$

Thus

$$f(x) = \frac{1}{2}(x-4)^2 + |x|.$$

This is a one-dimensional LASSO-type problem:

smooth quadratic loss + nonsmooth absolute-value penalty.

Exact Proximal-Gradient Step

Since

$$g'(x) = x - 4, \quad g''(x) = 1,$$

we have

$$L = 1, \quad \eta = \frac{1}{L} = 1.$$

The exact gradient step is

$$y^{k-1} = x^{k-1} - g'(x^{k-1}) = x^{k-1} - (x^{k-1} - 4) = 4.$$

Thus the exact proximal subproblem is

$$\min_x \left\{ |x| + \frac{1}{2}(x-4)^2 \right\}.$$

For $x > 0$,

$$|x| + \frac{1}{2}(x-4)^2 = x + \frac{1}{2}(x-4)^2.$$

Differentiating gives

$$1 + (x-4) = x - 3.$$

Hence the exact proximal update is

$$x^k = 3 = x^*.$$

Inexact Gradient and Inexact Proximal Step

Now include gradient error e_k . The gradient step becomes

$$y^{k-1} = x^{k-1} - \left(g'(x^{k-1}) + e_k \right).$$

Since $g'(x^{k-1}) = x^{k-1} - 4$,

$$y^{k-1} = x^{k-1} - \left(x^{k-1} - 4 + e_k \right) = 4 - e_k.$$

For $h(x) = |x|$, the proximal operator is soft-thresholding:

$$\text{prox}_{|\cdot|}(y) = \text{sign}(y) \max\{|y| - 1, 0\}.$$

Assuming $4 - e_k > 1$,

$$\text{prox}_{|\cdot|}(4 - e_k) = 3 - e_k.$$

Now allow the proximal solve to be inexact by a point error δ_k :

$$x^k = 3 - e_k + \delta_k.$$

So the distance from the true minimizer is

$$x^k - x^* = -e_k + \delta_k.$$

Choosing Errors That Satisfy Convergence

In reality, errors arise from rounding error, noisy data, stochastic gradients, finite precision, or approximation, however for this example, suppose we have:

$$e_k = \frac{1}{k^2}, \quad \delta_k = \frac{1}{2k^2}.$$

Then

$$x^k = 3 - \frac{1}{k^2} + \frac{1}{2k^2} = 3 - \frac{1}{2k^2}.$$

Hence

$$x^k \rightarrow 3 = x^*.$$

The gradient error condition is satisfied because

$$\sum_{k=1}^{\infty} \|e_k\| = \sum_{k=1}^{\infty} \frac{1}{k^2} < \infty.$$

For the proximal error, the proximal objective error is

$$\varepsilon_k = \frac{1}{2} \delta_k^2.$$

by plugging in

$$3 + \delta_k \text{ into } f(x) = \frac{1}{2}(x - 4)^2 + |x|$$

Using

$$\delta_k = \frac{1}{2k^2},$$

we get

$$\varepsilon_k = \frac{1}{2} \left(\frac{1}{2k^2} \right)^2 = \frac{1}{8k^4}.$$

Therefore,

$$\sqrt{\varepsilon_k} = \frac{1}{\sqrt{8}k^2}.$$

Since

$$\sum_{k=1}^{\infty} \frac{1}{k^2} < \infty,$$

we also have

$$\sum_{k=1}^{\infty} \sqrt{\varepsilon_k} < \infty.$$

Objective Error and Main Takeaway

For this objective, if $x = 3 + s$, then

$$f(3 + s) - f(3) = \frac{1}{2}s^2.$$

In our inexact method,

$$x^k = 3 - e_k + \delta_k,$$

so

$$s = x^k - 3 = -e_k + \delta_k.$$

Therefore,

$$f(x^k) - f(x^*) = \frac{1}{2}(-e_k + \delta_k)^2.$$

With

$$e_k = \frac{1}{k^2}, \quad \delta_k = \frac{1}{2k^2},$$

we get

$$-e_k + \delta_k = -\frac{1}{k^2} + \frac{1}{2k^2} = -\frac{1}{2k^2}.$$

Thus

$$f(x^k) - f(x^*) = \frac{1}{2} \left(-\frac{1}{2k^2} \right)^2 = \frac{1}{8k^4} \rightarrow 0.$$

So even though every step is inexact, the errors shrink quickly enough that the iterates still converge and the objective error goes to zero.

Example 13.3.2 (Proposed by Andrew Wang). The previous example shows that inexact proximal gradient descent can still converge when the errors decrease sufficiently fast. This example shows the opposite direction: if the error does not decay, then the method may fail to converge to the true minimizer.

Consider the one-dimensional optimization problem

$$\min_{x \in \mathbb{R}} f(x), \quad f(x) = g(x) + h(x),$$

where

$$g(x) = \frac{1}{2}x^2, \quad h(x) = 0.$$

The unique minimizer is

$$x^* = 0.$$

Since

$$g'(x) = x,$$

the gradient of g is 1-Lipschitz. Thus we may take the stepsize

$$\eta = 1.$$

In the exact proximal gradient method, since $h = 0$, the proximal operator is the identity map. Hence the update becomes

$$x^{k+1} = x^k - \eta g'(x^k) = x^k - x^k = 0.$$

Therefore the exact method reaches the minimizer in one step.

Now suppose that the gradient is computed with an additive error e_k . The inexact gradient step is

$$x^{k+1} = x^k - \eta \left(g'(x^k) + e_k \right).$$

Using $\eta = 1$ and $g'(x^k) = x^k$, we obtain

$$x^{k+1} = x^k - \left(x^k + e_k \right) = -e_k.$$

If the error is persistent, for example

$$e_k = \varepsilon \quad \text{for all } k,$$

where $\varepsilon \neq 0$, then

$$x^{k+1} = -\varepsilon \quad \text{for all } k.$$

Thus the iterates do not converge to the true minimizer $x^* = 0$. Instead, they stay at distance

$$|x^{k+1} - x^*| = |\varepsilon|$$

from the minimizer. The objective error is also nonzero:

$$f(x^{k+1}) - f(x^*) = \frac{1}{2}\varepsilon^2.$$

Moreover, the accumulated gradient error is infinite:

$$\sum_{k=1}^{\infty} |e_k| = \sum_{k=1}^{\infty} |\varepsilon| = +\infty.$$

This example explains why the summability condition on the gradient error is important. Inexactness is acceptable only when the errors are controlled. If the error persists and does not decay, the inexact proximal gradient method may converge only to a biased point, or may fail to converge to the true minimizer.

14 Lecture Notes 14 on Duality and Lagrangian

Chenghan Li summarizes: This lecture introduces duality and the Lagrangian method for constrained optimization. The guiding idea is that the order of choices matters: in general, a “min-max” problem and a “max-min” problem need not have the same value. We first recall the primal linear programming problem and derive its dual problem from the Lagrangian by introducing multipliers for the constraints. Then we extend this idea to a general constrained optimization problem with equality and inequality constraints. The dual objective is obtained by minimizing the Lagrangian over the primal variable, and it gives a lower bound on the primal optimal value. This leads to weak duality, which always holds and can be viewed as a minimax inequality. We also discuss strong duality, where the primal and dual optimal values coincide, and define the duality gap as the difference between them. The lecture ends by explaining the main gain from duality: dual feasible points provide certificates and bounds for the primal problem, and the duality gap can be used as a stopping criterion in computations.

Scribe Team: Prashast Singh, Justin Zhou

14.1 Lecture Notes

“Better to be the head of a chicken than the tail of a phoenix.”

–Old Chinese proverb

The mathematical point is that the order of decisions matters. If we first choose the “species” and then the “rank,” the result may be different from first choosing the rank and then the species. In optimization, this idea appears through the inequality

$$\min_x \max_{u,v} L(x, u, v) \geq \max_{u,v} \min_x L(x, u, v),$$

which is the basic form of weak duality. The left-hand side means that we first take the dual variables that best maximize the Lagrangian penalty, and then minimize over x . Please note that in practice, this means: first pick a x , and maximize with respect to dual variables u, v as an inner loop; then minimizer over primal variable x . The right-hand side means that we first minimize over x , then choose the best dual variables. In general these two operations cannot be exchanged freely.

In this lecture we introduce Lagrangian duality. The main purpose is to convert constrained optimization problems into unconstrained ones by adding the constraints into the objective function using Lagrange multipliers. This gives a systematic way to construct lower bounds for minimization problems.

14.1.1 Primal-Dual in linear programming problem

We *first* recall a standard linear programming problem in primal form:

$$\begin{aligned} \min_x \quad & c^\top x \\ \text{s.t.} \quad & Ax = b, \\ & Gx \leq h. \end{aligned}$$

where x is the primal variable and c is the coefficient vector. Here, the constraint $Ax = b$ is an equality, while the constraint $Gx \leq h$ is an inequality.

The idea is to use the Lagrangian to handle these constraints. For multipliers u and v , define the Lagrangian as

$$L(x, u, v) = c^\top x + u^\top (Ax - b) + v^\top (Gx - h).$$

The vectors u and v must match the dimensions of the corresponding constraints so that the terms above are inner products. The multiplier u corresponds to the equality constraint $Ax = b$, and v corresponds to the inequality constraint $Gx \leq h$.

Consider the inequality constraint. If x is feasible, then

$$Ax - b = 0.$$

Therefore,

$$u^\top (Ax - b) = 0$$

for any value of u . Thus the equality multiplier u is unrestricted.

Now consider the inequality constraint. If x is feasible, then

$$Gx - h \leq 0.$$

If we require

$$v \geq 0,$$

then

$$v^\top (Gx - h) \leq 0.$$

Therefore, for every feasible x , every unrestricted u , and every $v \geq 0$, we have

$$L(x, u, v) = c^\top x + u^\top (Ax - b) + v^\top (Gx - h) \leq c^\top x.$$

Equivalently,

$$c^\top x \geq L(x, u, v) \quad \text{for every feasible } x.$$

Remark. The sign condition $v \geq 0$ is crucial. The inequality term $Gx - h$ is nonpositive for feasible points. To make sure that $v^\top(Gx - h) \leq 0$, we need $v \geq 0$. On the other hand, the equality multiplier u can be anything because the equality term is exactly zero on feasible points.

Second, we derive the dual objective from the Lagrangian.

Let C denote the feasible set of the primal linear program. Since

$$c^\top x \geq L(x, u, v)$$

for every $x \in C$, we get

$$\min_{x \in C} c^\top x \geq \min_{x \in C} L(x, u, v).$$

Also, minimizing over all x gives an even smaller value than minimizing only over feasible x . Hence,

$$\min_{x \in C} c^\top x \geq \min_{x \in C} L(x, u, v) \geq \min_x L(x, u, v).$$

This motivates the definition of the dual objective, which we define as

$$g(u, v) := \inf_x L(x, u, v).$$

Remark. In lectures $g(x)$ is often written as $\min_x L(x, u, v)$ as shorthand, but the more precise way to refer to $g(x)$ is $\inf_x L(x, u, v)$, since the minimum may not always exist.

Thus for every unrestricted u and every $v \geq 0$,

$$\min_{x \in C} c^\top x \geq g(u, v).$$

This means $g(u, v)$ gives a lower bound on the optimal value of the primal minimization problem.

Finally, we derive the dual linear program.

Now compute $g(u, v)$ explicitly. Starting from

$$L(x, u, v) = c^\top x + u^\top(Ax - b) + v^\top(Gx - h),$$

we expand:

$$\begin{aligned} L(x, u, v) &= c^\top x + u^\top Ax - u^\top b + v^\top Gx - v^\top h \\ &= \langle c + A^\top u + G^\top v, x \rangle - \langle u, b \rangle - \langle v, h \rangle. \end{aligned}$$

Since x is unconstrained when we compute $\inf_x L(x, u, v)$, two cases follow.

Case 1. $c + A^\top u + G^\top v = 0$: The term involving x disappears, and

$$\inf_x L(x, u, v) = -\langle u, b \rangle - \langle v, h \rangle.$$

Case 2. $c + A^\top u + G^\top v \neq 0$: The linear function

$$\langle c + A^\top u + G^\top v, x \rangle$$

is unbounded below as x ranges over all of $\mathbb{R}^n$. Therefore,

$$\inf_x L(x, u, v) = -\infty.$$

Thus

$$g(u, v) = \begin{cases} -\langle u, b \rangle - \langle v, h \rangle, & c + A^\top u + G^\top v = 0, \\ -\infty, & c + A^\top u + G^\top v \neq 0. \end{cases}$$

Therefore, maximizing the dual objective is equivalent to the following dual linear program:

$$\begin{aligned} \max_{u, v} \quad & -\langle u, b \rangle - \langle v, h \rangle \\ \text{s.t.} \quad & c + A^\top u + G^\top v = 0, \\ & v \geq 0. \end{aligned}$$

Equivalently, the equality constraint can be written as

$$-A^\top u - G^\top v = c.$$

The variable u is free, while v is constrained to be nonnegative.

Remark. The key point is that the dual problem comes from first fixing the multipliers u, v , then minimizing the Lagrangian over the primal variable x . In other words, for fixed u, v , we search for

$$\bar{x} = x(u, v)$$

such that

$$L(x(u, v), u, v) = g(u, v).$$

Then we choose u, v to make this lower bound as large as possible.

14.1.2 General constrained optimization problem

The same construction works beyond linear programming. Consider a general constrained optimization problem

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m, \\ & l_j(x) = 0, \quad j = 1, \dots, r. \end{aligned}$$

The function f can even be nonconvex. The inequality constraints are $h_i(x) \leq 0$, and the equality constraints are $l_j(x) = 0$.

Remark. In the linear programming example above, u was used for the equality constraint and v for the inequality constraint. In the general notation below, we follow the common convention that u denotes the multipliers for inequality constraints and v denotes the multipliers for equality constraints.

For this general problem, define the Lagrangian

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x).$$

Here

$$u_i \geq 0$$

for the inequality constraints, while the equality multipliers v_j are unrestricted.

If x is feasible, then

$$h_i(x) \leq 0 \quad \text{and} \quad l_j(x) = 0.$$

Therefore, for $u_i \geq 0$,

$$\sum_{i=1}^m u_i h_i(x) \leq 0,$$

and

$$\sum_{j=1}^r v_j l_j(x) = 0.$$

Thus, for every feasible x ,

$$L(x, u, v) \leq f(x).$$

Equivalently,

$$f(x) \geq L(x, u, v) \quad \text{for every feasible } x, u \geq 0, \text{ and } v \text{ unrestricted.}$$

Definition 14.1 (Dual function). For the general constrained problem, the dual function is

$$g(u, v) := \inf_x L(x, u, v).$$

The dual problem is

$$\max_{u \geq 0, v} g(u, v).$$

Let

$$C = \{x : h_i(x) \leq 0, i = 1, \dots, m, \text{ and } l_j(x) = 0, j = 1, \dots, r\}$$

be the feasible set, and let

$$p^* := \inf_{x \in C} f(x)$$

be the primal optimal value. Since $f(x) \geq L(x, u, v)$ for every feasible x , we have

$$p^* = \inf_{x \in C} f(x) \geq \inf_{x \in C} L(x, u, v) \geq \inf_x L(x, u, v) = g(u, v).$$

Therefore,

$$p^* \geq g(u, v) \quad \text{for every } u \geq 0 \text{ and every } v.$$

Taking the supremum over all dual variables gives

$$p^* \geq \sup_{u \geq 0, v} g(u, v).$$

If we define the dual optimal value by

$$d^* := \sup_{u \geq 0, v} g(u, v),$$

then weak duality says

$$p^* \geq d^*.$$

Theorem 14.2 (Weak duality). For any constrained minimization problem with Lagrangian

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x),$$

where $u \geq 0$, the primal optimal value is always at least the dual optimal value:

$$p^* \geq d^*.$$

Proof. For every feasible x , every $u \geq 0$, and every v , we have

$$L(x, u, v) \leq f(x).$$

Taking the infimum over feasible x gives

$$\inf_{x \in C} f(x) \geq \inf_{x \in C} L(x, u, v).$$

Since minimizing over all x can only decrease the value,

$$\inf_{x \in C} L(x, u, v) \geq \inf_x L(x, u, v) = g(u, v).$$

Thus

$$p^* \geq g(u, v)$$

for every dual feasible pair (u, v) . Taking the supremum over all such (u, v) yields

$$p^* \geq d^*.$$

■

The weak duality inequality can also be viewed as a **minimax inequality**. Since the Lagrangian penalizes infeasible points, we can formally write

$$\inf_{x \in C} f(x) = \inf_x \sup_{u \geq 0, v} L(x, u, v).$$

Indeed, if x is feasible, then the inequality terms satisfy $h_i(x) \leq 0$, so the supremum over $u_i \geq 0$ is attained at $u_i = 0$, and the equality terms vanish. Thus the supremum gives $f(x)$. If x violates an equality constraint, then because v_j is unrestricted, the term $v_j l_j(x)$ can be made arbitrarily large. If x violates an inequality constraint, then for some i , $h_i(x) > 0$, and $u_i h_i(x)$ can be made arbitrarily large by taking $u_i \rightarrow +\infty$. Hence infeasible points are assigned value $+\infty$.

On the other hand,

$$d^* = \sup_{u \geq 0, v} \inf_x L(x, u, v).$$

Therefore weak duality is the statement

$$\inf_x \sup_{u \geq 0, v} L(x, u, v) \geq \sup_{u \geq 0, v} \inf_x L(x, u, v).$$

This inequality always holds. It says that

$$\min \max \geq \max \min.$$

Remark. This is where the phrase “Better be the head of a dog than the tail of a lion” is useful. The first choice and the second choice are not symmetric. If the outside choice is “species,” then lion is better than dog. If the inside choice is “rank,” then head is better than tail. Changing the order of the choices changes the meaning of the final result. Similarly, changing the order of min and max changes the value of the optimization problem.

14.1.3 Strong duality and duality gap

Weak duality always gives

$$p^* \geq d^*.$$

The difference

$$p^* - d^*$$

is called the duality gap. When

$$p^* = d^*,$$

we say that strong duality holds.

In terms of the Lagrangian, strong duality asks when we can exchange the order of minimization and maximization:

$$\inf_x \sup_{u \geq 0, v} L(x, u, v) = \sup_{u \geq 0, v} \inf_x L(x, u, v).$$

This equality does not always hold. Extra assumptions are needed, such as convexity and suitable constraint qualifications.

When strong duality holds, the primal optimizer and dual optimizer can often induce each other. More precisely, suppose $\bar{x}$ is primal optimal and $(\bar{u}, \bar{v})$ is dual optimal. Then one expects a saddle point relation:

$$L(\bar{x}, u, v) \leq L(\bar{x}, \bar{u}, \bar{v}) \leq L(x, \bar{u}, \bar{v}).$$

The right inequality says that $\bar{x}$ minimizes $L(x, \bar{u}, \bar{v})$ over x . The left inequality says that $(\bar{u}, \bar{v})$ maximizes $L(\bar{x}, u, v)$ over the dual variables.

Remark. In computations, the duality gap gives a useful stopping criterion. Suppose x^* is primal feasible and (u^*, v^*) is dual feasible. If

$$f(x^*) - g(u^*, v^*) \leq \varepsilon,$$

then the primal value and the dual lower bound are within ε . Since

$$g(u^*, v^*) \leq d^* \leq p^* \leq f(x^*),$$

a small gap certifies that x^* is nearly optimal.

Example 14.1.1 (quadratic programming). Consider the quadratic programming problem

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^\top Q x + c^\top x \\ \text{s.t.} \quad & A x = b, \\ & x \geq 0, \end{aligned}$$

where Q is positive definite. The inequality constraint $x \geq 0$ can be written as

$$-x \leq 0.$$

Let $u \geq 0$ be the multiplier for the constraint $-x \leq 0$, and let v be the unrestricted multiplier for the equality

constraint $Ax = b$. Using the convention $v^\top(Ax - b)$, the Lagrangian is

$$\begin{aligned} L(x, u, v) &= \frac{1}{2}x^\top Qx + c^\top x - u^\top x + v^\top(Ax - b) \\ &= \frac{1}{2}x^\top Qx + \langle c - u + A^\top v, x \rangle - b^\top v. \end{aligned}$$

If x is feasible, then $x \geq 0$ and $Ax - b = 0$. Thus

$$-u^\top x \leq 0 \quad \text{and} \quad v^\top(Ax - b) = 0.$$

Therefore,

$$f(x) \geq L(x, u, v)$$

for all feasible x , all $u \geq 0$, and all v .

To compute the dual function, minimize $L(x, u, v)$ with respect to x . Since Q is positive definite, this is a strictly convex quadratic function in x , so the minimizer is determined by the first-order condition:

$$\partial_x L(x, u, v) = 0.$$

We compute

$$\partial_x L(x, u, v) = Qx + c - u + A^\top v.$$

Setting this equal to zero gives

$$Qx + c - u + A^\top v = 0,$$

and hence

$$x(u, v) = -Q^{-1}(c - u + A^\top v).$$

Substituting this minimizer back into the Lagrangian gives

$$g(u, v) = -\frac{1}{2}(c - u + A^\top v)^\top Q^{-1}(c - u + A^\top v) - b^\top v.$$

Therefore, the dual problem is

$$\begin{aligned} \max_{u, v} \quad & -\frac{1}{2}(c - u + A^\top v)^\top Q^{-1}(c - u + A^\top v) - b^\top v \\ \text{s.t.} \quad & u \geq 0. \end{aligned}$$

Remark. The sign of the equality multiplier is a convention. If instead one writes the equality term as $v^\top(b - Ax)$, then the formulas above change by replacing v with $-v$. Since v is unrestricted, both conventions describe the same dual problem after a change of variables.

14.1.4 Main gain from duality

The main gain from duality is that the dual function is always concave in the dual variables. This is because

$$g(u, v) = \inf_x L(x, u, v)$$

is the pointwise infimum of affine functions of (u, v) . Therefore, even when the original primal problem is nonconvex, the dual problem

$$\max_{u \geq 0, v} g(u, v)$$

has a concave objective maximized over a convex constraint set. In the language of convex optimization, this means the dual problem is a convex optimization problem.

Thus duality gives both theoretical and computational advantages:

- it gives lower bounds on the primal optimal value;
- it provides the weak duality inequality $p^* \geq d^*$;
- it gives a stopping criterion through the duality gap;
- and under additional assumptions, strong duality lets us recover primal and dual optimizers from each other.

We now explain the concavity of the dual objective more explicitly. Recall that for the general constrained problem,

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m, \\ & l_j(x) = 0, \quad j = 1, \dots, r, \end{aligned}$$

the Lagrangian is

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x),$$

where $u_i \geq 0$ for all inequality constraints and v_j is unrestricted for equality constraints. The dual objective is

$$g(u, v) = \inf_x L(x, u, v).$$

When the infimum is attained, we may write this as

$$g(u, v) = \min_x L(x, u, v).$$

A useful trick is to rewrite this infimum as the negative of a supremum:

$$\begin{aligned} g(u, v) &= \inf_x L(x, u, v) \\ &= -\sup_x (-L(x, u, v)) \\ &= -\sup_x \left(-f(x) - \sum_{i=1}^m u_i h_i(x) - \sum_{j=1}^r v_j l_j(x) \right). \end{aligned}$$

Thus the dual function is the negative of a pointwise supremum.

Proposition 14.1.2 (Pointwise maximum of convex functions). Let $\{F_s\}_{s \in S}$ be a family of convex functions, where x is the input variable and s is an index. If

$$G(x) = \sup_{s \in S} F_s(x),$$

then G is convex.

Proof. Let x, y be in the domain and let $t \in [0, 1]$. Since each F_s is convex, for every index s ,

$$F_s((1-t)x + ty) \leq (1-t)F_s(x) + tF_s(y).$$

Taking the supremum over s on the left-hand side gives

$$\begin{aligned} G((1-t)x + ty) &= \sup_{s \in S} F_s((1-t)x + ty) \\ &\leq \sup_{s \in S} ((1-t)F_s(x) + tF_s(y)) \\ &\leq (1-t) \sup_{s \in S} F_s(x) + t \sup_{s \in S} F_s(y) \\ &= (1-t)G(x) + tG(y). \end{aligned}$$

Therefore G is convex. ■

Remark. Geometrically, the pointwise maximum of convex functions is the upper envelope of a family of convex curves. In the lecture sketch, the red curve represented this upper envelope. Since it lies above all the convex curves and is formed by taking the maximum at each point, it remains convex.

Now apply this proposition to the dual objective. For each fixed x , define

$$\varphi_x(u, v) = -f(x) - \sum_{i=1}^m u_i h_i(x) - \sum_{j=1}^r v_j l_j(x).$$

Here x is treated as an index, while (u, v) are the variables. For fixed x , the function $\varphi_x(u, v)$ is affine in (u, v) . In particular, it is both convex and concave in (u, v) .

Using the previous rewriting of g , we have

$$g(u, v) = -\sup_x \varphi_x(u, v).$$

Since each $\varphi_x(u, v)$ is affine, it is convex in (u, v) . Therefore,

$$(u, v) \mapsto \sup_x \varphi_x(u, v)$$

is convex by the pointwise maximum property. Hence its negative is concave:

$$g(u, v) = -\sup_x \varphi_x(u, v) \quad \text{is concave in } (u, v).$$

Therefore, the dual problem

$$\max_{u \geq 0, v} g(u, v)$$

is a concave maximization problem over the convex feasible set

$$\{(u, v) : u \geq 0\}.$$

Equivalently, one may minimize the convex function $-g(u, v)$ over the same feasible set. This is why the dual problem is considered a convex optimization problem.

Remark. This conclusion is important because it does not require the original primal problem to be convex. Even if f , h_i , or l_j are nonconvex in x , the dual objective $g(u, v)$ is always concave in the dual variables (u, v) . Thus duality can produce a convex optimization problem from a primal problem that may not itself be convex.

14.2 Exercises

Exercise. Solve

$$\begin{aligned} \min_{x_1, x_2} \quad & \frac{1}{2}x_1^2 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Proof. *Proof by Tao Xu, proofread by Ben Pearson*

Since

$$x_1 + x_2 = 1$$

Then

$$x_2 = 1 - x_1$$

Since $x_1, x_2 \geq 0$

$$x_1 \geq 0, \quad 1 - x_1 \geq 0$$

Hence

$$0 \leq x_1 \leq 1$$

Substituting $x_2 = 1 - x_1$ into the objective function gives

$$f(x_1) = \frac{1}{2}x_1^2 + 1 - x_1, \quad 0 \leq x_1 \leq 1$$

Then

$$f'(x_1) = x_1 - 1, \quad f''(x_1) = 1 > 0$$

Thus f is convex. The critical point satisfies

$$x_1 - 1 = 0$$

Therefore,

$$x_1^* = 1$$

Then

$$x_2^* = 1 - x_1^* = 0$$

The optimal value is

$$p^* = \frac{1}{2}(1)^2 + 0 = \frac{1}{2}$$

Hence

$$x^* = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad p^* = \frac{1}{2}$$

■

Exercise. For the constrained quadratic problem

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^\top Qx + c^\top x \\ \text{s.t.} \quad & Ax = b, x \geq 0. \end{aligned}$$

for some positive semi-definite matrix $Q \succeq 0$. Rewrite the Exercise 14.2.1 above into this form.

Proof. *Proof by Tao Xu, proofread by Ben Pearson*

$$\begin{aligned} \min_{x_1, x_2} \quad & \frac{1}{2}x_1^2 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Let

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

We need rewrite the objective function in the form

$$\frac{1}{2}x^\top Qx + c^\top x$$

Since the quadratic part is only

$$\frac{1}{2}x_1^2$$

we choose

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$x^\top Qx = (x_1 \ x_2) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2$$

Therefore,

$$\frac{1}{2}x^\top Qx = \frac{1}{2}x_1^2$$

The linear part of the objective function is

$$x_2$$

Thus we choose

$$c = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Then

$$c^\top x = (0 \ 1) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_2$$

Hence

$$\frac{1}{2}x^\top Qx + c^\top x = \frac{1}{2}x_1^2 + x_2.$$

Now consider the equality constraint

$$x_1 + x_2 = 1$$

This can be written as

$$Ax = b$$

where

$$A = \begin{pmatrix} 1 & 1 \end{pmatrix} \quad b = 1$$

The constraint

$$x_1, x_2 \geq 0$$

is exactly

$$x \geq 0$$

Therefore

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^\top Qx + c^\top x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

where

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad b = 1$$

Finally, for any

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

we have

$$z^\top Qz = z_1^2 \geq 0$$

Thus

$$Q \succeq 0$$

■

Exercise. Show that the Lagrange dual function of the quadratic optimization problem is given by

$$g(u, v) = \begin{cases} -\frac{1}{2}(c - u + A^\top v)^\top Q^+(c - u + A^\top v) - b^\top v & c - u + A^\top v \perp \text{null}(Q); \\ -\infty & \text{otherwise.} \end{cases}$$

Combine the solution in Exercise 14.2.1, what is the definition of Q^+ here?

Proof. Drafted by Hannah Nguyen, Proofread by Frank Reyes Felsner

From the earlier example, for all feasible x and all $u, v \geq 0$, the Lagrangian is given by:

$$L(x, u, v) = \frac{1}{2}x^\top Qx + \langle c - u + A^\top v, x \rangle - v^\top b$$

It also follows that to compute the dual function, we minimize $L(x, u, v)$ with respect to x :

$$\begin{aligned} \partial_x L(x, u, v) &= Qx + c - u + A^\top v \\ &= 0 \end{aligned}$$

Then, this implies

$$Qx = -(c - u + A^\top v)$$

If Q is positive definite, its inverse Q^{-1} exist and the minimizer is unique. But since Q is positive semidefinite, this falls into two cases, if Q is positive definite, its inverse Q^{-1} exists and the minimizer is unique. But since Q is positive semidefinite, Q may be singular. Therefore, we have two cases.

Let

$$y = c - u + A^\top v.$$

Then

$$L(x, u, v) = \frac{1}{2}x^\top Qx + y^\top x - b^\top v.$$

The stationarity condition is

$$Qx + y = 0,$$

or equivalently,

$$Qx = -y.$$

Since Q is symmetric positive semidefinite, the equation $Qx = -y$ has a solution if and only if

$$-y \in \text{range}(Q).$$

Because Q is symmetric,

$$\text{range}(Q) = \text{null}(Q)^\perp.$$

Therefore, the minimizer exists if and only if

$$y \perp \text{null}(Q).$$

In terms of the original variables, this condition is

$$c - u + A^\top v \perp \text{null}(Q).$$

Under this condition, one minimizer is

$$x^* = -Q^+ y = -Q^+(c - u + A^\top v),$$

where Q^+ is the Moore–Penrose pseudoinverse of Q .

Substituting x^* into the Lagrangian gives

$$\begin{aligned} g(u, v) &= \inf_x L(x, u, v) \\ &= L(x^*, u, v) \\ &= \frac{1}{2}(x^*)^\top Q x^* + y^\top x^* - b^\top v. \end{aligned}$$

Since $Q x^* = -y$, we have

$$y^\top x^* = -(x^*)^\top Q x^*.$$

Therefore,

$$\begin{aligned} g(u, v) &= \frac{1}{2}(x^*)^\top Q x^* - (x^*)^\top Q x^* - b^\top v \\ &= -\frac{1}{2}(x^*)^\top Q x^* - b^\top v. \end{aligned}$$

Using $x^* = -Q^+ y$, this becomes

$$g(u, v) = -\frac{1}{2}y^\top Q^+ y - b^\top v.$$

Substituting back $y = c - u + A^\top v$, we obtain

$$g(u, v) = -\frac{1}{2}(c - u + A^\top v)^\top Q^+(c - u + A^\top v) - b^\top v.$$

If instead

$$c - u + A^\top v \notin \text{null}(Q),$$

then the linear term has a component in a direction where the quadratic term is zero, so the Lagrangian is unbounded below. Hence,

$$g(u, v) = -\infty.$$

Thus,

$$g(u, v) = \begin{cases} -\frac{1}{2}(c - u + A^\top v)^\top Q^+(c - u + A^\top v) - b^\top v, & c - u + A^\top v \perp \text{null}(Q), \\ -\infty, & \text{otherwise.} \end{cases}$$

■

Exercise. Use 0-order definition to show that $g(u, v)$ is jointly convex.

Proof. Drafted by Eric Wong, proofread by Hannah Nguyen

Recall the Lagrangian

$$L(x, u, v) = f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j l_j(x),$$

so that

$$g(u, v) = \inf_x L(x, u, v).$$

Let (u_1, v_1) and (u_2, v_2) be any two dual feasible points, and let $t \in [0, 1]$. We must show

$$g(tu_1 + (1-t)u_2, tv_1 + (1-t)v_2) \geq t g(u_1, v_1) + (1-t) g(u_2, v_2).$$

For any fixed x , the Lagrangian is affine (hence linear) in (u, v) :

$$L(x, tu_1 + (1-t)u_2, tv_1 + (1-t)v_2) = t L(x, u_1, v_1) + (1-t) L(x, u_2, v_2).$$

Taking the infimum over x on both sides,

$$\begin{aligned} g(tu_1 + (1-t)u_2, tv_1 + (1-t)v_2) &= \inf_x [t L(x, u_1, v_1) + (1-t) L(x, u_2, v_2)] \\ &\geq t \inf_x L(x, u_1, v_1) + (1-t) \inf_x L(x, u_2, v_2) \\ &= t g(u_1, v_1) + (1-t) g(u_2, v_2), \end{aligned}$$

where the inequality uses the standard fact that $\inf(f + g) \geq \inf f + \inf g$. Since (u_1, v_1) , (u_2, v_2) , and $t \in [0, 1]$ were arbitrary, the definition of convexity is satisfied, and $g(u, v)$ is jointly convex in (u, v) . ■

Exercise. Consider the constrained optimization problem

$$\min_{x \in \mathbb{R}} x^2$$

subject to

$$x \geq 1.$$

1. Construct the Lagrangian.
2. Compute the dual function.
3. Formulate and solve the dual problem.
4. Verify that strong duality holds.

Proof. *Drafted by Charles Yang, proofread by Bradley Nakagawa*

We rewrite the constraint as

$$1 - x \leq 0.$$

The Lagrangian is

$$L(x, \lambda) = x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

The dual function is

$$g(\lambda) = \inf_x L(x, \lambda).$$

Differentiating with respect to x ,

$$\frac{\partial L}{\partial x} = 2x - \lambda.$$

Setting the derivative equal to zero gives

$$x = \frac{\lambda}{2}.$$

Substituting into the Lagrangian yields

$$g(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \lambda\left(1 - \frac{\lambda}{2}\right) = \lambda - \frac{\lambda^2}{4}.$$

Therefore the dual problem is

$$\max_{\lambda \geq 0} \left(\lambda - \frac{\lambda^2}{4} \right).$$

Differentiating,

$$g'(\lambda) = 1 - \frac{\lambda}{2}.$$

Hence

$$\lambda^* = 2.$$

The optimal dual value is

$$d^* = g(2) = 2 - 1 = 1.$$

For the primal problem, the minimum occurs at

$$x^* = 1,$$

with objective value

$$p^* = 1.$$

Thus

$$p^* = d^* = 1,$$

and the duality gap is zero.

Therefore strong duality holds. ■

14.3 More Examples

Example 14.3.1 (Proposed by Frank Reyes Felsner).

Consider the primal problem

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 = 1, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Rewrite the nonnegativity constraints as

$$-x_1 \leq 0, \quad -x_2 \leq 0.$$

Let $v \in \mathbb{R}$ be the multiplier for the equality constraint, and let $u_1, u_2 \geq 0$ be the multipliers for the inequality constraints. The Lagrangian is

$$\begin{aligned} L(x, u, v) &= x_1 + 2x_2 + v(x_1 + x_2 - 1) - u_1x_1 - u_2x_2 \\ &= (1 + v - u_1)x_1 + (2 + v - u_2)x_2 - v. \end{aligned}$$

To compute

$$g(u, v) = \inf_x L(x, u, v),$$

we need the coefficients of x_1 and x_2 to vanish. Otherwise the infimum is $-\infty$. Thus

$$1 + v - u_1 = 0, \quad 2 + v - u_2 = 0.$$

When these equations hold,

$$g(u, v) = -v.$$

Therefore the dual problem is

$$\begin{aligned} \max_{u_1, u_2, v} \quad & -v \\ \text{s.t.} \quad & 1 + v - u_1 = 0, \\ & 2 + v - u_2 = 0, \\ & u_1, u_2 \geq 0. \end{aligned}$$

Equivalently,

$$u_1 = 1 + v, \quad u_2 = 2 + v.$$

Since $u_1, u_2 \geq 0$, we need

$$v \geq -1, \quad v \geq -2.$$

Thus

$$v \geq -1.$$

The dual objective is to maximize $-v$, so we choose the smallest feasible v :

$$v^* = -1.$$

Hence

$$d^* = 1.$$

The primal optimum is obtained by choosing $x_1 = 1, x_2 = 0$, giving

$$p^* = 1.$$

Thus

$$p^* = d^* = 1.$$

Example 14.3.2 (Proposed by Ritaja Subrahmanya). The idea of the Lagrangian is deeply connected to physics. For instance, the dual problem is one which appears natural in physical contexts. The idea is that a physical system wants to be at its minimal energy state while being constrained to a region or curve. For instance, we can represent the lagrangian as

$$L(x, \lambda) = f(x) + \lambda g(x)$$

where $f(x)$ represents the energy of the system and $g(x)$ is the constraint on the boundary, so that λ penalizes violating $g(x)$. In mechanics, λ is often seen as some sort of normal or tension force.

We can further extend this notion to variational problems. Here, rather than looking working over a finite dimensional vector, we now are minimizing an entire function. The function (typically representing the energy) can be written in the form

$$J[u] = \int_a^b F(x, u(x), u'(x)) dx$$

We wish to find the function u such that minimizes $J[u]$.

Consider a small change to the function $u_\epsilon = u + \epsilon v$ where v is some arbitrary test function. The first variation of this new function thus becomes

$$\delta J[u] = \frac{d}{d\epsilon} J[u + \epsilon v] \Big|_{\epsilon=0}.$$

Furthermore, by computing the derivatives, a minimizer must satisfy $\delta J[u](v) = 0$ for all possible v .

Thus, by expanding out and solving the derivative expression, we find that

$$\delta J[u](v) = \int_a^b \left(\frac{\partial F}{\partial u} - \frac{d}{dx} \left(\frac{\partial F}{\partial u'} \right) \right) v dx.$$

Note that this must hold for all smooth test functions v and hence, we have that

$$\frac{\partial F}{\partial u} - \frac{d}{dx} \left(\frac{\partial F}{\partial u'} \right) = 0$$

The characterization of the optimality condition is called the *Euler Lagrange Equation*.

Hence, we see that the min-max nature of these problems are naturally encoded in the Euler-Lagrange equation. These equations are further useful because they work regardless of the coordinate system, hence generalizing physical laws into one equation that helps us gain insight into the physical motion of the object. Furthermore, recall from MATH 132A that the KKT conditions provide the first derivative conditions for a solution u to be optimal. Hence, we see that the Euler-Lagrange equations produce an infinite-dimensional analogue to the KKT conditions in the variational context.

The main takeaway from this example is to understand how the Lagrangian method has a physical meaning. Many equations of mathematical physics are fundamentally saddle-point conditions where the behavior of the physical system can be modeled as a function with respect to some constraints. In this sense, we can see how our mathematical theory follows physical principles by searching for the "path of least resistance" as motivated by Hamilton's principle in classical mechanics.

Example 14.3.3 (Proposed by Cade Camacho: Langrange multipliers as soft constraints). In many situations, it is not necessary to strictly enforce constraints. Instead of requiring exact feasibility, one may allow penalized violations in the objective function. This leads to an idea of soft constraints, where conditions are encouraged rather than enforced. Consider a simple problem where we want to minimize a function while staying close to a point x_0 :

$$\min_x f(x) \quad \text{s.t. } x \approx x_0$$

A rigid interpretation of this constraint is the equality constraint ($x = x_0$). The corresponding constrained problem is:

$$\min_x f(x) \quad \text{s.t. } x = x_0,$$

whose solution ($x^* = x_0$, $p^* = f(x_0)$) is trivial and fixed, since equality is enforced. Instead of requiring the constraint, we can penalize deviations, effectively relaxing it. This leads to the unconstrained problem

$$\min_x f(x) + \lambda \|x - x_0\|^2$$

where $\lambda \geq 0$ controls how strongly deviations from x_0 are penalized.

Interpretation: The parameter λ determines the trade-off between minimizing the objective and staying close to x_0 :

- If $\lambda = 0$, the constraint is entirely ignored.
- If $0 < \lambda < \infty$, deviations are possible but penalized appropriately.
- As $\lambda \rightarrow \infty$, the solution is forced towards x_0 .

Lagrangian Perspective: Consider the exact constrained problem with $x, x_0 \in \mathbb{R}^d$:

$$\min_x f(x) \quad \text{s.t. } x = x_0.$$

Its Lagrangian is:

$$L(x, v) = f(x) + v^\top (x - x_0),$$

where $v \in \mathbb{R}^d$ is the Lagrangian multiplier associated with the equality constraint. The dual function is

$$g(v) = \inf_x (f(x) + v^\top (x - x_0))$$

This shows that the dual variable v acts as a linear penalty on violations of the constraint. Rather than demanding $x = x_0$ directly, the Lagrangian allows violations but assigns them a cost determined by v .

Dual variables can be interpreted as prices on constraints, while penalty methods replace hard constraints with costs for violation. This provides a useful bridge between exact constrained optimization and regularized optimization problems.

Example 14.3.4 (Proposed by Charles Yang, Dual Variables as Shadow Prices). One of the main benefits of Lagrangian duality is that the dual variables often have useful interpretations. In resource-allocation problems, a dual variable can be understood as a *shadow price*: it measures how much the optimal value changes when a constraint is slightly tightened or relaxed.

Consider the convex optimization problem

$$p^* = \inf_{x_1, x_2} \left\{ x_1^2 + x_2^2 : x_1 + x_2 \geq 10 \right\}.$$

We may interpret x_1 and x_2 as the production quantities of two products. The objective

$$x_1^2 + x_2^2$$

represents a cost with increasing marginal cost, while the constraint

$$x_1 + x_2 \geq 10$$

requires total production to meet demand.

To form the Lagrangian, rewrite the constraint as

$$10 - x_1 - x_2 \leq 0.$$

Then the Lagrangian is

$$L(x_1, x_2, \lambda) = x_1^2 + x_2^2 + \lambda(10 - x_1 - x_2), \quad \lambda \geq 0.$$

The dual function is

$$g(\lambda) = \inf_{x_1, x_2} L(x_1, x_2, \lambda).$$

For a fixed $\lambda \geq 0$, we minimize the Lagrangian over x_1 and x_2 . Differentiating gives

$$\frac{\partial L}{\partial x_1} = 2x_1 - \lambda, \quad \frac{\partial L}{\partial x_2} = 2x_2 - \lambda.$$

Setting these equal to zero gives

$$x_1 = x_2 = \frac{\lambda}{2},$$

and substituting this back into the Lagrangian yields

$$g(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \left(\frac{\lambda}{2}\right)^2 + \lambda\left(10 - \frac{\lambda}{2} - \frac{\lambda}{2}\right).$$

Therefore,

$$g(\lambda) = 10\lambda - \frac{\lambda^2}{2}.$$

The dual problem is

$$d^* = \sup_{\lambda \geq 0} \left(10\lambda - \frac{\lambda^2}{2}\right).$$

Since

$$g'(\lambda) = 10 - \lambda,$$

the maximizer is

$$\lambda^* = 10.$$

The optimal dual value is

$$d^* = g(10) = 100 - \frac{100}{2} = 50.$$

For the primal problem, the objective is symmetric in x_1 and x_2 . Since the cost increases as production increases, the demand constraint is active at the optimum:

$$x_1 + x_2 = 10.$$

The symmetric minimizer is

$$x_1^* = x_2^* = 5.$$

Thus

$$p^* = 5^2 + 5^2 = 50.$$

Hence

$$p^* = d^* = 50,$$

so the duality gap is zero.

Now consider the interpretation of the optimal dual variable. Suppose the demand requirement is changed from

$$10$$

to

$$10 + \varepsilon.$$

Duality theory suggests that, for small changes in the constraint, the optimal value changes approximately by

$$\lambda^* \varepsilon.$$

Since

$$\lambda^* = 10,$$

one additional unit of required production increases the minimum cost by approximately 10 units. For example, if the demand increases from 10 to 11, then the new symmetric solution is

$$x_1 = x_2 = 5.5.$$

The new objective value is

$$5.5^2 + 5.5^2 = 60.5.$$

The actual increase in cost is

$$60.5 - 50 = 10.5.$$

This is close to the first-order prediction

$$\lambda^* \cdot 1 = 10.$$

The discrepancy occurs because the shadow price gives a local, first-order approximation. This example shows that dual variables are not merely algebraic objects. They can also measure the marginal value of a constraint, which is why they are often called shadow prices.

15 Lecture Notes 15 on Duality in the Space of Probability measures

Prashast Singh summarizes: We introduce duality for optimisation over probability measures through the optimal transport problem. First we review Lagrangian duality where we replace hard constraints with duality functions, and show how dual variables indeed enforce feasibility. In the optimal transport problem, we are, in essence, tasked to seek a joint distribution with marginals prescribed that minimise transportation cost, which is interpreted as a moving mass between distributions. Here, the direct gradient descent is unsuitable because updates can leave the probability simplex. We thus derive the dual formulation by relaxing the marginal constraints with Lagrange multipliers. The resulting dual optimises functions instead of probability measures thereby yielding a convex and often easier problem.

Scribe Team: Ben Pearson, Bradley Nakagawa

15.1 Lecture Notes

15.1.1 Second View of Lagrangian Duality

Recall the general constrained minimization problem:

$$\begin{cases} \min_x f(x) \\ \text{s.t. } h_i(x) \leq 0, i = 1, \dots, m, \\ \ell_j(x) = 0, j = 1, \dots, r. \end{cases}$$

A useful alternative perspective encodes the constraints via an indicator function. Let C denote the feasible set and define

$$I_C(x) = \begin{cases} 0 & x \in C, \\ +\infty & x \notin C. \end{cases}$$

Then

$$\min_{x \in C} f(x) = \min_x [f(x) + I_C(x)] = \min_x \begin{cases} f(x) & x \in C, \\ +\infty & x \notin C, \end{cases}$$

which simply equals $\min_{x \in C} f(x)$. The indicator I_C acts as an exact penalty: feasible points incur no penalty, and infeasible points are assigned $+\infty$.

The indicator is hard to optimize directly. The Lagrangian replaces I_C with a soft penalty that is linear in the multipliers. For a fixed parameter u , consider the linear penalty $P(x) = u^\top x$. This motivates minimizing

$$\min_x [f(x) + \langle u, x \rangle],$$

which potentially allows the minimizer $\bar{x}$ to lie outside C .

The full Lagrangian encodes both inequality and equality constraints. For each fixed x , define

$$R_x(u, v) := u h(x) + v \ell(x),$$

which is linear in (u, v) and belongs to a family indexed by x . Then

$$\max_{u \geq 0, v} R_x(u, v) = \begin{cases} 0 & x \in C \quad [h(x) \leq 0, \ell(x) = 0], \\ +\infty & x \notin C \quad [h(x) > 0 \text{ or } \ell(x) \neq 0]. \end{cases}$$

(To verify: if $h(x) > 0$, take $u \rightarrow +\infty$ with $v = 0$ so that $u h(x) \rightarrow +\infty$ and $v \ell(x) = 0$. Thus, $R_x(u, v) = u h(x) + v \ell(x) \rightarrow +\infty$. If $\ell(x) \neq 0$, take $v = +\infty \cdot \text{sign}(\ell(x))$ with $u = 0$ so that $v \ell(x) \rightarrow +\infty$. Thus, $R_x(u, v) = u h(x) + v \ell(x) \rightarrow +\infty$.)

Remark. This recovers the indicator: $\max_{u \geq 0, v} R_x(u, v) = I_C(x)$. The key insight is that the maximization over (u, v) acts as an exact feasibility check. For the minimization problem, we want to use $+\infty$ as the penalty (infeasible points are excluded); for a maximization problem, the penalty would instead be $-\infty$.

15.1.2 The Optimal Transport Problem

Let Ω be a domain and let $\mu, \nu \in \mathcal{P}(\Omega)$ be two probability measures. The primal optimal transport problem seeks the best joint distribution $Q(x, y)$ solving

$$\min_{Q \in \Gamma(\mu, \nu)} \int_{X \times Y} |x - y|^2 dQ(x, y),$$

where the *transport plan set* (couplings) is

$$\Gamma(\mu, \nu) := \left\{ Q \left| \int_{A \times Y} Q(x, y) dx dy = \int_A \mu(x) dx \text{ for any Borel set } A, \quad \int_{X \times B} Q(x, y) dx dy = \int_B \nu(y) dy \text{ for any Borel set } B \right\}.$$

The marginal conditions say μ is the x -marginal of Q and ν is the y -marginal of Q . The quantity $\int |x - y|^2 dQ$ is the squared transport distance times mass, i.e., energy/action.

Remark. This formulation is known as the Kantorovich Problem and earned Leonid Kantorovich the 1975 Nobel Prize in Economics.

- The optimization is over the *space of probability measures*, which is more complex than a Euclidean or Hilbert space — it lives on a curved Riemannian manifold.
- The objective $C(Q) = \int |x - y|^2 dQ(x, y)$ is linear (hence convex) in Q :

$$C((1-t)Q_1 + tQ_2) = (1-t)C(Q_1) + tC(Q_2), \quad C : \mathcal{P}(\Omega \times \Omega) \rightarrow [0, +\infty).$$
- As shown below, the constraint set $\Gamma(\mu, \nu)$ is also convex. Therefore, the primal OT problem is a convex optimization problem.

Theorem 15.1 (Convexity of the Transport Plan Set). For any two probability measures $\mu, \nu \in \mathcal{P}(\Omega)$, the transport plan set $\Gamma(\mu, \nu)$ is a convex set.

Proof.

Let $Q_1, Q_2 \in \Gamma(\mu, \nu)$ and $t \in [0, 1]$. We define the convex combination:

$$Q_t := (1-t)Q_1 + tQ_2.$$

We show $Q_t \in \Gamma(\mu, \nu)$ by verifying that it satisfies both marginal conditions.

Using the linearity of integration, the x -marginal condition evaluates to:

$$\begin{aligned} \int_{A \times Y} Q_t(x, y) dx dy &= \int_{A \times Y} [(1-t)Q_1(x, y) + tQ_2(x, y)] dx dy \\ &= (1-t) \int_{A \times Y} Q_1(x, y) dx dy + t \int_{A \times Y} Q_2(x, y) dx dy. \end{aligned}$$

Since Q_1 and Q_2 are both valid transport plans in $\Gamma(\mu, \nu)$, their x -marginals are equal to $\mu(x)$:

$$\int_{A \times Y} Q_t(x, y) dx dy = (1-t) \int_A \mu(x) dx + t \int_A \mu(x) dx = \int_A \mu(x) dx.$$

By an identical calculation over the region $X \times B$, the y -marginal condition satisfies:

$$\int_{X \times B} Q_t(x, y) dx dy = \int_B \nu(y) dy.$$

Because the convex combination Q_t preserves both prescribed marginal distributions, $Q_t \in \Gamma(\mu, \nu)$. Thus, the set $\Gamma(\mu, \nu)$ is convex. ■

A natural question is: *why not apply gradient descent (GD) directly on Q ?* The update rule is:

$$Q^{k+1} = Q^k - \eta \nabla C(Q^k) = \begin{bmatrix} Q_{11} & \cdots & Q_{1n} \\ & \ddots & \\ Q_{n1} & \cdots & Q_{nn} \end{bmatrix} - \eta \begin{bmatrix} \Delta Q_{11} & \cdots & \Delta Q_{1n} \\ & \ddots & \\ \Delta Q_{n1} & \cdots & \Delta Q_{nn} \end{bmatrix}.$$

The problem is that the update will escape the probability simplex: after subtracting the gradient, entries Q_{ij} can become negative, violating the nonnegativity constraint. One remedy is Projected Gradient Descent (project back onto the simplex after each step), but this suffers from slow convergence. Instead, we turn to the dual problem.

15.1.3 Dual Problem

We derive the dual by relaxing the marginal constraints using Lagrange multipliers. We write the primal equivalently as

$$\min_{\substack{Q \in \mathcal{M}_+ \\ Q(x, y) \geq 0}} \int |x - y|^2 dQ(x, y) + \text{penalty}(Q),$$

where $\mathcal{M}_+$ denotes nonnegative Radon measures and

$$\text{penalty}(Q) = \begin{cases} 0 & Q \in \Gamma(\mu, \nu) \quad (Q \text{ has } \mu, \nu \text{ as marginals}), \\ +\infty & Q \notin \Gamma(\mu, \nu). \end{cases}$$

Remark. A nonnegative Radon measure is a function that takes a set as an input and outputs a nonnegative real number, or $+\infty$. It effectively assigns a value to every region of a domain and adds up the values associated with each region included in the input. It also has a few special properties. For example, every point in the domain must have at least one neighborhood where the measure of that neighborhood is finite. Also, the measure of any open set is equal to the supremum of the measures of the compact subsets of the open set.

Using the Lagrangian framework, the marginal penalty can be written as a supremum over dual functions $f, g \in C^1(\Omega)$:

$$\text{penalty}(Q) = \sup_{\substack{f \in C^1(\Omega) \\ g \in C^1(\Omega)}} \left[\int_X f(x) d\mu(x) + \int_Y g(y) d\nu(y) - \int_{X \times Y} (f(x) + g(y)) dQ(x, y) \right].$$

Remark. For simplicity of notation, let $\Phi(Q; f, g) := \int_X f \, d\mu + \int_Y g \, d\nu - \int_{X \times Y} (f(x) + g(y)) \, dQ$.

- If $Q \in \Gamma(\mu, \nu)$: Since Q has marginals μ and ν ,

$$\int_{X \times Y} (f(x) + g(y)) \, dQ = \int_X f \, d\mu + \int_Y g \, d\nu,$$

so $\Phi(Q; f, g) = 0$ for all f, g . Hence $\sup_{f, g} \Phi = 0 = \text{penalty}$.

- If $Q \notin \Gamma(\mu, \nu)$, $Q(x, y) \geq 0$: If the x -marginal of Q is not μ , choose $g \equiv 0$ and $f \not\equiv 0$ so that $\int f \, d\mu - \int f \, dQ = +\infty$ (taking $f(x) < 0$ forces $Q(x, y) = +\infty$ when $\mu = 0$, $\nu = +\infty \cdot \text{sign}(\ell(x))$). Hence $\sup_{f, g} \Phi = +\infty = \text{penalty}$.

All ideas here follow directly from Lagrangian methods.

Substituting the penalty expression gives the min-max formulation:

$$\begin{aligned} \min_{Q \in \Gamma(\mu, \nu)} \int |x - y|^2 \, dQ &= \min_{Q(x, y) \geq 0} \int |x - y|^2 \, dQ + \max_{f, g} \left[\int f(x) \, d\mu + \int g(y) \, d\nu - \int (f(x) + g(y)) \, dQ \right] \\ &= \min_{Q(x, y) \geq 0} \max_{f, g} \left[\int f(x) \, d\mu + \int g(y) \, d\nu + \int (|x - y|^2 - f(x) - g(y)) \, dQ \right] \\ &\geq \max_{f, g} \min_{Q(x, y) \geq 0} \left[\int f(x) \, d\mu + \int g(y) \, d\nu + \int (|x - y|^2 - f(x) - g(y)) \, dQ \right]. \end{aligned}$$

The last inequality is weak duality (min max $\geq$ max min). The inner minimization over $Q(x, y) \geq 0$ evaluates to

$$\min_{Q(x, y) \geq 0} \int [|x - y|^2 - f(x) - g(y)] \, dQ = \begin{cases} 0 & \text{feasible set: } f(x) + g(y) \leq |x - y|^2 \, \forall x, y, \\ -\infty & \text{nonfeasible set: } f(x) + g(y) > |x - y|^2 \text{ for some } x, y. \end{cases}$$

The dual of the optimal transport problem is

$$\max_{\substack{f, g \\ f(x) + g(y) \leq |x - y|^2}} \int_X f(x) \, d\mu(x) + \int_Y g(y) \, d\nu(y).$$

Theorem 15.2 (Convexity of the Dual Feasible Set). The dual constraint set of the optimal transport problem is a convex set.

Proof. Let (f_1, g_1) and (f_2, g_2) be two feasible dual pairs s.t.

$$f_i(x) + g_i(y) \leq |x - y|^2 \quad \text{for all } x, y \in \Omega, \quad i = 1, 2.$$

Let $t \in [0, 1]$ and define the convex combination

$$f_t := (1 - t)f_1 + tf_2, \quad g_t := (1 - t)g_1 + tg_2.$$

We must show that (f_t, g_t) is also feasible, i.e., that $f_t(x) + g_t(y) \leq |x - y|^2$ for all $x, y \in \Omega$.

Fix any $x, y \in \Omega$. Multiplying the feasibility inequality for (f_1, g_1) by $(1 - t) \geq 0$ and the inequality for (f_2, g_2) by $t \geq 0$, then adding, gives

$$f_t(x) + g_t(y) = (1 - t)(f_1(x) + g_1(y)) + t(f_2(x) + g_2(y)) \leq (1 - t)|x - y|^2 + t|x - y|^2 = |x - y|^2.$$

Since $x, y \in \Omega$ were arbitrary, (f_t, g_t) satisfies the dual constraint, so the dual constraint set is convex. $\blacksquare$

Remark. The dual problem optimizes over functions f and g , not over measures. Functions live in a Hilbert space (flat geometry), which is an easier optimization landscape than the space of probability measures.

In the discrete setting, $\mu = (u_1, \dots, u_m)$ and $\nu = (v_1, \dots, v_m)$ lie in the probability simplex, $Q \in \mathbb{R}^{m \times m}$, and $f = (f_1, \dots, f_m) \in \mathbb{R}^m$, $g = (g_1, \dots, g_m) \in \mathbb{R}^m$. Because of this, Q takes up much more storage than f and g combined when m is large.

Remark. One potential benefit of finding the dual problem is that it switches the amount of variables and constraints you have. For example, our primal in the previous problem had m^2 variables and only $2m$ constraints, making it difficult to solve. The dual problem, on the other hand, had $2m$ variables to solve with m^2 constraints, which makes it much easier to find a solution if there is one.

15.2 Exercises

Exercise (Proposed by Ben Thomas). Let $\mu = (\frac{1}{2}, \frac{1}{2})$, $\nu = (\frac{1}{2}, \frac{1}{2})$, and cost matrix $C_{ij} = |i - j|^2$ for $i, j \in \{1, 2\}$. Find the primal optimal plan $Q^* \in \Gamma(\mu, \nu)$ and dual potentials $f, g \in \mathbb{R}^2$ achieving the same objective value, thereby confirming that the duality gap is zero.

Proof. Drafted by Ben Thomas, Proofread by

Primal. The cost matrix is $C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so the objective $\sum_{ij} C_{ij} Q_{ij} = Q_{12} + Q_{21}$ is minimized by placing all mass on the diagonal:

$$Q^* = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \quad W_2^2(\mu, \nu) = 0.$$

Dual. Take $f = g = (0, 0)$. The constraints $f_i + g_j \leq C_{ij}$ reduce to $0 \leq C_{ij}$, which holds for all i, j . The dual objective equals

$$\frac{1}{2}f_1 + \frac{1}{2}f_2 + \frac{1}{2}g_1 + \frac{1}{2}g_2 = 0 = W_2^2(\mu, \nu).$$

Since the primal and dual values agree, the duality gap is zero. ■

15.3 More Examples

Example 15.3.1 (Proposed by Jake Jurek). **Monte Carlo optimal transport between empirical measures.** This example computes an optimal transport cost directly from *samples*, tying the Kantorovich problem of this lecture to the empirical measure $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$, a central object in statistics. We take the squared cost $|x - y|^2$, so the optimal value is the squared Wasserstein-2 distance W_2^2 .

Setup. Let $\mu = N(0, 1)$ and $\nu = N(3, 1)$. Draw i.i.d. samples $X_1, \dots, X_n \sim \mu$ and $Y_1, \dots, Y_n \sim \nu$, and form the empirical measures

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}, \quad \nu_n = \frac{1}{n} \sum_{i=1}^n \delta_{Y_i}.$$

The primal problem solves by sorting. The discrete Kantorovich problem

$$W_2^2(\mu_n, \nu_n) = \min_{Q \in \Gamma(\mu_n, \nu_n)} \sum_{i,j} |X_i - Y_j|^2 Q_{ij}$$

is, in one dimension with a convex cost, solved by the *monotone (sorting)* coupling: pairing the k -th smallest source point with the k -th smallest target point is optimal. Writing $X_{(1)} \leq \dots \leq X_{(n)}$ and $Y_{(1)} \leq \dots \leq Y_{(n)}$ for the order statistics,

$$W_2^2(\mu_n, \nu_n) = \frac{1}{n} \sum_{i=1}^n |X_{(i)} - Y_{(i)}|^2.$$

This is the discrete instance of the quantile formula $W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(q) - F_\nu^{-1}(q)|^2 dq$. The sorting solution sidesteps exactly the difficulty noted in Subsection 15.1.3: a gradient step on the coupling matrix Q leaves the probability simplex, whereas sorting stays on it for free. Figure 24 shows one draw.

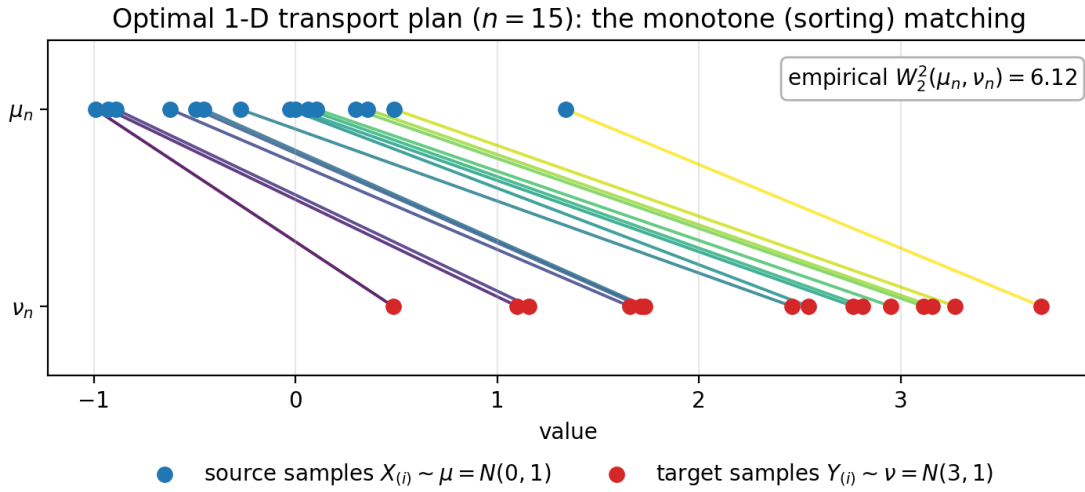


Figure 24: The optimal 1-D transport plan for one draw of size $n = 15$. The optimal coupling is the monotone sorting matching; the segments do not cross. The empirical cost shown is $W_2^2(\mu_n, \nu_n) = \frac{1}{n} \sum_i |X_{(i)} - Y_{(i)}|^2$.

Closed-form ground truth. For Gaussians, $W_2^2(N(m_1, \sigma_1^2), N(m_2, \sigma_2^2)) = (m_1 - m_2)^2 + (\sigma_1 - \sigma_2)^2$, so here

$$W_2^2(\mu, \nu) = (0 - 3)^2 + (1 - 1)^2 = 9.$$

Convergence. As $n \rightarrow \infty$ the empirical quantile functions converge to the true quantile functions, hence $W_2^2(\mu_n, \nu_n) \rightarrow W_2^2(\mu, \nu) = 9$. Figure 25 runs this experiment over 60 random seeds. The left panel shows the empirical cost converging to 9 with a percentile band that narrows as n grows; the right panel shows the fluctuation $|W_2^2(\mu_n, \nu_n) - 9|$ decaying on a log-log scale at the familiar Monte Carlo rate $\propto n^{-1/2}$ (fitted slope ≈ -0.50).

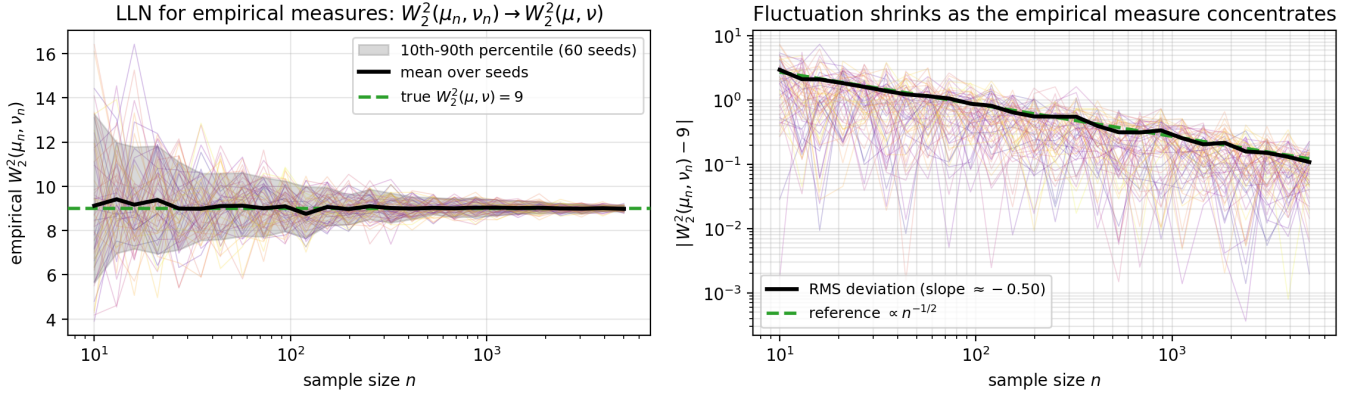


Figure 25: $\mu = N(0, 1)$, $\nu = N(3, 1)$, squared cost $|x - y|^2$. *Left*: the empirical transport cost $W_2^2(\mu_n, \nu_n)$ over 60 seeds (faint paths), their mean (bold), and the 10th–90th percentile band, converging to the true value 9 (dashed). *Right*: the deviation from 9 shrinks at the Monte Carlo rate $n^{-1/2}$ (fitted slope ≈ -0.50).

Example 15.3.2 (Proposed by Prashast Singh). **Why Gradient Descent Does Not Preserve Feasibility.**

Consider the discrete optimal transport problem with cost matrix

$$C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Let the current transport plan be

$$Q = \begin{pmatrix} 0.5 & 0.5 \\ 0 & 0 \end{pmatrix}.$$

This is a valid coupling since all entries are nonnegative and

$$\sum_{i,j} Q_{ij} = 1.$$

For the transport objective

$$J(Q) = \sum_{i,j} C_{ij} Q_{ij},$$

the gradient is simply

$$\nabla J(Q) = C.$$

Taking a gradient descent step with learning rate $\eta = 0.4$ gives

$$Q^+ = Q - \eta \nabla J(Q) = \begin{pmatrix} 0.5 & 0.5 \\ 0 & 0 \end{pmatrix} - 0.4 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0.1 & -0.3 \\ -1.2 & -1.6 \end{pmatrix}.$$

Several entries of Q^+ are now negative, so Q^+ is no longer a probability measure and therefore is not a feasible transport plan.

The issue is not specific to this example. The feasible set of transport plans is constrained by

$$Q_{ij} \geq 0,$$

together with row-sum and column-sum conditions that enforce the prescribed source and target marginals. A standard gradient step does not automatically preserve these constraints.

Example 15.3.3 (Proposed by Prashast Singh). **Discrete Earth-Mover Interpretation.**

Optimal transport can be viewed as the problem of moving probability mass between locations. Consider the discrete space

$$X = \{1, 2, 3\},$$

with transportation cost given by the squared distance

$$c(i, j) = |i - j|^2.$$

Let

$$\mu = (1, 0, 0), \quad \nu = (0, 1, 0),$$

so that all mass initially sits at location 1, while the target distribution places all mass at location 2.

The only feasible transport plan moves one unit of mass from location 1 to location 2. Therefore,

$$W_2^2(\mu, \nu) = |1 - 2|^2 = 1.$$

Now consider the target distribution

$$\nu' = (0, 0, 1).$$

Again, there is only one feasible transport plan: all mass must move from location 1 to location 3. The transport cost is

$$W_2^2(\mu, \nu') = |1 - 3|^2 = 4.$$

Although both target distributions differ from μ by the same amount of probability mass, the second transport problem is more expensive because the mass must travel farther.

It is thus noteworthy that optimal transport measures not only *how much* probability mass must be moved, but also *how far* it must move. This geometric feature distinguishes optimal transport from divergences that compare distributions pointwise without accounting for the geometry of the space.

Remark (A dual certificate). The sorted matching can be certified by the Kantorovich dual of this lecture: find potentials $f = (f_1, \dots, f_n)$, $g = (g_1, \dots, g_n)$ maximizing $\frac{1}{n} \sum_i f_i + \frac{1}{n} \sum_j g_j$ subject to $f_i + g_j \leq |X_i - Y_j|^2$ for all i, j . By weak duality this lower-bounds the primal cost, and at optimality complementary slackness forces $f_i + g_{\sigma(i)} = |X_i - Y_{\sigma(i)}|^2$ on the optimal (sorted) pairs σ , so the dual value equals $\frac{1}{n} \sum_i |X_{(i)} - Y_{(i)}|^2$ and the duality gap is zero. Thus the primal sorted matching and the dual potentials agree, illustrating strong duality on a concrete finite instance.

Example 15.3.4 (Proposed by Pablo Gómez). A generalization of the Kantorovich Dual Problem

Let X, Y be separable completely metrizable topological spaces (that is, they are each homeomorphic to some complete metric space that possess some countable dense subset), let μ be a probability measure on X , and let ν be a probability measure on Y . Let $c : X \times Y \mapsto [0, +\infty]$ be a lower semi-continuous cost function (that is to say, $c(x, y) \leq \liminf_{d((x, y), (x', y')) \rightarrow 0} c(x', y')$ for every $(x, y) \in X \times Y$). Whenever π is a probability measure on $X \times Y$ and $(\phi, \psi) \in L^1(d\mu) \times L^1(d\nu)$, define

$$I[\pi] = \int_{X \times Y} c(x, y) d\pi(x, y) \quad \text{and} \quad J(\phi, \psi) = \int_X \phi(x) d\mu(x) + \int_Y \psi(y) d\nu(y).$$

If we denote by $\Pi(\mu, \nu)$ the set of Borel probability measures on $X \times Y$ such that for all measurable $A \subset X$ and $B \subset Y$ we have

$$\pi[A \times Y] = \mu(A) \quad , \quad \pi[X \times B] = \nu(B)$$

and by Φ_c the set of measurable functions $(\phi, \psi) \in L^1(d\mu) \times L^1(d\nu)$ which are cost-competitive (i.e. $\phi(x) + \psi(y) \leq c(x, y)$ for $d\mu$ -a.e. in X and $d\nu$ -a.e. in Y), then

$$\inf_{\Pi(\mu, \nu)} I[\pi] = \sup_{\Phi_c} J(\phi, \psi).$$

Furthermore, this infimum is attained by some $\pi \in \Pi(\mu, \nu)$ and we can restrict ϕ, ψ to be continuous and bounded.

We ought to care about this formulation because it generalizes our original problem in a very appealing way. It subjects the cost function to looser conditions, tells us that our L^1 functions ϕ, ψ can be taken to be "nice," and allows us to work over more abstract spaces that possess (only) the necessary properties for our results to hold.

Let's also consider what the consequences are of this result for applications. A classical one runs as follows: suppose Alice are the owner of a chain of coffee shops, and wants to transfer some quantity of coffee from your warehouses to your shops. Alice can hire trucks to do this for you, and they will charge $c(x, y)$ to transport a unit of coffee from warehouse x to store y . The amount of coffee at each warehouse and the amount of coffee each store should receive are fixed.

Then sufficiently clever mathematicians (such as the students in this class) can propose a beneficial arrangement, saying "I'll handle all of this! I'll charge you $\phi(x)$ to load a unit of coffee at x and $\psi(y)$ to unload it at y in such a way that $\phi(x) + \psi(y) \leq c(x, y)$. In fact, to ensure that this is the case I am even willing to give financial *compensation* for some places in the form of negative prices." Naturally, Alice ought to accept such a deal.

Our intuitions tell us that we might be noticeably limiting out winnings from such an arrangement, but this result affirms the opposite: in fact, we can find a way to charge an amount arbitrarily close to what Alice was going to pay originally and still be a preferable alternative to the original plan.

Remark. In the case that $X = Y$ (still separable completely metrizable) and $c(x, y)$ is a metric on $X \times X$ (possibly different from the one that induces the topology on X), we actually have another result as well:

Theorem 15.3 (Kantorovich-Rubenstein Theorem). If

$$\mathcal{T}_d(\mu, \nu) = \inf_{\pi \in \Pi(\mu \times \nu)} \int_{X \times X} d(x, y) d\pi(x, y)$$

is the cost of optimal transportation for the cost function $c(x, y) = d(x, y)$ and $\text{Lip}(X)$ is the space of Lipschitz functions on X with the norm

$$\|\phi\|_{\text{Lip}} := \sup_{x \neq y} \frac{|\phi(x) - \phi(y)|}{d(x, y)},$$

then

$$\mathcal{T}_d(\mu, \nu) = \sup \left\{ \int_X \phi d(\mu - \nu) \mid \phi \in L^1(d|\mu - \nu|); \quad \|\phi\|_{\text{Lip}} \leq 1 \right\}.$$

Moreover, this supremum remains the same if ϕ is taken to be bounded.

This tells us that when the cost function is a metric, the total cost only depends on the difference $\mu - \nu$. Indeed, in this case the Kantorovich OT problem is equivalent to

$$\inf \{ I[\pi] \mid \pi[A \times X] - \pi[X \times A] = (\mu - \nu)[A] \}.$$

For a general cost function, these problems are *not* equivalent (this is a very relaxed version).

References

- [Bec17] Amir Beck. *First-order methods in optimization*, volume 25 of *MOS-SIAM Series on Optimization*. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA; Mathematical Optimization Society, Philadelphia, PA, 2017.
- [BV04] Stephen Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge University Press, Cambridge, 2004.
- [San17] Filippo Santambrogio. {Euclidean, metric, and Wasserstein} gradient flows: an overview. *Bull. Math. Sci.*, 7(1):87–154, 2017.
- [Vil09] C. Villani. *Optimal transport: old and new*, volume 338 of *Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. 2009.